

Models of Market Liquidity

Applications to Traditional Markets and Automated Market Makers



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Abstract

The thesis studies algorithmic trading problems for liquidity takers (LTs) and liquidity providers (LPs) in traditional electronic markets and in decentralised trading platforms that use Automated Market Makers (AMMs). AMMs are a new paradigm in the design of trading venues and are based on liquidity pooling. Chapter 1 introduces the literature on algorithmic trading in traditional markets. Chapter 2 solves a multi-asset optimal trading problem in LOBs where the agent adopts a CARA utility and the prices follow multivariate Ornstein-Uhlenbeck dynamics. These dynamics account for the presence of cointegration between the assets' prices. We showcase the use of the model to enhance the performance of portfolio execution programmes and to build statistical arbitrage strategies using data from the Equity and FX markets. Chapter 3 uses stochastic control tools to solve a general differential Riccati equation (DRE) with an indefinite quadratic coefficient. We use the result to provide solutions to two algorithmic portfolio trading problems; one is optimal trading in LOBs with Bayesian learning of the drift using signals and past prices, the second is multi-asset market making in OTC markets with three important features: (i) dynamics that incorporate cointegration and predictive signals, (ii) external hedging and client tiering, and (iii) computationally efficient methods to obtain quotes for large portfolios. Chapter 4 introduces AMMs and describes how liquidity takers and providers interact in liquidity pools. Chapter 5 studies optimal execution in a constant function market maker (CFMM), a popular type of AMMs. In particular, we use the convexity of the trading function as an approximation for the execution costs incurred by LTs. Chapter 6 introduces a new comprehensive metric of predictable loss (PL) for LPs in CFMMs. PL compares the value of the LP's holdings in the liquidity pool with that of a self-financing portfolio that replicates the LP's holdings and invests in a risk-free account. We show that the losses stem from two sources: the convexity cost, which depends on liquidity taking activity and the convexity of the pool's trading function; the opportunity cost, which is due to locking the LP's assets in the pool. Finally, Chapter 7 derives continuous-time dynamics for the wealth of LPs and provides a closed-form dynamic liquidity provision strategy in constant product market makers (CPMMs) with concentrated liquidity. The strategy dynamically adjusts the liquidity range around the exchange rate as a function of market trend, volatility, and liquidity taking activity in the pool. We prove that the profitability of liquidity

provision depends on the tradeoff between PL and fee income. Chapter 5, 6, and 7 use Uniswap v3 data to showcase the superior out-of-sample performance of our strategies.

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Multi-Asset Models for Liquidity Taking and Liquidity Provision in Traditional Markets

Chapter 1

Introduction

1.1 Optimal execution

In many traditional markets, the demand and supply for assets are cleared through matching mechanisms in limits order books (LOBs). In LOBs, prices are discretised in price increments called ticks. Traders post limit orders that indicate at which price they are ready to buy or sell a certain quantity of assets. Their orders sit in the LOB until they are either (partially or fully) filled against a market order, or cancelled. A market order is matched against limit orders at the best available prices until it is completely filled.

Investors and market operators such as institutional traders, mutual funds, and brokers, regularly interact with LOBs to buy or sell large amounts of assets. If the quantity of an order represents a significant portion of the LOB's trading volume, then executing all assets in one trade is costly or impossible. As a result, the optimal split of large blocks of assets (meta orders) into smaller trades (child orders) has become a classical problem in the quantitative finance literature and in practice. Financial agents wish to minimise adverse price movements which are a consequence of their trading activity by trading slowly, while mitigating market price risk by trading rapidly.

The optimal execution literature formulates models to help agents control their overall trading costs. The results of these models are trading schedules that optimally balance between trading slowly to minimise execution costs, measured as the difference between a reference price and the average price obtained for a trade, and trading rapidly to minimise inventory risk. The first approaches are by [Bertsimas and Lo \(1998\)](#), [Almgren and Chriss \(1999\)](#), and [Almgren and Chriss \(2001\)](#), and various extensions and models for the optimal execution of large orders have been proposed in the last two decades. At present, almost all practitioners slice their large orders into child orders according to optimised trading schedules.

The first Almgren-Chriss model is a discrete-time model where the agent posts market orders to maximise a mean-variance objective function. Many extensions of this seminal model have been proposed. Some works extend the modeling framework. [Forsyth et al. \(2012\)](#) examine the use of quadratic variation rather than variance in the objective function, [Schied and Schöneborny \(2009\)](#) use stochastic optimal control tools to characterise and find optimal strategies for a Von Neumann–Morgenstern investor, and [Guéant \(2015\)](#) provides results for optimal liquidation within a Von Neumann–Morgenstern expected utility framework with general market impact functions and derives subsequent results for block trade pricing.

Other works extend the assumptions on model parameters. [Almgren \(2003\)](#) studies the case of random execution costs, [Almgren \(2009, 2012\)](#) consider stochastic liquidity and volatility, [Lehalle \(2008\)](#) discusses the statistical aspects of the variability of estimators

of the main exogenous variables such as volumes or volatilities in the optimisation phase, [Cartea and Jaimungal \(2016b\)](#) provide a closed-form strategy incorporating order flows from all agents, and [Cartea et al. \(2018b\)](#) analyse how model misspecification of order arrival rates, limit order fill probability, and asset price dynamics affect the agent's optimal trading strategy. Furthermore, [Cartea et al. \(2021b\)](#) and [Cartea and Sánchez-Betancourt \(2021, 2022b\)](#) propose models that incorporate latency to improve trading performance. Finally, [Cartea et al. \(2022b\)](#) and [Cartea et al. \(2022c,d,a\)](#) investigate algorithmic collusion.

Furthermore, numerous market impact and limit order book (LOB) models have also been studied. For instance, [Lorenz and Schied \(2013\)](#) investigate the stability of optimal strategies in the presence of transient price impact with exponential decay and general dynamics of a drift in the underlying price process accounting for the price impact of other agents. [Obizhaeva and Wang \(2013\)](#), later generalised in [Alfonsi et al. \(2008\)](#), propose a single-asset market impact model where price dynamics are derived from a dynamic LOB model with resilience. [Alfonsi and Schied \(2010\)](#) derive explicit optimal execution strategies in a discrete-time LOB model with general shape functions and an exponentially decaying price impact. [Gatheral \(2010\)](#) uses the no-dynamic-arbitrage principle to address the viability of market impact models. [Gatheral et al. \(2012\)](#) obtain explicit optimal strategies with a transient market impact in an expected cost minimisation setup. Finally, [Cont et al. \(2022, 2023\)](#) use data-driven models to characterise cross-impact of order flow.

Several works consider other types of order and execution strategies than those of the original Almgren-Chriss framework, which focuses on orders of the Implementation Shortfall (IS) type with MOs only. Other execution strategies have been studied in the literature, like Volume-Weighted Average Price (VWAP) orders in [Konishi \(2002\)](#), [Frei and Westray \(2015\)](#), [Guéant and Royer \(2014\)](#), and [Cartea and Jaimungal \(2016a\)](#) but also Target Close (TC) orders and Percentage of Volume (POV) orders, in [Guéant \(2016\)](#). Besides, several models focusing on optimal execution with limit orders have been proposed, as in [Bayraktar and Ludkovski \(2014\)](#), but also in [Guéant et al. \(2012\)](#) and [Guéant and Lehalle \(2015\)](#). The literature also addresses the existence of various trading venues. The case of optimal splitting of orders across different liquidity pools has been addressed in [Laruelle et al. \(2011\)](#) and [Cartea et al. \(2015\)](#).

Finally, a few works in the algorithmic trading literature use model free and data-driven approaches which better scale to complex models of the environment. [Guéant and Manziuk \(2019\)](#) use a model-based deep actor-critic Reinforcement Learning (RL) algorithm to approximate the optimal market making strategy in a high-dimensional setup, [Cartea et al. \(2021a\)](#) use double deep Q network learning to derive statistical arbitrage strategies, and [Ning et al. \(2021\)](#) use a model free Deep Q-Learning algorithm with LOB states as input features to derive execution strategies; see also [Scalzo et al. \(2021\)](#); [Arroyo et al. \(2022\)](#); [Duran-Martin et al. \(2022\)](#); [Borde et al. \(2023\)](#); [Arroyo et al. \(2023\)](#). In [Vuletić et al. \(2023\)](#); [Cont et al. \(2023\)](#); [Zhang et al. \(2023\)](#); [Lu et al.](#)

(2023), the authors study data driven approaches to forecast financial time-series of returns and volatility.

The work in this thesis relates to generalisations of price dynamics, and to the literature which incorporates signals, uncertainty, and learning in execution. [Cartea and Jaimungal \(2016b\)](#) consider order flow as a driver of asset prices, [Cartea et al. \(2018a\)](#) study volume imbalance as a price predictor and considers an optimal trading framework that incorporates order book signals, and [Cartea et al. \(2019\)](#) use multivariate Ornstein-Uhlenbeck (multi-OU) dynamics for the prices. [Laruelle et al. \(2013\)](#) derive a strategy where the agents learns the parameters of a jump process, [Cartea et al. \(2017\)](#) incorporate model uncertainty in execution, and [Casgrain and Jaimungal \(2019\)](#) derive strategies with learning of latent state distribution upon which prices depend. Similar to the problem addressed in Chapter 3, [Bhudisaksang and Cartea \(2021a,b\)](#) introduce an adaptive control framework which is robust to model misspecification, where the agent continuously learns the drift, and where her uncertainty follows a jump-diffusion. Recently, [Cartea and Sánchez-Betancourt \(2022a\)](#) derive closed-form strategies where a broker provides liquidity to an informed trader and uses the toxic flow to extract the trend signal.

The results of Chapters 3, 5, and 7 contribute to the stream of the optimal execution literature that adds predictive signals of future price changes. Typical examples of these signals include a drift in asset prices, order book imbalances, forecasts of the future order flow of market participants, and other price-based technical indicators. The usual formalism in the literature with predictive signals is to consider Brownian or Black-Scholes dynamics, along with independent mean-reverting Markov signals. The case of Ornstein-Uhlenbeck-type signals is of special interest as it usually leads to closed-form formulae for the optimal strategies. The pioneering work that considers general dynamics for signals that drive the asset prices and that derives closed-form optimal strategies is in [Cartea and Jaimungal \(2016b\)](#). [Belak et al. \(2018\)](#) consider optimal execution with general Markov signals and an application to “target zone models”. [Lehalle and Neuman \(2019\)](#) and [Neuman and Voß \(2020\)](#) provide an optimal trading framework incorporating Markov signals and a transient market impact. Other execution models that use signals are in [Forde et al. \(2022\)](#); [Fouque et al. \(2021\)](#).

Multi-asset optimal execution

In practice, operators routinely face the problem of having to execute simultaneously large orders regarding various assets, such as in block trading for funds facing large subscriptions or withdrawals, or when considering multi-asset trades in statistical arbitrage trading strategies. More generally, banks and market makers manage their (il)liquidity and market risk, when it comes to executing trades, in the context of a central risk book; hence the need for multi-asset models. However, in contrast to the single-asset case, the existing literature on the joint execution of large orders in multiple assets, or a single asset within a multi-asset portfolio, is limited. Besides, most existing papers

consider correlated Brownian motions when modelling the joint dynamics of prices. The problem of using single-asset models or unsuitable multivariate models for portfolio trading is that the resulting trading curves of individual assets do not balance well execution costs and price impact with price risk at the portfolio or strategy level.

The first paper to build multi-asset trading curves in an optimised way is [Almgren and Chriss \(2001\)](#). Almgren and Chriss considered, in an appendix of their seminal paper, a multi-asset extension of their discrete-time model. A few extensions to this model have been proposed in the literature. [Lehalle \(2009\)](#) considers adding an inventory constraint to balance the different portfolio lines during the portfolio execution process. [Schied et al. \(2010\)](#) show that when prices follow Bachelier dynamics, deterministic strategies are optimal for a trader with an exponential utility objective function. [Cartea et al. \(2015\)](#) use stochastic control tools to derive multi-asset optimal execution strategies such as optimal entry/exit times and cointegration-based statistical arbitrage. [Bismuth et al. \(2019\)](#) address optimal portfolio liquidation (along with other problems) by coupling Bayesian learning and stochastic control to derive optimal strategies under uncertainty on model parameters in the Almgren-Chriss framework. [Emschwiller et al. \(2021\)](#) extend optimal trading with Markovian predictors to the multi-asset case, with linear trading costs, using a mean-field approach that reduces the problem to a single-asset one.

The next section introduces a basic multi-asset model inspired by the Almgren-Chriss framework, where the asset prices follow correlated Brownian dynamics and the agent adopts a constant absolute risk aversion utility. The work in this thesis extends the model of Section 1.1.1 to take into account cointegration, uncertain drifts, and predictive signals.

1.1.1 A multi-asset model

Consider a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P}; \mathbb{F} = (\mathcal{F}_t)_{t \in [0, T]})$, with $T > 0$, satisfying the usual conditions. We assume this probability space to be large enough to support all the processes we introduce.

We consider a market with $d \in \mathbb{N}^*$ assets,¹ and an agent that liquidates her holdings in a portfolio over a trading window $[0, T]$. Let $q_0 \in \mathbb{R}^d$ be the initial holdings of the agent in every asset. The inventory process $(q_t)_{t \in [0, T]} = (q_t^1, \dots, q_t^d)_{t \in [0, T]}^\top$ of the agent evolves with the trading speed $(v_t)_{t \in [0, T]} = (v_t^1, \dots, v_t^d)_{t \in [0, T]}^\top$ for each asset, and we write²

$$dq_t = v_t dt.$$

¹ We denote by $\mathbb{N}^*$ the set $\mathbb{N}^* := \mathbb{N} \setminus \{0\}$ of positive integers.

² The superscript $^\top$ designates the transpose operator. It transforms here a line vector into a column vector.

We consider in this section the problem of multi-asset optimal execution in the case where prices are correlated arithmetic Brownian motions. The prices $(S_t)_{t \in [0, T]} = (S_t^1, \dots, S_t^d)_{t \in [0, T]}^\top$ of the d assets follow the dynamics

$$dS_t = V dW_t,$$

with $S_0 \in \mathbb{R}^d$ known, the matrix $V \in \mathcal{M}_d(\mathbb{R})$ captures the covariance and we define the covariation matrix $\Sigma = VV^\top$, and $(W_t)_{t \in [0, T]} = (W_t^1, \dots, W_t^d)_{t \in [0, T]}^\top$ is a d -dimensional standard Brownian motion with independent coordinates.³

Here, we consider that the temporary price impact and/or the execution costs of the agent's trading activity have a positive-definite quadratic form, which is a common assumption in the optimal execution literature; see [Cartea et al. \(2015\)](#); [Guéant \(2016\)](#). Thus, the value $(\tilde{X}_t)_{t \in [0, T]}$ of the agent's cash account throughout the trading window follows the dynamics

$$dX_t = -v_t^\top S_t dt - v_t^\top \eta v_t dt,$$

where the initial cash $X_0 \in \mathbb{R}$ is known, and $\eta \in \mathcal{S}_d^{++}(\mathbb{R})$ is the temporary market impact matrix.⁴

In optimal trading problems solved using stochastic optimal control tools, the agent derives the optimal strategy by maximising the expected utility of her terminal wealth. The objective functions considered in the literature are numerous. In this thesis, we use the CARA utility function as in [Guéant \(2016\)](#) and the risk-adjusted expectation objective function as in [Cartea et al. \(2015\)](#). In this section, we solve the multi-asset optimal execution problem when the agent adopts a CARA utility function.

In the original Almgren-Chriss model, market impact is permanent and temporary. The permanent component durably impacts the asset prices with no resilience, and the temporary price impact captures the execution costs and the cost of walking the book. [Gatheral \(2010\)](#) shows that the condition of absence of dynamic arbitrage ensures that the permanent price impact is linear. Thus, the impacted price process $(\tilde{S}_t)_{t \in [0, T]} = (\tilde{S}_t^1, \dots, \tilde{S}_t^d)_{t \in [0, T]}^\top$ follows the dynamics:

$$d\tilde{S}_t = dS_t + K v_t dt,$$

where $K \in \mathcal{S}_d(\mathbb{R})$ is the matrix that quantifies the linear permanent impact of the agent's trading activity on the prices. In the case of the CARA utility and Brownian dynamics, [Schied et al. \(2010\)](#) prove that one can reduce the set of admissible strategies to that of absolutely continuous deterministic execution strategies. In particular, the linear permanent price impact does not play a role in the optimisation problem. In the remainder of this section, we do not take permanent price impact into account.

³ We denote by $\mathcal{M}_{d,k}(\mathbb{R})$ the set of $d \times k$ real matrices and by $\mathcal{M}_d(\mathbb{R}) := \mathcal{M}_{d,d}(\mathbb{R})$ the set of $d \times d$ real square matrices. The set of real symmetric $d \times d$ matrices is denoted by $\mathcal{S}_d(\mathbb{R})$.

⁴ The subset of positive-definite and positive semi-definite matrices of $\mathcal{S}_d(\mathbb{R})$ are respectively denoted by $\mathcal{S}_d^{++}(\mathbb{R})$ and $\mathcal{S}_d^+(\mathbb{R})$.

The terminal wealth of the agent is the sum of the terminal cash X_T and the value of any outstanding terminal inventory q_T valued at the terminal price S_T . To enforce liquidation by the terminal time T , one can constrain the optimisation problem by enforcing $q_T = 0$. This hard constraint features a singular boundary condition that makes the optimisation problem difficult to solve, in particular, one cannot use the classical tools of optimal control. Thus, it is a common practice to add a penalisation term in the objective function. The penalisation term acts as a discount applied to the Mark-to-Market value $q_T^\top S_T$ of the remaining assets and penalises any non-zero terminal position. In this thesis, we only consider the case where the final penalty is a positive-definite quadratic form. Thus, we value the terminal inventory q_T of the agent at $q_T^\top S_T - q_T^\top \Gamma q_T$ where $\Gamma \in \mathcal{S}_d^{++}(\mathbb{R})$.

The agent maximises the performance criterion

$$\mathbb{E} \left[-\exp \left(-\gamma \left(X_T + q_T^\top S_T - q_T^\top \Gamma q_T \right) \right) \right], \quad (1.1)$$

where $\gamma > 0$ is the absolute risk aversion parameter of the agent.

The Hamilton-Jacobi-Bellman (HJB) equation associated with the problem (1.1) is

$$\begin{aligned} 0 = & \partial_t w(t, x, q, S) + \sup_{v \in \mathbb{R}^d} \left(-(v^\top S + v^\top \eta v) \partial_x w(t, x, q, S) + v^\top \nabla_q w(t, x, q, S) \right) \\ & + \frac{1}{2} \text{Tr} \left(\Sigma D_{SS}^2 w(t, x, q, S) \right), \end{aligned}$$

for all $(t, x, q, S) \in [0, T) \times \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^d$ with terminal condition

$$w(T, x, q, S) = -\exp \left(-\gamma (x + q^\top S - q^\top \Gamma q) \right) \quad \forall (x, q, S) \in \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^d.$$

We use the ansatz

$$w(t, x, q, S) = -\exp \left(-\gamma (x + q^\top S + \theta(t, q)) \right) \quad \forall (t, x, q, S) \in [0, T] \times \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^d,$$

to obtain the new HJB in θ

$$0 = \partial_t \theta(t, q) + \sup_{v \in \mathbb{R}^d} \left(v^\top \nabla_q \theta(t, q) - v^\top \eta v \right) - \frac{\gamma}{2} q^\top \Sigma q, \quad (1.2)$$

with terminal condition

$$\theta(T, q) = -q^\top \Gamma q \quad \forall q \in \mathbb{R}^d.$$

The supremum in (1.2) is reached with the optimal control in feedback form

$$v_s^* = \frac{1}{2} \eta^{-1} \nabla_q \theta(s, q),$$

for all $s \in [t, T]$. Consequently, we obtain the HJB

$$0 = \partial_t \theta(t, q) + \frac{1}{4} \nabla_q \theta(t, q)^\top \eta^{-1} \nabla_q \theta(t, q) - \frac{\gamma}{2} q^\top \Sigma q,$$

with terminal condition

$$\theta(T, q) = -q^\top \Gamma q \quad \forall q \in \mathbb{R}^d.$$

We use the second ansatz

$$\theta(t, q) = q^\top A(t) q + B(t)^\top q + C(t),$$

for all $(t, q) \in [0, T] \times \mathbb{R}^d$, and find that the problem reduces to finding $A \in C^1([0, T], \mathcal{S}_d(\mathbb{R}))$, $B \in C^1([0, T], \mathbb{R}^d)$ and $C \in C^1([0, T], \mathbb{R})$ that solve the following ODE system

$$\begin{cases} A'(t) = \frac{\gamma}{2} \Sigma - A(t) \eta^{-1} A(t) \\ B'(t) = -A(t) \eta^{-1} B(t) \\ C'(t) = -\frac{1}{4} B(t)^\top \eta^{-1} B(t), \end{cases}$$

with terminal conditions

$$A(T) = -\Gamma, \quad B(T) = C(T) = 0.$$

The solutions for B and C are $B = C = 0$. Therefore, the problem reduces to finding $A \in C^1([0, T], \mathcal{S}_d(\mathbb{R}))$ solution of the following terminal value problem:

$$\begin{cases} A'(t) = \frac{\gamma}{2} \Sigma - A(t) \eta^{-1} A(t) \\ A(T) = -\Gamma. \end{cases} \quad (1.3)$$

We show that when $\Sigma \in \mathcal{S}_d^{++}(\mathbb{R})$, A can be found in closed-form. We introduce the change of variables

$$a(t) = \eta^{-\frac{1}{2}} A(t) \eta^{-\frac{1}{2}} \quad \forall t \in [0, T],$$

and notice that (1.3) is equivalent to the terminal value problem

$$\begin{cases} a'(t) = \hat{A}^2 - a(t)^2 \\ a(T) = -C, \end{cases} \quad (1.4)$$

where

$$\hat{A} = \sqrt{\frac{\gamma}{2}} \left(\eta^{-\frac{1}{2}} \Sigma \eta^{-\frac{1}{2}} \right)^{\frac{1}{2}} \in \mathcal{S}_d^{++}(\mathbb{R}),$$

and

$$C = \eta^{-\frac{1}{2}} \Gamma \eta^{-\frac{1}{2}} \in \mathcal{S}_d^+(\mathbb{R}).$$

To solve (1.4) we use a classical trick for Riccati equations in the next result.

Proposition 1.1.1 Let $\xi : [0, T] \rightarrow \mathcal{S}_d(\mathbb{R})$ defined as

$$\xi(t) = -\frac{\hat{A}^{-1}}{2} \left(I - e^{-2\hat{A}(T-t)} \right) - e^{-\hat{A}(T-t)} \left(C + \hat{A} \right)^{-1} e^{-\hat{A}(T-t)} \quad (1.5)$$

be the unique solution of the linear ODE

$$\begin{cases} \xi'(t) = \hat{A}\xi(t) + \xi(t)\hat{A} + I_d \\ \xi(T) = -\left(C + \hat{A}\right)^{-1}. \end{cases} \quad (1.6)$$

Then $\forall t \in [0, T]$, $\xi(t)$ is invertible and

$$a : t \in [0, T] \rightarrow \hat{A} + \xi(t)^{-1} \in \mathcal{S}_d(\mathbb{R})$$

is the unique solution of (1.4). $\square$

Proof 1.1.1 First, we verify that ξ , defined in (1.5), is the solution of the linear ODE (1.6). Next, for all $t \in [0, T]$, $-\xi(t)$ is the sum of

$$\frac{\hat{A}^{-1}}{2} \left(I - e^{-2\hat{A}(T-t)} \right) \in \mathcal{S}_d^+(\mathbb{R})$$

and

$$e^{-\hat{A}(T-t)} \left(C + \hat{A} \right)^{-1} e^{-\hat{A}(T-t)} \in \mathcal{S}_d^{++}(\mathbb{R}),$$

so $-\xi(t) \in \mathcal{S}_d^{++}(\mathbb{R})$ and $\xi(t)$ is invertible. We also note that

$$\begin{aligned} a'(t) &= -\xi(t)^{-1} \xi'(t) \xi(t)^{-1} \\ &= -\xi(t)^{-1} \hat{A} - \hat{A} \xi(t)^{-1} - \xi(t)^{-2} \\ &= \hat{A}^2 - \left(\hat{A} + \xi(t)^{-1} \right)^2 \\ &= \hat{A}^2 - a(t)^2, \end{aligned}$$

and $a(T) = -C$, hence the result. $\square$

We deduce the following corollary:

Corollary 1.1.1

$$\forall t \in [0, T], \quad A(t) = \eta^{\frac{1}{2}} \left(\hat{A} - \left(\frac{\hat{A}^{-1}}{2} \left(I - e^{-2\hat{A}(T-t)} \right) + e^{-\hat{A}(T-t)} \left(C + \hat{A} \right)^{-1} e^{-\hat{A}(T-t)} \right)^{-1} \right) \eta^{\frac{1}{2}}. \quad (1.7)$$

The problem admits a classical solution. The results in Chapter 2 apply to this problem (when $R = 0$) and rigorously characterise the strategy. In particular,

$$w(t, x, q, S) = -\exp \left(-\gamma (x + q^\top S + q^\top A(t) q) \right),$$

where A is in (1.7), is the value function associated with the optimisation problem, and the optimal trading speed is

$$v_s^* = \eta^{-1} q_s^\top A(s).$$

1.1.2 Contributions

Multi-asset execution with multi-OU dynamics

Chapter 2 solves a multi-asset optimal execution problem when the agent adopts a CARA utility and the prices follow multi-OU dynamics which account for the presence of cointegration between the assets' prices. We showcase the use of the model to enhance the performance of portfolio execution programmes and to build statistical arbitrage strategies using real data from the Equity and FX markets.

Multi-OU dynamics account for the information in the prices of all assets when trading individual assets. Moreover, the dynamics can incorporate information from external assets or signals that share common stochastic trends with the traded assets and which act as predictors of the future prices. This information enhances the trading performance of execution programmes and can serve as a basis to execute statistical arbitrages.

In practice, market operators such as brokers or mutual funds are in charge of assets from a specific sector or asset class. Thus, the assets prices usually share common stochastic trends and co-movements. For example, in equity markets, a large number of shares from a specific economic sector have cointegrated dynamics as they share common sources of systemic risk. Often, when trading these assets simultaneously, price movements of a single asset that are adverse when considered independently can become profitable in the portfolio case. Operators need to account for the joint dynamics that assets exhibit.

The first paper to consider multi-OU dynamics for optimal trading is [Cartea et al. \(2019\)](#), where an agent maximises the expected PnL minus a running risk penalty. The problem reduces to a Riccati ODE for which they provide global existence and uniqueness, and subsequently characterise the optimal strategy with a standard verification argument. In the model of Chapter 2, the CARA utility function leads to a Riccati ODE where the quadratic term has both positive and negative eigenvalues. Thus, we do not rely on classical existence theorems. However, we are able to rigorously characterise the optimal trading strategy by using comparison arguments that use the original control problem rather than the Riccati ODE itself. This type of Riccati ODEs, which we generalise in Chapter 3, arise in various financial decision problems.

The advantages of multi-OU dynamics for practitioners are numerous. Considering single-asset execution within a portfolio allows to manage risk across a basket of assets. Agents can hold securities on their balance sheets for longer, reducing market impact and execution costs. Moreover, from a regulation point of view, multivariate optimal execution models that naturally offset risks in a portfolio are of great interest. The new FRTB (Fundamental Review of the Trading Book) regulation will lead practitioners to assess liquidity risks within a centralised risk book for capital requirements. In this context, our model can reduce the liquidity risk of the execution process by taking into account the joint dynamics of the assets.

Portfolio execution with Bayesian learning

Section 3.3 of Chapter 3 solves the problem of optimal execution for an agent in charge of a portfolio who uses signals and past observations of prices to learn the drift of the assets composing her portfolio. The estimated drift is continuously updated throughout the trading programme to enhance the agent's trading performance. [Casgrain and Jaimungal \(2019\)](#) propose a framework where the drift of the midprice stems from a diffusive component and a pure jump component that are both driven by state variables and a latent continuous time Markov chain that modulates their dynamics. The authors derive an SDE for the filter process and obtain optimal trading strategies where the objective function is the expected terminal PnL minus a running penalty. Section 3.3 of Chapter 3 considers a similar problem, where now the agent learns the latent drift from the observed prices and a set of predictive signals, and the objective function is CARA. A simpler version of this model has been proposed, but not solved, in [Bismuth et al. \(2019\)](#).

In most execution models, the parameters governing the dynamics of assets are fixed. Consequently, in practice, market operators estimate these parameters prior to the trading window. The downsides of this approach are twofold. First, if the estimates are correct but the dynamics change during the execution process, or if the estimates are incorrect, the strategy derived by the agent becomes non-optimal. Second, the optimal strategy does not take into account the uncertainty about the true value of the parameters.

These issues are commonly faced by market participants in practice and they are studied in the literature. In particular, some address this issue by using filtering and learning techniques in a framework with partial information, or mix online learning with optimal control. However, combining online learning with optimisation in mathematical finance is mainly explored in the context of the optimal portfolio choice literature; see [Bismuth et al. \(2019\)](#) and references therein.

Chapter 3 (Section 3.3) uses Bayesian learning where the goal is to encode beliefs about the parameters of a given model, and later update their values by observing data. In our case, the prices follow drifted Bachelier dynamics, and the drifts are unknown. The agent has a multivariate Gaussian prior over the future distribution of the drifts of the assets in her portfolio. The prior states that the values of the drifts will fall within some range according to some probability distribution. Throughout the execution programme, the agent uses signals and price observations to update her estimates and improve the overall trading performance. At any point in time during the execution programme, the agent's latest estimate of the true value of the drifts is the expectation of the drifts conditionally on the filtration generated by the prices and the signals that she observes.

The model combines online learning with stochastic optimal control to derive strategies in a partial information framework. Note that online learning is forward in time and dynamic programming uses backward induction. The approach is powerful for two

reasons. First, the strategy adapts to the rapidly changing market trends by using the information in the prices and in a set of signals. Second, the strategy incorporates the uncertainty about the future value of the drift vector, and the fact that the agent continues learning its posterior distribution. A first version of the model has been introduced in [Bismuth et al. \(2019\)](#) along with other portfolio problems. However, the authors fail to provide an existence result for the nonlinear ordinary differential equation characterizing the problem. Our model incorporates signals and rigorously characterises the optimal trading strategy using results on the solvability of indefinite differential Riccati equations that we derive in Section 3.1.

1.2 Optimal market making in OTC markets

Most electronic exchanges clear the demand and supply of liquidity in LOBs, where a large panel of market participants such as institutional or individual investors, dealers, brokers, and arbitrageurs can interact. Alternatively, OTC markets are off-exchange “quote-driven” markets that are based on a network of market makers that set prices at which liquidity takers can trade. To set prices, liquidity providers in OTC markets constantly stream bid and ask quotes at which they are ready to immediately trade, or they respond to requests made by their clients; the so-called RFQs (Request For Quote).⁵ In this thesis, we consider a market maker in an OTC market, in charge of a large portfolio comprising multiple assets.

OTC market makers faces a complex problem. They provide bid and ask quotes for various assets that exhibit complex joint dynamics without seeing the full depth of price and clients. Consequently, it is key to properly account for risks at the portfolio level. However, a large proportion of multi-asset market making models in the literature only consider correlated Brownian dynamics. Additionally, multi-asset market making is challenging due to high dimensionality and the resulting numerical challenges to obtain the optimal quotes.

To derive the optimal bid and ask quotes that they provide to market participants, market makers usually use a utility maximisation framework. One of the first approaches to mathematically tackle this problem are in [Ho and Stoll \(1981\)](#) and [Ho and Stoll \(1983\)](#) where the authors derive a quoting strategy in a multi-period sequential bidding setup. The work in [Glosten and Milgrom \(1985\)](#) analyses the market maker’s decision problem within a model with informed and uninformed traders, giving rise to numerous models that oppose insiders and liquidity consumers who interact through a market maker that sets prices. The literature has later been revived after the seminal work in [Avellaneda and Stoikov \(2008\)](#) who considered a market with a reference price in the form of a Brownian motion, and modelled the number of orders at the bid and the ask as a counting process whose intensity depends on the distance of the quotes from the

⁵ For a better understanding of how the system of RFQs operates, we refer the interested reader to [Fermanian et al. \(2016\)](#).

reference price. Although the work in [Avellaneda and Stoikov \(2008\)](#) and its extensions were intended to solve optimal market making within an LOB, the approach is better suited for an OTC market maker that receives RFQs, unless the quoted spreads in the LOB are large in terms of the tick size.

The model in [Avellaneda and Stoikov \(2008\)](#) gave rise to an extensive literature on optimal market making. [Guéant et al. \(2013\)](#) provide a rigorous analysis of the problem, for which they consider the exponential intensity functions suggested in [Avellaneda and Stoikov \(2008\)](#). The authors use the tools of stochastic optimal control to prove that the problem reduces to a system of linear ODEs under inventory constraints. They also analyse the asymptotic regime of the optimal quotes when the time horizon T tends to $+\infty$, and propose closed-form approximations using results from spectral analysis. In addition, Cartea and his co-authors also study optimal market making models. [Cartea et al. \(2015\)](#) use stochastic control tools to rigorously characterise the optimal market making strategy that uses an objective function given by the expectation of the P&L minus a running penalty. [Cartea et al. \(2017\)](#) study the ambiguity in the specification of fill probabilities and the dynamics of intensities and prices. Finally, [Cartea and Wang \(2020\)](#) incorporate signals in the market making strategy.

More recently, other features have also been considered in the literature, such as general intensity shapes depending on unobservable factors in [Campi and Zabaljauregui \(2020\)](#), persistence in the order flow in [Jusselin \(2021\)](#), variability in the request sizes in [Bergault and Guéant \(2021\)](#), and client tiering and access to liquidity pools in [Barzykin et al. \(2021\)](#). The case of options market making is studied in the literature starting with [Stoikov and Sağlam \(2009\)](#) where the authors use a mean-variance framework. In [El Aoud and Abergel \(2015\)](#), the authors extend the problem by considering a stochastic volatility model for prices.

Multi-asset market making

The models of multi-asset market making suffer from the curse of dimensionality because the number of equations to solve grows exponentially with the number of assets. Multiple approaches are proposed by Guéant and various co-authors to study the issue of the dimensionality in market making problems. [Guéant \(2017\)](#) proposes a general model reconciling the approaches in [Guéant et al. \(2013\)](#) and [Cartea et al. \(2014\)](#) in a multi-asset model, and most importantly obtains interpretable approximations for the optimal quotes. [Guéant and Manziuk \(2019\)](#) use neural networks instead of grids to obtain numerical solutions of the problem in high dimension. They use a model-based actor-critic algorithm for approximating the value function and the optimal strategies within a framework similar to that of [Avellaneda and Stoikov \(2008\)](#). [Bergault and Guéant \(2021\)](#) propose an approach for approximating the optimal quotes by using a low-dimensional factor space on which market risk is projected. Finally, [Bergault et al. \(2021\)](#) propose an approach where the Hamilton-Jacobi equation of the problem is

regarded as a perturbation of another simpler quadratic one, whose solution in their case can be obtained in closed form using Linear-Quadratic-Gaussian tools.

Section 1.2.1 describes a basic multi-asset model which considers correlated Brownian dynamics for the asset prices, and where the market maker adopts a CARA utility. Additional details and proofs of the results in Section 1.2.1 are in [Guéant \(2017\)](#).

1.2.1 A multi-asset model

Here, a market maker that operates in an OTC market is in charge of a portfolio containing $d \in \mathbb{N}^*$ assets. The joint dynamics of the asset prices $(S_t)_{t \in [0, T]} = (S_t^1, \dots, S_t^d)_{t \in [0, T]}^\top$ follow a d -dimensional Brownian process:

$$dS_t = V dW_t,$$

where the initial values $S_0 \in \mathbb{R}^d$ are known and $(W_t)_{t \in [0, T]}$ is a d -dimensional standard Brownian motion with independent coordinates. We denote by $\Sigma = V^\top V$ the variance-covariance matrix associated with S .

Throughout a trading window $[0, T]$, where $T > 0$, the market maker continuously chooses the prices at which she is ready to buy or sell the d assets. We define the processes $(S_t^{i,b})_{t \in [0, T]}$ and $(S_t^{i,a})_{t \in [0, T]}$ as the bid and ask prices of asset i , respectively. We denote by the processes $(\delta_t^{i,b})_{t \in [0, T]}$ and $(\delta_t^{i,a})_{t \in [0, T]}$ the distances of the quotes from the reference price S^i for, respectively, the bid and the ask and we write

$$\delta_t^{i,b} = S_t^i - S_t^{i,b} \quad \text{and} \quad \delta_t^{i,a} = S_t^{i,a} - S_t^i, \quad \forall t \in [0, T].$$

For $i \in \{1, \dots, d\}$, two point processes $(N_t^{i,b})_{t \in [0, T]}$ and $(N_t^{i,a})_{t \in [0, T]}$ model the number of transactions at the bid and ask side for asset i (see [Cartea \(2013\)](#) and [Cartea et al. \(2018c\)](#) for the use of point processes in financial decision problems). We assume that the market maker receives RFQs for a constant size Δ^i for the i -th asset and that she ceases to trade asset i if her (absolute) inventory in asset i exceeds a risk limit Q^i following a trade.

The inventory $(q_t)_{t \in [0, T]} = (q_t^1, \dots, q_t^d)_{t \in [0, T]}^\top$ follows the dynamics

$$\forall i \in \{1, \dots, d\}, \quad dq_t^i = \Delta^i dN_t^{i,b} - \Delta^i dN_t^{i,a},$$

where the initial inventory q_0^i in the i -th asset is known. The point processes are independent of the Brownian W and have respective intensity processes $(\lambda_t^{i,b})_{t \in [0, T]}$ and $(\lambda_t^{i,a})_{t \in [0, T]}$ given by

$$\lambda_t^{i,b} = \Lambda^{i,b} \left(\delta_t^{i,b} \right) \mathbb{1}_{q_{t-}^i + \Delta^i < Q^i} \quad \text{and} \quad \lambda_t^{i,a} = \Lambda^{i,a} \left(\delta_t^{i,a} \right) \mathbb{1}_{q_{t-}^i - \Delta^i > -Q^i},$$

for two functions $\Lambda^{i,b}, \Lambda^{i,a} : \mathbb{R} \mapsto \mathbb{R}_+$. The intensity of the arrival of orders at the bid and the ask decreases with the distance of the quotes from the reference price. Thus, we consider that the functions $\Lambda^{i,b}$ and $\Lambda^{i,a}$ are decreasing and go to zero when the quotes go to $+\infty$.

Finally, the cash process $(X_t)_{t \in [0, T]}$ of the market maker evolves as

$$\begin{aligned} dX_t &= \sum_{i=1}^d S_t^{i,a} \Delta^i dN_t^{i,a} - S_t^{i,b} \Delta^i dN_t^{i,b} \\ &= \sum_{i=1}^d \left(\delta_t^{i,b} \Delta^i dN_t^{i,b} + \delta_t^{i,a} \Delta^i dN_t^{i,a} \right) - \sum_{i=1}^d S_t^i dq_t^i. \end{aligned}$$

We consider, as in [Avellaneda and Stoikov \(2008\)](#), that the market maker maximises the expected value of a CARA utility function with absolute risk aversion parameter $\gamma > 0$. Thus the market maker maximises the performance criterion

$$\mathbb{E} \left[-\exp \left(-\gamma \left(X_T + q_T^\top S_T - q_T^\top \Gamma q_T \right) \right) \right]. \quad (1.8)$$

Let $\{e^1, \dots, e^d\}$ be the canonical basis of $\mathbb{R}^d$, i.e. for all $i \in \{1, \dots, d\}$, e^i is a vector of zeros except for the i th element which is equal to 1. The HJB equation associated with (1.8) is

$$\begin{aligned} 0 &= \partial_t w(t, x, q, S) + \frac{1}{2} \text{Tr} \left(\Sigma D_{SS}^2 w(t, x, q, S) \right) \\ &+ \sum_{i=1}^d \mathbb{1}_{\{q^i + \Delta^i \leq Q^i\}} \sup_{\delta^{i,b}} \Lambda^{i,b}(\delta^{i,b}) \left(w \left(t, x - \Delta^i \left(S^i - \delta^{i,b} \right), q + \Delta^i e^i, S \right) - w(t, x, q, S) \right) \\ &+ \sum_{i=1}^d \mathbb{1}_{\{q^i - \Delta^i \geq -Q^i\}} \sup_{\delta^{i,a}} \Lambda^{i,a}(\delta^{i,a}) \left(w \left(t, x + \Delta^i \left(S^i + \delta^{i,a} \right), q - \Delta^i e^i, S \right) - w(t, x, q, S) \right), \end{aligned} \quad (1.9)$$

for all $(t, x, q, S) \in [0, T) \times \mathbb{R} \times \mathcal{Q} \times \mathbb{R}^d$, with terminal condition

$$w(T, x, q, S) = -\exp \left(-\gamma \left(x + q^\top S - q^\top \Gamma q \right) \right), \quad \forall (x, q, S) \in \mathbb{R} \times \mathcal{Q} \times \mathbb{R}^d, \quad (1.10)$$

where $\mathcal{Q} = \prod_{i=1}^d (\Delta^i \mathbb{Z} \cap [-Q^i, Q^i])$.⁶

To reduce the dimensionality of the problem, we use the ansatz

$$w(t, x, q, S) = -\exp \left(-\gamma \left(x + q^\top S + \theta(t, q) \right) \right), \quad (1.11)$$

and obtain the following HJB in θ :

⁶ We denote by $\Delta \mathbb{Z}$ the set of multiples of Δ , i.e. $\Delta \mathbb{Z} = \{\dots, -2\Delta, -\Delta, 0, \Delta, 2\Delta, \dots\}$, where $\Delta \in \mathbb{R}_+^*$ is a positive number.

$$\begin{aligned}
0 = & \partial_t \theta(t, q, S) - \frac{\gamma}{2} q^\top \Sigma q \\
& + \sum_{i=1}^d \mathbb{1}_{\{q^i + \Delta^i \leq Q^i\}} \Delta^i H^{i,b} \left(\frac{\theta(t, q, S) - \theta(t, q + \Delta^i e^i, S)}{\Delta^i} \right) \\
& + \sum_{i=1}^d \mathbb{1}_{\{q^i - \Delta^i \geq -Q^i\}} \Delta^i H^{i,a} \left(\frac{\theta(t, q, S) - \theta(t, q - \Delta^i e^i, S)}{\Delta^i} \right), \quad (1.12)
\end{aligned}$$

on $[0, T] \times \mathcal{Q}$ with terminal condition

$$\theta(T, q) = -q^\top \Gamma q, \quad \forall q \in \mathcal{Q}, \quad (1.13)$$

where the Hamiltonian functions $H^{i,b}$ and $H^{i,a}$ are

$$\begin{aligned}
H^{i,b}(p) &= \sup_{\delta} \frac{\Lambda^{i,b}(\delta)}{\gamma \Delta^i} (1 - \exp(-\gamma \Delta^i (\delta - p))) \\
H^{i,a}(p) &= \sup_{\delta} \frac{\Lambda^{i,a}(\delta)}{\gamma \Delta^i} (1 - \exp(-\gamma \Delta^i (\delta - p))),
\end{aligned}$$

The following results solve the optimal multi-asset market making problem and are proved in [Guéant \(2017\)](#).

Theorem 1.2.1 There exists a unique function $\theta : [0, T] \times \mathcal{Q} \mapsto \mathbb{R}$, C^1 in time, solution of the HJB (1.12) with terminal condition (1.13). $\square$

Theorem 1.2.2 Let $\theta : \mathbb{R}^+ \times \mathcal{Q} \rightarrow \mathbb{R}$ be the unique solution to the HJB (1.12) with terminal condition (1.13). Then, the function u defined in (1.11) is the unique solution to (1.9) with terminal condition (1.10). Moreover, for all $i \in \{1, \dots, d\}$, the optimal bid and ask quotes

$$S_t^{i,b} = S_t^i - \delta_t^{i,b*} \quad \text{and} \quad S_t^{i,a} = S_t^i + \delta_t^{i,a*}$$

are obtained by

$$\begin{aligned}
\delta_t^{i,b*} &= \tilde{\delta}^{i,b*} \left(\frac{\theta(t, q_{t-}) - \theta(t, q_{t-} + \Delta^i e^i)}{\Delta^i} \right) \quad \text{when } q_{t-} + \Delta^i e^i \in \Pi_{i=1}^d [-Q^i, Q^i], \\
\delta_t^{i,a*} &= \tilde{\delta}^{i,a*} \left(\frac{\theta(t, q_{t-}) - \theta(t, q_{t-} - \Delta^i e^i)}{\Delta^i} \right) \quad \text{when } q_{t-} - \Delta^i e^i \in \Pi_{i=1}^d [-Q^i, Q^i], \quad (1.14)
\end{aligned}$$

where the functions $\tilde{\delta}^{i,b*}(\cdot)$ and $\tilde{\delta}^{i,a*}(\cdot)$ are defined as

$$\begin{aligned}
\tilde{\delta}^{i,b*}(p) &= \Lambda^{i,b^{-1}} \left(\gamma \Delta^i H^{i,b}(p) - H^{i,b'}(p) \right), \\
\tilde{\delta}^{i,a*}(p) &= \Lambda^{i,a^{-1}} \left(\gamma \Delta^i H^{i,a}(p) - H^{i,a'}(p) \right).
\end{aligned}$$

1.2.2 Contributions

Section 3.2 of Chapter 3 introduces a comprehensive framework for multi-asset market making in OTC markets with three important features : (i) dynamics that incorporate cointegration and predictive signals, (ii) external hedging and client tiering, and (iii) computationally efficient methods to obtain quotes for large portfolios.

The main contributions of the model are twofold. First, the strategy uses dynamics that incorporate price predictors to improve trading performance and instruments from an external trading venue to offset inventory risk. More precisely, the strategy uses multi-OU dynamics to model the joint dynamics of these variables and those of the assets quoted by the market maker. Second, the model uses an approximation technique similar to that in [Bergault et al. \(2021\)](#) to derive an approximate strategy that reduces to a Riccati ODE which can be efficiently solved using ODE numerical solving methods. In particular, we characterise the approximate strategy and provide an existence and uniqueness result for the Riccati ODE using the results derived in Section 3.1.

The advantages of our framework are numerous. In practice, OTC market makers are usually in charge of multiple assets from a specific asset class. This is for instance the case of market makers in OTC bond markets who provide liquidity for multiple bonds issued by a single issuer or by similar issuers. Multi-OU dynamics for asset prices are more suitable in these cases. In the single-bond case, it is key to take into account the mean-reversion effects that are inherent to the bond market. For portfolios of bonds, multi-OU dynamics capture the complex cointegration features and can encode the market maker's views on the future behavior of prices. More precisely, multi-OU dynamics specify (i) the long-term unconditional mean of prices, i.e., the level of the prices on the long-run, (ii) the speed at which prices converge to this mean, and (ii) the cointegration factors governing the dynamics of the different assets. Thus, this process offers a convenient model for agents to encode their beliefs or to estimate these parameters from data. Finally, the multi-OU dynamics in our strategy incorporate hedging instruments and predictive signals to enhance trading performance.

In these cases, our strategy leads to efficient quotes, efficient hedging, and computational efficiency. First, quotes are more efficient because it is often the case that skewing the quotes for a particular asset for which the market maker already holds inventory may be unnecessary if an opposite position is held in a correlated and/or cointegrated asset, or in a combination of assets that offsets the risk. Second, hedging is more efficient because positions can be held for longer and hedging may not be necessary when the market maker has opposing positions in correlated and/or cointegrated assets. On the other hand, the market maker's positions may need to be reduced or hedged rapidly when multiple positions share the same directional risk. Finally, the strategy of Section 3.2 can be used for high-dimensional portfolios and multiple predictive signals and can be computed efficiently, which is key for market makers in high frequency markets.

The model of Section 3.2 uses a similar framework to that in [Avellaneda and Stoikov \(2008\)](#) and the rigorous framework in [Guéant et al. \(2013\)](#), for a market maker in charge

of a portfolio where prices follow multi-OU dynamics. The market maker interacts with N clients, and receives RFQs with variable sizes. Moreover, she has access to an external trading venue where she can hedge risk at a cost. We derive the optimal quotes in feedback form, and then use the approximation method in [Bergault et al. \(2021\)](#) to derive an approximation strategy. We demonstrate that the approximation problem simplifies the initial HJB equation into a system of ODEs which involves a matrix differential Riccati equation. This equation is a special case of the Riccati ODEs solved in Section 3.1.

Chapter 2

Multi-asset optimal execution and statistical arbitrage under OU dynamics

The results of this chapter have been obtained in collaboration with Philippe Bergault and Olivier Guéant and are in [Bergault et al. \(2022\)](#).

A classical model for the multivariate dynamics of financial variables that goes beyond that of correlated Brownian motions is the multivariate Ornstein-Uhlenbeck (multi-OU) model. It is especially attractive because it is parsimonious, and yet general enough to cover a wide spectrum of multi-dimensional dynamics. Multi-OU dynamics offer a large coverage since particular cases include correlated Brownian motions but also cointegrated dynamics which are heavily used in statistical arbitrage.

When trading an asset, the dynamics of another asset within or outside the portfolio can be regarded as a predictive signal that can enhance the execution process. Thus, the stream of the literature that considers predictive signals in the asset prices dynamics is closely related to the topic of multi-asset optimal execution with multi-OU dynamics. The work in [Cartea et al. \(2019\)](#) is the pioneering work in the use of the multi-OU model for the price dynamics in a multi-asset optimal execution problem. Indeed, the authors proposed an interesting model where the asset prices have multi-OU dynamics and the agent maximises an objective function given by the expectation of the Profit and Loss (PnL) minus a running penalty related to the instantaneous variance of the portfolio. In their approach, the problem boils down to a system of ODEs involving a Matrix Riccati ODE for which the classical existence theorems related to linear-quadratic control theory apply.

In this chapter, we propose a model similar to the one in [Cartea et al. \(2019\)](#), but where the objective function is of the Von Neumann-Morgenstern type: an expected exponential utility of the PnL.¹ By using classical stochastic optimal control tools we show that the problem boils down to solving a system of ODEs involving a Matrix Riccati ODE. However, unlike the results in [Cartea et al. \(2019\)](#), the use of an expected

¹ Our model accounts therefore for the risk in a different manner than the model presented in [Cartea et al. \(2019\)](#). Comparisons are difficult to carry out as risk aversion parameters in the two models have different meanings.

exponential utility framework to account for the risk leads to a Matrix Riccati ODE for which classical existence theorems do not apply. By using *a priori* estimates obtained thanks to optimal control tools, we nevertheless prove an existence and uniqueness result for the latter ODE and obtain a verification theorem that provides a rigorous solution to the execution problem.

The main contribution of this chapter is therefore to propose a model for multi-asset portfolio execution under multi-OU price dynamics in an expected utility framework that accounts for the overall risk associated with the execution process. We focus on the problem where an agent is in charge of unwinding a large portfolio, but also illustrate the use of the results for multi-asset statistical arbitrage strategies.

The remainder of this chapter is organized as follows. Section 2.1 presents the optimal execution problem in the form of a stochastic optimal control problem and shows that solving the associated Hamilton-Jacobi-Bellman (HJB) equation reduces to solving a system of ODEs involving a Matrix Riccati ODE. We then prove a global existence result for that ODE and eventually provide a solution to the initial stochastic optimal control problem thanks to a verification argument. Section 2.2 illustrates the results on both real data, from the foreign exchange and stock markets, and simulated data. Our examples focus on optimal liquidation but we also illustrate and discuss the use of our results for statistical arbitrage strategies. This chapter uses the results of the model in Subsection 1.1.1 and Appendix 2.2.2.2. Subsection 1.1.1 is dedicated to the special case where the multi-OU dynamics reduce to simple correlated Brownian dynamics. Appendix 2.2.2.2 is dedicated to a limit case of our model where execution costs and terminal penalty are ignored; this limit case is useful to obtain *a priori* estimates for our general problem.

2.1 The optimal liquidation problem

2.1.1 Modelling framework and notations

In this chapter, we consider a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P}; \mathbb{F} = (\mathcal{F}_t)_{t \in [0, T]})$ satisfying the usual conditions. We assume this probability space to be large enough to support all the processes we introduce.

We consider a market with $d \in \mathbb{N}^*$ assets, and a trader wishing to liquidate her portfolio over a period of time $[0, T]$, with $T > 0$. Her inventory process $(q_t)_{t \in [0, T]} = (q_t^1, \dots, q_t^d)_{t \in [0, T]}^\top$ evolves as

$$dq_t = v_t dt,$$

with $q_0 \in \mathbb{R}^d$ given, where $(v_t)_{t \in [0, T]} = (v_t^1, \dots, v_t^d)_{t \in [0, T]}^\top$ represents the trading rate of the trader for each asset.

The fundamental prices of the d assets are modelled as a d -dimensional Ornstein-Uhlenbeck process $(S_t)_{t \in [0, T]} = (S_t^1, \dots, S_t^d)_{t \in [0, T]}^\top$:

$$dS_t = R(\bar{S} - S_t)dt + VdW_t,$$

and we introduce the market price process $(\tilde{S}_t)_{t \in [0, T]} = (\tilde{S}_t^1, \dots, \tilde{S}_t^d)_{t \in [0, T]}^\top$ with dynamics:

$$d\tilde{S}_t = dS_t + Kv_t dt,$$

with $S_0 = \tilde{S}_0 \in \mathbb{R}^d$ given, where $\bar{S} \in \mathbb{R}^d$, $R \in \mathcal{M}_d(\mathbb{R})$, $V \in \mathcal{M}_{d,k}(\mathbb{R})$, $K \in \mathcal{S}_d(\mathbb{R})$, and $(W_t)_{t \in [0, T]} = (W_t^1, \dots, W_t^k)_{t \in [0, T]}^\top$ is a k -dimensional standard Brownian motion (with independent coordinates) for some $k \in \mathbb{N}^*$. In these dynamics, the matrix R steers the deterministic part of the process, $\bar{S}$ represents the unconditional long-term expectation of $(S_t)_{t \in [0, T]}$, and V drives the dispersion (for what follows, we introduce $\Sigma = VV^\top$ the covariation matrix of the process). The matrix K represents the linear permanent impact the agent has on the prices.² More precisely, since

$$\begin{aligned} d\tilde{S}_t &= dS_t + Kv_t dt = R(\bar{S} - S_t)dt + Kv_t dt + VdW_t \\ &= R(\bar{S} + K(q_t - q_0) - \tilde{S}_t)dt + Kv_t dt + VdW_t, \end{aligned}$$

trading impacts both current market prices and long-term expectations.

Ornstein-Uhlenbeck processes are well suited when prices exhibit mean reversion and/or when there exist one or several linear combinations of asset prices that are stationary. In the latter case, we say that the assets involved in the linear combinations are cointegrated (a situation often encountered in statistical arbitrage). For more details on cointegration in continuous time, we refer to [Comte \(1999\)](#).

Finally, the process $(\tilde{X}_t)_{t \in [0, T]}$ modelling the trader's cash account has the dynamics

$$d\tilde{X}_t = -v_t^\top \tilde{S}_t dt - L(v_t) dt,$$

with $\tilde{X}_0 \in \mathbb{R}$ given, where $L : \mathbb{R}^d \rightarrow \mathbb{R}_+$ is a function representing the temporary market impact of trades and/or the execution costs incurred by the trader; see [Guéant \(2016\)](#) for an introduction to this type of models. In this chapter, we only consider the case where L is a positive-definite quadratic form, i.e.³

$$L(v) = v^\top \eta v \quad \text{with} \quad \eta \in \mathcal{S}_d^{++}(\mathbb{R}).$$

The trader aims at maximising the expected utility of her wealth at the end of the trading window $[0, T]$. This wealth is the sum of the amount $\tilde{X}_T$ on the cash account at time T and the value of the remaining inventory evaluated here at $q_T^\top \tilde{S}_T - \tilde{\ell}(q_T)$, where the discount term $\tilde{\ell}(q_T)$ applied to the Mark-to-Market (MtM) value of the remaining assets ($q_T^\top \tilde{S}_T$) penalises any non-zero terminal position. In this chapter, we

² It is assumed symmetric to avoid price manipulation.

³ The subset of positive-definite and positive semi-definite matrices of $\mathcal{S}_d(\mathbb{R})$ are respectively denoted by $\mathcal{S}_d^{++}(\mathbb{R})$ and $\mathcal{S}_d^+(\mathbb{R})$.

only consider the case where $\tilde{\ell}$ is a positive-definite quadratic form, i.e. $\tilde{\ell}(q) = q^\top \tilde{\Gamma} q$ with $\tilde{\Gamma} \in \mathcal{S}_d^{++}(\mathbb{R})$ (see below for a stronger assumption on $\tilde{\Gamma}$).

To define the set of admissible controls $\mathcal{A}$, we first introduce a notion of “linear growth” relevant in our context.

Definition 2.1.1 Let $t \in [0, T]$. An $\mathbb{R}^d$ -valued, $\mathbb{F}$ -adapted process $(\zeta_s)_{s \in [t, T]}$ is said to satisfy a linear growth condition on $[t, T]$ with respect to $(S_s)_{s \in [t, T]}$ if there exists a constant $C_{t, T} > 0$ such that for all $s \in [t, T]$,

$$\|\zeta_s\| \leq C_{t, T} \left(1 + \sup_{\tau \in [t, s]} \|S_\tau\| \right)$$

almost surely.⁴

□

We then define for all $t \in [0, T]$:

$$\mathcal{A}_t = \left\{ (v_s)_{s \in [t, T]}, \mathbb{R}^d\text{-valued, } \mathbb{F}\text{-adapted, satisfying} \right. \\ \left. \text{a linear growth condition with respect to } (S_s)_{s \in [t, T]} \right\},$$

and take $\mathcal{A} := \mathcal{A}_0$.⁵

Mathematically, the trader therefore wants to solve the dynamic optimisation problem

$$\sup_{v \in \mathcal{A}} \mathbb{E} \left[-\exp \left(-\gamma \left(\tilde{X}_T + q_T^\top \tilde{S}_T - \tilde{\ell}(q_T) \right) \right) \right], \quad (2.1)$$

where $\gamma > 0$ is the absolute risk aversion parameter of the trader.

Notice that

$$\begin{aligned} \tilde{X}_T + q_T^\top \tilde{S}_T - \tilde{\ell}(q_T) &= \tilde{X}_0 + q_0^\top \tilde{S}_0 + \int_0^T q_t^\top d\tilde{S}_t - \int_0^T L(v_t) dt - \tilde{\ell}(q_T) \\ &= X_0 + q_0^\top S_0 + \int_0^T q_t^\top dS_t + \int_0^T q_t^\top K v_t dt - \int_0^T L(v_t) dt - \tilde{\ell}(q_T) \\ &= X_T + q_T^\top S_T - \tilde{\ell}(q_T) + \frac{1}{2} q_T^\top K q_T - \frac{1}{2} q_0^\top K q_0, \end{aligned}$$

where $X_0 = \tilde{X}_0$ and the process $(X_t)_{t \in [0, T]}$ has dynamics

$$dX_t = -v_t^\top S_t dt - L(v_t) dt.$$

Let us now define the penalty function $\ell : \mathbb{R}^d \rightarrow \mathbb{R}$ by:

⁴ In this chapter, $\|\cdot\|$ denotes a fixed norm on $\mathbb{R}^d$ (for instance, the Euclidean norm).

⁵ We restrict our analysis to linear growth strategies for mathematical convenience, but we expect the candidate control to be optimal among a larger class of processes.

$$\ell(q) = \tilde{\ell}(q) - \frac{1}{2}q^\top Kq = q^\top \tilde{\Gamma}q - \frac{1}{2}q^\top Kq \quad \forall q \in \mathbb{R}^d.$$

In what follows, we assume that ℓ is a positive semi-definite quadratic form, i.e. $\Gamma = \tilde{\Gamma} - \frac{1}{2}K \in \mathcal{S}_d^+(\mathbb{R})$.

Remark 2.1.1 The assumption on $\tilde{\Gamma}$ is not restrictive as in practice, $\tilde{\Gamma}$ (and therefore Γ) is chosen arbitrarily large to enforce liquidation. $\square$

It is then straightforward to see that Problem (2.1) is equivalent to the following problem:

$$\sup_{v \in \mathcal{A}} \mathbb{E} \left[-\exp \left(-\gamma \left(X_T + q_T^\top S_T - \ell(q_T) \right) \right) \right]. \quad (2.2)$$

It is natural to use the tools of stochastic optimal control to solve the above dynamic optimisation problem. Let us define the value function of the problem $u : [0, T] \times \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ as

$$u(t, x, q, S) = \sup_{v \in \mathcal{A}_t} \mathbb{E} \left[-\exp \left(-\gamma \left(X_T^{t,x,S,v} + (q_T^{t,q,v})^\top S_T^{t,S} - \ell(q_T^{t,q,v}) \right) \right) \right],$$

where, for $(t, x, q, S) \in [0, T] \times \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^d$ and $v \in \mathcal{A}_t$, the processes $(q_s^{t,q,v})_{s \in [t, T]}$, $(S_s^{t,S})_{s \in [t, T]}$, and $(X_s^{t,x,S,v})_{s \in [t, T]}$ have respective dynamics

$$\begin{aligned} dq_s^{t,q,v} &= v_s ds, \\ dS_s^{t,S} &= R(\bar{S} - S_s^{t,S}) ds + V dW_s, \\ dX_s^{t,x,S,v} &= -v_s^\top S_s^{t,S} ds - L(v_s) ds, \end{aligned}$$

with $S_t^{t,S} = S$, $q_t^{t,q,v} = q$, and $X_t^{t,x,S,v} = x$.

2.1.2 Hamilton-Jacobi-Bellman equation

The HJB equation associated with Problem (2.2) is given by⁶

$$\begin{aligned} 0 = & \partial_t w(t, x, q, S) + \sup_{v \in \mathbb{R}^d} \left(-(v^\top S + L(v)) \partial_x w(t, x, q, S) + v^\top \nabla_q w(t, x, q, S) \right) \\ & + (\bar{S} - S)^\top R^\top \nabla_S w(t, x, q, S) + \frac{1}{2} \text{Tr} \left(\Sigma D_{SS}^2 w(t, x, q, S) \right), \end{aligned} \quad (2.3)$$

for all $(t, x, q, S) \in [0, T] \times \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^d$ with the terminal condition

$$w(T, x, q, S) = -\exp \left(-\gamma (x + q^\top S - \ell(q)) \right) \quad \forall (x, q, S) \in \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^d. \quad (2.4)$$

⁶ u will be solution to that equation, but as we do not know it yet, we write the equation with an unknown function w .

In order to study (2.3), we use the following ansatz:

$$\begin{aligned} w(t, x, q, S) &= -\exp(-\gamma(x + q^\top S + \theta(t, q, S))) \\ \forall(t, x, q, S) &\in [0, T] \times \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^d. \end{aligned} \quad (2.5)$$

The interest of this ansatz is based on the following proposition:

Proposition 2.1.1 Let $\tau < T$. If there exists $\theta \in C^{1,1,2}([\tau, T] \times \mathbb{R}^d \times \mathbb{R}^d, \mathbb{R})$ solution to

$$\begin{aligned} 0 &= \partial_t \theta(t, q, S) + \sup_{v \in \mathbb{R}^d} (v^\top \nabla_q \theta(t, q, S) - L(v)) + \frac{1}{2} \text{Tr} \left(\Sigma D_{SS}^2 \theta(t, q, S) \right) \\ &\quad - \frac{\gamma}{2} (q + \nabla_S \theta(t, q, S))^\top \Sigma (q + \nabla_S \theta(t, q, S)) + (\bar{S} - S)^\top R^\top (\nabla_S \theta(t, q, S) + q) \end{aligned} \quad (2.6)$$

on $[\tau, T] \times \mathbb{R}^d \times \mathbb{R}^d$, with terminal condition

$$\theta(T, q, S) = -\ell(q) \quad \forall(q, S) \in \mathbb{R}^d \times \mathbb{R}^d, \quad (2.7)$$

then the function $w : [\tau, T] \times \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ defined by

$$w(t, x, q, S) = -\exp(-\gamma(x + q^\top S + \theta(t, q, S))) \quad \forall(t, x, q, S) \in [\tau, T] \times \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^d$$

is a solution to (2.3) on $[\tau, T] \times \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^d$ with terminal condition (2.4). $\square$

Proof 2.1.1 Let $\theta \in C^{1,1,2}([\tau, T] \times \mathbb{R}^d \times \mathbb{R}^d, \mathbb{R})$ be a solution to (2.6) on $[\tau, T] \times \mathbb{R}^d \times \mathbb{R}^d$ with terminal condition (2.7), then we have for all $(t, x, q, S) \in [\tau, T] \times \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^d$:

$$\begin{aligned} &\partial_t w(t, x, q, S) + \sup_{v \in \mathbb{R}^d} (-(v^\top S + L(v)) \partial_x w(t, x, q, S) + v^\top \nabla_q w(t, x, q, S)) \\ &\quad + (\bar{S} - S)^\top R^\top \nabla_S w(t, x, q, S) + \frac{1}{2} \text{Tr} \left(\Sigma D_{SS}^2 w(t, x, q, S) \right) \\ &= \sup_{v \in \mathbb{R}^d} (\gamma(v^\top S + L(v)) w(t, x, q, S) - \gamma v^\top (\nabla_q \theta(t, q, S) + S) w(t, x, q, S)) \\ &\quad - \gamma \partial_t \theta(t, q, S) w(t, x, q, S) + \frac{\gamma^2}{2} \text{Tr} (\Sigma (q + \nabla_S \theta(t, q, S)) (q + \nabla_S \theta(t, q, S))^\top w(t, x, q, S)) \\ &\quad - \gamma (\bar{S} - S)^\top R^\top (\nabla_S \theta(t, q, S) + q) w(t, x, q, S) - \frac{1}{2} \text{Tr} \left(\gamma \Sigma D_{SS}^2 \theta(t, q, S) w(t, x, q, S) \right) \\ &= -\gamma w(t, x, q, S) \left(\partial_t \theta(t, q, S) + \sup_{v \in \mathbb{R}^d} (v^\top \nabla_q \theta(t, q, S) - L(v)) + \frac{1}{2} \text{Tr} \left(\Sigma D_{SS}^2 \theta(t, q, S) \right) \right. \\ &\quad \left. - \frac{\gamma}{2} (q + \nabla_S \theta(t, q, S))^\top \Sigma (q + \nabla_S \theta(t, q, S)) + (\bar{S} - S)^\top R^\top (\nabla_S \theta(t, q, S) + q) \right) \\ &= 0. \end{aligned}$$

As it is straightforward to verify that w satisfies the terminal condition (2.4), the result is proved. $\square$

The above result does not rely on the quadratic assumptions for L and ℓ . In the quadratic case we consider in this chapter, θ can be found in almost closed form. To prove this point, the first thing to notice is that the Legendre-Fenchel transform of L writes

$$H : p \in \mathbb{R}^d \mapsto \sup_{v \in \mathbb{R}^d} v^\top p - L(v) = \sup_{v \in \mathbb{R}^d} v^\top p - v^\top \eta v = \frac{1}{4} p^\top \eta^{-1} p, \quad (2.8)$$

as the supremum is reached at $v^* = \frac{1}{2} \eta^{-1} p$.

Consequently, we get the following HJB equation for θ :

$$\begin{aligned} 0 = & \partial_t \theta(t, q, S) + \frac{1}{4} \nabla_q \theta(t, q, S)^\top \eta^{-1} \nabla_q \theta(t, q, S) + \frac{1}{2} \text{Tr} \left(\Sigma D_{SS}^2 \theta(t, q, S) \right) \\ & - \frac{\gamma}{2} (q + \nabla_S \theta(t, q, S))^\top \Sigma (q + \nabla_S \theta(t, q, S)) + (\bar{S} - S)^\top R^\top (\nabla_S \theta(t, q, S) + q), \end{aligned} \quad (2.9)$$

with terminal condition

$$\theta(T, q, S) = -q^\top \Gamma q \quad \forall (q, S) \in \mathbb{R}^d \times \mathbb{R}^d. \quad (2.10)$$

To further study (2.9), we introduce a second ansatz and look for a solution θ of the following form:

$$\theta(t, q, S) = q^\top A(t) q + q^\top B(t) S + S^\top C(t) S + D(t)^\top q + E(t)^\top S + F(t) \quad (2.11)$$

for all $(t, q, S) \in [0, T] \times \mathbb{R}^d \times \mathbb{R}^d$, or equivalently

$$\theta(t, q, S) = \begin{pmatrix} q \\ S \end{pmatrix}^\top P(t) \begin{pmatrix} q \\ S \end{pmatrix} + \begin{pmatrix} D(t) \\ E(t) \end{pmatrix}^\top \begin{pmatrix} q \\ S \end{pmatrix} + F(t),$$

where $P : [0, T] \rightarrow \mathcal{S}_{2d}(\mathbb{R})$ is defined as

$$P(t) = \begin{pmatrix} A(t) & \frac{1}{2} B(t) \\ \frac{1}{2} B(t)^\top & C(t) \end{pmatrix}.$$

The interest of this ansatz is stated in the following proposition.

Proposition 2.1.2 Let $\tau < T$. Assume there exist $A \in C^1([\tau, T], \mathcal{S}_d(\mathbb{R}))$, $B \in C^1([\tau, T], \mathcal{M}_d(\mathbb{R}))$, $C \in C^1([\tau, T], \mathcal{S}_d(\mathbb{R}))$, $D \in C^1([\tau, T], \mathbb{R}^d)$, $E \in C^1([\tau, T], \mathbb{R}^d)$, $F \in C^1([\tau, T], \mathbb{R})$ satisfying the system of ODEs

$$\begin{cases} A'(t) = \frac{\gamma}{2} (B(t) + I_d) \Sigma (B(t)^\top + I_d) - A(t) \eta^{-1} A(t) \\ B'(t) = (B(t) + I_d) R + 2\gamma (B(t) + I_d) \Sigma C(t) - A(t) \eta^{-1} B(t) \\ C'(t) = R^\top C(t) + C(t) R + 2\gamma C(t) \Sigma C(t) - \frac{1}{4} B(t)^\top \eta^{-1} B(t) \\ D'(t) = -(B(t) + I_d) R \bar{S} + \gamma (B(t) + I_d) \Sigma E(t) - A(t) \eta^{-1} D(t) \\ E'(t) = -2C(t) R \bar{S} + R^\top E(t) + 2\gamma C(t) \Sigma E(t) - \frac{1}{2} B(t)^\top \eta^{-1} D(t) \\ F'(t) = -\bar{S}^\top R^\top E(t) - \text{Tr}(\Sigma C(t)) + \frac{\gamma}{2} E(t)^\top \Sigma E(t) - \frac{1}{4} D(t)^\top \eta^{-1} D(t), \end{cases} \quad (2.12)$$

where I_d denotes the identity matrix in $\mathcal{M}_d(\mathbb{R})$, with terminal conditions

$$A(T) = -\Gamma, \quad B(T) = C(T) = D(T) = E(T) = F(T) = 0. \quad (2.13)$$

Then θ defined by (2.11) satisfies (2.9) on $[\tau, T] \times \mathbb{R}^d \times \mathbb{R}^d$ with terminal condition (2.10). $\square$

Proof 2.1.2 Let $A \in C^1([\tau, T], \mathcal{S}_d(\mathbb{R}))$, $B \in C^1([\tau, T], \mathcal{M}_d(\mathbb{R}))$, $C \in C^1([\tau, T], \mathcal{S}_d(\mathbb{R}))$, $D \in C^1([\tau, T], \mathbb{R}^d)$, $E \in C^1([\tau, T], \mathbb{R}^d)$, $F \in C^1([\tau, T], \mathbb{R})$ verify (2.12) on $[\tau, T]$ with terminal condition (2.13). Consider $\theta : [\tau, T] \times \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ defined by (2.11). Then we obtain for all $(t, q, S) \in [\tau, T] \times \mathbb{R}^d \times \mathbb{R}^d$:

$$\begin{aligned} & \partial_t \theta(t, q, S) + \frac{1}{4} \nabla_q \theta(t, q, S)^\top \eta^{-1} \nabla_q \theta(t, q, S) + \frac{1}{2} \text{Tr} \left(\Sigma D_{SS}^2 \theta(t, q, S) \right) \\ & - \frac{\gamma}{2} (q + \nabla_S \theta(t, q, S))^\top \Sigma (q + \nabla_S \theta(t, q, S)) + (\bar{S} - S)^\top R^\top (\nabla_S \theta(t, q, S) + q), \\ = & q^\top A'(t) q + q^\top B'(t) S + S^\top C'(t) S + D'(t)^\top q + E'(t)^\top S + F'(t) \\ & + q^\top A(t) \eta^{-1} A(t) q + q^\top A(t) \eta^{-1} B(t) S + \frac{1}{4} S^\top B(t)^\top \eta^{-1} B(t) S \\ & + D(t)^\top \eta^{-1} A(t) q + \frac{1}{2} (D(t))^\top \eta^{-1} B(t) S + \frac{1}{4} D(t)^\top \eta^{-1} D(t) \\ & + \text{Tr}(\Sigma C(t)) - \frac{\gamma}{2} (q + B(t)^\top q + 2C(t)S + E(t))^\top \Sigma (q + B(t)^\top q + 2C(t)S + E(t)) \\ & + \bar{S}^\top R^\top q + \bar{S}^\top R^\top (B(t)^\top q + 2C(t)S + E(t)) \\ & - S^\top R^\top q - S^\top R^\top (B(t)^\top q + 2C(t)S + E(t)) \\ = & q^\top \left(A'(t) - \frac{\gamma}{2} (B(t) + I_d) \Sigma (B(t)^\top + I_d) + \frac{1}{4} (2A(t)) \eta^{-1} (2A(t)) \right) q \\ & + q^\top \left(B'(t) - (I_d + B(t)) R - 2\gamma (B(t) + I_d) \Sigma C(t) + A(t) \eta^{-1} B(t) \right) S \\ & + S^\top \left(C'(t) - R^\top C(t) - C(t) R - 2\gamma C(t) \Sigma C(t) + \frac{1}{4} B(t)^\top \eta^{-1} B(t) \right) S \\ & + \left(D'(t) + (B(t) + I_d) R \bar{S} - \gamma (B(t) + I_d) \Sigma E(t) + A(t) \eta^{-1} D(t) \right)^\top q \\ & + \left(E'(t) + 2C(t) R \bar{S} - R^\top E(t) - 2\gamma C(t) \Sigma E(t) + \frac{1}{2} B(t)^\top \eta^{-1} D(t) \right)^\top S \\ & + \left(F'(t) + \bar{S}^\top R^\top E(t) + \text{Tr}(\Sigma C(t)) - \frac{\gamma}{2} E(t)^\top \Sigma E(t) + \frac{1}{4} D(t)^\top \eta^{-1} D(t) \right) \\ = & 0. \end{aligned}$$

As it is straightforward to verify that θ satisfies the terminal condition (2.10), the result is proved. $\square$

Remark 2.1.2 Two remarks can be made on the system of ODEs (2.12):

- This system of ODEs can clearly be decomposed into three groups of equations: the first three ODEs for A , B and C are independent of the others and can be solved as a first step; once we know A , B , and C we can solve the linear ODEs for D and E , and finally F can be obtained with a simple integration;
- When $R = 0$ (i.e. in the case where the prices S of the d assets are correlated arithmetic Brownian motions), there is a trivial solution to the last five equations which is $B = C = D = E = F = 0$. The function A can then be found using classical techniques (as shown in Subsection 1.1.1). $\square$

It is noteworthy that the first system, i.e.

$$\begin{cases} A'(t) = \frac{\gamma}{2}(B(t) + I_d)\Sigma(B(t)^\top + I_d) - A(t)\eta^{-1}A(t) \\ B'(t) = (B(t) + I_d)R + 2\gamma(B(t) + I_d)\Sigma C(t) - A(t)\eta^{-1}B(t) \\ C'(t) = R^\top C(t) + C(t)R + 2\gamma C(t)\Sigma C(t) - \frac{1}{4}B(t)^\top \eta^{-1}B(t) \end{cases} \quad (2.14)$$

boils down to the following Matrix Riccati ODE in $P = \begin{pmatrix} A & \frac{1}{2}B \\ \frac{1}{2}B & C \end{pmatrix}$:

$$P'(t) = Q + Y^\top P(t) + P(t)Y + P(t)UP(t), \quad (2.15)$$

where

$$Q = \frac{1}{2} \begin{pmatrix} \gamma \Sigma R \\ R^\top 0 \end{pmatrix} \in \mathcal{S}_{2d}(\mathbb{R}), \quad Y = \begin{pmatrix} 0 & 0 \\ \gamma \Sigma R \end{pmatrix} \in \mathcal{M}_{2d}(\mathbb{R}), \quad U = \begin{pmatrix} -\eta^{-1} & 0 \\ 0 & 2\gamma \Sigma \end{pmatrix} \in \mathcal{S}_{2d}(\mathbb{R}),$$

and the terminal condition writes

$$P(T) = \begin{pmatrix} -\Gamma & 0 \\ 0 & 0 \end{pmatrix} \in \mathcal{S}_{2d}(\mathbb{R}). \quad (2.16)$$

When compared to the Matrix Riccati ODEs arising in the linear-quadratic optimal control literature, the distinctive aspect of our equation is that the matrix U characterizing the quadratic term in the Riccati equation has both positive and negative eigenvalues. In particular, we cannot rely on existing results (see for instance Theorem 3.5 of Freiling (2002)) to prove that there exists a solution to (2.15) with terminal condition (2.16). In this chapter, we address the existence of a solution by using *a priori* estimates for the value function.⁷

Regarding the set of equations (2.14), there exists a unique local solution by Cauchy-Lipschitz theorem. In the following section, we therefore first state a verification theorem that solves the problem on an interval $[\tau, T]$, and use that very result to address global existence and uniqueness of a solution on $[0, T]$.

⁷ Surprisingly maybe, the inequalities we derive do not seem to derive in a direct manner from a purely analytic argument such as the classical comparison principle for matrix Riccati equations – see Theorem 3.4 of Freiling (2002).

2.1.3 Main mathematical results

Theorem 2.1.1 Let $\tau < T$. Let the functions $A \in C^1([\tau, T], \mathcal{S}_d(\mathbb{R}))$, $B \in C^1([\tau, T], \mathcal{M}_d(\mathbb{R}))$, $C \in C^1([\tau, T], \mathcal{S}_d(\mathbb{R}))$, $D \in C^1([\tau, T], \mathbb{R}^d)$, $E \in C^1([\tau, T], \mathbb{R}^d)$, $F \in C^1([\tau, T], \mathbb{R})$ be a solution to the system (2.12) on $[\tau, T)$ with terminal condition (2.13), and consider the function θ defined by (2.11) and the associated function w defined by (2.5).

Then for all $(t, x, q, S) \in [\tau, T] \times \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^d$ and $v = (v_s)_{s \in [t, T]} \in \mathcal{A}_t$, we have

$$\mathbb{E} \left[-e^{-\gamma \left(X_T^{t, x, S, v} + (q_T^{t, q, v})^\top S_T^{t, S} - \ell(q_T^{t, q, v}) \right)} \right] \leq w(t, x, q, S). \quad (2.17)$$

Moreover, equality is obtained in (2.17) by taking the optimal control $(v_s^*)_{s \in [t, T]} \in \mathcal{A}_t$ given by the closed-loop feedback formula

$$v_s^* = \frac{1}{2} \eta^{-1} \left(2A(s)q_s^{t, q, v} + B(s)S_s^{t, S} + D(s) \right). \quad (2.18)$$

In particular, $w = u$ on $[\tau, T] \times \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^d$. □

Proof 2.1.3 Let $t \in [\tau, T)$, we first prove that $(v_s^*)_{s \in [t, T]} \in \mathcal{A}_t$ (i.e., $(v_s^*)_{s \in [t, T]}$ is well-defined and admissible). Let us consider the Cauchy initial value problem

$$\forall s \in [t, T], \quad \frac{d\tilde{q}_s}{ds} = \frac{1}{2} \eta^{-1} \left(2A(s)\tilde{q}_s + B(s)S_s^{t, S} + D(s) \right), \quad \tilde{q}_t = q.$$

The unique solution of that Cauchy problem writes

$$\tilde{q}_s = \exp \left(\int_t^s \phi(\varrho) d\varrho \right) \left(q + \int_t^s \psi \left(\varrho, S_\varrho^{t, S} \right) \exp \left(- \int_t^\varrho \phi(\varsigma) d\varsigma \right) d\varrho \right),$$

where ϕ and ψ are defined by

$$\begin{aligned} \phi : s \in [t, T] &\mapsto \eta^{-1} A(s), \\ \psi : (s, S) \in [t, T] \times \mathbb{R}^d &\mapsto \frac{1}{2} \eta^{-1} (B(s)S + D(s)). \end{aligned}$$

Then v^* can be written as

$$\begin{aligned} v_s^* &= \frac{d\tilde{q}_s}{ds} \\ &= \phi(s) \exp \left(\int_t^s \phi(\varrho) d\varrho \right) \left(q + \int_t^s \psi \left(\varrho, S_\varrho^{t, S} \right) \exp \left(- \int_t^\varrho \phi(\varsigma) d\varsigma \right) d\varrho \right) + \psi \left(s, S_s^{t, S} \right). \end{aligned}$$

We see from the definition of ϕ and the affine form of ψ in S that v^* satisfies a linear growth condition, and is therefore in $\mathcal{A}_t$.

Let us consider $(t, x, q, S) \in [\tau, T] \times \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^d$ and $v = (v_s)_{s \in [t, T]} \in \mathcal{A}_t$. We now prove that

$$\mathbb{E} \left[w \left(T, X_T^{t,x,S,v}, q_T^{t,q,v}, S_T^{t,S} \right) \right] \leq w(t, x, q, S).$$

We use the following notations for readability:

$$\begin{aligned} \forall s \in [t, T], \quad w(s, X_s^{t,x,S,v}, q_s^{t,q,v}, S_s^{t,S}) &= w_s^{t,x,q,S,v}, \\ \forall s \in [t, T], \quad \theta(s, q_s^{t,q,v}, S_s^{t,S}) &= \theta_s^{t,q,S,v}. \end{aligned}$$

By Itô's formula, we have $\forall s \in [\tau, T]$

$$dw_s^{t,x,q,S,v} = \mathcal{L}^v w_s^{t,x,q,S,v} ds + \left(\nabla_S w_s^{t,x,q,S,v} \right)^\top V dW_s,$$

where

$$\begin{aligned} \mathcal{L}^v w_s^{t,x,q,S,v} &= \partial_t w_s^{t,x,q,S,v} - (v^\top S + v^\top \eta v) \partial_x w_s^{t,x,q,S,v} + v^\top \nabla_q w_s^{t,x,q,S,v} \\ &\quad + (\bar{S} - S)^\top R^\top \nabla_S w_s^{t,x,q,S,v} + \frac{1}{2} \text{Tr} \left(\Sigma D_{SS}^2 w_s^{t,x,q,S,v} \right). \end{aligned}$$

From (2.5) and (2.11) we have

$$\begin{aligned} \nabla_S w_s^{t,x,q,S,v} &= -\gamma w_s^{t,x,q,S,v} \left(q_s^{t,q,v} + \nabla_S \theta_s^{t,q,S,v} \right) \\ &= -\gamma w_s^{t,x,q,S,v} \left(q_s^{t,q,v} + B(s)^\top q_s^{t,q,v} + 2C(s) S_s^{t,S} + E(s) \right). \end{aligned}$$

We define $\forall s \in [t, T]$,

$$\begin{aligned} \kappa_s^{q,S,v} &= -\gamma \left(q_s^{t,q,v} + B(s)^\top q_s^{t,q,v} + 2C(s) S_s^{t,S} + E(s) \right), \\ \xi_{t,s}^{q,S,v} &= \exp \left(\int_t^s \kappa_\varrho^{q,S,v} V dW_\varrho - \frac{1}{2} \int_t^s \kappa_\varrho^{q,S,v} \Sigma \kappa_\varrho^{q,S,v} d\varrho \right). \end{aligned}$$

We then have

$$d \left(w_s^{t,x,q,S,v} \left(\xi_{t,s}^{q,S,v} \right)^{-1} \right) = \left(\xi_{t,s}^{q,S,v} \right)^{-1} \mathcal{L}^v w_s^{t,x,q,S,v} ds.$$

By definition of w , $\mathcal{L}^v w_s^{t,x,q,S,v} \leq 0$. Moreover, equality holds for the control reaching the supremum in (2.8). This supremum is reached for the unique value

$$\begin{aligned} v_s &= \frac{1}{2} \eta^{-1} \nabla_q \theta_s^{t,q,S,v} \\ &= \frac{1}{2} \eta^{-1} \left(2A(s) q_s^{t,q,v} + B(s) S_s^{t,S} + D(s) \right), \end{aligned}$$

As a consequence, $\left(w_s^{t, x, q, S, v} \left(\xi_{t, s}^{q, S, v} \right)^{-1} \right)_{s \in [t, T]}$ is nonincreasing and therefore

$$w \left(T, X_T^{t, x, S, v}, q_T^{t, q, v}, S_T^{t, S} \right) \leq w(t, x, q, S) \xi_{t, T}^{q, S, v},$$

with equality when $(v_s)_{s \in [t, T]} = (v_s^*)_{s \in [t, T]}$. Taking expectations we get

$$\mathbb{E} \left[w \left(T, X_T^{t, x, S, v}, q_T^{t, q, v}, S_T^{t, S} \right) \right] \leq w(t, x, q, S) \mathbb{E} \left[\xi_{t, T}^{q, S, v} \right].$$

We proceed to prove that $\mathbb{E} \left[\xi_{t, T}^{q, S, v} \right]$ is equal to 1. To do so, we use that $\xi_{t, t}^{q, S, v} = 1$ and prove that $(\xi_{t, s}^{q, S, v})_{s \in [t, T]}$ is a martingale under $(\mathbb{P}; \mathbb{F} = (\mathcal{F}_s)_{s \in [t, T]})$.

We know that $(q_s^{t, q, v})_{s \in [t, T]}$ satisfies a linear growth condition with respect to $(S_s^{t, S})_{s \in [t, T]}$ since v is an admissible control. Given the form of κ , there exists a constant C such that, almost surely, for all $s < \tau \leq T$,

$$\sup_{r \in [s, \tau]} \|\kappa_r^{q, S, v}\|^2 \leq C \left(1 + \sup_{r \in [s, \tau]} \|W_r - W_s\|^2 \right).$$

By using classical properties of the Brownian motion, we prove that

$$\exists \epsilon > 0, \forall s \in [t, T], \quad \mathbb{E} \left[\exp \left(\frac{1}{2} \int_s^{(s+\epsilon) \wedge T} \left(\kappa_\varrho^{q, S, v} \right)^\top \Sigma \kappa_\varrho^{q, S, v} d\varrho \right) \right] < +\infty.$$

Using a classical trick due to Beneš (see [Karatzas and Shreve \(2014\)](#), Chapter 5), we see that $(\xi_{t, s}^{q, S, v})_{s \in [t, T]}$ is a martingale under $(\mathbb{P}; \mathbb{F} = (\mathcal{F}_s)_{s \in [t, T]})$.

We obtain

$$\mathbb{E} \left[w \left(T, X_T^{t, x, S, v}, q_T^{t, q, v}, S_T^{t, S} \right) \right] \leq w(t, x, q, S),$$

with equality when $(v_s)_{s \in [t, T]} = (v_s^*)_{s \in [t, T]}$.

We conclude that

$$\begin{aligned} u(t, x, q, S) &= \sup_{(v_s)_{s \in [t, T]} \in \mathcal{A}_t} \mathbb{E} \left[-\exp \left(-\gamma \left(X_T^{t, x, S, v} + \left(q_T^{t, q, v} \right)^\top S_T^{t, S} - \ell \left(q_T^{t, q, v} \right) \right) \right) \right] \\ &= \mathbb{E} \left[-\exp \left(-\gamma \left(X_T^{t, x, S, v^*} + \left(q_T^{t, q, v^*} \right)^\top S_T^{t, S} - \ell \left(q_T^{t, q, v^*} \right) \right) \right) \right] \\ &= w(t, x, q, S). \end{aligned}$$

We will next proceed to prove existence and uniqueness of a solution to the system of ODEs (2.12) on $[0, T]$ with terminal condition (2.13), or equivalently to (2.15) with terminal condition (2.16).⁸

Theorem 2.1.2 There exists a unique solution $A \in C^1([0, T], \mathcal{S}_d(\mathbb{R}))$, $B \in C^1([0, T], \mathcal{M}_d(\mathbb{R}))$, $C \in C^1([0, T], \mathcal{S}_d(\mathbb{R}))$, $D \in C^1([0, T], \mathbb{R}^d)$, $E \in C^1([0, T], \mathbb{R}^d)$, $F \in C^1([0, T], \mathbb{R})$ to the system of ODEs (2.12) on $[0, T]$ with terminal condition (2.13). $\square$

Proof 2.1.4 To prove Theorem 2.1.2, it is enough, as explained in Remark 2.1.2, to show existence and uniqueness for $A \in C^1([0, T], \mathcal{S}_d(\mathbb{R}))$, $B \in C^1([0, T], \mathcal{M}_d(\mathbb{R}))$, and $C \in C^1([0, T], \mathcal{S}_d(\mathbb{R}))$, or equivalently, existence and uniqueness on $[0, T]$ of a solution $P \in C^1([0, T], \mathcal{S}_{2d}(\mathbb{R}))$ to (2.15) with terminal condition (2.16).

By Cauchy-Lipschitz theorem, there exists a unique maximal solution⁹ (A, B, C) to the system of ODEs (2.14) with terminal condition (2.13) defined on an open interval $(t_{\min}, t_{\max}) \ni T$, and by Theorem 2.1.1, the associated function w defined by (2.5) corresponds to the value function of the problem restricted to $[\tau, T]$ for all $\tau \in (t_{\min}, T)$.

To prove our result, we need to show that $t_{\min} = -\infty$. For that purpose, our strategy consists in proving that the matrix $P(t) = \begin{pmatrix} A(t) & \frac{1}{2}B(t) \\ \frac{1}{2}B(t)^\top & C(t) \end{pmatrix}$ cannot blow up in finite time. This is proved by, first, finding (thanks to the control problem) lower and upper bounds for the function θ which are, like θ , polynomials of degree at most 2 in (q, S) , and then converting these bounds into bounds for $P(t)$ in the sense of the natural order on symmetric matrices.¹⁰

By contradiction, let us assume that $t_{\min} \in (-\infty, T)$ and let $\tau \in (t_{\min}, T)$. Let $(t, x, q, S) \in [\tau, T] \times \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^d$ and let us consider the sub-optimal strategy $v = (0)_{s \in [t, T]} \in \mathcal{A}_t$ for which $\forall s \in [t, T], q_s^{t, q, v} = q$ and

$$\begin{aligned} & \mathbb{E} \left[-\exp \left(-\gamma \left(X_T^{t, x, S, v} + (q_T^{t, q, v})^\top S_T^{t, S} - \ell(q_T^{t, q, v}) \right) \right) \right] \\ &= \mathbb{E} \left[-\exp \left(-\gamma \left(x + q^\top S + q^\top \left(S_T^{t, S} - S \right) - q^\top \Gamma q \right) \right) \right]. \end{aligned}$$

Since $(S_s^{t, S})_{s \in [t, T]}$ follows multivariate Ornstein-Uhlenbeck dynamics, we know that

$$S_T^{t, S} - S = \left(I - e^{-R(T-t)} \right) (\bar{S} - S) + \int_t^T e^{-R(T-s)} V dW_s.$$

Then $S_T^{t, S} - S \sim \mathcal{N} \left(\left(I - e^{-R(T-t)} \right) (\bar{S} - S), \Sigma_t \right)$, where the covariance matrix is defined by

$$\Sigma_t = \int_t^T e^{-R(T-s)} \Sigma e^{-R^\top(T-s)} ds.$$

⁸ The result in fact holds on $(-\infty, T]$ as the initial time plays no role.

⁹ The fact that A and C are symmetric is itself a consequence of Cauchy-Lipschitz theorem since (A, B, C) and $(A^\top, B, C^\top)$ are solution of the same Cauchy problem.

¹⁰ For $\underline{M}, \bar{M} \in \mathcal{S}_d(\mathbb{R})$, $\underline{M} \leq \bar{M}$ if and only if $\bar{M} - \underline{M} \in \mathcal{S}_d^+(\mathbb{R})$.

Then,

$$\begin{aligned} & \mathbb{E} \left[-\exp \left(-\gamma \left(X_T^{t,x,S,v} + (q_T^{t,q,v})^\top S_T^{t,S} - \ell(q_T^{t,q,v}) \right) \right) \right] \\ &= -\exp(-\gamma(x + q^\top S)) \exp \left(-\gamma \left(q^\top \left(I - e^{-R(T-t)} \right) (\bar{S} - S) - q^\top \Gamma q - \frac{1}{2} \gamma q^\top \Sigma_t q \right) \right). \end{aligned}$$

Since the strategy is sub-optimal, if we consider θ defined as in (2.11), we have by Theorem 2.1.1

$$\begin{aligned} & -\exp(-\gamma(x + q^\top S + \theta(t, q, S))) \\ & \geq -\exp(-\gamma(x + q^\top S)) \exp \left(-\gamma \left(q^\top \left(I - e^{-R(T-t)} \right) (\bar{S} - S) - q^\top \Gamma q - \frac{1}{2} \gamma q^\top \Sigma_t q \right) \right). \end{aligned}$$

We conclude that for all $(t, q, S) \in [\tau, T] \times \mathbb{R}^d \times \mathbb{R}^d$,

$$\begin{aligned} \theta(t, q, S) &= \begin{pmatrix} q \\ S \end{pmatrix}^\top P(t) \begin{pmatrix} q \\ S \end{pmatrix} + \begin{pmatrix} D(t) \\ E(t) \end{pmatrix}^\top \begin{pmatrix} q \\ S \end{pmatrix} + F(t) \\ &\geq \begin{pmatrix} q \\ S \end{pmatrix}^\top \begin{pmatrix} -\frac{\gamma}{2} \Sigma_t - \Gamma & -\frac{1}{2} \left(I - e^{-R(T-t)} \right) \\ -\frac{1}{2} \left(I - e^{-R^\top(T-t)} \right) & 0 \end{pmatrix} \begin{pmatrix} q \\ S \end{pmatrix} + \bar{S}^\top \left(I - e^{-R^\top(T-t)} \right) q. \end{aligned}$$

We therefore necessarily have, for the natural order on symmetric matrices,

$$\forall t \in [\tau, T], \quad P(t) \geq \begin{pmatrix} -\frac{\gamma}{2} \Sigma_t - \Gamma & -\frac{1}{2} \left(I - e^{-R(T-t)} \right) \\ -\frac{1}{2} \left(I - e^{-R^\top(T-t)} \right) & 0 \end{pmatrix}.$$

Now, for $(t, x, q, S) \in [\tau, T] \times \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^d$, we have

$$\begin{aligned} & \sup_{v \in \mathcal{A}_t} \mathbb{E} \left[-\exp \left(-\gamma \left(X_T^{t,x,S,v} + (q_T^{t,q,v})^\top S_T^{t,S} - (q_T^{t,q,v})^\top \Gamma q_T^{t,q,v} \right) \right) \right] \\ &= \sup_{v \in \mathcal{A}_t} \mathbb{E} \left[-\exp \left(-\gamma \left(x + q^\top S + \int_t^T (q_s^{t,q,v})^\top dS_s - \int_t^T L(v_s) ds - (q_T^{t,q,v})^\top \Gamma q_T^{t,q,v} \right) \right) \right] \\ &\leq \exp(-\gamma(x + q^\top S)) \sup_{v \in \mathcal{A}_t} \mathbb{E} \left[-\exp \left(-\gamma \left(\int_t^T (q_s^{t,q,v})^\top dS_s \right) \right) \right], \end{aligned}$$

If $(v_s)_{s \in [t, T]} \in \mathcal{A}_t$, it is straightforward to see that the process $(q_s^{t,q,v})_{s \in [t, T]}$ is in the space of admissible controls $\mathcal{A}_t^{\text{Merton}}$ defined by (2.20) in Appendix 2.2.2.2, in which we study a Merton problem that can be regarded as a limit case of ours when the execution costs and terminal costs vanish. Therefore,

$$\sup_{v \in \mathcal{A}_t} \mathbb{E} \left[-\exp \left(-\gamma \left(X_T^{t,x,S,v} + (q_T^{t,q,v})^\top S_T^{t,S} - (q_T^{t,q,v})^\top \Gamma q_T^{t,q,v} \right) \right) \right]$$

$$\leq \exp(-\gamma(x + q^\top S)) \sup_{(q_s)_{s \in [t, T]} \in \mathcal{A}_t^{\text{Merton}}} \mathbb{E} \left[-\exp \left(-\gamma \left(\int_t^T q_s^\top dS_s \right) \right) \right]. \quad (2.19)$$

As shown in Appendix 2.2.2.2, inequality (2.19) writes

$$-\exp(-\gamma(x + q^\top S + \theta(t, q, S))) \leq -\exp(-\gamma(x + q^\top S + \hat{\theta}(t, S))),$$

where

$$\hat{\theta}(t, S) = S^\top \hat{C}(t) S + \hat{E}(t)^\top S + \hat{F}(t),$$

with $\hat{C} \in C^1([\tau, T], \mathcal{S}_d(\mathbb{R}))$, $\hat{E} \in C^1([\tau, T], \mathbb{R}^d)$, $\hat{F} \in C^1([\tau, T], \mathbb{R})$ defined by

$$\begin{cases} \hat{C}(t) = \frac{1}{2\gamma}(T-t)R^\top \Sigma^{-1}R, \\ \hat{E}(t) = -\frac{1}{\gamma}(T-t)R^\top \Sigma^{-1}R\bar{S}, \\ \hat{F}(t) = \frac{1}{4\gamma}(T-t)^2 \text{Tr}(R^\top \Sigma^{-1}R\Sigma) + \frac{1}{2\gamma}(T-t)\bar{S}^\top R^\top \Sigma^{-1}R. \end{cases}$$

We conclude that for all $(t, q, S) \in [\tau, T] \times \mathbb{R}^d \times \mathbb{R}^d$,

$$\begin{aligned} \theta(t, q, S) &= \begin{pmatrix} q \\ S \end{pmatrix}^\top P(t) \begin{pmatrix} q \\ S \end{pmatrix} + \begin{pmatrix} D(t) \\ E(t) \end{pmatrix}^\top \begin{pmatrix} q \\ S \end{pmatrix} + F(t) \\ &\leq \begin{pmatrix} q \\ S \end{pmatrix}^\top \begin{pmatrix} 0 & 0 \\ 0 & \hat{C}(t) \end{pmatrix} \begin{pmatrix} q \\ S \end{pmatrix} + \begin{pmatrix} 0 \\ \hat{E}(t) \end{pmatrix}^\top \begin{pmatrix} q \\ S \end{pmatrix} + \hat{F}(t). \end{aligned}$$

Therefore,

$$\forall t \in [\tau, T], \quad P(t) \leq \begin{pmatrix} 0 & 0 \\ 0 & \hat{C}(t) \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & \frac{1}{2\gamma}(T-t)R^\top \Sigma^{-1}R \end{pmatrix}.$$

We have therefore $\forall \tau \in (t_{\min}, T)$, $\forall t \in [\tau, T]$:

$$\begin{pmatrix} -\frac{\gamma}{2}\Sigma_t - \Gamma & -\frac{1}{2}(I - e^{-R(T-t)}) \\ -\frac{1}{2}(I - e^{-R^\top(T-t)}) & 0 \end{pmatrix} \leq P(t) \leq \begin{pmatrix} 0 & 0 \\ 0 & \frac{1}{2\gamma}(T-t)R^\top \Sigma^{-1}R \end{pmatrix}.$$

As $t_{\min}$ is supposed to be finite, there exists $\underline{M}, \bar{M} \in \mathcal{S}_d(\mathbb{R})$ with $\underline{M} \leq \bar{M}$ such that $\forall t \in [t_{\min}, T]$, $P(t)$ stays in the compact set $\{M \in \mathcal{S}_d(\mathbb{R}) \mid \underline{M} \leq M \leq \bar{M}\}$. This contradicts the maximality of the solution, hence $t_{\min} = -\infty$. $\square$

Theorem 2.1.2 implies that Theorem 2.1.1 can be applied with $\tau = 0$. In particular, our optimal execution problem is solved and the optimal strategy is given by the closed-loop feedback control (2.18). In the next section, we illustrate our results with simulations of prices and numerical approximations of the optimal strategies.

2.2 Performance analysis

In this section, we present several applications of our results. We first exemplify the use of the optimal strategy derived in the above section by a trader wishing to unwind a single-asset portfolio. In particular, we show that the optimal liquidation strategy in our model with mean reversion is really different from that derived in the Almgren-Chriss model. We also demonstrate the usefulness of our results for statistical arbitrage purposes in the one-asset case. The one-asset examples are based on data from the foreign exchange (FX) market. We then illustrate our results in the multi-asset case by considering a pair of two cointegrated French stocks. We start with a two-asset portfolio liquidation problem and compare the optimal liquidation strategy in our model with that obtained in a multi-asset Almgren-Chriss model. We then illustrate the use of our results for statistical arbitrage purposes (a pair trading strategy in our case).¹¹

In the Almgren-Chriss model used for carrying out comparisons, the price dynamics is of the form $dS_t = V_{AC}dW_t$, where $V_{AC} \in \mathcal{M}_{d,k}(\mathbb{R})$, i.e. a simple Bachelier dynamics (with correlations). This dynamics differs from that of the OU model we use throughout Chapter 2 (i.e. $dS_t = R(\bar{S} - S_t)dt + VdW_t$) when $R \neq 0$. In particular, if prices exhibit mean reversion or a cointegrated behavior, as is the case in our examples, the classical Almgren-Chriss model shall not properly take the true multivariate dynamics of prices into account, with sometimes important consequences in terms of risk management.

2.2.1 Single-asset case

In order to illustrate the use of the optimal strategies we derived in the above section, let us start with a single-asset case. For that purpose, we use data from the FX market, in which asset prices often exhibit mean reversion. More precisely, we consider a FX futures contract (hereafter CDU1) on the currency pair Canadian Dollar (CAD) / US Dollar (USD) that is exchanged on the Chicago Mercantile Exchange. The contract specifications are given in Table 2.1.

Underlying asset	Canadian Dollar
Quotation currency	US Dollar
Contract size	CAD 100000
Expiry date	September 14, 2021

Table 2.1: CDU1 contract specifications

¹¹ Throughout this section, optimal trading strategies are computed by approximating the solution of the Riccati ODEs using implicit Euler schemes.

We plot in Figure 2.1 the mid-price of CDU1,¹² sampled every 60 seconds during the regular trading hours (02:00-16:00 Central Time),¹³ over the three following trading days: August 11, August 12, and August 13, 2021.

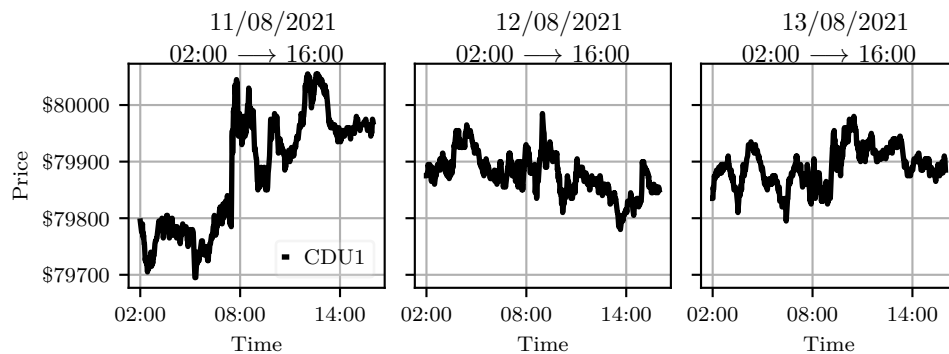


Fig. 2.1: Mid-price of CDU1 sampled every 60 seconds during the regular trading hours (02:00-16:00 CT). Top left: August 11, 2021. Top right: August 12, 2021. Bottom: August 13, 2021.

2.2.1.1 Liquidation problem

We consider the case of a trader wishing to unwind a long position in 2250 contracts¹⁴ during the third day, i.e. on August 13, 2021.

To exemplify the use of our strategies, we first estimate Ornstein-Uhlenbeck parameters using prices from the two preceding trading days: August 11 and August 12, 2021. Coefficients are classically estimated using least squares regression.¹⁵ In order to set the value of the execution cost / temporary market impact parameter η , we use a similar argument as in [Almgren and Chriss \(2001\)](#): we suppose that the additional cost incurred per contract when trading a given volume is proportional to the participation rate to the market. More precisely, for each percent of participation rate (in practice we consider a flat volume curve that matches the average daily volume), a cost corresponding to half the bid-ask spread¹⁶ is incurred. Rounding values, this results in setting $\eta = 5 \cdot 10^{-3} \$ \cdot \text{day}$. For the terminal penalty parameter Γ , we set a high value to enforce

¹² CDU1 is usually quoted in USD cents per CAD. However, to use our model, the price must take account of the contract size and be the contract value in USD.

¹³ Although CDU1 is quoted continuously with a 60-minute break each day beginning at 16:00 CT, we only consider the trading hours between 02:00 and 16:00 CT because the contract is only liquid during these hours corresponding to European and American market activity.

¹⁴ This represents roughly 5% of the average daily traded volume over the period considered in this example.

¹⁵ A time discretisation of an Ornstein-Uhlenbeck model gives rise to an Auto-Regressive model of order 1, or AR(1). The parameters of an AR(1) model are classically estimated by using least squares regression. Conversion of AR(1) coefficients into their continuous-time counterparts is straightforward.

¹⁶ The average bid-ask spread is close to the tick value equal to \$5 per contract.

complete liquidation by the end of the trading day. For the risk aversion parameter γ , we choose an intermediate value that does not neutralize any of the financial effects our model could illustrate. Choosing a too high value of the risk aversion parameter would force the trader to liquidate quickly with no illustration of the impact of mean reversion. On the contrary, choosing a too low value of the risk aversion parameter would result in the trader accumulating unrealistically large positions to benefit from mean reversion at the expense of the original liquidation problem.

The resulting values used to run our algorithms are given in Table 2.2.¹⁷

Parameter	Value
T	1 day
q_0	2250
S_0	\$79835
R	5.1 day ⁻¹
$\bar{S}$	\$79887
σ	243.67 \$ · day ^{-$\frac{1}{2}$}
η	5 · 10 ⁻³ \$ · day
Γ	\$100
γ	2 · 10 ⁻⁵ \$ ⁻¹

Table 2.2: Value of the parameters.

We plot in Figure 2.2 the asset price trajectory $(S_t)_{t \in [0, T]}$ on August 13, 2021 and the inventory process $(q_t)_{t \in [0, T]}$ corresponding to the use of the optimal liquidation strategy derived in the previous section.¹⁸ For comparison purposes, we also plot the inventory process when using a classical Almgren-Chriss (AC) strategy.¹⁹

The results shown in Figure 2.2 deserve several remarks. First, the optimal liquidation strategy in our model with mean reversion is different from that derived in the Almgren-Chriss model. In particular, the liquidation process is significantly faster in the latter case because the process of unwinding the portfolio appears far riskier to a trader who believes that the price evolves as a Brownian motion than to another one who believes in a mean-reverting Ornstein-Uhlenbeck dynamics. Second, in the case of the ACOU strategy, the trader progressively unwinds her long position over the trading day but also takes advantage of mean reversion. When the price is below $\bar{S}$, the trader tends

¹⁷ In the one-asset case, $\Sigma = VV^\top$ is a scalar. We classically write it as σ^2 and document the value of σ .

¹⁸ In what follows, this strategy is often referred to as ACOU (Almgren-Chriss under Ornstein-Uhlenbeck dynamics) strategy.

¹⁹ To compute the AC strategy, we estimate the parameter V_{AC} of the Bachelier dynamics. This parameter is a scalar in our one-asset case and we denote it by σ_{AC} instead of V_{AC} . A simple estimation based on price increments leads to $\sigma_{AC} = 244.02$ \$ · day^{- $\frac{1}{2}$} which slightly differs from σ because the drift term in the OU model captures part of the variance.

to reduce the pace of her selling process or even buys some contracts. Symmetrically, when the price is above $\bar{S}$, the trader tends to sell at a faster pace. One exception to the above should nevertheless be noticed, close to time T . Indeed, because of the high value of the final penalty, close to time T the trader focuses more on liquidating her portfolio and cares less about price oscillations. In particular, she buys back some contracts to end up flat because she previously went short to bet on the reversion of the price towards $\bar{S}$.

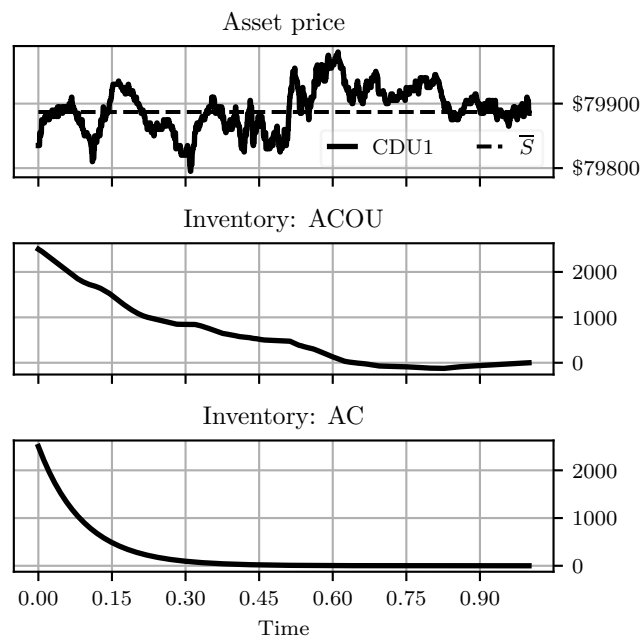


Fig. 2.2: Top: CDU1 price trajectory on August 13, 2021 – $(S_t)_{t \in [0, T]}$. Middle: Trajectory of the inventory when using the optimal strategy corresponding to the estimated Ornstein-Uhlenbeck process – $(q_t)_{t \in [0, T]}$. Bottom: Trajectory of the inventory when using the optimal strategy corresponding to a Brownian motion (Bachelier) model for the price (classical Almgren-Chriss strategy).

2.2.1.2 Impact of mean reversion

We have just seen that mean reversion plays a key role in the characteristics of the strategy. In order to further study the impact of the mean-reversion parameter R , we consider the same parameters as in Table 2.2 except that we force the mean-reversion parameter R to have the following values: $R = 0 \text{ day}^{-1}$, $R = 3 \text{ day}^{-1}$, and $R = 10 \text{ day}^{-1}$. Moreover, unlike what we did previously, we consider now a price trajectory in line with the choice of the parameter R . For that purpose, we simulate price trajectories by using the same path of the Brownian motion but different values of R .

In Figure 2.3, we plot an instance of the (simulated) trajectories of the price process $(S_t)_{t \in [0, T]}$ for the different values of R and the corresponding inventory processes

$(q_t)_{t \in [0, T]}$ – along with the associated trading volume (or trading speed) processes $(v_t)_{t \in [0, T]}$ – when using the optimal execution strategy.

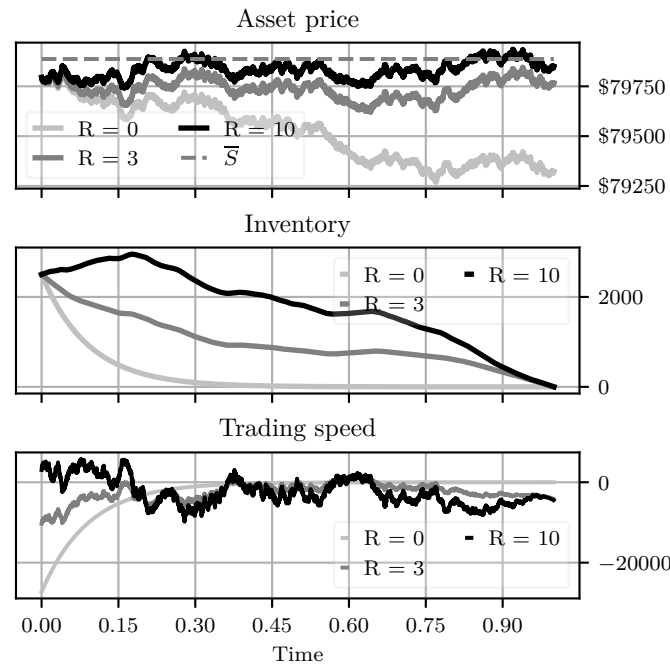


Fig. 2.3: Top: Simulated trajectories of the asset price for the different values of R – $(S_t)_{t \in [0, T]}$. Middle: Corresponding inventory processes when using the optimal execution strategy – $(q_t)_{t \in [0, T]}$. Bottom: Trajectories of the corresponding trading volume (or trading speed) processes – $(v_t)_{t \in [0, T]}$.

The results of Figure 2.3 confirm that the way the optimal strategy handles risk depends strongly on the mean-reversion parameter R . In particular, when R is large, the trader acts almost²⁰ as if she was performing a VWAP/TWAP²¹ execution strategy plus a mean-reverting statistical arbitrage strategy. In particular, in the case where $R = 10 \text{ day}^{-1}$, the process $(v_t)_{t \in [0, T]}$ oscillates around its average (which is of course fixed by the total number of contracts to sell). These oscillations are highly correlated with those of $(S_t)_{t \in [0, T]}$: the trader sells faster when the price is above $\bar{S}$ and slower (she even buys sometimes) when it is below $\bar{S}$. These observations also illustrate that the optimal strategy is strongly dependent on the specification of the parameter R , thus model risk can be significant. Recall that this parameter can be obtained through estimation or through encoding one's beliefs on the future deterministic behaviour of the price. It is thus key for agents that use our model to carefully select the value of R .

²⁰ Close to time T , because of the high value of Γ , the trader focuses more on unwinding her portfolio and cares less about price oscillations, as can clearly be seen in Figure 2.3.

²¹ VWAP and TWAP respectively mean Volume-Weighted Average Price and Time-Weighted Average Price. VWAP and TWAP strategies are commonly used by traders to execute orders at a price as close as possible to the average price of all transactions over a given period.

2.2.1.3 Statistical arbitrage

Given the observations of the previous subsection, it is natural to illustrate how our model can be used to build a statistical arbitrage strategy. For that purpose, we consider a trader with no initial inventory who starts trading the futures contract CDU1 on August 13, 2021 and wants to maximise the expected utility of her PnL at the end of the day (with no final penalty).

To run our algorithm we use the same parameters as in Table 2.2 except that $\Gamma = 0$ and $q_0 = 0$. The results are plotted in Figure 2.4: the price process $(S_t)_{t \in [0, T]}$, the inventory process $(q_t)_{t \in [0, T]}$ when using the optimal strategy, and the associated trajectory of the PnL, i.e. the process $(X_t + q_t S_t)_{t \in [0, T]}$.

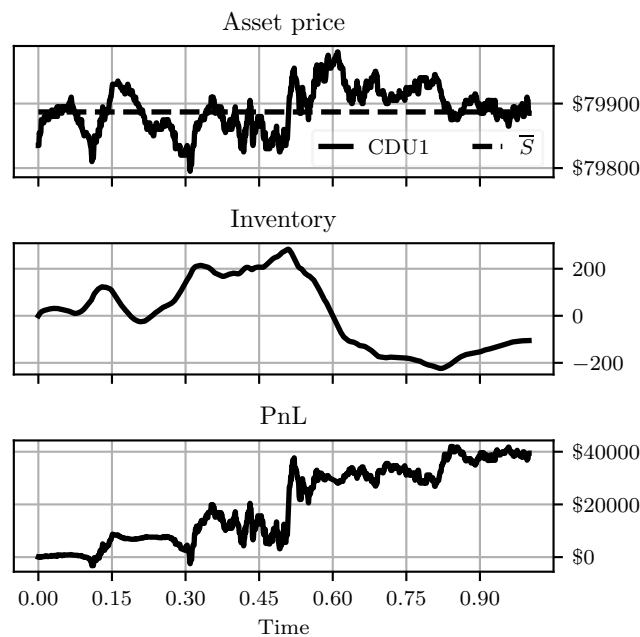


Fig. 2.4: Top: CDU1 price trajectory on August 13, 2021 – $(S_t)_{t \in [0, T]}$. Middle: Trajectory of the inventory when using the optimal strategy starting from $q_0 = 0$ – $(q_t)_{t \in [0, T]}$. Bottom: Corresponding trajectory of the PnL – $(X_t + q_t S_t)_{t \in [0, T]}$.

Because the trader believes that the price mean reverts around $\bar{S}$, her optimal strategy, in the absence of execution costs, would consist in having a long position when the price is below $\bar{S}$ and a short position when the price is above $\bar{S}$. The optimal strategy when execution costs are taken into account is however more complex because the control of execution costs introduces inertia in the position of the trader. It consists instead in trading progressively to target a long position when the price is below $\bar{S}$ and a short position when the price is above $\bar{S}$. However, because of inertia, it happens that the position remains long while the price is still far above $\bar{S}$, as exemplified in Figure 2.4. We see nevertheless that the strategy would have been profitable to the trader.

Another interesting experiment for assessing the performance of the strategy, when used for pure statistical arbitrage, consists in testing it on simulated price trajectories (using the same Ornstein-Uhlenbeck parameters as above for both simulating prices

and computing the optimal strategies). We plot in Figure 2.5, the distribution of the final PnL after 1500 simulations of the price process. We see that our strategy allows to make money by taking advantage of the mean reversion: we get a positive final profit of \$107698 on average, with a standard deviation of \$57791, and the distribution looks skewed to the right (towards profits rather than losses).

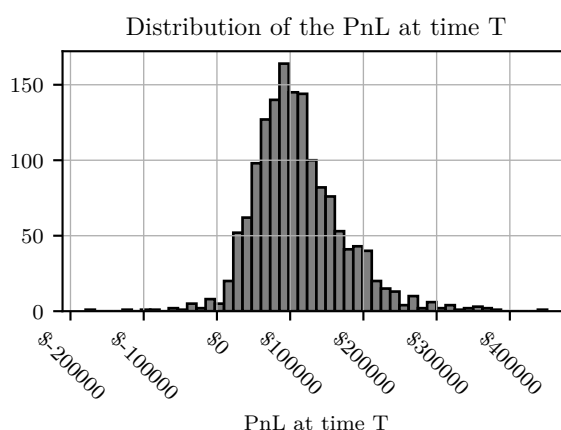


Fig. 2.5: Distribution of the final PnL for the statistical arbitrage strategy on CDU1 (for 1500 simulations).

2.2.2 Multi-asset case

We now come to the use of our optimal strategies in the multi-asset case when asset prices exhibit a cointegrated behaviour that can be modeled by a multi-OU process. For that purpose, we use data from two French stocks within the banking sector: BNP Paribas (hereafter BNP) and Société Générale (hereafter GLE).

We plot in Figure 2.6 the mid-prices of BNP and GLE sampled every 60 seconds during the regular trading hours (09:00-17:30) over the week August 09-August 13, 2021. We clearly see that the stock prices of the two companies are driven by the same factors and should be cointegrated.²²

2.2.2.1 Portfolio liquidation in the presence of cointegration

We consider the case of a trader wishing to unwind a portfolio with 75000 shares of BNP and 75000 shares of GLE on August 13, 2021.²³

Similarly to our one-asset example, we consider that the trader estimates the parameters of a multi-OU model using prices from the four preceding trading days, here August 9, 10, 11, and 12, 2021. These parameters are estimated using classical linear regression

²² The existence of a cointegration vector is confirmed by a Johansen's cointegration test (see below).

²³ This represents roughly 5% of the average daily traded volume over the period considered in this example.

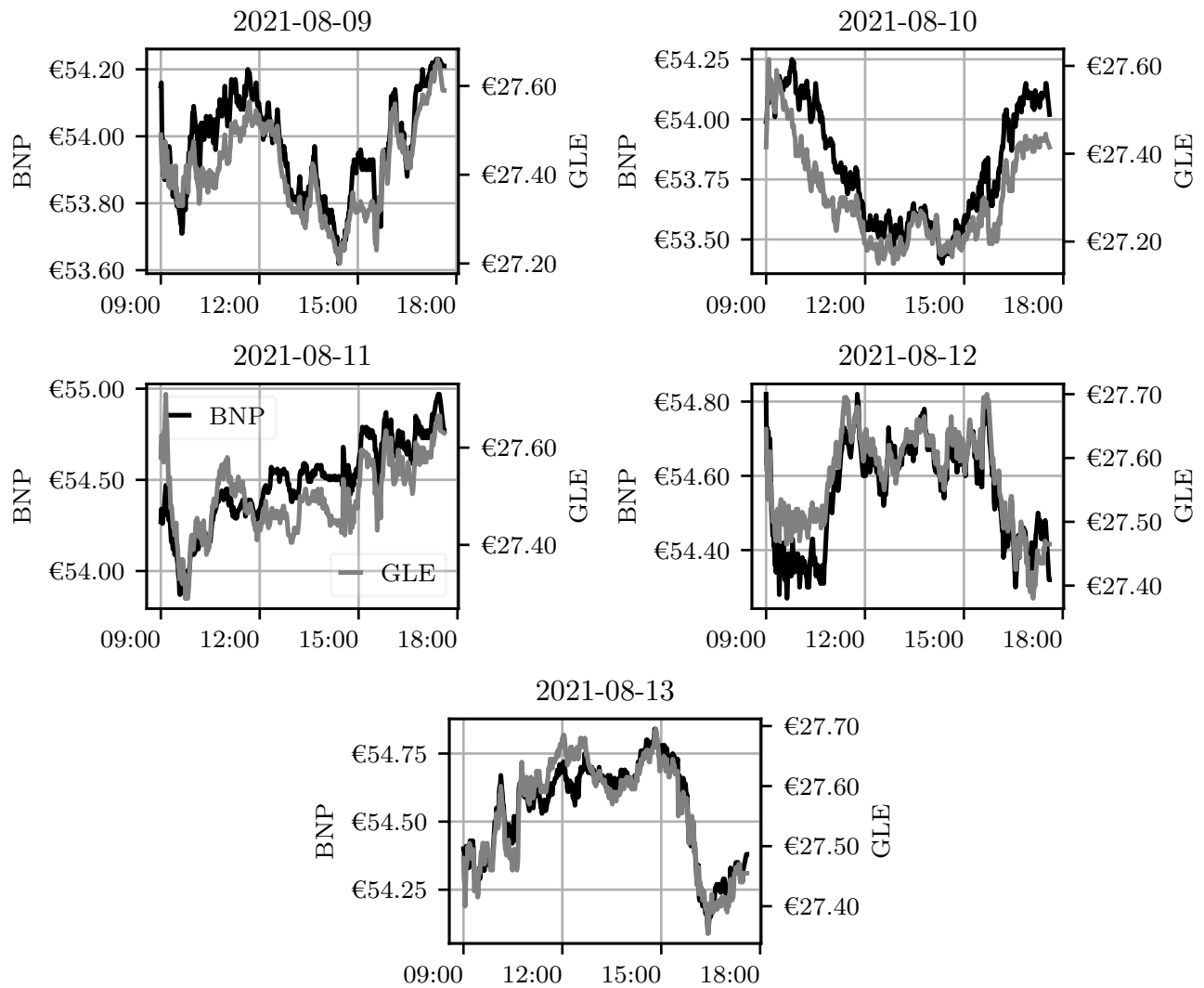


Fig. 2.6: Mid-prices of BNP (left axis) and GLE (right axis) sampled every 60 seconds during the regular trading hours (09:00-17:30) over the week August 09-August 13, 2021.

techniques.²⁴ Referring to BNP and GLE by respectively using the superscripts 1 and 2, the estimated values of the parameters are given in Table 2.3.

²⁴ A time discretisation of a multi-OU model gives rise to a Vector Auto-Regressive model of order 1, or VAR(1). The parameters of a VAR(1) model are classically estimated by using least squares regression. Conversion of VAR(1) coefficients into their continuous-time multi-OU counterparts is straightforward.

Parameter	Estimate
R	$\begin{pmatrix} 0.33 & 3.95 \\ -2.52 & 10.23 \end{pmatrix} \text{ day}^{-1}$
$\bar{S}$	$(\bar{S}^1, \bar{S}^2) = (\text{€}54.23, \text{€}27.45)$
Σ	$\begin{pmatrix} 0.47 & 0.20 \\ 0.20 & 0.14 \end{pmatrix} \text{€}^2 \cdot \text{day}^{-1}$

Table 2.3: Multi-OU estimated parameters for the pair (BNP, GLE).

The use of a Johansen's cointegration test²⁵ rejects the hypothesis of no cointegration but does not reject a cointegration rank $r = 1$ for the pair (BNP, GLE).²⁶ Furthermore, the estimated value of the matrix R suggests that the space of cointegration vectors is spanned by $(1, -3.46)$.

We give in Table 2.4 the detailed results for the Johansen's cointegration Trace test.

Null Hypothesis	Trace statistics	Critical Value	Conclusion
$r \leq 0$	16.77	15.49	Rejected
$r \leq 1$	2.293	3.841	Not rejected

Table 2.4: Johansen's cointegration Trace test results (critical values are given for a significance level of 95%).

To illustrate the use of our strategies and illustrate the different effects in the multi-asset case, we run our algorithms for two different values of the risk aversion parameter γ . More precisely, we consider the parameters stated in Table 2.5 (values of the multi-OU parameters are not recalled, see Table 2.3).²⁷

²⁵ We use the Trace test and not the Maximum Eigenvalue test in what follows.

²⁶ Johansen's approach is based on the VAR(1) formulation $\Delta S_t = a + \Pi S_{t-1} + \epsilon_t$, where $a \in \mathbb{R}^d$, $\Pi \in \mathcal{M}_d(\mathbb{R})$ and ϵ is normally distributed. In a nutshell, it iteratively tests the null hypothesis $\text{rank}(\Pi) \leq k$ (corresponding to the existence of at most k linearly independent cointegration vectors) for different values of k using likelihood ratio statistics that follow tabulated distributions (see [Johansen \(1991\)](#) for more details). In practice, the cointegration rank retained is the first value of k for which the null hypothesis is not rejected.

²⁷ The same logic as in the one-asset case has been applied for the choice of the parameters.

Parameter	Value
T	1 day
q_0	(75000, 75000)
S_0	$(S_0^1, S_0^2) = (\text{€}54.4, \text{€}27.48)$
η	$\begin{pmatrix} 4 \cdot 10^{-7} & 0 \\ 0 & 2 \cdot 10^{-7} \end{pmatrix} \text{€} \cdot \text{day}$
Γ	$\text{€}100 \times I_2$
γ	$2 \cdot 10^{-5} \text{€}^{-1}$ or $2 \cdot 10^{-3} \text{€}^{-1}$

Table 2.5: Value of the parameters.

We plot in Figures 2.7 and 2.8 the trajectory on August 13, 2021 of the price process $(S_t)_{t \in [0, T]}$ and the spread process $((S_t^1 - \bar{S}^1) - 3.46(S_t^2 - \bar{S}^2))_{t \in [0, T]}$ corresponding to the cointegration vector, along with the inventory process $(q_t)_{t \in [0, T]}$ corresponding to the use of the optimal liquidation strategy with $\gamma = 2 \cdot 10^{-5} \text{€}^{-1}$ and $\gamma = 2 \cdot 10^{-3} \text{€}^{-1}$ respectively.²⁸ For comparison purposes, we also plot the inventory process when using a classical Almgren-Chriss (AC) strategy.²⁹

We clearly see, for both values of γ that the ACOU strategy is different from the AC one³⁰ although they both succeed in unwinding the portfolio. This is easily understandable: because of the presence of the matrix R in the multi-OU model, there is a drift in the dynamics of the prices that can be exploited to make money (while still controlling market risk).

It is interesting to understand the difference between what happens when γ is small versus what happens when γ is large because the trading curves exhibit very different properties. When γ is small, we observe that the trader oversells the two stocks (before buying back close to time T to unwind the portfolio) and speculates therefore on the reversion of the two stock prices towards $\bar{S}^1$ and $\bar{S}^2$ respectively – because prices are above these values. On the contrary, when γ is large, shorting the two stocks simultaneously appears too risky and, once the portfolio has been partially liquidated, the trader uses instead a long/short strategy. In that case, the trader seems in fact to apply a statistical arbitrage strategy related to the spread process $((S_t^1 - \bar{S}^1) - 3.46(S_t^2 - \bar{S}^2))_{t \in [0, T]}$: for $t \geq 0.35$, when the spread process is below 0, the trader is long BNP and short GLE – she buys the spread – while it becomes (slightly) long GLE and (slightly)

²⁸ As above, this strategy is referred to as ACOU (Almgren-Chriss under multivariate Ornstein-Uhlenbeck dynamics) strategy.

²⁹ To compute the AC strategy, we estimate the parameter V_{AC} of the Bachelier dynamics. A simple estimation based on price increments leads to $\Sigma_{AC} = V_{AC} V_{AC}^\top = \begin{pmatrix} 0.48 & 0.19 \\ 0.19 & 0.13 \end{pmatrix} \text{€}^2 \cdot \text{day}^{-1}$ which slightly differs from Σ because the drift term in the OU model captures part of the variance.

³⁰ It is noteworthy that the trading curves for BNP and GLE are almost the same in the AC case. Because of the value of S_0 and η this is unsurprising.

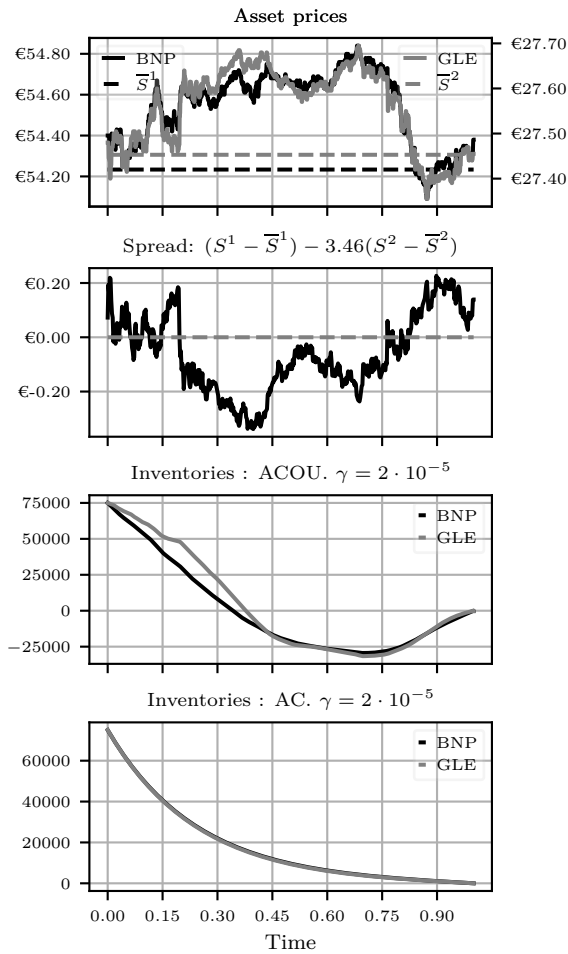


Fig. 2.7: Top two plots: BNP and GLE price trajectories on August 13, 2021 – $(S_t)_{t \in [0, T]}$ – and trajectory of the spread corresponding to the cointegration vector on August 13, 2021 – $((S_t^1 - \bar{S}^1) - 3.46(S_t^2 - \bar{S}^2))_{t \in [0, T]}$. Bottom two plots: Trajectory of the inventories when using the optimal strategy corresponding to the estimated multi-OU process with $\gamma = 2 \cdot 10^{-5} \text{ €}^{-1}$ – $(q_t)_{t \in [0, T]}$ – and when using the optimal strategy corresponding to correlated Brownian motions for the stock prices (classical Almgren-Chriss strategy with $\gamma = 2 \cdot 10^{-5} \text{ €}^{-1}$).

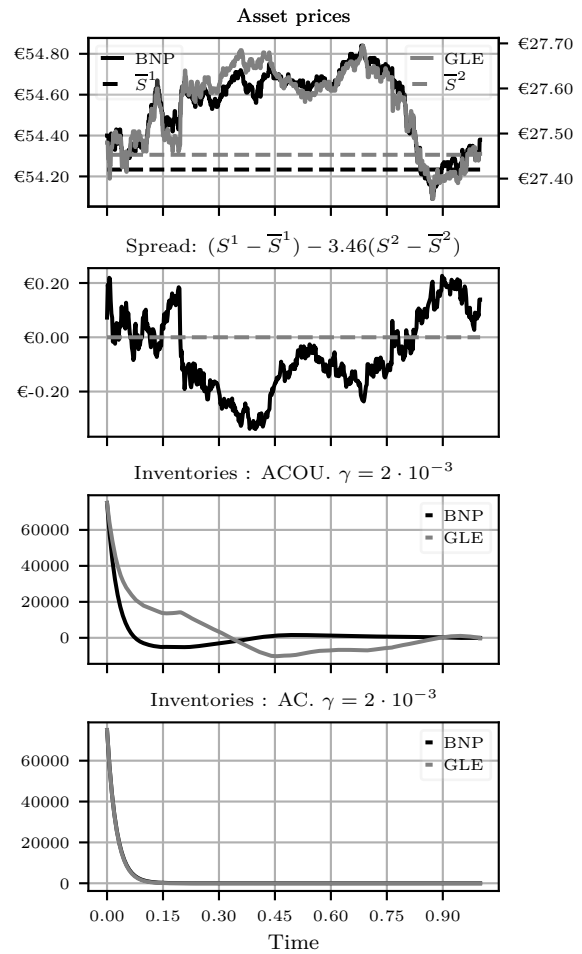


Fig. 2.8: Top two plots: BNP and GLE price trajectories on August 13, 2021 – $(S_t)_{t \in [0, T]}$ –, and trajectory of the spread corresponding to the cointegration vector on August 13, 2021 – $((S_t^1 - \bar{S}^1) - 3.46(S_t^2 - \bar{S}^2))_{t \in [0, T]}$. Bottom two plots: Trajectory of the inventories when using the optimal strategy corresponding to the estimated multi-OU process with $\gamma = 2 \cdot 10^{-3} \text{ €}^{-1}$ – $(q_t)_{t \in [0, T]}$ –, and when using the optimal strategy corresponding to correlated Brownian motions for the stock prices (classical Almgren-Chriss strategy with $\gamma = 2 \cdot 10^{-3} \text{ €}^{-1}$).

short BNP – she shorts the spread – at the very end of the period when the spread becomes positive. The fact that the trader “trades the spread” will appear even more clearly in what follows as we focus on statistical arbitrage strategies.

2.2.2.2 Statistical arbitrage

Given the previous remarks, it is natural to illustrate the use of our model in the context of pure statistical arbitrage. For that purpose, we consider a trader with no initial inventory who starts trading on August 13, 2021 and wants to maximise the expected utility of her PnL at the end of the day (with no final penalty).

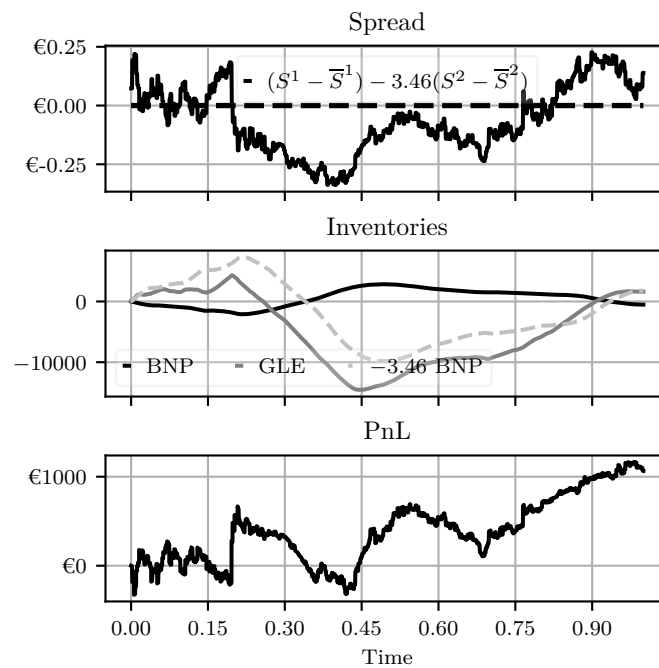


Fig. 2.9: Top: Trajectory of the spread on August 13, 2021 – $((S_t^1 - \bar{S}^1) - 3.46(S_t^2 - \bar{S}^2))_{t \in [0, T]}$. Middle: Trajectory of the inventories when using the optimal strategy corresponding to the estimated multi-OU process with $\gamma = 2 \cdot 10^{-3} \text{ €}^{-1} - (q_t)_{t \in [0, T]}$. Bottom: Trajectory of the PnL – $(X_t + q_t^T S_t)_{t \in [0, T]}$.

To run our algorithm we use the same parameters as in Tables 2.3 and 2.5 with $\gamma = 2 \cdot 10^{-3} \text{ €}^{-1}$ except that $\Gamma = 0$ and $q_0 = (0, 0)$. The results are plotted in Figure 2.9: the spread process $((S_t^1 - \bar{S}^1) - 3.46(S_t^2 - \bar{S}^2))_{t \in [0, T]}$, the inventory process $(q_t)_{t \in [0, T]}$ when using the optimal strategy, and the associated trajectory of the PnL, i.e. the process $(X_t + q_t^T S_t)_{t \in [0, T]}$.

We clearly see that for $\gamma = 2 \cdot 10^{-3} \text{ €}^{-1}$ the optimal strategy is a long/short strategy. As can be seen in Figure 2.9 (middle plot), the process $(-3.46q_t^1)_{t \in [0, T]}$ appears to be in line with $(q_t^2)_{t \in [0, T]}$. This confirms that the strategy consists mainly in “buying or selling the spread” depending on the sign of the spread process $((S_t^1 - \bar{S}^1) - 3.46(S_t^2 - \bar{S}^2))_{t \in [0, T]}$.

Finally, as in the previous subsection, we test our optimal strategy on simulated price trajectories (using the same multi-OU parameters as above for both simulating prices and computing the optimal strategies). We plot in Figure 2.10, the distribution of the final PnL after 1500 simulations of the price process when using the optimal strategy with the parameters of Table 2.5 (with $\gamma = 2 \cdot 10^{-3} \text{ €}^{-1}$) except that $\Gamma = 0$ and $q_0 = (0, 0)$ (because we focus on statistical arbitrage). We see that our strategy allows to make money by taking advantage of the price dynamics: we get a positive final profit of €2230 on average, with a standard deviation of €1145, and the distribution is, as above, skewed towards profits rather than losses.³¹

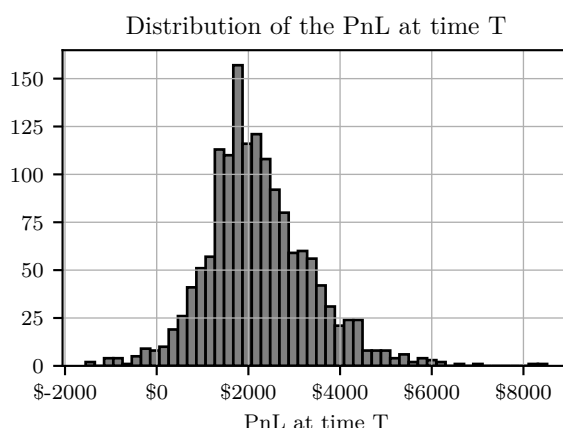


Fig. 2.10: Distribution of the final PnL for the statistical arbitrage strategy on BNP and GLE (for 1500 simulations).

Conclusions

In this chapter, we have shown how to account for cross-asset co-movements when executing trades in multiple assets. In our model, the agent has an exponential utility and the prices have multivariate Ornstein-Uhlenbeck dynamics, capturing the complex cross-asset dynamics of prices better than correlated Brownian motions only. The advantage of our approach is twofold: (i) it better accounts for risk at the portfolio level, and (ii) it is versatile and can be used for basket execution and statistical arbitrage.

³¹ The PnLs are smaller in absolute value than in the one-asset example because the value of γ is higher here.

Appendix: Merton problem under OU dynamics and exponential utility

2.2.3 Modelling framework

We study in this appendix a Merton problem where prices have multivariate Ornstein-Uhlenbeck dynamics. It is closely related to our problem and can be seen as some form of limit case corresponding to no execution costs (i.e. $L = 0$) and no terminal penalty (i.e. $\ell = 0$).

The results obtained in this appendix are essential in our proof of existence of a solution to the system of ODEs (2.12) on $[0, T]$ with terminal condition (2.13) (see Theorem 2.1.2).

As in Chapter 2,³² we consider a model with d assets, whose prices are modelled by a d -dimensional stochastic process $(S_t)_{t \in [0, T]} = (S_t^1, \dots, S_t^d)_{t \in [0, T]}^\top$ with dynamics

$$dS_t = R(\bar{S} - S_t)dt + VdW_t,$$

where $\bar{S} \in \mathbb{R}^d$, $R \in \mathcal{M}_d(\mathbb{R})$, $V \in \mathcal{M}_{d,k}(\mathbb{R})$, and $(W_t)_{t \in [0, T]} = (W_t^1, \dots, W_t^k)_{t \in [0, T]}^\top$ is a k -dimensional standard Brownian motion (with independent coordinates), for some $k \in \mathbb{N}^*$. As before, we write $\Sigma = VV^\top$.

We consider a trader optimizing her portfolio over the period $[0, T]$ by controlling at each time the number of each asset in her portfolio, i.e. she controls the d -dimensional process $(q_t)_{t \in [0, T]} = (q_t^1, \dots, q_t^d)_{t \in [0, T]}^\top$, where q_t^i denotes the number of assets i in the portfolio at time t , for each $i \in \{1, \dots, d\}$ ($t \in [0, T]$).³³ The process $(q_t)_{t \in [0, T]}$ lies in the space of admissible controls $\mathcal{A}_0^{Merton}$, where for $t \in [0, T]$, the set $\mathcal{A}_t^{Merton}$ is defined as

$$\mathcal{A}_t^{Merton} := \left\{ (q_s)_{s \in [t, T]}, \mathbb{R}^d\text{-valued, } \mathbb{F}\text{-adapted,} \right. \quad (2.20)$$

$$\left. \text{satisfying a linear growth condition with respect to } (S_s)_{s \in [t, T]} \right\}.$$

We introduce the process $(\mathcal{V}_t)_{t \in [0, T]}$ modelling the MtM value of the trader's portfolio, i.e.

$$\forall t \in [0, T], \quad \mathcal{V}_t = \mathcal{V}_0 + \int_0^t q_s^\top dS_s, \quad \mathcal{V}_0 \in \mathbb{R} \text{ given.}$$

For a given $\gamma > 0$, the trader aims at maximizing the following objective function:

³² We consider no permanent market impact in this Appendix.

³³ Unlike what happens in the model of Section 2, our control variable is here the number of assets and not the volume traded. It is only when execution costs are incurred by the trader that trading rates / trading volumes are indeed the relevant control variables.

$$\mathbb{E} \left[-e^{-\gamma \mathcal{V}_T} \right], \quad (2.21)$$

over the set of admissible controls $(q_t)_{t \in [0, T]} \in \mathcal{A}_0^{Merton}$. We define her value function $\hat{u} : [0, T] \times \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}$ as

$$\hat{u}(t, \mathcal{V}, S) = \sup_{(q_s)_{s \in [t, T]} \in \mathcal{A}_t^{Merton}} \mathbb{E} \left[-e^{-\gamma \mathcal{V}_T^{t, \mathcal{V}, S, q}} \right] \quad \forall (t, \mathcal{V}, S) \in [0, T] \times \mathbb{R} \times \mathbb{R}^d,$$

where $(\mathcal{V}_s^{t, \mathcal{V}, S, q})_{s \in [t, T]}$ denotes the process defined by

$$d\mathcal{V}_s^{t, \mathcal{V}, S, q} = q_s^\top dS_s^{t, S}, \quad \mathcal{V}_t^{t, \mathcal{V}, S, q} = \mathcal{V}$$

with

$$dS_s^{t, S} = R(\bar{S} - S_s^{t, S})ds + VdW_s, \quad S_t^{t, S} = S.$$

2.2.4 HJB equation

The HJB equation associated with Problem (2.21) is given by

$$\begin{aligned} 0 = & \partial_t \hat{w}(t, \mathcal{V}, S) + \nabla_S \hat{w}(t, \mathcal{V}, S)^\top R(\bar{S} - S) + \frac{1}{2} \text{Tr} \left(\Sigma D_{SS}^2 \hat{w}(t, \mathcal{V}, S) \right) \\ & + \sup_{q \in \mathbb{R}^d} \left\{ \partial_{\mathcal{V}} \hat{w}(t, \mathcal{V}, S) q^\top R(\bar{S} - S) + \frac{1}{2} \partial_{\mathcal{V}\mathcal{V}}^2 \hat{w}(t, \mathcal{V}, S) q^\top \Sigma q + \partial_{\mathcal{V}} \nabla_S \hat{w}(t, \mathcal{V}, S)^\top \Sigma q \right\} \end{aligned} \quad (2.22)$$

for all $(t, \mathcal{V}, S) \in [0, T] \times \mathbb{R} \times \mathbb{R}^d$, with terminal condition

$$\hat{w}(T, \mathcal{V}, S) = -e^{-\gamma \mathcal{V}} \quad \forall (\mathcal{V}, S) \in \mathbb{R} \times \mathbb{R}^d. \quad (2.23)$$

To solve the above HJB equation, we use the ansatz

$$\hat{w}(t, \mathcal{V}, S) = -e^{-\gamma(\mathcal{V} + \hat{\theta}(t, S))}.$$

Indeed, we have the following proposition:

Proposition 2.2.1 If there exists $\hat{\theta} \in C^{1,2}([0, T] \times \mathbb{R}^d, \mathbb{R})$ solution to

$$0 = \partial_t \hat{\theta}(t, S) + \frac{1}{2} \text{Tr} \left(\Sigma D_{SS}^2 \hat{\theta}(t, S) \right) + \frac{1}{2\gamma} (\bar{S} - S)^\top R^\top \Sigma^{-1} R (\bar{S} - S) \quad (2.24)$$

on $[0, T] \times \mathbb{R}^d$, with terminal condition

$$\hat{\theta}(T, S) = 0 \quad \forall S \in \mathbb{R}^d, \quad (2.25)$$

then the function $\hat{w} : [0, T] \times \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}$ defined by

$$\hat{w}(t, \mathcal{V}, S) = -e^{-\gamma(\mathcal{V} + \hat{\theta}(t, S))} \quad \forall (t, \mathcal{V}, S) \in [0, T] \times \mathbb{R} \times \mathbb{R}^d$$

is a solution to (2.22) on $[0, T] \times \mathbb{R} \times \mathbb{R}^d$ with terminal condition (2.23). $\square$

Proof 2.2.1 Let $\hat{\theta} : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}$ be a solution to (2.24) on $[0, T] \times \mathbb{R}^d$ with terminal condition (2.25), then we have for all $(t, \mathcal{V}, S) \in [0, T] \times \mathbb{R} \times \mathbb{R}^d$:

$$\begin{aligned}
 & \partial_t \hat{w}(t, \mathcal{V}, S) + \nabla_S \hat{w}(t, \mathcal{V}, S)^\top R(\bar{S} - S) + \frac{1}{2} \text{Tr} \left(\Sigma D_{SS}^2 \hat{w}(t, \mathcal{V}, S) \right) \\
 & + \sup_{q \in \mathbb{R}^d} \left\{ \partial_{\mathcal{V}} \hat{w}(t, \mathcal{V}, S) q^\top R(\bar{S} - S) + \frac{1}{2} \partial_{\mathcal{V}\mathcal{V}}^2 \hat{w}(t, \mathcal{V}, S) q^\top \Sigma q + \partial_{\mathcal{V}} \nabla_S \hat{w}(t, \mathcal{V}, S)^\top \Sigma q \right\} \\
 = & -\gamma \partial_t \hat{\theta}(t, S) \hat{w}(t, \mathcal{V}, S) - \gamma \nabla_S \hat{\theta}(t, S) R(\bar{S} - S) \hat{w}(t, \mathcal{V}, S) \\
 & - \gamma \hat{w}(t, \mathcal{V}, S) \frac{1}{2} \text{Tr} \left(\Sigma D_{SS}^2 \hat{\theta}(t, S) \right) + \frac{\gamma^2}{2} \hat{w}(t, \mathcal{V}, S) \nabla_S \hat{\theta}(t, S)^\top \Sigma \nabla_S \hat{\theta}(t, S) \\
 & + \sup_{q \in \mathbb{R}^d} \left\{ -\gamma \hat{w}(t, \mathcal{V}, S) q^\top R(\bar{S} - S) + \frac{\gamma^2}{2} \hat{w}(t, \mathcal{V}, S) q^\top \Sigma q \right. \\
 & \quad \left. + \gamma^2 \hat{w}(t, \mathcal{V}, S) \nabla_S \hat{\theta}(t, \mathcal{V}, S)^\top \Sigma q \right\} \\
 = & -\gamma \hat{w}(t, \mathcal{V}, S) \left(\partial_t \hat{\theta}(t, S) + \nabla_S \hat{\theta}(t, S) R(\bar{S} - S) + \frac{1}{2} \text{Tr} \left(\Sigma D_{SS}^2 \hat{\theta}(t, S) \right) \right. \\
 & \quad \left. - \frac{\gamma}{2} \nabla_S \hat{\theta}(t, S)^\top \Sigma \nabla_S \hat{\theta}(t, S) + \sup_{q \in \mathbb{R}^d} \left\{ q^\top \left(R(\bar{S} - S) - \gamma \Sigma \nabla_S \hat{\theta}(t, S) \right) - \frac{\gamma}{2} q^\top \Sigma q \right\} \right).
 \end{aligned}$$

The supremum in the above equation is reached for $q = q^*(t, S) = \frac{1}{\gamma} \Sigma^{-1} R(\bar{S} - S) - \nabla_S \hat{\theta}(t, S)$, and we obtain therefore after simplifications:

$$\begin{aligned}
 & \partial_t \hat{w}(t, \mathcal{V}, S) + \nabla_S \hat{w}(t, \mathcal{V}, S)^\top R(\bar{S} - S) + \frac{1}{2} \text{Tr} \left(\Sigma D_{SS}^2 \hat{w}(t, \mathcal{V}, S) \right) \\
 & + \sup_{q \in \mathbb{R}^d} \left\{ \partial_{\mathcal{V}} \hat{w}(t, \mathcal{V}, S) q^\top R(\bar{S} - S) + \frac{1}{2} \partial_{\mathcal{V}\mathcal{V}}^2 \hat{w}(t, \mathcal{V}, S) q^\top \Sigma q + \partial_{\mathcal{V}} \nabla_S \hat{w}(t, \mathcal{V}, S)^\top \Sigma q \right\} \\
 = & -\gamma \hat{w}(t, \mathcal{V}, S) \left(\partial_t \hat{\theta}(t, S) + \frac{1}{2} \text{Tr} \left(\Sigma D_{SS}^2 \hat{\theta}(t, S) \right) + \frac{1}{2\gamma} (\bar{S} - S)^\top R^\top \Sigma^{-1} R(\bar{S} - S) \right) \\
 = & 0.
 \end{aligned}$$

As $\hat{w}$ satisfies the terminal condition (2.23), the result is proved. $\square$

We now use a second ansatz and look for a function $\hat{\theta}$ solution to (2.24) on $[0, T] \times \mathbb{R}^d$ with terminal condition (2.25) of the following form:

$$\hat{\theta}(t, S) = S^\top \hat{C}(t) S + \hat{E}(t)^\top S + \hat{F}(t). \quad (2.26)$$

We have indeed the following proposition:

Proposition 2.2.2 Assume there exists $\hat{C} \in C^1([0, T], \mathcal{S}_d(\mathbb{R}))$, $\hat{E} \in C^1([0, T], \mathbb{R}^d)$, $\hat{F} \in C^1([0, T], \mathbb{R})$ satisfying the system of ODEs

$$\begin{cases} \hat{C}'(t) &= -\frac{1}{2\gamma} R^\top \Sigma^{-1} R \\ \hat{E}'(t) &= \frac{1}{\gamma} R^\top \Sigma^{-1} R \bar{S} \\ \hat{F}'(t) &= -\text{Tr}(\hat{C}(t) \Sigma) - \frac{1}{2\gamma} \bar{S}^\top R^\top \Sigma^{-1} R \bar{S}, \end{cases} \quad (2.27)$$

with terminal condition

$$\hat{C}(T) = \hat{E}(T) = \hat{F}(T) = 0. \quad (2.28)$$

Then the function $\hat{\theta}$ defined by (2.26) satisfies (2.24) on $[0, T] \times \mathbb{R}^d$ with terminal condition (2.25). $\square$

Proof 2.2.2 Let us consider $\hat{C} \in C^1([0, T], \mathcal{S}_d(\mathbb{R}))$, $\hat{E} \in C^1([0, T], \mathbb{R}^d)$, $\hat{F} \in C^1([0, T], \mathbb{R})$ verifying (2.27) on $[0, T]$ with terminal condition (2.28). Let us consider $\hat{\theta} : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}$ defined by (2.26). Then we obtain for all $(t, S) \in [0, T] \times \mathbb{R}^d$:

$$\begin{aligned} & \partial_t \hat{\theta}(t, S) + \frac{1}{2} \text{Tr}(\Sigma D_{SS}^2 \hat{\theta}(t, S)) + \frac{1}{2\gamma} (\bar{S} - S)^\top R^\top \Sigma^{-1} R (\bar{S} - S) \\ &= S^\top \hat{C}'(t) S + \hat{E}'(t)^\top S + \hat{F}'(t) + \text{Tr}(\hat{C}(t) \Sigma) + \frac{1}{2\gamma} (\bar{S} - S)^\top R^\top \Sigma^{-1} R (\bar{S} - S) \\ &= 0. \end{aligned}$$

As it is straightforward to verify that $\hat{\theta}$ satisfies the terminal condition (2.25), the result is proved. $\square$

It is straightforward to see that there exists a unique solution $\hat{C} \in C^1([0, T], \mathcal{S}_d(\mathbb{R}))$, $\hat{E} \in C^1([0, T], \mathbb{R}^d)$, $\hat{F} \in C^1([0, T], \mathbb{R})$ to (2.27) with terminal condition (2.28). We can then prove the following verification theorem.

Theorem 2.2.1 Let $\hat{C} \in C^1([0, T], \mathcal{S}_d(\mathbb{R}))$, $\hat{E} \in C^1([0, T], \mathbb{R}^d)$, $\hat{F} \in C^1([0, T], \mathbb{R})$ be the solutions to (2.27) with terminal condition

$$\hat{C}(T) = \hat{E}(T) = \hat{F}(T) = 0,$$

i.e. for all $t \in [0, T]$,

$$\begin{cases} \hat{C}(t) = \frac{1}{2\gamma} (T-t) R^\top \Sigma^{-1} R, \\ \hat{E}(t) = -\frac{1}{\gamma} (T-t) R^\top \Sigma^{-1} R \bar{S}, \\ \hat{F}(t) = \frac{1}{4\gamma} (T-t)^2 \text{Tr}(R^\top \Sigma^{-1} R \Sigma) + \frac{1}{2\gamma} (T-t) \bar{S}^\top R^\top \Sigma^{-1} R. \end{cases}$$

We consider the function $\hat{\theta}$ defined by

$$\hat{\theta}(t, S) = S^\top \hat{C}(t) S + \hat{E}(t)^\top S + \hat{F}(t),$$

and the associated function $\hat{w}$ defined by

$$\hat{w}(t, \mathcal{V}, S) = -e^{-\gamma(\mathcal{V} + \hat{\theta}(t, S))}.$$

For all $(t, \mathcal{V}, S) \in [0, T] \times \mathbb{R} \times \mathbb{R}^d$ and $q = (q_s)_{s \in [t, T]} \in \mathcal{A}_t^{Merton}$, we have

$$\mathbb{E} \left[-e^{-\gamma \mathcal{V}_T^{t, \mathcal{V}, S, q}} \right] \leq \hat{w}(t, \mathcal{V}, S). \quad (2.29)$$

Moreover, equality is obtained in (2.29) by taking the optimal control $(q_s^*)_{s \in [t, T]} \in \mathcal{A}_t^{Merton}$ given by the closed-loop feedback formula

$$q_s^* = \frac{1}{\gamma} (I_d + (T - s)R^\top) \Sigma^{-1} R (\bar{S} - S_s^{t, S}).$$

In particular, $\hat{w} = \hat{u}$. □

Proof 2.2.3 It is obvious that $(q_s^*)_{s \in [t, T]} \in \mathcal{A}_t^{Merton}$ (i.e., $(q_s^*)_{s \in [t, T]}$ is well defined and admissible):

$$\exists C_{t, T} > 0, \forall s \in [t, T], \quad \|q_s^*\| \leq C_{t, T} \left(1 + \sup_{\tau \in [t, s]} \|S_\tau\| \right).$$

Let us consider $(t, \mathcal{V}, S) \in [0, T] \times \mathbb{R} \times \mathbb{R}^d$ and $q = (q_s)_{s \in [t, T]} \in \mathcal{A}_t^{Merton}$. We now prove that

$$\mathbb{E} \left[\hat{w} \left(T, \mathcal{V}_T^{t, \mathcal{V}, S}, S_T^{t, S} \right) \right] \leq \hat{w}(t, \mathcal{V}, S).$$

We use the following notations for readability

$$\forall s \in [t, T], \quad \hat{w} \left(s, \mathcal{V}_s^{t, \mathcal{V}, S, q}, S_s^{t, S} \right) = \hat{w}_s^{t, \mathcal{V}, S, q},$$

$$\forall s \in [t, T], \quad \hat{\theta}(s, S_s^{t, S}) = \hat{\theta}_s^{t, S}.$$

By Itô's formula, we have $\forall s \in [0, T]$

$$d\hat{w}_s^{t, \mathcal{V}, S, q} = \mathcal{L}^q \hat{w}_s^{t, \mathcal{V}, S, q} ds + \left(\partial_{\mathcal{V}} \hat{w}_s^{t, \mathcal{V}, S, q} q_s + \nabla_S \hat{w}_s^{t, \mathcal{V}, S, q} \right)^\top V dW_s,$$

where

$$\begin{aligned} \mathcal{L}^q \hat{w}_s^{t, \mathcal{V}, S, q} &= \partial_t \hat{w}_s^{t, \mathcal{V}, S, q} + \left(\nabla_S \hat{w}_s^{t, \mathcal{V}, S, q} \right)^\top R (\bar{S} - S) + \partial_{\mathcal{V}} \hat{w}_s^{t, \mathcal{V}, S, q} q_s^\top R (\bar{S} - S) \\ &\quad + \frac{1}{2} \text{Tr} \left(\Sigma D_{SS}^2 \hat{w}_s^{t, \mathcal{V}, S, q} \right) + \frac{1}{2} \partial_{\mathcal{V}\mathcal{V}}^2 \hat{w}_s^{t, \mathcal{V}, S, q} q_s^\top \Sigma q_s + \left(\partial_{\mathcal{V}} \nabla_S \hat{w}_s^{t, \mathcal{V}, S, q} \right)^\top \Sigma q_s. \end{aligned}$$

We have

$$\nabla_S \hat{w}_s^{t, \mathcal{V}, S, q} = -\gamma \hat{w}_s^{t, \mathcal{V}, S, q} \nabla_S \theta_s^{t, S}$$

$$= -\gamma \hat{w}_s^{t,\mathcal{V},S,q} \left(2\hat{C}(s)S_s^{t,S} + \hat{E}(s) \right),$$

and

$$\partial_{\mathcal{V}} \hat{w}_s^{t,\mathcal{V},S,q} = -\gamma \hat{w}_s^{t,\mathcal{V},S,q}.$$

We define for all $s \in [t, T]$

$$\begin{aligned} \kappa_s^q &= -\gamma \left(q_s + 2\hat{C}(s)S_s^{t,S} + \hat{E}(s) \right), \\ \xi_{t,s}^q &= \exp \left(\int_t^s \kappa_{\varrho}^{q\top} V dW_{\varrho} - \frac{1}{2} \int_t^s \kappa_{\varrho}^{q\top} \Sigma \kappa_{\varrho}^q d\varrho \right). \end{aligned}$$

We then have

$$d \left(\hat{w}_s^{t,\mathcal{V},S,q} \left(\xi_{t,s}^q \right)^{-1} \right) = \left(\xi_{t,s}^q \right)^{-1} \mathcal{L}^q \hat{w}_s^{t,\mathcal{V},S,q} ds.$$

By definition of $\hat{w}$, $\mathcal{L}^q \hat{w}_s^{t,\mathcal{V},S,q} \leq 0$.

Moreover, equality holds for the control reaching the supremum in (2.22). It is easy to see that the supremum is reached for the unique value

$$\begin{aligned} q_s &= \frac{1}{\gamma} \Sigma^{-1} R(\bar{S} - S_s^{t,S}) - \nabla_S \hat{\theta}(t, S_s^{t,S}) \\ &= \frac{1}{\gamma} \Sigma^{-1} R(\bar{S} - S_s^{t,S}) - 2\hat{C}(s)S_s^{t,S} - \hat{E}(s) \\ &= \frac{1}{\gamma} (I_d + (T-s)R^\top) \Sigma^{-1} R(\bar{S} - S_s^{t,S}) \end{aligned}$$

which corresponds to $(q_s)_{s \in [t, T]} = (q_s^*)_{s \in [t, T]}$.

As a consequence, $\left(\hat{w}_s^{t,\mathcal{V},S,q} \left(\xi_{t,s}^q \right)^{-1} \right)_{s \in [t, T]}$ is nonincreasing and therefore

$$\hat{w} \left(T, \mathcal{V}_T^{t,\mathcal{V},S,q}, S_T^{t,S} \right) \leq \hat{w}(t, \mathcal{V}, S) \xi_{t,T}^q,$$

with equality when $(q_s)_{s \in [t, T]} = (q_s^*)_{s \in [t, T]}$.

Taking expectation we get

$$\mathbb{E} \left[\hat{w} \left(T, \mathcal{V}_T^{t,\mathcal{V},S,q}, S_T^{t,S} \right) \right] \leq \hat{w}(t, \mathcal{V}, S) \mathbb{E} \left[\xi_{t,T}^q \right].$$

We proceed to prove that $\mathbb{E} \left[\xi_{t,T}^q \right]$ is equal to 1. To do so, we use that $\xi_{t,t}^q = 1$ and prove that $(\xi_{t,s}^q)_{s \in [t, T]}$ is a martingale under $(\mathbb{P}; \mathbb{F} = (\mathcal{F}_s)_{s \in [t, T]})$.

We know that $(q_s^{t,q})_{s \in [t,T]}$ satisfies a linear growth condition with respect to $(S_s^{t,S})_{s \in [t,T]}$. Given the form of κ one can easily show that there exists a constant C such that

$$\sup_{s \in [t,T]} \|\kappa_s^q\|^2 \leq C \left(1 + \sup_{s \in [t,T]} \|W_s - W_t\|^2 \right).$$

By using classical properties of the Brownian motion, we prove that

$$\exists \epsilon > 0, \forall s \in [t, T], \quad \mathbb{E} \left[\exp \left(\frac{1}{2} \int_s^{(s+\epsilon) \wedge T} \kappa_\varrho^q \top \Sigma \kappa_\varrho^q d\varrho \right) \right] < +\infty.$$

Using a classical trick due to Beneš (see [Karatzas and Shreve \(2014\)](#), Chapter 5), we see that $(\xi_{t,s}^q)_{s \in [t,T]}$ is a martingale under $(\mathbb{P}; \mathbb{F} = (\mathcal{F}_s)_{s \in [t,T]})$.

We obtain

$$\mathbb{E} \left[\hat{w} \left(T, \mathcal{V}_T^{t,\mathcal{V},S,q}, S_T^{t,S} \right) \right] \leq \hat{w}(t, \mathcal{V}, S),$$

with equality when $(q_s)_{s \in [t,T]} = (q_s^*)_{s \in [t,T]}$. We conclude that

$$\begin{aligned} \hat{u}(t, \mathcal{V}, S) &= \sup_{(q_s)_{s \in [t,T]} \in \mathcal{A}_t^{Merton}} \mathbb{E} \left[-\exp \left(-\gamma V_T^{t,\mathcal{V},S,q} \right) \right] \\ &= \mathbb{E} \left[-\exp \left(-\gamma V_T^{t,\mathcal{V},S,q^*} \right) \right] \\ &= \hat{w}(t, \mathcal{V}, S). \end{aligned}$$

Chapter 3

Solvability of Riccati equations and applications to algorithmic trading

The results of this chapter are in [Drissi \(2022\)](#).

Differential Riccati equations (DREs) with indefinite coefficients arise in a wide class of control problems such as linear-quadratic zero-sum games, linear-quadratic Gaussian (LQG) control problems with indefinite cost, and linear-exponential-quadratic Gaussian (LEQG) control problems. Most results on these equations provide a statement of equivalence between the existence of solutions to a DRE and the solvability of a control problem. In this chapter, we provide an existence and uniqueness result for the solution to a type of DRE with indefinite deterministic matrix coefficients which arises in various control setups. Special cases of the DRE are relevant for algorithmic trading problems in LEQG setups, where the cost is an exponential of a quadratic functional of the state and the controls. In particular, the quadratic term matrix coefficient has both positive and negative eigenvalues, so classical existence results are not readily available.

We use the result to provide solutions to two algorithmic portfolio trading problems; one is market making with signals and hedging, the second is optimal trading with Bayesian learning of the drift. In both problems, the optimal strategies take the form of a linear feedback control whose feedback gain is obtained from the solution to a special case of the DRE that we study. Efficient numerical solving techniques for such equations have been extensively studied and provide computationally efficient methods to market participants, for whom speed and precision are key for better performance.

As a first application for liquidity providers, we introduce a comprehensive framework for multi-asset market making in OTC markets with three important features : (i) portfolio dynamics that incorporate signals, (ii) external hedging and client tiering, and (iii) computationally efficient methods to obtain quotes for large portfolios.

OTC markets are off-exchange and quote-driven markets based on a network of market makers that allow clients to buy and sell securities. In these markets, quotes are

either streamed to clients or clients request for quotes (RFQs).¹ In practice, OTC market makers provide liquidity for assets within a specific sector or asset class. Thus, the price dynamics of assets for which they make quotes usually share common stochastic trends and co-movements. For example, in OTC bond markets, a large number of bonds from the same issuer are traded and one needs to account for the joint dynamics that they exhibit; appropriate dynamics are necessary to obtain quotes that balance risk at the portfolio level.

Furthermore, market makers often use price predictors to improve trading performance, and use instruments that offset their inventory risk to hedge their positions. Our framework uses multivariate Ornstein–Uhlenbeck (multi–OU) dynamics to model the joint dynamics of these variables. We use an approximation technique similar to that introduced in [Bergault et al. \(2021\)](#) and we solve the optimal quote and optimal hedging approximation problem using the DRE existence and uniqueness result.

As a second application for liquidity takers, we combine the tools of stochastic control and online learning to derive an optimal portfolio trading strategy. The agent uses observed prices and signals to continuously update her estimate of the drift in the prices. The strategy uses signals either to improve the performance of execution programmes or to execute statistical arbitrages. We show that the problem reduces to a DRE for which existence and uniqueness of a solution is a straightforward application of our result.

In most execution strategies in the literature, model parameters are estimated and fixed prior to the start of the trading window. There are two downsides to this approach. First, parameter misspecification makes the strategy non-optimal. Second, the strategy does not incorporate uncertainty about the parameters. This work builds on the work in [Bismuth et al. \(2019\)](#) and proposes a model where the agent uses a prior distribution over the drift parameter to encode her uncertainty, and continuously uses price and signal observations to update her estimate during the execution programme.

Classical results on LQG problems and the associated DREs are in [Abou-Kandil et al. \(2012\)](#); [Fleming and Rishel \(2012\)](#). [Jacobson \(1973\)](#) uses optimal control to solve an LEQG control problem and obtains optimal controls that are equivalent to those obtained for quadratic differential games. [Whittle \(1990\)](#) and [Lim and Zhou \(2005\)](#) use the stochastic maximum principle to obtain similar results. Our existence and uniqueness result generalises that of [Bergault et al. \(2022\)](#); in particular, the matrix coefficients are deterministic time-dependent matrix functions and the quadratic term in the cost functional depends on the observed system. The existence and uniqueness of solutions in general remains an open problem for matrix DREs with indefinite coefficients. This work relates to the literature that addresses the solvability of these equations in special cases; see [Chen et al. \(1998\)](#); [Rami et al. \(2001\)](#); [Hu and Zhou \(2003\)](#). Finally, matrix DREs arise in various algorithmic trading problems; see [Cartea et al. \(2019, 2022g\)](#); [Barzykin et al. \(2022b\)](#),

¹ For a better understanding of the RFQ system, we refer to [Fermanian et al. \(2016\)](#).

The remainder of this chapter proceeds as follows. Section 3.1 introduces a family of DREs with indefinite coefficients and provides existence and uniqueness results based on a set of assumptions. Section 3.2 derives a portfolio market making strategy that uses signals to improve trading performance and uses a trading venue to hedge risk. The joint dynamics of the assets in the portfolio, the hedging instruments, and the signals follow multi-OU dynamics. We show that the approximation of the optimal quoting and hedging strategies reduces to a DRE that is a special case of that studied in Section 3.1. Finally, Section 3.3 solves optimal portfolio trading with Bayesian learning of the drift, using signals and past observations of the price to improve performance. We show that the problem reduces to a DRE that is also a special case of that studied in Section 3.1.

3.1 A general differential matrix Riccati equation

DREs with indefinite matrix coefficients arise in LQ games (Jacobson, 1973), LQG problems with indefinite cost (Chen et al., 1998), and LEQG problems (Whittle, 1990). Their solvability is also key in practical financial decision problems, such as portfolio optimisation (Zhou and Li, 2000), option pricing (Kohlmann and Zhou, 2000), and algorithmic trading (Bergault et al., 2022). Most existing results show that a control problem is well-posed if there are solutions to the DRE from which one obtains the optimal feedback control. In this section, we study a special type of DREs and provide conditions for the existence and uniqueness of a solution.

We motivate the DRE that we study by the type of DREs that arise in algorithmic trading problems where an agent adopts a CARA utility and where the dynamics of the asset prices are (linearly) driven by signals and past observations of prices. Section 3.2 and Section 3.3 give two examples of such problems.

We study the matrix DRE²

$$\dot{P}(t) = Q(t) + Y(t)^\top P(t) + P(t)Y(t) + P(t)U(t)P(t), \quad \forall t \in [0, T], \quad (3.1)$$

with terminal condition $P(T)$, where $P \in \mathcal{S}_{r+d}(\mathbb{R})$, $r \leq d$, and $(r, d) \in \mathbb{N}^* \times \mathbb{N}^*$.³ Let $\gamma > 0$, the matrix coefficients $Q \in C^0([0, T], \mathcal{S}_{r+d}(\mathbb{R}))$, $Y \in C^0([0, T], \mathcal{M}_{r+d}(\mathbb{R}))$, and $U \in C^0([0, T], \mathcal{S}_{r+d}(\mathbb{R}))$ are⁴

$$Q(t) = \begin{pmatrix} Q_{11}(t) & 0 \\ 0 & 0 \end{pmatrix}, \quad Y(t) = \begin{pmatrix} Y_{11}(t) & 0 \\ Y_{21}(t) & 0 \end{pmatrix}, \quad U(t) = \begin{pmatrix} \gamma U_{11}(t) & 0 \\ 0 & -U_{22}(t) \end{pmatrix},$$

and the terminal condition is

² The superscript $^\top$ stands for the transpose operator.

³ $\mathbb{N}^*$ is the set of positive integers.

⁴ $\mathcal{M}_{d,r}(\mathbb{R})$ is the set of $d \times r$ real matrices and $\mathcal{M}_d(\mathbb{R}) := \mathcal{M}_{d,d}(\mathbb{R})$ is the set of $d \times d$ real square matrices. We also denote the set of real symmetric $d \times d$ matrices by $\mathcal{S}_d(\mathbb{R})$. Finally, the subset of positive-definite and positive semi-definite matrices of $\mathcal{S}_d(\mathbb{R})$ are respectively denoted by $\mathcal{S}_d^{++}(\mathbb{R})$ and $\mathcal{S}_d^+(\mathbb{R})$.

$$P(T) = - \begin{pmatrix} \Psi & \frac{1}{2}\Upsilon \\ \frac{1}{2}\Upsilon^\top & \Gamma \end{pmatrix}, \quad (3.2)$$

where $\Psi \in \mathcal{S}_d^+(\mathbb{R})$, $\Gamma \in \mathcal{S}_r(\mathbb{R})$, $\Upsilon \in \mathcal{M}_{d,r}(\mathbb{R})$, $Q_{11} \in C^0([0, T], \mathcal{S}_d(\mathbb{R}))$, $Y_{11} \in C^0([0, T], \mathcal{M}_d(\mathbb{R}))$, $Y_{21} \in C^0([0, T], \mathcal{M}_{r,d}(\mathbb{R}))$, $U_{11} \in C^0([0, T], \mathcal{S}_d^+(\mathbb{R}))$, and $U_{22} \in C^0([0, T], \mathcal{S}_r^{++}(\mathbb{R}))$.

We show that a solution to the DRE (3.1) with terminal condition (3.2) exists and is unique when the coefficients satisfy a set of assumptions. First, Subsection 3.1.1 shows that the solution to the DRE corresponds to the solution to an LEQG control problem. Next, Subsection 3.1.2 uses the control problem and our assumptions to provide a-priori bounds to the solution to the DRE, and subsequently obtains existence and uniqueness of a solution.

3.1.1 An equivalent exponential utility control problem

Here, we show that the DRE (3.1) corresponds to an LEQG control problem, which then becomes the key tool to prove the main result of existence and uniqueness of a solution in Subsection 3.1.2. Let $(\Omega, \mathcal{F}, \mathbb{P}; \mathbb{F} = (\mathcal{F}(t))_{t \in [0, T]})$ be a filtered probability space that satisfies the usual conditions and which supports all the processes defined in this section. Consider the following deterministic matrix functions:

$$\begin{cases} A(t) = U_{22}(t)^{-1}, \\ B(t) = 2U_{22}(t)Y_{21}(t), \\ C(t) = Q_{11}(t) + Y_{21}(t)^\top U_{22}(t)Y_{21}(t), \end{cases} \quad \text{and} \quad \begin{cases} R(t) = -Y_{11}(t), \\ \Sigma(t) = \frac{1}{2}U_{11}(t). \end{cases} \quad (3.3)$$

Note that $A \in C^0([0, T], \mathcal{S}_r^{++}(\mathbb{R}))$, $B \in C^0([0, T], \mathcal{M}_{r,d}(\mathbb{R}))$, $C \in C^0([0, T], \mathcal{S}_d(\mathbb{R}))$, $R \in C^0([0, T], \mathcal{M}_d(\mathbb{R}))$, and $\Sigma \in C^0([0, T], \mathcal{S}_d^+(\mathbb{R}))$. Furthermore, let $j \in \mathbb{N}^*$ be the rank of the matrix $\Sigma(t)$ and consider a matrix $V \in C^0([0, T], \mathcal{M}_{d,j}(\mathbb{R}))$ such that $\Sigma(t) = V(t)V(t)^\top$ for all $t \in [0, T]$. The change of variables in (3.3) leads to the following matrix coefficients for the DRE (3.1):

$$Q(t) = \begin{pmatrix} C(t) - \frac{1}{4}B(t)^\top A(t)B(t) & 0 \\ 0 & 0 \end{pmatrix}, \quad Y(t) = \begin{pmatrix} -R(t) & 0 \\ \frac{-1}{2}A(t)B(t) & 0 \end{pmatrix}, \quad U(t) = \begin{pmatrix} 2\gamma\Sigma(t) & 0 \\ 0 & -A(t)^{-1} \end{pmatrix}.$$

Consider an observable system described by a d -dimensional state processes $(x_t)_{t \in [0, T]}$ with linear dynamics

$$dx_t = R(t)x_t dt + V(t)dW_t,$$

where $x_0 \in \mathbb{R}^d$ is known and $T > 0$ is fixed. The matrix function R encodes the linear dependence of the system to its previous state, V scales the variance of the noise, and

$(W_t)_{t \in [0, T]} = \left(W_t^1, \dots, W_t^j \right)_{t \in [0, T]}^\top$ is a j -dimensional standard Brownian motion with independent coordinates.

An agent controls an r -dimensional process $(u_t)_{t \in [0, T]}$ that affects two measured outputs $(y_t)_{t \in [0, T]}$ and $(z_t)_{t \in [0, T]}$. The set of admissible controls is

$$\mathcal{A}_t = \left\{ (u_s)_{s \in [t, T]}^\top, \mathbb{R}^r\text{-valued, } \mathbb{F}\text{-adapted,} \right. \\ \left. \text{satisfying a linear growth condition w.r.t } (x_s)_{s \in [t, T]} \right\},$$

and we write $\mathcal{A} := \mathcal{A}_0$.⁵ The output y is affected linearly and follows the dynamics

$$dy_t = u_t dt,$$

where $y_0 \in \mathbb{R}^r$ is known. On the other hand, the output z takes a quadratic form in the system and the controls and follows the dynamics

$$dz_t = - \begin{pmatrix} u_t \\ x_t \end{pmatrix}^\top \begin{pmatrix} A(t) & \frac{1}{2}B(t) \\ \frac{1}{2}B(t)^\top & C(t) \end{pmatrix} \begin{pmatrix} u_t \\ x_t \end{pmatrix} dt,$$

where $z_0 \in \mathbb{R}$ is known. Note that the condition $r \leq d$ above guarantees that the dimension of the agent's control u is smaller than that of the system x . In our problem, the output y measures an objective that the agents targets at the terminal time T , and the output z measures an objective that the agent wishes to maximise or a cost that she wishes to minimise by the terminal time T .

The performance criterion of the agent associated with a control $u \in \mathcal{A}$, is

$$\mathbb{E} \left[-\exp \left(-\gamma \left(z_T - \begin{pmatrix} x_T \\ y_T \end{pmatrix}^\top \begin{pmatrix} \Psi & \frac{1}{2}\Upsilon \\ \frac{1}{2}\Upsilon^\top & \Gamma \end{pmatrix} \begin{pmatrix} x_T \\ y_T \end{pmatrix} \right) \right) \right], \quad (3.4)$$

where γ measures risk aversion.

We introduce the following assumptions that are subsequently used in the remainder of this section:

Assumption 3.1.1

- (i) $C(t) = Q_{11}(t) + Y_{21}(t)^\top U_{22}(t) Y_{21}(t) \in \mathcal{S}_d^+$ for all $t \in [0, T]$,
- (ii) $B(t) \Sigma(t) B(t)^\top \in \mathcal{S}_d^{++}$, for all $t \in [0, T]$,

⁵ Let $t \in [0, T]$. An $\mathbb{R}^r$ -valued, $\mathbb{F}$ -adapted process $(\zeta_s)_{s \in [t, T]}$ is said to satisfy a linear growth condition on $[t, T]$ with respect to $(x_s)_{s \in [t, T]}$ if there exists a constant $C_{t,T} > 0$ such that for all $s \in [t, T]$, $\|\zeta_s\| \leq C_{t,T} \left(1 + \sup_{\tau \in [t, s]} \|x_\tau\| \right)$ almost surely. Here, $\|\cdot\|$ is a fixed norm on $\mathbb{R}^d$.

(iii) $Y_{11}(s)$ and $Y_{11}(t)$ commute, for all $(s, t) \in [0, T] \times [0, T]$.

Assumption 3.1.2 $\forall t \in [0, T], C(t) = \Psi = 0$. □

In the optimisation problem (3.4), Assumption 3.1.1-(i) ensures that the term in the performance criterion that is quadratic in the state x is a cost and not a gain. Assumptions 3.1.1-(ii) and 3.1.1-(iii) are technical conditions to obtain an upper bound for the value function which results in an upper bound for the solution to the DRE; see the proof of Theorem 3.1.2. Finally, when $r = d$, when the Assumptions 3.1.1 and 3.1.2 hold, and when the matrix coefficients in (3.1) are constant, the DRE (3.1) is similar to that in [Bergault et al. \(2022\)](#).

The agent maximises her performance (3.4) and her value function $\vartheta : [0, T] \times \mathbb{R}^d \times \mathbb{R}^r \times \mathbb{R} \rightarrow \mathbb{R}$ is

$$\vartheta(t, x, y, z) = \sup_{u \in \mathcal{A}_t} \mathbb{E} \left[-\exp \left(-\gamma \left(z_T^{t,x,z,u} - \begin{pmatrix} x_T^{t,x} \\ y_T^{t,y,u} \end{pmatrix}^\top \begin{pmatrix} \Psi & \frac{1}{2} \Upsilon \\ \frac{1}{2} \Upsilon^\top & \Gamma \end{pmatrix} \begin{pmatrix} x_T^{t,x} \\ y_T^{t,y,u} \end{pmatrix} \right) \right) \right], \quad (3.5)$$

where the processes $(x_s^{t,x})_{s \in [t, T]}$, $(y_s^{t,y,u})_{s \in [t, T]}$ and $(z_s^{t,x,z,u})_{s \in [t, T]}$ follow the dynamics

$$dz_s^{t,x,z,u} = - \begin{pmatrix} u_s \\ x_s^{t,x} \end{pmatrix}^\top \begin{pmatrix} A(s) & \frac{1}{2} B(s) \\ \frac{1}{2} B(s)^\top & C(s) \end{pmatrix} \begin{pmatrix} u_s \\ x_s^{t,x} \end{pmatrix} ds \quad \text{and} \quad \begin{cases} dx_s^{t,x} = R(s) x_s^{t,x} ds + V(s) dW_s, \\ dy_s^{t,y,u} = u_s ds, \end{cases}$$

for $(t, x, y, z) \in [0, T] \times \mathbb{R}^d \times \mathbb{R}^r \times \mathbb{R}$ and $u \in \mathcal{A}_t$, with $x_t^{t,x} = x$, $y_t^{t,y,u} = y$ and $z_t^{t,x,z,u} = z$.

Next, we state the following helpful result.

Proposition 3.1.1 Let Assumption 3.1.1 hold. If Assumption 3.1.2 also holds, then $\vartheta(t, x, y, z) > -\infty$. Otherwise, if Assumption 3.1.2 does not hold, then $\exists \bar{\gamma} > 0$ such that $\forall \gamma \in (0, \bar{\gamma}]$, $\vartheta(t, x, y, z) > -\infty$. □

Proof 3.1.1 Take $(t, x, y, z) \in [0, T] \times \mathbb{R}^d \times \mathbb{R}^r \times \mathbb{R}$, and consider the control $\underline{u} = (0)_{s \in [t, T]}$. The dynamics of the problem starting at t become

$$\begin{cases} y_s^{t,y,\underline{u}} &= y \\ dz_s^{t,x,z,\underline{u}} &= - (x_s^{t,x})^\top C(s) x_s^{t,x} ds \\ dx_s^{t,x} &= R(s) x_s^{t,x} ds + V(s) dW_s, \end{cases}$$

and the performance criterion for the strategy $\underline{u}$ becomes

$$\mathbb{E} \left[-\exp \left(-\gamma \left(z - \int_t^T (x_s^{t,x})^\top C(s) x_s^{t,x} ds + y^\top \Upsilon^\top x_T^{t,x} - (x_T^{t,x})^\top \Psi x_T^{t,x} - y^\top \Gamma y \right) \right) \right].$$

In the remainder of the proof, the controlled process notation is dropped, i.e., we write $x_s = x_s^{t,x}$, $y_s = y_s^{t,y,\underline{u}}$, and $z_s = z_s^{t,x,z,\underline{u}}$, for all $s \in [t, T]$. Next, write the solution of the SDE for the dynamics of x for all $s \in [t, T]$ as

$$x_s = \underline{R}(t, s)x + \int_t^s \underline{V}(u, s) dW_u, \quad (3.6)$$

where

$$\begin{cases} \underline{R}(t, s) &= \exp\left(\int_t^s R(u) du\right), \\ \underline{V}(u, s) &= \exp\left(\int_u^s R(\pi) d\pi\right) V(u). \end{cases}$$

To prove this point, consider the integrator $E_s = \exp\left(-\int_t^s R(u) du\right)x_s$, and notice that

$$dE_s = \exp\left(-\int_t^s R(u) du\right) V(s) dW_s,$$

where Assumption 3.1.1-(iii) is used.⁶ Next, integrate both sides to find the dynamics in (3.6). Notice that the solution is normal and write

$$x_s \sim \mathcal{N}(\underline{R}(s)x, \underline{\Sigma}(t, s)),$$

where

$$\underline{\Sigma}(t, s) = \int_t^s \exp\left(\int_u^s R(\pi) d\pi\right) \Sigma(u) \exp\left(\int_u^s R(\pi)^\top d\pi\right) du,$$

for all $s \in [t, T]$.

Use the solution (3.6) to write the following equalities in the case of the control $\underline{u}$:

$$\begin{cases} \int_t^T x_s^\top C(s) x_s dt &= x^\top \tilde{R}(t, T)x + 2x^\top \tilde{W}(t, T) + \tilde{C}(t, T), \\ x_T^\top \Psi x_T &= x^\top \underset{\vee}{R}(t, T)x + 2x^\top \underset{\vee}{W}(t, T) + \underset{\vee}{C}(t, T), \end{cases} \quad (3.7)$$

where

$$\begin{cases} \tilde{R}(t, T) &= \int_t^T \underline{R}(t, s)^\top C(s) \underline{R}(t, s) ds, \\ \tilde{W}(t, T) &= \int_t^T \int_t^s \underline{R}(t, s)^\top C(s) \underline{V}(t, u) dW_u ds, \\ \tilde{C}(t, T) &= \int_t^T \left(\int_t^s \underline{V}(t, u) dW_u\right)^\top C(s) \left(\int_t^s \underline{V}(t, u) dW_u\right) ds, \\ \underset{\vee}{R}(t, T) &= \underline{R}(t, T)^\top \Psi \underline{R}(t, T), \\ \underset{\vee}{W}(t, T) &= \underline{R}(t, T)^\top \Psi \int_t^T \underline{V}(t, s) dW_s, \\ \underset{\vee}{C}(t, T) &= \left(\int_t^T \underline{V}(t, s) dW_s\right)^\top \Psi \left(\int_t^T \underline{V}(t, s) dW_s\right). \end{cases}$$

Use Fubini for $\tilde{W}$ and write

⁶ In particular, when the matrices $R(u)$ and $R(s)$ commute for all $(u, s) \in [t, T]$, so do the matrices $R(u)$ and $\int_t^s R(\pi) d\pi$, and so do the matrices $R(u)$ and $\exp\left(\int_t^s R(\pi) d\pi\right)$.

$$\begin{cases} \tilde{W}(t, T) \sim \mathcal{N}\left(0, \tilde{\Sigma}(t, T)\right), \\ \underline{W}_{\vee}(t, T) \sim \mathcal{N}\left(0, \underline{\Sigma}_{\vee}(t, T)\right), \end{cases}$$

where

$$\begin{cases} \tilde{\Sigma}(t, T) = \int_t^T \left(\int_s^T \underline{R}(t, s)^\top C(s) \underline{V}(t, u) du \right)^\top \left(\int_s^T \underline{R}(t, s)^\top C(s) \underline{V}(t, u) du \right) ds, \\ \underline{\Sigma}_{\vee}(t, T) = \int_t^T \underline{R}(t, T) \Psi \underline{V}(t, s) \underline{V}(t, s)^\top \Psi^\top \underline{R}(t, T)^\top ds. \end{cases}$$

Finally, use the definition of the value function in (3.5), equations (3.7), and Hölder inequality to write

$$\begin{aligned} & \vartheta(t, x, y, z) \tag{3.8} \\ & \geq \mathbb{E} \left[-\exp \left(-\gamma \left(z - y^\top \Gamma y - \int_t^T (x_s^{t,x})^\top C(s) x_s^{t,x} ds - (x_T^{t,x})^\top \Psi x_T^{t,x} - y^\top \Upsilon^\top x_T^{t,x} \right) \right) \right] \\ & = -\mathbb{E} \left[\exp \left(-\gamma \left(-2x^\top \tilde{W}(t, T) - \tilde{C}(t, T) - 2x^\top \underline{W}_{\vee}(t, T) - \underline{C}_{\vee}(t, T) - y^\top \Upsilon^\top x_T^{t,x} \right) \right) \right] \\ & \quad \exp \left(-\gamma \left(z - y^\top \Gamma y - x^\top \tilde{R}(t, T) x - x^\top \underline{R}_{\vee}(t, T) x \right) \right) \\ & \geq -\mathbb{E} \left[\exp \left(8\gamma x^\top \tilde{W}(t, T) \right) \right]^{1/4} \mathbb{E} \left[\exp \left(8\gamma x^\top \underline{W}_{\vee}(t, T) \right) \right]^{1/4} \\ & \quad \mathbb{E} \left[\exp \left(4\gamma y^\top \Upsilon^\top x_T^{t,x} \right) \right]^{1/4} \mathbb{E} \left[\exp \left(4\gamma \tilde{C}(t, T) + 4\gamma \underline{C}_{\vee}(t, T) \right) \right]^{1/4} \\ & \quad \exp \left(-\gamma \left(z - y^\top \Gamma y - x^\top \tilde{R}(t, T) x - x^\top \underline{R}_{\vee}(t, T) x \right) \right) \\ & = -\exp \left(-\gamma z + x^\top \left(8\gamma^2 \tilde{\Sigma}(t, T) + 8\gamma^2 \underline{\Sigma}_{\vee}(t, T) + \gamma \tilde{R}(t, T) + \gamma \underline{R}_{\vee}(t, T) \right) x \right) \\ & \quad \exp \left(y^\top \left(2\gamma^2 \Upsilon^\top \underline{\Sigma}(t, s) \Upsilon + \gamma \Gamma \right) y + y^\top \left(\gamma \Upsilon^\top \underline{R}(t, T) \right) x \right) \\ & \quad \mathbb{E} \left[\exp \left(4\gamma \tilde{C}(t, T) + 4\gamma \underline{C}_{\vee}(t, T) \right) \right]^{1/4}. \end{aligned}$$

If Assumption 3.1.2 holds, then the term $\mathbb{E} \left[\exp \left(4\gamma \tilde{C}(t, T) + 4\gamma \underline{C}_{\vee}(t, T) \right) \right]^{1/4}$ in the above inequality is one and thus $\vartheta(t, x, y, z) > -\infty$. If Assumption 3.1.2 does not hold, then there exists a small enough $\bar{\gamma} > 0$, such that for all $0 < \gamma < \bar{\gamma}$, $\vartheta(t, x, y, z) > -\infty$. $\square$

The next theorem shows that the solution to the optimisation problem (3.4) is associated with that of the DRE (3.1).

Theorem 3.1.1 Assume there exist functions $M_1 \in C^1([0, T], \mathcal{S}_d(\mathbb{R}))$, $M_2 \in C^1([0, T], \mathcal{M}_{d,r}(\mathbb{R}))$, and $M_3 \in C^1([0, T], \mathcal{S}_r(\mathbb{R}))$ that satisfy the DRE (3.1) in

$$P(t) = \begin{pmatrix} M_1(t) & \frac{1}{2}M_2(t) \\ \frac{1}{2}M_2(t)^\top & M_3(t) \end{pmatrix},$$

where the coefficients are in (3.3). Next, let $M_4 \in C^1([0, T], \mathbb{R}^d)$ and $M_5 \in C^1([0, T], \mathbb{R}^r)$ be the unique solution to the linear ODE system

$$\begin{cases} \dot{M}_4(t) = -\frac{1}{2}(M_2(t) + B(t)^\top)A(t)^{-1}M_5(t) - R(t)^\top M_4(t) + 2\gamma M_1(t)\Sigma(t)M_4(t), \\ \dot{M}_5(t) = -M_3(t)^\top A(t)^{-1}M_5(t) + \gamma M_2(t)^\top \Sigma(t)M_4(t), \end{cases} \quad (3.9)$$

with terminal condition

$$M_4(T) = M_5(T) = 0, \quad (3.10)$$

and define $M_6 \in C^1([0, T], \mathbb{R})$ as

$$M_6(t) = \int_0^t \left(\frac{1}{4}M_5(s)^\top A(s)^{-1}M_5(s) - \text{Tr}(\Sigma(s)M_1(s)) + \frac{\gamma}{2}M_4(s)^\top \Sigma(s)M_4(s) \right) ds. \quad (3.11)$$

Let the function $\theta : [0, T] \times \mathbb{R}^d \times \mathbb{R}^r \mapsto \mathbb{R}$ and the associated function $w : [0, T] \times \mathbb{R}^d \times \mathbb{R}^r \times \mathbb{R} \mapsto \mathbb{R}$ be

$$\begin{aligned} \theta(t, x, y) = & x^\top M_1(t)x + x^\top M_2(t)y + y^\top M_3(t)y \\ & + x^\top M_4(t) + y^\top M_5(t) + M_6(t), \end{aligned} \quad (3.12)$$

$$w(t, y, x, z) = -\exp(-\gamma(z + \theta(t, x, y))). \quad (3.13)$$

Then for all $(t, x, y, z) \in [0, T] \times \mathbb{R}^d \times \mathbb{R}^r \times \mathbb{R}$ and $u = (u_s)_{s \in [t, T]} \in \mathcal{A}_t$, we have

$$\mathbb{E} \left[-\exp \left(-\gamma \left(z_T^{t, x, z, u} - \begin{pmatrix} x_T^{t, x} \\ y_T^{t, y, u} \end{pmatrix}^\top \begin{pmatrix} \Psi & \frac{1}{2}\Upsilon \\ \frac{1}{2}\Upsilon^\top & \Gamma \end{pmatrix} \begin{pmatrix} x_T^{t, x} \\ y_T^{t, y, u} \end{pmatrix} \right) \right) \right] \leq w(t, x, y, z). \quad (3.14)$$

Let Assumption 3.1.1 hold. If Assumption 3.1.2 also holds, then equality is obtained in (3.14) with the optimal control $(u_s^*)_{s \in [t, T]} \in \mathcal{A}_t$ in feedback form

$$u_s^* = \frac{1}{2}A(s)^{-1}(M_2(s)^\top x_s + 2M_3(s)y_s + M_5(s) - B(s)x_s). \quad (3.15)$$

Otherwise, if Assumption 3.1.2 does not hold, then $\exists \bar{\gamma} > 0$ such that $\forall \gamma \in (0, \bar{\gamma}]$, equality is obtained in (3.14) with the optimal control (3.15). Finally, in both cases, $w = \vartheta$ on $[0, T] \times \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^r$. $\square$

Proof 3.1.2 From HJB to Riccati

First, the HJB equation associated with the control problem (3.4) is

$$0 = \partial_t w + \sup_u \left\{ u^\top O(t)^\top \nabla_y w - u^\top A(t)u \partial_z w - u^\top B(t)x \partial_z w \right\} \quad (3.16)$$

$$-x^\top C(t)x \partial_z w + x^\top R(t)^\top \nabla_x w + \frac{1}{2} \text{Tr} \left(\Sigma(t) D_{xx}^2 w \right),$$

for all $(t, x, y, z) \in [0, T) \times \mathbb{R}^d \times \mathbb{R}^r \times \mathbb{R}$, with terminal condition

$$w(T, x, y, z) = -\exp \left(-\gamma \left(z - \begin{pmatrix} x \\ y \end{pmatrix}^\top \begin{pmatrix} \Psi & \frac{1}{2}\Upsilon \\ \frac{1}{2}\Upsilon^\top & \Gamma \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \right) \right). \quad (3.17)$$

Consider the function θ defined in (3.12) and notice that it solves the HJB

$$0 = \partial_t \theta + \sup_u \left\{ u^\top \nabla_y \theta - u^\top A(t)u - u^\top B(t)x \right\} \quad (3.18)$$

$$-x^\top C(t)x + x^\top R(t)^\top \nabla_x \theta + \frac{1}{2} \text{Tr} \left(\Sigma(t) D_{xx}^2 \theta \right) - \frac{\gamma}{2} \nabla_x \theta^\top \Sigma \nabla_x \theta,$$

with terminal condition

$$\theta(T, x, y) = -\begin{pmatrix} x \\ y \end{pmatrix}^\top \begin{pmatrix} \Psi & \frac{1}{2}\Upsilon \\ \frac{1}{2}\Upsilon^\top & \Gamma \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}, \quad \forall (x, y) \in \mathbb{R}^d \times \mathbb{R}^r. \quad (3.19)$$

The supremum in (3.18) is attained for the optimal control

$$u_s^\star = \frac{1}{2} A(s)^{-1} (\nabla_y \theta(s, x, y) - B(s)x_s), \quad \forall s \in [t, T]. \quad (3.20)$$

Replace in (3.18) to obtain the new HJB

$$0 = \partial_t \theta + \frac{1}{4} (\nabla_y \theta - B(t)x)^\top A(t)^{-1} (\nabla_y \theta - B(t)x) \quad (3.21)$$

$$-x^\top C(t)x + x^\top R(t)^\top \nabla_x \theta + \frac{1}{2} \text{Tr} \left(\Sigma(t) D_{xx}^2 \theta \right) - \frac{\gamma}{2} \nabla_x \theta^\top \Sigma(t) \nabla_x \theta.$$

Next, use the definition of θ in (3.12), replace in the HJB (3.21), and identify the terms of the polynomial to obtain the ODE system

$$\begin{cases} \dot{M}_1(t) = -\frac{1}{4} (M_2(t) - B(t)^\top) A(t)^{-1} (M_2(t)^\top - B(t)) + C(t) \\ \quad - R(t)^\top M_1(t) - M_1(t) R(t) + 2\gamma M_1(t)^\top \Sigma(t) M_1(t), \\ \dot{M}_2(t) = -(M_2(t) - B(t)^\top) A(t)^{-1} M_3(t) - R(t)^\top M_2(t) + 2\gamma M_1(t)^\top \Sigma(t) M_2(t), \\ \dot{M}_3(t) = -M_3(t)^\top A(t)^{-1} M_3(t) + \frac{\gamma}{2} M_2(t)^\top \Sigma(t) M_2(t), \\ \dot{M}_4(t) = -\frac{1}{2} (M_2(t) + B(t)^\top) A(t)^{-1} M_5(t) - R(t)^\top \dot{M}_4(t) + 2\gamma M_1(t)^\top \Sigma(t) M_4(t), \\ \dot{M}_5(t) = -M_3(t)^\top A(t)^{-1} M_5(t) + \gamma M_2(t)^\top \Sigma(t) M_4(t), \\ \dot{M}_6(t) = \frac{1}{4} M_5(t)^\top A(t)^{-1} M_5(t) - \text{Tr}(\Sigma(t) M_1(t)) + \frac{\gamma}{2} M_4(t)^\top \Sigma(t) M_4(t), \end{cases} \quad (3.22)$$

with terminal conditions $M_1(T) = -\Psi$, $M_2(T) = -\Upsilon$, $M_3(T) = -\Gamma$, and $M_4(T) = M_5(T) = M_6(T) = 0$.

Theorem (3.1.1) assumes the Riccati ODE (3.1) with terminal condition (3.2) admits a unique solution. Let $M_1 \in C^1([0, T], \mathcal{S}_d(\mathbb{R}))$, $M_2 \in C^1([0, T], \mathcal{M}_{r,d}(\mathbb{R}))$, and $M_3 \in C^1([0, T], \mathcal{S}_r(\mathbb{R}))$ be that solution. The ODE system (3.9) with terminal condition (3.10) is thus linear and admits a unique solution $M_4 \in C^1([0, T], \mathbb{R}^d)$, $M_5 \in C^1([0, T], \mathbb{R}^r)$, and $M_6 \in C^1([0, T], \mathbb{R})$ is obtained in (3.11). Consequently, the ODE system (3.22) admits a unique solution, and the candidate optimal control in feedback is

$$u_s^\star = \frac{1}{2} A(s)^{-1} (M_2(s)^\top x_s + 2M_3(s) y_s + M_5(s) - B(s) x_s), \quad \forall s \in [t, T]. \quad (3.23)$$

Define θ as in (3.12) and notice it solves the HJB equation (3.21) with terminal condition (3.19). Finally, define w as in (3.13) and notice it solves the HJB equation (3.16) with terminal condition (3.17), which is the HJB associated to our control problem. Thus, w is a candidate solution for the control problem (3.5).

Admissibility of the optimal control

To prove that $(u_s^\star)_{s \in [t, T]}$ in (3.23) is well-defined and admissible, consider the Cauchy initial value problem, $\forall s \in [t, T]$,

$$\begin{aligned} \frac{d\tilde{y}_s}{ds} &= \frac{1}{2} A(s)^{-1} (M_2(s)^\top x_s + 2M_3(s) y_s + M_5(s) - B(s) x_s) \\ &= \phi(s) y_s + \Phi(s, x_s), \end{aligned} \quad (3.24)$$

where $\tilde{y}_t = y$, $\phi(s) = A(s)^{-1} M_3(t)$, and

$$\Phi(s, x_s) = \frac{1}{2} A(s)^{-1} (M_2(s)^\top x_s + M_5(s) - B(s) x_s).$$

The unique solution to (3.24) is

$$\tilde{y}_s = \exp\left(\int_t^s \phi(r) dr\right) \left(y + \int_t^s \Phi(r, x_r) \exp\left(-\int_t^r \phi(\epsilon) d\epsilon\right) dr\right)$$

and the optimal control is

$$v_s^\star = \frac{d\tilde{y}_s}{ds} = \phi(s) \exp\left(\int_t^s \phi(r) dr\right) \left(y + \int_t^s \Phi(r, x_r) \exp\left(-\int_t^r \phi(\epsilon) d\epsilon\right) dr\right) + \Phi(s, x_s).$$

Note from the affine form of Φ in x that $u^\star$ satisfies a linear growth condition in x and is therefore admissible.

Verification

Fix $(t, x, y, z) \in [0, T] \times \mathbb{R}^d \times \mathbb{R}^r \times \mathbb{R}$ and $(v_s)_{s \in [t, T]} \in \mathcal{A}_t$. To prove that

$$\mathbb{E} \left[w \left(T, x_T^{t,x}, y_T^{t,u}, z_T^{t,x,z,u} \right) \right] \leq w(t, x, y, z),$$

use Ito's lemma for $s \in [t, T]$ and write

$$dw_s^u = \mathcal{L}^u w_s^u ds + (\nabla_x w_s^u)^\top V(s) dW_s, \quad (3.25)$$

where the notation $w(s, x_s^{t,x}, y_s^{t,u}, z_s^{t,x,z,u}) = w_s^u$ is used and

$$\begin{aligned} \mathcal{L}^u w_s^u = & \partial_t w_s^u - (x^\top C(s)x + u^\top A(s)u + x^\top B(s)^\top u) \partial_z w_s^u \\ & + u^\top \nabla_y w_s^u + x^\top R(s)^\top \nabla_x w_s^u + \frac{1}{2} \text{Tr} \left(\Sigma D_{xx}^2 w_s^u \right). \end{aligned}$$

Next, use (3.12) and (3.13) to see that

$$\nabla_x w_s^u = -\gamma w_s^u \nabla_x \theta_s^u = -\gamma w_s^u (2M_1(s)x_s + M_2(s)y_s + M_4(s)) = w_s^u \kappa_s^u,$$

where $\kappa_s^u = -\gamma (2M_1(s)x_s + M_2(s)y_s + M_4(s))$. The equality in (3.25) thus becomes

$$dw_s^u = \mathcal{L}^u w_s^u ds + \kappa_s^{u\top} (w_s^u)^\top V(s) dW_s. \quad (3.26)$$

Next, define

$$\chi_{t,s}^u = \exp \left(\int_t^s \kappa_r^{u\top} V(r) dW_r - \frac{1}{2} \int_t^s \kappa_r^{u\top} \Sigma(r) \kappa_r^u dr \right), \quad (3.27)$$

and use (3.26) and (3.27) and Ito's lemma to write

$$\begin{aligned} d \left(w_s^u (\chi_{t,s}^u)^{-1} \right) &= dw_s^u (\chi_{t,s}^u)^{-1} + d(\chi_{t,s}^u)^{-1} w_s^u - (\chi_{t,s}^u)^{-1} \kappa_s^{u\top} \Sigma(s) \nabla_x w_s^u ds \\ &= (\chi_{t,s}^u)^{-1} (dw_s^u - w_s^u \kappa_s^{u\top} V(s) dW_s + w_s^u \kappa_s^{u\top} \Sigma(s) \kappa_s^u ds - \kappa_s^{u\top} \Sigma(s) \nabla_x w_s^u ds) \\ &= (\chi_{t,s}^u)^{-1} (\mathcal{L}^u w_s^u ds). \end{aligned}$$

By the definition of w , we know that $\mathcal{L}^u w_s^u \leq 0$, and that equality holds for the control in (3.20) which corresponds to the optimal control (3.15). Thus, $\left(w_s^u (\chi_{t,s}^u)^{-1} \right)_{s \in [t, T]}$ is a nonincreasing process, consequently

$$\begin{aligned} w \left(T, x_T^{t,x}, y_T^{t,u}, z_T^{t,x,z,u} \right) \left(\chi_{t,T}^u \right)^{-1} &\leq w(t, x, y, z) \\ \implies w \left(T, x_T^{t,x}, y_T^{t,u}, z_T^{t,x,z,u} \right) &\leq w(t, x, y, z) \chi_{t,T}^u \\ \implies \mathbb{E} \left[w \left(T, x_T^{t,x}, y_T^{t,u}, z_T^{t,x,z,u} \right) \right] &\leq w(t, x, y, z) \mathbb{E} \left[\chi_{t,T}^u \right]. \end{aligned}$$

The next step is to prove that $\mathbb{E} \left[\chi_{t,T}^u \right] = 1$. Note that κ_s^u is affine in x and y , and because $(y_s^{t,u})_{s \in [t, T]}$ satisfies a linear growth condition with respect to $(x_s^{t,u})_{s \in [t, T]}$, there exists a constant C^k such that, almost surely for all $s < \tau \leq T$,

$$\sup_{r \in [s, \tau]} \|\kappa_r^u\|^2 \leq C^k \left(1 + \sup_{r \in [s, \tau]} \|W_r - W_s\|^2 \right).$$

Use the classical properties of the Brownian motion to write

$$\exists \varepsilon > 0, \forall s \in [t, T], \quad \mathbb{E} \left[\exp \left(\frac{1}{2} \int_s^{(s+\varepsilon) \wedge T} (\kappa_r^u)^\top \Sigma \kappa_r^u dr \right) \right] < +\infty,$$

and use the results in [Karatzas and Shreve \(2014\)](#) to obtain that $(\xi_{t,s}^u)_{s \in [t, T]}$ is a martingale. Thus, $\mathbb{E} \left[\chi_{t,T}^u \right] = 1$ because $\chi_{t,t}^u = 1$. Finally, conclude that

$$\vartheta(t, x, y, z) = \mathbb{E} \left[-\exp \left(-\gamma \left(z_T^{t,x,z,u^*} - \left(x_T^{t,x} \right)^\top \begin{pmatrix} \Psi & \frac{1}{2} \Upsilon \\ \frac{1}{2} \Upsilon^\top & \Gamma \end{pmatrix} \begin{pmatrix} x_T^{t,x} \\ y_T^{t,y,u^*} \end{pmatrix} \right) \right) \right] = w(t, x, y, z).$$

Theorem 3.1.1 solves the control problem (3.5) when the DRE (3.1) with terminal condition (3.2) admits a unique solution on $[0, T]$. The optimal control is a linear feedback with gains obtained from P . Subsection 3.1.2 uses the equivalent control problem to find a-priori bounds and prove existence and uniqueness of a solution to the DRE (3.1).

3.1.2 Existence and uniqueness of a solution

The next theorem provides conditions for the existence and uniqueness of a solution to the DRE (3.1) with terminal condition (3.2).

Theorem 3.1.2 Let Assumption 3.1.1 hold. If Assumption 3.1.2 also holds, then the DRE (3.1) with terminal condition (3.2) has a unique symmetric solution. Otherwise, if Assumption 3.1.2 does not hold, then $\exists \bar{\gamma} > 0$ such that the DRE (3.1) with terminal condition (3.2) has a unique symmetric solution for all $\gamma \in (0, \bar{\gamma}]$.

Proof 3.1.3 First, notice that if a unique solution to the Riccati ODE exists, then the ODE system (3.9) with terminal condition (3.10) is linear and M_6 is in (3.11), hence existence and uniqueness of a solution is immediate. It is thus enough to prove existence of a solution to the DRE. Cauchy-Lipschitz theorem gives existence and uniqueness of a left-maximal solution on the set $(T - \xi, T]$. Theorem 3.1.1 ensures that the associated function w defined in (3.12) is the value function of the problem on any interval $[\tau, T]$, where $\tau \in (T - \xi, T)$.

The proof consists in establishing by contradiction that ξ cannot be finite. This is achieved by providing a-priori bounds for a solution P of (3.1) with terminal condition (3.2). This proves that the solution does not blow up in finite time.⁷ The proof uses the original control problem to find bounds for the function θ in the form of polynomials of degree 2 in (x, y) , which translate into bounds for the quadratic term of θ , i.e., P .

⁷ Bounds are in the sense of the natural order on symmetric matrices: for $\underline{M}, \overline{M} \in \mathcal{S}_d(\mathbb{R})$, $\underline{M} \leq \overline{M}$ if and only if $\overline{M} - \underline{M} \in \mathcal{S}_d^+(\mathbb{R})$.

Lower bound

Fix $\tau \in (T - \xi, T)$, take $(t, x, y, z) \in [\tau, T] \times \mathbb{R}^d \times \mathbb{R}^r \times \mathbb{R}$, and consider the control $\underline{u} = (0)_{s \in [t, T]}$. The same arguments as in the proof of Proposition 3.1.1 lead to the inequality in (3.8) which translates into

$$\begin{aligned} \theta(t, x, y) &= \begin{pmatrix} x_t \\ y_t \end{pmatrix}^\top P(t) \begin{pmatrix} x_t \\ y_t \end{pmatrix} + \begin{pmatrix} x_t \\ y_t \end{pmatrix}^\top \begin{pmatrix} M_4(t) \\ M_5(t) \end{pmatrix} + M_6(t) \\ &\geq \begin{pmatrix} x_t \\ y_t \end{pmatrix}^\top \begin{pmatrix} \underline{M}_1(t) & \underline{M}_2(t) \\ \underline{M}_2(t)^\top & \underline{M}_3(t) \end{pmatrix} \begin{pmatrix} x_t \\ y_t \end{pmatrix} + \underline{M}_6(t), \end{aligned}$$

where

$$\begin{cases} \underline{M}_1(t) = & R(t, T) - 2\gamma \left(\tilde{\Sigma}(t, T) + \tilde{\Sigma}_V(t, T) \right) + \underline{R}(t, T)^\top \Psi \underline{R}(t, T), \\ \underline{M}_2(t) = & -\frac{\gamma}{2} \Upsilon^\top \underline{R}(t, T), \\ \underline{M}_3(t) = & -2\gamma^2 \Upsilon^\top \underline{\Sigma}(t, s) \Upsilon - \gamma \Gamma, \\ \underline{M}_6(t) = & -\frac{1}{4\gamma} \log \left(\mathbb{E} \left[\exp \left(4\gamma \tilde{C}(t, T) + 4\gamma \tilde{C}_V(t, T) \right) \right] \right). \end{cases}$$

Note that $\underline{M}_6$ exists in both cases of Theorem 3.1.2. The above order translates into the same order for the quadratic components of both polynomials of degree 2. In particular, one obtains a lower bound for P in the form

$$P(t) \geq \begin{pmatrix} \underline{M}_1(t) & \underline{M}_2(t) \\ \underline{M}_2(t)^\top & \underline{M}_3(t) \end{pmatrix}.$$

Upper bound

Fix $\tau \in (T - \xi, T)$, take $(t, x, y, z) \in [\tau, T] \times \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}$, and notice that

$$\begin{aligned} &\mathbb{E} \left[-\exp \left(-\gamma \left(z_T^{t,x,z,u} - \begin{pmatrix} x_T^{t,x} \\ y_T^{t,y} \end{pmatrix}^\top \begin{pmatrix} \Psi & \frac{1}{2} \Upsilon \\ \frac{1}{2} \Upsilon^\top & \Gamma \end{pmatrix} \begin{pmatrix} x_T^{t,x} \\ y_T^{t,y} \end{pmatrix} \right) \right) \right] \\ &\leq \mathbb{E} \left[-\exp \left(-\gamma \left(z_T^{t,x,z,u} - y_T^{t,y} \Upsilon^\top x_T^{t,x} \right) \right) \right] \\ &\leq \mathbb{E} \left[-\exp \left(-\gamma \left(z - \int_t^T u_s^\top B(s) x_s^{t,x} ds - y_T^{t,y} \Upsilon^\top x_T^{t,x} \right) \right) \right], \end{aligned} \tag{3.28}$$

because A, C, Ψ , and Γ are positive matrices. In the remainder of the proof for the upper bound, the controlled process notation is dropped for readability, i.e., we write $x_s = x_s^{t,x}$, $y_s = y_s^{t,y,u}$, and $z_s = z_s^{t,x,z,u}$, for all $s \in [t, T]$.

Next, use integration by parts to obtain

$$\int_t^T u_s^\top B(s) x_s ds = y_T^\top B(T) x_T - y^\top B(t) x$$

$$-\int_t^T y_s^\top (B'(s) + B(s)R(s))x_s ds - \int_t^T y_s^\top B(s)V(s)dW_s,$$

and use (3.28) to write

$$\begin{aligned} & \sup_{u \in \mathcal{A}_t} \mathbb{E} \left[-\exp \left(-\gamma \left(z_T - \begin{pmatrix} x_T \\ y_T \end{pmatrix}^\top \begin{pmatrix} \Psi & \frac{1}{2}\Upsilon \\ \frac{1}{2}\Upsilon^\top & \Gamma \end{pmatrix} \begin{pmatrix} x_T \\ y_T \end{pmatrix} \right) \right) \right] \\ & \leq \exp(-\gamma(z + y^\top B(t)x)) \\ & \quad \sup_{u \in \mathcal{A}_t} \mathbb{E} \left[-\exp \left(-\gamma \left(\int_t^T y_s \left(\bar{R}(s)^\top x_s ds + y_s^\top B(s)V(s)dW_s \right) - y_T^\top \bar{\Upsilon}^\top x_T \right) \right) \right] \\ & \leq \exp(-\gamma(z + y^\top B(t)x)) \sup_{y \in \mathcal{B}_t} \mathbb{E} \left[-\exp \left(-\gamma \left(\mathcal{V}_T - y_T^\top \bar{\Upsilon}^\top x_T \right) \right) \right], \end{aligned} \quad (3.29)$$

where $\bar{R}(s) = B'(s) + B(s)R(s)$, $\bar{\Upsilon} = B(T)^\top + \Upsilon$, the set $\mathcal{B}_t$ is

$$\begin{aligned} \mathcal{B}_t = & \left\{ (y_s)_{s \in [t, T]}, \mathbb{R}^d\text{-valued } \mathcal{F}^S\text{-adapted,} \right. \\ & \left. \text{satisfying a linear growth condition w.r.t } (x_s)_{s \in [t, T]} \right\}, \end{aligned}$$

the process $(\mathcal{V}_s)_{s \in [t, T]}$ has dynamics

$$d\mathcal{V}_s = y_s^\top \bar{R}(s)x_s ds + y_s^\top B(s)V(s)dW_s,$$

and the last inequality holds because when $(u_s)_{s \in [t, T]} \in \mathcal{A}_t$, then $(y_s)_{s \in [t, T]} \in \mathcal{B}_t$.

Next, to solve the optimisation problem

$$\sup_{y \in \mathcal{B}_t} \mathbb{E} \left[-\exp \left(-\gamma \left(\mathcal{V}_T - y_T^\top \bar{\Upsilon}^\top x_T \right) \right) \right], \quad (3.30)$$

define the associated value function $\bar{\vartheta} : [0, T] \times \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}$ as

$$\bar{\vartheta}(t, \mathcal{V}, x) = \sup_{y \in \mathcal{B}_t} \mathbb{E} \left[-\exp \left(-\gamma \left(\mathcal{V}_T^{t, \mathcal{V}, y, x} - y_T^\top \bar{\Upsilon}^\top x_T^{t, x} \right) \right) \right].$$

□

The value function $\bar{\vartheta}$ is the unique solution to the HJB equation

$$\begin{aligned} 0 = & \partial_t \bar{w} + x^\top R(t)^\top \nabla_x \bar{w} + \frac{1}{2} \text{Tr} \left(\Sigma(t) D_{xx}^2 \bar{w} \right) \\ & + \sup_{y \in \mathcal{B}} \left\{ x^\top \bar{R}(t)^\top y \partial_{\mathcal{V}} \bar{w} + \frac{1}{2} y^\top B(t) \Sigma(t) B(t)^\top y \partial_{\mathcal{V} \mathcal{V}} \bar{w} + \Sigma(t) B(t)^\top y \partial_{\mathcal{V}} \nabla_x \bar{w} \right\}, \end{aligned}$$

with terminal condition $\bar{w}(T, \mathcal{V}, x) = -\exp(-\gamma \mathcal{V})$. Next, use the ansatz

$$\bar{w}(t, \mathcal{V}, x) = -\exp \left(-\gamma \left(\mathcal{V} + \bar{\theta}(t, x) \right) \right)$$

to obtain the HJB

$$0 = \partial_t \bar{\theta} + x^\top R(t)^\top \nabla_x \bar{\theta} + \frac{1}{2} \text{Tr} \left(\Sigma(t) D_{xx}^2 \bar{\theta} \right) - \frac{\gamma}{2} \nabla_x \bar{\theta}^\top \Sigma(t) \nabla_x \bar{\theta} \\ + \sup_{y \in \mathcal{B}} \{ y^\top \left(\bar{R}(t)x - \gamma B(t) \Sigma(t) \nabla_x \bar{\theta} \right) - \frac{\gamma}{2} y^\top B(t) \Sigma(t) B(t)^\top y \}, \quad (3.32)$$

with terminal condition $\bar{\theta}(T, x) = 0$. So the HJB (3.32) becomes

$$0 = \partial_t \bar{\theta} + \frac{1}{2} \text{Tr} \left(\Sigma(t) D_{xx}^2 \bar{\theta} \right) + \frac{1}{2\gamma} x^\top \check{R}(t)x - x^\top \tilde{R}(t) \nabla_x \bar{\theta},$$

where

$$\begin{cases} \check{R}(t) = \bar{R}(t)^\top (B(t) \Sigma(t) B(t)^\top)^{-1} \bar{R}(t) \\ \tilde{R}(t) = -R(t)^\top + \bar{R}(t)^\top (B(t) \Sigma(t) B(t)^\top)^{-1} B(t) \Sigma(t), \end{cases}$$

where Assumption 3.1.1-(ii) ensures that $B(t) \Sigma(t) B(t)^\top$ is invertible. The optimal control in feedback form is

$$y_s^\star = \frac{1}{\gamma} (B(s) \Sigma(s) B(s)^\top)^{-1} \left(\bar{R}(s)x - \gamma B(s) \Sigma(s) \nabla_x \bar{\theta}(s, x) \right). \quad (3.33)$$

Next, use the ansatz $\bar{\theta}(t, x) = x^\top \bar{M}_1(t)x + x^\top \bar{M}_4(t) + \bar{M}_6(t)$ to obtain the system of ODEs

$$\begin{cases} \dot{\bar{M}}_1(t) = -\frac{1}{2\gamma} \check{R}(t) + \tilde{R}(t) \bar{M}_1(t) + \bar{M}_1(t) \tilde{R}(t) \\ \dot{\bar{M}}_4(t) = \tilde{R}(t) \bar{M}_4(t) \\ \dot{\bar{M}}_6(t) = -\text{Tr} \left(\Sigma(t) \bar{M}_1(t) \right), \end{cases} \quad (3.34)$$

with terminal condition $\bar{M}_1(T) = \bar{M}_4(T) = \bar{M}_6(T) = 0$. The system of ODEs (3.34) admits a unique solution because the equation in $\bar{M}_1$ is a linear system. Moreover, $\bar{M}_4 = 0$ and $\bar{M}_6$ is obtained by a simple integration. The optimal control in feedback form in (3.33) becomes

$$y_s^\star = \frac{1}{\gamma} (B(s) \Sigma(s) B(s)^\top)^{-1} \left(\bar{R}(s)x - 2\gamma B(s) \Sigma(s) \bar{M}_1(s)x - \gamma B(s) \Sigma(s) \bar{M}_4(s) \right).$$

Note that we obtain a classical solution to the optimisation problem (3.30) and a similar verification argument as in Theorem 3.1.1 ensures that the solution to the HJB corresponds to the value function $\bar{\vartheta}$. Thus, the inequality (3.29) becomes

$$-\exp(-\gamma(z + \theta(t, x, y))) \leq -\exp\left(-\gamma\left(z + y^\top B(t)x + \bar{\theta}(t, x)\right)\right),$$

thus

$$\begin{aligned}\theta(t, x, y) &= \begin{pmatrix} x_t \\ y_t \end{pmatrix}^\top P(t) \begin{pmatrix} x_t \\ y_t \end{pmatrix} + \begin{pmatrix} x_t \\ y_t \end{pmatrix}^\top \begin{pmatrix} M_4(t) \\ M_5(t) \end{pmatrix} + M_6(t) \\ &\leq \begin{pmatrix} x_t \\ y_t \end{pmatrix}^\top \begin{pmatrix} \overline{M}_1(t) & \frac{1}{2}B(t)^\top \\ \frac{1}{2}B(t) & 0 \end{pmatrix} \begin{pmatrix} x_t \\ y_t \end{pmatrix} + \overline{M}_6(t) .\end{aligned}$$

The above order translates into the same order for the quadratic components of both polynomials of degree 2. In particular, one obtains an upper bound for P in the form

$$P(t) \leq \begin{pmatrix} \overline{M}_1(t) & \frac{1}{2}B(t)^\top \\ \frac{1}{2}B(t) & 0 \end{pmatrix} .$$

Existence

As ξ is considered finite, there exists $\underline{M}, \overline{M} \in \mathcal{S}_d(\mathbb{R})$ with $\underline{M} \leq \overline{M}$ such that $\forall t \in [T - \xi, T]$, $P(t)$ stays in the compact set $\{M \in \mathcal{S}_d(\mathbb{R}) \mid \underline{M} \leq M \leq \overline{M}\}$. This contradicts the maximality of the solution, hence $\xi = \infty$, proving the result.

The following results extend Theorem 3.1.2 and shows that solutions to the DRE (3.1) provide solutions to larger families of DREs.

Corollary 3.1.1 Assume P is the unique solution to (3.1), then the DRE

$$\dot{\tilde{P}}(t) = \tilde{Q}(t) + \tilde{Y}(t)^\top \tilde{P}(t) + \tilde{P}(t) \tilde{Y}(t) + \tilde{P}(t) U(t) \tilde{P}(t)$$

with terminal condition $\tilde{P}(T) = P(T) - K$ admits a unique solution, where $K \in \mathcal{M}_{r+d}(\mathbb{R})$, and the coefficients are $\tilde{Y}(t) = Y(t) + U(t)K$, and $\tilde{Q}(t) = Q(t) + K U(t)K + Y(t)^\top K + K Y(t)$. $\square$

Corollary 3.1.2 Assume P is the unique solution to (3.1) and let

$$Z \in C^1([0, T], \mathcal{M}_{r+d}^{++}(\mathbb{R})) ,$$

then the DRE

$$\dot{\tilde{P}}(t) = \tilde{Q}(t) + \tilde{Y}(t)^\top \tilde{P}(t) + \tilde{P}(t) \tilde{Y}(t) + \tilde{P}(t) U(t) \tilde{P}(t)$$

with terminal condition $\tilde{P}(T) = Z(T)^\top P(T) Z(T)$ admits a unique solution, where

$$\begin{cases} \tilde{Y}(t) = Z(t)^{-1} Y(t) Z(t) + Z(t)^{-1} Z'(t) , \\ \tilde{U}(t) = Z(t)^{-1} U(t) (Z(t)^\top)^{-1} , \\ \tilde{Q}(t) = Z(t)^\top Q(t) Z(t) . \end{cases}$$

In the next sections, we study two algorithmic portfolio trading problems which are solved using the result in Theorem 3.1.2. First, we extend multi-asset market making to dynamics that incorporate signals and hedging instruments. Next, we solve optimal portfolio trading with Bayesian learning of the drift.

3.2 Multi-asset optimal market making

We consider a market maker in charge of quoting multiple assets in an OTC market. She receives RFQs from different clients with variable sizes, and decides on the optimal quotes at any point in time. The market maker trades assets that she quotes and other hedging instruments in a dealer-to-dealer venue to hedge risk. The joint dynamics of the quoted asset prices, the hedging instruments prices, and a set of signals follow a multi-OU process. We use the approximation method introduced in [Bergault et al. \(2021\)](#) to obtain a quoting and hedging strategy that can efficiently be computed by solving the DRE (3.1).

3.2.1 The model

Let $(\Omega, \mathcal{F}, \mathbb{P}; \mathbb{F} = (\mathcal{F}(t))_{t \in [0, T]})$ be a standard filtered probability space that supports all the processes we introduce and satisfies the usual conditions. A market maker operates in an OTC market and is in charge of a portfolio containing $r \in \mathbb{N}^*$ assets. She uses a set of $k \in \mathbb{N}$ signals and we model the joint dynamics of prices and signals by a d -dimensional multi-OU process $(S_t)_{t \in [0, T]} = (S_t^1, \dots, S_t^d)_{t \in [0, T]}^\top$ with dynamics:

$$dS_t = R(\bar{S} - S_t)dt + VdW_t, \quad (3.35)$$

where $d = r + k$, the initial values $S_0 \in \mathbb{R}^d$ are known, $\bar{S} \in \mathbb{R}^d$, $R \in \mathcal{M}_d(\mathbb{R})$, $V \in \mathcal{M}_{d,j}(\mathbb{R})$, and $(W_t)_{t \in [0, T]} = (W_t^1, \dots, W_t^j)_{t \in [0, T]}^\top$ is a j -dimensional standard Brownian motion with independent coordinates for some $j \in \mathbb{N}^*$. Without loss of generality, we assume that the first r elements of the vector S correspond to the asset prices.

Multi-OU dynamics are effective when the market maker uses the predictive power of signals, or for hedging purposes when there exists a linear combination of assets that reduces the overall risk. In the dynamics (3.35), the parameter $\bar{S}$ is the unconditional expectation of prices and signals in the long-run, and the matrix R drives the joint deterministic part in the dynamics. We denote by $\Sigma = VV^\top$ the variance-covariance matrix associated with the process $(S_t)_{t \in [0, T]}$, and Σ^r the variance-covariance matrix associated with the first r elements of the process $(S_t)_{t \in [0, T]}$.

In practice, market makers hedge their positions with securities that they quote and with other liquid instruments for which they do not offer quotes, but which offset their risk. For example, a market maker in charge of corporate bonds hedges rates risk with bond futures, and further hedges credit risk with credit derivatives. Moreover, market makers use latent factor models and other market variables with predictive power to enhance performance. Our framework is designed for these cases.

To incorporate hedging instruments in our model, for which the market maker does not propose quotes, it is enough to consider zero RFQ arrival intensities for these assets. To incorporate signals, we consider joint dynamics of prices and signals, but only consider inventory in the assets; see [Cartea et al. \(2019\)](#) for a similar setup.

3.2.2 The dynamics

Throughout the trading window $[0, T]$, the market maker interacts with N clients and chooses the prices at which she is ready to buy or sell the quoted assets.

Bid and ask quotes

The market maker quotes depend on the client and the size $z > 0$ of the request. We define the bid and ask prices of asset i for client's tier n , as the $\mathcal{P} \otimes \mathcal{B}(\mathbb{R}_+^*)$ -measurable maps $S^{i,n,b}, S^{i,n,a} : \Omega \times [0, T] \times \mathbb{R}_+^* \mapsto \mathbb{R}$, where $\mathcal{P}$ is the σ -algebra of predictable subsets of $\Omega \times [0, T]$, and $\mathcal{B}(\mathbb{R}_+^*)$ are the Borelian sets of $\mathbb{R}_+^*$.

For $i \in \{1, \dots, r\}$ and $n \in \{1, \dots, N\}$, the maps $(\delta_t^{i,n,b}(\cdot))_{t \in [0, T]}$ and $(\delta_t^{i,n,a}(\cdot))_{t \in [0, T]}$ are the distances of the quotes from the reference price S^i for, respectively, the bid and the ask and we write

$$\delta_t^{i,n,b}(z) = S_t^i - S_t^{i,n,b}(z) \quad \text{and} \quad \delta_t^{i,n,a} = S_t^{i,n,a}(z) - S_t^i, \quad \forall t \in [0, T].$$

Number of transactions and RFQ sizes

For $i \in \{1, \dots, r\}$ and $n \in \{1, \dots, N\}$, we use two cadlag $\mathbb{R}_+^*$ -marked point processes $N^{i,n,b}$ and $N^{i,n,a}$ to model the number of transactions at the bid and ask side, for asset i and tier n . We also define the probability measures $\psi^{i,n,b}$ and $\psi^{i,n,a}$ that correspond to the distributions of the requested sizes at the bid and ask side, respectively, of client's tier n for asset i , and which are used to integrate over the distribution of sizes. Furthermore, we consider that the point processes $N^{i,n,b}$ and $N^{i,n,a}$ have respective intensity kernels $(\lambda_t^{i,n,b}(dz))_{t \in \mathbb{R}_+}$ and $(\lambda_t^{i,n,a}(dz))_{t \in \mathbb{R}_+}$ given by⁸

$$\lambda_t^{i,n,b} = \Lambda^{i,n,b}(\delta_t^{i,n,b}(z))\psi^{i,n,b}(dz) \quad \text{and} \quad \lambda_t^{i,n,a} = \Lambda^{i,n,a}(\delta_t^{i,n,a}(z))\psi^{i,n,a}(dz).$$

Recall that the intensities corresponding to the hedging instruments are set to zero.

Inventory and internalisation / externalisation

For $i \in \{1, \dots, r\}$, the inventory process of the market maker for asset i is denoted by $(q_t^i)_{t \in [0, T]}$. We denote by $(q_t)_{t \in [0, T]}$ the vector process $(q_t^1, \dots, q_t^r)_{t \in [0, T]}^\top$. The market maker uses another trading venue to trade the assets for which she proposes quotes, but

⁸ Moreover, for each asset $i \in \{1, \dots, r\}$ and tier $n \in \{1, \dots, N\}$, we assume that the functions $\Lambda^{i,n,\cdot}$ for the bid and the ask satisfy the following properties, where $\Lambda^{i,n,\cdot'}$ and $\Lambda^{i,n,\cdot''}$ are respectively the first and second derivative of $\Lambda^{i,n,\cdot}$: (i) $\Lambda^{i,n,\cdot}$ is twice continuously differentiable, (ii) $\Lambda^{i,n,\cdot''}(\delta) < 0$, (iii) $\lim_{\delta \rightarrow +\infty} \Lambda^{i,n,\cdot}(\delta) = 0$, and (iv) $\sup_{\delta} \frac{\Lambda^{i,n,\cdot}(\delta)\Lambda^{i,n,\cdot''}(\delta)}{(\Lambda^{i,n,\cdot'}(\delta))^2} < 2$.

also to trade other hedging instruments. After an RFQ is received, the market maker can either internalise, i.e., keep an inventory while waiting for an offsetting order, or she can hedge, with a cost, her risk. We use the process $(v_t)_{t \in \mathbb{R}_+} = (v_t^1, \dots, v_t^r)_{t \in \mathbb{R}_+}^\top$ to model the execution speed on the trading venue. Two types of costs are incurred while trading: a temporary price impact incorporated in the agent's wealth and considered quadratic in the speed, and a permanent impact incorporated in the price, and usually considered linear in the speed. Here, we drop the linear permanent impact component. Finally, the inventory of the market maker in asset $i \in \{1, \dots, r\}$ has dynamics combining the results of her market making and hedging activity:

$$dq_t^i = v_t^i dt + \sum_{n=1}^N \int_{\mathbb{R}_+^*} z N^{i,n,b}(dt, dz) - \sum_{n=1}^N \int_{\mathbb{R}_+^*} z N^{i,n,a}(dt, dz), \quad q_0^i \text{ given.}$$

Cash process

We use the process $(X_t)_{t \in [0, T]}$ to model the amount on the market maker's cash account and which follows the dynamics

$$\begin{aligned} dX_t = & -S_t^\top \mathcal{J}^\top dq_t - v_t^\top \eta v_t dt \\ & + \sum_{i=1}^r \sum_{n=1}^N \int_{\mathbb{R}_+^*} z \left(\delta_t^{i,n,b}(z) N^{i,n,b}(dt, dz) + \delta_t^{i,n,a}(z) N^{i,n,a}(dt, dz) \right), \end{aligned}$$

with $X_0 \in \mathbb{R}$ given, $\eta \in \mathcal{S}_r^{++}(\mathbb{R})$ quantifies the execution costs from externalisation, and $\mathcal{J}$ is an $r \times d$ matrix with $\mathcal{J}_{ij} = \mathbb{1}_{i=j}$. The matrix $\mathcal{J}$ maps the first r elements of the d -dimensional price S to an r -dimensional vector.

3.2.3 The optimisation problem

The market maker maximises the expected exponential utility of her final wealth by the end of the trading window $[0, T]$:

$$\mathbb{E} \left[-\exp \left(-\gamma \left(X_T + q_T^\top \mathcal{J} S_T - q_T^\top \Gamma q_T \right) \right) \right], \quad (3.36)$$

where $\mathcal{A} = \mathcal{A}_0$ is the set of admissible strategies, and $\mathcal{A}_t$ is defined as

$$\begin{aligned} \mathcal{A}_t = & \left\{ \left(\delta_s^b, \delta_s^a, v_s \right)_{s \in [t, T]} : \Omega \times [t, T] \times \mathbb{R}_+^* \rightarrow \mathbb{R}^2 \times \mathbb{R}^r \quad \text{s.t.} \quad \left(\delta^b, \delta^a \right) \text{ is} \right. \\ & \mathcal{P} \otimes \mathcal{B}(\mathbb{R}_+^*)\text{-measurable and bounded from below,} \\ & \left. \text{and } v \text{ is } \mathcal{P}\text{-measurable and bounded} \right\}. \end{aligned}$$

The final wealth in the performance criterion (3.36) is the sum of the final cash amount X_T and the remaining inventory valued at $q_T^\top \mathcal{J} S_T - q_T^\top \Gamma q_T$, where $\Gamma \in \mathcal{S}_r^{++}(\mathbb{R})$. The final penalty $q_T^\top \Gamma q_T$ is a discount applied to the terminal mark-to-market value of the assets which penalises any non-zero final inventory.

The value function $\vartheta : [0, T] \times \mathbb{R} \times \mathbb{R}^r \times \mathbb{R}^d \rightarrow \mathbb{R}$ of the market maker is

$$\vartheta(t, x, q, S) = \sup_{(\delta^b, \delta^a, v) \in \mathcal{A}_t} \mathbb{E} \left[-e^{-\gamma \left(X_T^{t, x, S, \delta^b, \delta^a, v} + \left(q_T^{t, q, \delta^b, \delta^a, v} \right)^\top \mathcal{J} S_T^{t, S} - \left(q_T^{t, q, \delta^b, \delta^a, v} \right)^\top \Gamma q_T^{t, q, \delta^b, \delta^a, v} \right)} \right],$$

where the processes $\left(q_T^{t, q, \delta^b, \delta^a, v} \right)_{s \geq t}$, $\left(X_T^{t, x, S, \delta^b, \delta^a, v} \right)_{s \geq t}$ and $\left(S_T^{t, S} \right)_{s \geq t}$ start at t with values $(x, S, \delta^b, \delta^a, v)$ for the controls and the state variables. The Hamilton–Jacobi–Bellman (HJB) equation associated with the market making problem is

$$\begin{aligned} 0 = & \partial_t w(t, x, q, S) + \left(\bar{S} - S \right)^\top R^\top \nabla_S w(t, x, q, S) + \frac{1}{2} \text{Tr} \left(\Sigma D_{SS}^2 w(t, x, q, S) \right) \\ & + \sup_v \left\{ v^\top \nabla_q w(t, x, q, S) - (v^\top \mathcal{J} S + v^\top \eta v) \partial_x w(t, x, q, S) \right\} \\ & + \sum_{i=1}^r \sum_{n=1}^N \int_{\mathbb{R}_+^*} \sup_{\delta^{i, n, b}} \Lambda^{i, n, b} \left(\delta_t^{i, n, b}(z) \right) \\ & \quad \left(w \left(t, x - z(S^i - \delta_t^{i, n, b}(z)), q + z e^i, S \right) - w(t, x, q, S) \right) \psi^{i, n, b}(dz) \\ & + \sum_{i=1}^r \sum_{n=1}^N \int_{\mathbb{R}_+^*} \sup_{\delta^{i, n, a}} \Lambda^{i, n, a} \left(\delta_t^{i, n, a}(z) \right) \\ & \quad \left(w \left(t, x + z(S^i + \delta_t^{i, n, a}(z)), q - z e^i, S \right) - w(t, x, q, S) \right) \psi^{i, n, a}(dz), \end{aligned} \quad (3.37)$$

where $\{e^i\}$ is the canonical basis of $\mathbb{R}^r$, with terminal condition

$$w(T, x, q, S) = -\exp(-\gamma(x + q^\top \mathcal{J} S - q^\top \Gamma q)), \quad \forall (x, q, S) \in \mathbb{R} \times \mathbb{R}^r \times \mathbb{R}^d. \quad (3.38)$$

In the next proposition, for which a proof is straightforward, we use the ansatz

$$w(t, x, q, S) = -\exp(-\gamma(x + q^\top \mathcal{J} S + \theta(t, q, S))).$$

Proposition 3.2.1 Assume there is a solution $\theta \in C^{1,1,2}([0, T] \times \mathbb{R}^r \times \mathbb{R}^d, \mathbb{R})$ to the equation:

$$\begin{aligned} 0 = & \partial_t \theta(t, q, S) + \left(\bar{S} - S \right)^\top R^\top (\mathcal{J}^\top q + \nabla_S \theta(t, q, S)) + \frac{1}{2} \text{Tr} \left(\Sigma D_{SS}^2 \theta(t, q, S) \right) \\ & - \frac{\gamma}{2} (\mathcal{J}^\top q + \nabla_S \theta(t, q, S))^\top \Sigma (\mathcal{J}^\top q + \nabla_S \theta(t, q, S)) + \sup_v \left\{ v^\top \nabla_q \theta(t, q, S) - v^\top \eta v \right\} \end{aligned}$$

$$\begin{aligned}
& + \sum_{i=1}^r \sum_{n=1}^N \int_{\mathbb{R}_+^*} z H^{i,n,b} \left(z, \frac{\theta(t, q, S) - \theta(t, q + z e^i, S)}{z} \right) \psi^{i,n,b}(dz) \\
& + \sum_{i=1}^r \sum_{n=1}^N \int_{\mathbb{R}_+^*} z H^{i,n,a} \left(z, \frac{\theta(t, q, S) - \theta(t, q - z e^i, S)}{z} \right) \psi^{i,n,a}(dz),
\end{aligned} \tag{3.39}$$

on $[0, T) \times \mathbb{R}^r \times \mathbb{R}^d$, with terminal condition

$$\theta(T, q, S) = -q^\top \Gamma q, \quad \forall (q, S) \in \mathbb{R}^r \times \mathbb{R}^d,$$

where we define the Hamiltonian functions $H^{i,n,b}$ and $H^{i,n,a}$ respectively as

$$\begin{aligned}
H^{i,n,b}(z, p) &= \sup_{\delta} \frac{\Lambda^{i,n,b}(\delta)}{\gamma z} (1 - \exp(-\gamma z(\delta - p))), \\
H^{i,n,a}(z, p) &= \sup_{\delta} \frac{\Lambda^{i,n,a}(\delta)}{\gamma z} (1 - \exp(-\gamma z(\delta - p))).
\end{aligned} \tag{3.40}$$

Then, the function $w : [0, T] \times \mathbb{R} \times \mathbb{R}^r \times \mathbb{R}^d \rightarrow \mathbb{R}$, defined by

$$w(t, x, q, S) = -\exp(-\gamma(x + q^\top \mathcal{J} S + \theta(t, q, S))), \quad \forall (t, x, q, S) \in [0, T] \times \mathbb{R} \times \mathbb{R}^r \times \mathbb{R}^d,$$

is a solution to (3.37) with terminal condition (3.38). $\square$

To obtain the optimal trading (externalisation) speed, we write the Legendre-Fenchel transform

$$L : p \in \mathbb{R}^d \mapsto \sup_{v \in \mathbb{R}^d} v^\top p - v^\top \eta v = \frac{1}{4} p^\top \eta^{-1} p,$$

of the quadratic execution cost, so the supremum is reached at

$$v_s^{*\top} = \frac{1}{2} \nabla_q \theta(s, q_s, S_s)^\top \eta^{-1}.$$

Second, to obtain the optimal quotes, similar arguments as in [Guéant \(2017\)](#) lead to the optimal bid and ask quotes $S_t^{i,n,b}(z) = S_t^i - \delta_t^{i,n,b*}(z)$ and $S_t^{i,n,a}(z) = S_t^i + \delta_t^{i,n,a*}(z)$ for an RFQ of size z , where

$$\begin{aligned}
\delta_t^{i,n,b*}(z) &= \tilde{\delta}^{i,n,b*} \left(z, \frac{\theta(t, q_{t-}, S_t) - \theta(t, q_{t-} + z e^i, S_t)}{z} \right), \\
\delta_t^{i,n,a*}(z) &= \tilde{\delta}^{i,n,a*} \left(z, \frac{\theta(t, q_{t-}, S_t) - \theta(t, q_{t-} - z e^i, S_t)}{z} \right),
\end{aligned} \tag{3.41}$$

and the functions $\tilde{\delta}^{i,n,b*}(\cdot, \cdot)$ and $\tilde{\delta}^{i,n,a*}(\cdot, \cdot)$ are

$$\begin{aligned}\tilde{\delta}^{i,n,b*}(z,p) &= \Lambda^{i,n,b^{-1}} \left(\gamma z H^{i,n,b}(z,p) - H^{i,n,b'}(z,p) \right), \\ \tilde{\delta}^{i,n,a*}(z,p) &= \Lambda^{i,n,a^{-1}} \left(\gamma z H^{i,n,a}(z,p) - H^{i,n,a'}(z,p) \right),\end{aligned}$$

and $H^{i,n,\cdot'}(z,p)$ denotes the derivative of $H^{i,n,\cdot}(z,p)$ with respect to p . It is beyond the scope of this chapter to provide a formal solution to the optimal quote problem and we focus on the approximation technique (see Barzykin et al., 2022a, for a rigorous solution to a similar problem).

3.2.4 Quadratic approximation of the value function

One usually employs numerical methods to approximate the value function and hence the optimal quotes (3.41). These methods suffer from the curse of dimensionality and do not usually scale up to the multi-asset case, because the number of dimensions grows exponentially with the number of assets. In this subsection, we approximate the solution θ of the HJB (3.39) characterising the optimal quotes by the solution $\tilde{\theta}$ of another HJB for which computations are easier to carry out.

For small RFQ sizes, we use the approximation

$$\frac{\theta(t,q,S) - \theta(t,q + ze^i, S)}{z} \approx - \frac{\theta(t,q,S) - \theta(t,q - ze^i, S)}{z}.$$

Thus, the term we wish to approximate in (3.39) becomes a function of the form $p \rightarrow H^{i,n,b}(p) + H^{i,n,a}(-p)$. Moreover, the functions $H^{i,n,b}$ and $H^{i,n,a}$ in (3.40) are positive and decreasing in p , so we use a quadratic approximation. Furthermore, a quadratic approximation simplifies the problem to finding the solution to a DRE of the form (3.1). More precisely, for all $i \in \{1, \dots, r\}$ and $n \in \{1, \dots, N\}$, we approximate the Hamiltonian term

$$H^{i,n,b} \left(z, \frac{\theta(t,q,S) - \theta(t,q + ze^i, S)}{z} \right) + H^{i,n,a} \left(z, \frac{\theta(t,q,S) - \theta(t,q - ze^i, S)}{z} \right)$$

by the sum of the two quadratic functions

$$\begin{cases} \tilde{H}^{i,n,a} : (z,p) \mapsto \alpha_0^{i,n,a}(z) + \alpha_1^{i,n,a}(z)p + \frac{1}{2}\alpha_2^{i,n,a}(z)p^2, \\ \tilde{H}^{i,n,b} : (z,p) \mapsto \alpha_0^{i,n,b}(z) + \alpha_1^{i,n,b}(z)p + \frac{1}{2}\alpha_2^{i,n,b}(z)p^2. \end{cases}$$

The values of the approximation coefficients for each request size z are obtained using the Taylor expansion around $p = 0$:

$$\begin{cases} \alpha_0^{i,n,b}(z) &= H^{i,n,b}(z,0) \\ \alpha_0^{i,n,a}(z) &= H^{i,n,a}(z,0), \end{cases} \quad \text{and} \quad \begin{cases} \alpha_1^{i,n,b}(z) &= H^{i,n,b'}(z,0) \\ \alpha_1^{i,n,a}(z) &= H^{i,n,a'}(z,0), \end{cases}$$

and

$$\begin{cases} \alpha_0^{i,n,b}(z) &= H^{i,n,b''}(z, 0) \\ \alpha_0^{i,n,a}(z) &= H^{i,n,a''}(z, 0). \end{cases}$$

Next, we denote the approximation of the function θ by $\tilde{\theta}$, which verifies the HJB

$$\begin{aligned} 0 = & \partial_t \tilde{\theta}(t, q, S) + \left(\bar{S} - S \right)^\top R^\top \left(\mathcal{J}^\top q + \nabla_S \tilde{\theta}(t, q, S) \right) \\ & - \frac{\gamma}{2} \left(\mathcal{J}^\top q + \nabla_S \tilde{\theta}(t, q, S) \right)^\top \Sigma \left(\mathcal{J}^\top q + \nabla_S \tilde{\theta}(t, q, S) \right) \\ & + \frac{1}{2} \text{Tr} \left(\Sigma D_{SS}^2 \tilde{\theta}(t, q, S) \right) + \frac{1}{4} \nabla_q \tilde{\theta}(s, q_s, S_s)^\top \eta^{-1} \nabla_q \tilde{\theta}(s, q_s, S_s) \\ & + \sum_{i=1}^r \sum_{n=1}^N \int_{\mathbb{R}_+^*} z \alpha_0^{i,n,b}(z) \psi^{i,n,b}(dz) + \sum_{i=1}^r \sum_{n=1}^N \int_{\mathbb{R}_+^*} z \alpha_0^{i,n,a}(z) \psi^{i,n,a}(dz) \\ & + \sum_{i=1}^r \sum_{n=1}^N \int_{\mathbb{R}_+^*} \alpha_1^{i,n,b}(z) \left(\tilde{\theta}(t, q, S) - \tilde{\theta}(t, q + ze^i, S) \right) \psi^{i,n,b}(dz) \\ & + \sum_{i=1}^r \sum_{n=1}^N \int_{\mathbb{R}_+^*} \alpha_1^{i,n,a}(z) \left(\tilde{\theta}(t, q, S) - \tilde{\theta}(t, q + ze^i, S) \right) \psi^{i,n,a}(dz) \\ & + \sum_{i=1}^r \sum_{n=1}^N \int_{\mathbb{R}_+^*} \frac{1}{2z} \alpha_2^{i,n,b}(z) \left(\tilde{\theta}(t, q, S) - \tilde{\theta}(t, q + ze^i, S) \right)^2 \psi^{i,n,b}(dz) \\ & + \sum_{i=1}^r \sum_{n=1}^N \int_{\mathbb{R}_+^*} \frac{1}{2z} \alpha_2^{i,n,a}(z) \left(\tilde{\theta}(t, q, S) - \tilde{\theta}(t, q + ze^i, S) \right)^2 \psi^{i,n,a}(dz), \quad (3.42) \end{aligned}$$

with terminal condition

$$\tilde{\theta}(T, q, S) = -q^\top \Gamma q, \quad \forall (q, S) \in \mathbb{R}^r \times \mathbb{R}^d. \quad (3.43)$$

To further simplify the above HJB, use the ansatz

$$\tilde{\theta}(t, q, S) = q^\top A(t)q + q^\top B(t)S + S^\top C(t)S + D(t)^\top q + E(t)^\top S + F(t), \quad (3.44)$$

based on the next proposition whose proof is omitted.

Proposition 3.2.2 Assume there exist $A \in C^1([0, T], \mathcal{S}_r(\mathbb{R}))$, $B \in C^1([0, T], \mathcal{M}_{r,d}(\mathbb{R}))$, $C \in C^1([0, T], \mathcal{S}_d(\mathbb{R}))$, $D \in C^1([0, T], \mathbb{R}^r)$, $E \in C^1([0, T], \mathbb{R}^d)$, $F \in C^1([0, T], \mathbb{R})$ satisfying the system of ODEs

$$\begin{cases}
-\dot{A}(t) = -\frac{\gamma}{2} (B(t) - \mathcal{J}) \Sigma (B(t)^\top - \mathcal{J}) + A(t) \left(2V_{2,1}^b + 2V_{2,1}^a + \eta^{-1} \right) A(t) \\
-\dot{B}(t) = (\mathcal{J} - B(t)) (-R + 2\gamma \Sigma C(t)) + A(t) \left(2V_{2,1}^b + 2V_{2,1}^a + \eta^{-1} \right) B(t) \\
-\dot{C}(t) = R^\top C(t) + C(t) R - 2\gamma C(t) \Sigma C(t) + \frac{1}{4} B(t)^\top \left(2V_{2,1}^b + 2V_{2,1}^a + \eta^{-1} \right) B(t) \\
-\dot{D}(t) = (\mathcal{J} - B(t)) \left(R \bar{S} + \gamma \Sigma E(t) \right) + 2A(t) \left(v_{1,1}^b - v_{1,1}^a \right) \\
\quad + 2A(t) \left(V_{2,2}^b - V_{2,2}^a \right) \mathcal{D}(A(t)) + 2A(t) \left(V_{2,1}^b + V_{2,1}^a \right) D(t) + A(t)^\top \eta^{-1} D(t) \\
-\dot{E}(t) = -2C(t) R \bar{S} + R^\top E(t) - 2\gamma C(t) \Sigma E(t) + B(t)^\top \left(v_{1,1}^b - v_{1,1}^a \right) \\
\quad + B(t)^\top \left(V_{2,2}^b - V_{2,2}^a \right) \mathcal{D}(A(t)) + B(t)^\top \left(V_{2,1}^b + V_{2,1}^a \right) D(t) + \frac{1}{2} B(t)^\top \eta^{-1} D(t) \\
-\dot{F}(t) = -\bar{S}^\top R^\top E(t) - \frac{\gamma}{2} E(t)^\top \Sigma E(t) - \text{Tr}(\Sigma C(t)) + \text{Tr} \left(V_{0,1}^b + V_{0,1}^a \right) \\
\quad + \left(v_{1,1}^b - v_{1,1}^a \right)^\top D(t) + \frac{1}{2} D(t)^\top \left(V_{2,1}^b + V_{2,1}^a \right) D(t) \\
\quad + \text{Tr} \left(\left(V_{1,2}^b + V_{1,2}^a \right) A(t) \right) + \frac{1}{2} \mathcal{D}(A(t))^\top \left(V_{2,3}^b + V_{2,3}^a \right) \mathcal{D}(A(t)) \\
\quad + \mathcal{D}(A(t))^\top \left(V_{2,2}^b - V_{2,2}^a \right) D(t) + \frac{1}{4} D(t)^\top \eta^{-1} D(t),
\end{cases} \quad (3.45)$$

with terminal conditions

$$A(T) = -\Gamma, \quad B(T) = 0, \quad C(T) = 0, \quad D(T) = 0, \quad E(T) = 0, \quad F(T) = 0,$$

where $\mathcal{D}$ is the linear operator mapping a matrix onto the vector of its diagonal coefficients, and for $i \in \{1, \dots, r\}$, $n \in \{1, \dots, N\}$, $j \in \{0, 1, 2\}$, and $m \in \mathbb{N}$:

$$\begin{cases} \Delta_{j,m}^{i,n,b} = \int_{\mathbb{R}_+^*} z^m \alpha_j^{i,n,b}(z) \psi^{i,n,b}(dz) \\ \Delta_{j,m}^{i,n,a} = \int_{\mathbb{R}_+^*} z^m \alpha_j^{i,n,a}(z) \psi^{i,n,b}(dz), \end{cases} \quad \text{and} \quad \begin{cases} v_{j,m}^b = \left(\sum_{n=1}^N \Delta_{j,m}^{i,n,b} \right)_{i \in \{1, \dots, d\}}^\top \\ v_{j,m}^a = \left(\sum_{n=1}^N \Delta_{j,m}^{i,n,a} \right)_{i \in \{1, \dots, d\}}^\top, \end{cases}$$

and

$$\begin{cases} V_{j,m}^b = \text{diag}(v_{j,m}^b) \\ V_{j,m}^a = \text{diag}(v_{j,m}^a). \end{cases}$$

Then, $\tilde{\theta}$ defined by (3.44) satisfies (3.42) with terminal condition (3.43).

Notice that the system of ODEs (3.45) can be solve sequentially. First, solve the subsystem in (A, B, C) , then solve the linear subsystem in (D, E) , and F is obtained by an integration. Thus, obtaining the approximated $\tilde{\theta}$ reduces to solving the subsystem in (A, B, C) . Define $P : [0, T] \rightarrow \mathcal{S}_{r+d}(\mathbb{R})$ as

$$P(t) = \begin{pmatrix} A(t) & \frac{1}{2} B(t) \\ \frac{1}{2} B(t)^\top & C(t) \end{pmatrix},$$

and observe that the ODE system in (A, B, C) is equivalent to the DRE

$$\dot{P}(t) = Q + Y^\top P(t) + P(t)Y + P(t)UP(t), \quad (3.46)$$

with terminal condition

$$P(T) = \begin{pmatrix} -\Gamma & 0 \\ 0 & 0 \end{pmatrix}, \quad (3.47)$$

where

$$Q = \frac{1}{2} \begin{pmatrix} \gamma \Sigma^r & \mathcal{J}R \\ R^\top \mathcal{J}^\top & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 0 \\ \gamma \Sigma \mathcal{J}^\top & R \end{pmatrix}, \quad \text{and} \quad U = \begin{pmatrix} -2(V_{2,1}^b + V_{2,1}^a) - \eta^{-1} & 0 \\ 0 & 2\gamma \Sigma \end{pmatrix}.$$

Assume that $2V_{2,1}^b + 2V_{2,1}^a + \eta^{-1} \in \mathcal{S}_r^+(\mathbb{R})$. Here, Assumptions 3.1.1 and 3.1.2 hold, thus the results of Section 3.1 apply to the DRE (3.46) with terminal condition (3.47).

In particular, use Corollary 3.1.1 with $K = -\frac{1}{2} \begin{pmatrix} 0 & \mathcal{J} \\ \mathcal{J}^\top & 0 \end{pmatrix}$ and $Y = \mathcal{J}$ in the terminal condition (3.2) to obtain the DRE (3.46).

The approximation method of this section leads to the DRE (3.46) for which a solution exists and is unique, which can be approximated rapidly using existent efficient solving techniques. Once the approximated value function is obtained, the approximated quotes are obtained using the equations in (1.14), by replacing θ with $\tilde{\theta}$. Moreover, the trading strategy associated with the approximated value function is

$$\bar{v}_t^* = \frac{1}{2} \eta^{-1} (2A(t)q_t + B(t)S_t + D(t)).$$

The case of symmetric exponential intensity functions

If the intensity functions for the bid and ask sides are exponential, which is a commonplace assumption in the market making literature, then for all $i \in \{1, \dots, r\}$ and $n \in \{1, \dots, N\}$, we write

$$\Lambda^{i,n,b}(\delta) = \Lambda^{i,n,b}(\delta) = A^{i,n} e^{-k^{i,n}\delta},$$

where $A^{i,n}$ and $k^{i,n}$ are positive constants. Thus, the Hamiltonian functions are

$$H^{i,n,b}(z, p) = H^{i,n,a}(z, p) = \frac{A^{i,n}}{k^{i,n}} \left(1 + \frac{\gamma z}{k^{i,n}} \right)^{-\left(1 + \frac{k^{i,n}}{\gamma z^{i,n}}\right)} e^{-k^{i,n}p},$$

and the approximated quotes are

$$\begin{aligned} \bar{\delta}^{i,n,b}(z) &= -2q_{t-}^\top A(t)e^i - ze^{i^\top} A(t)e^i - S^\top B(t)^\top e^i - e^{i^\top} D(t) + \frac{1}{\gamma z} \log \left(1 + \frac{\gamma z}{k^{i,n}} \right), \\ \bar{\delta}^{i,n,a}(z) &= 2q_{t-}^\top A(t)e^i - ze^{i^\top} A(t)e^i + S^\top B(t)^\top e^i + e^{i^\top} D(t) + \frac{1}{\gamma z} \log \left(1 + \frac{\gamma z}{k^{i,n}} \right). \end{aligned}$$

3.3 Optimal execution with Bayesian learning of the drift

In this section, we propose a second application of the existence result of Section 3.1. Consider an agent who is in charge of a portfolio. The agent is uncertain about the drifts in the prices of the assets in the portfolio, and has a prior Gaussian distribution. The model we introduce combines stochastic optimal control, to obtain an optimal trading schedule, with online learning of the drift from observed prices and signals, to update the estimate of the drift while trading. The strategy we derive uses information from prices and signals to enhance trading performance for execution programmes, or to execute statistical arbitrages based on price predictors.

In our model, the latest estimate at time t of the true value of the drift is the expectation of the drift conditional on the filtration generated by the prices and signals up until time t . The approach is powerful in two ways. First, it uses online learning to continuously adjust the drift to the information in recent observations of prices and predictive signals. Second, it incorporates the uncertainty on the future value of the drift.

First, we recall the Bayesian learning approach introduced in [Bismuth et al. \(2019\)](#). Next, we show that the optimal trading problem reduces to solving a DRE similar to that in (3.1).

3.3.1 Bayesian learning

We consider a market with $r \in \mathbb{N}^*$ assets, and an agent who uses $k \in \mathbb{N}$ signals, and let $d = k + r$. We introduce the d -dimensional Brownian motion $(W_t)_{t \in [0, T]} = (W_t^1, \dots, W_t^d)_{t \in [0, T]}^\top$ adapted to the filtration $(\mathcal{F}(t))_{t \in [0, T]}$. The joint dynamics of the r prices and the k signals are modelled by a d -dimensional drifted Bachelier process $(S_t)_{t \in [0, T]} = (S_t^1, \dots, S_t^d)_{t \in [0, T]}^\top$ with dynamics

$$dS_t = \mu dt + V dW_t, \quad (3.48)$$

where $V \in \mathcal{M}_d(\mathbb{R})$ and we denote the variance-covariance matrix by $\Sigma = V V^\top$.

The drifts μ are unknown to the agent but she has a Gaussian prior distribution that we denote by m_μ . The drift μ and the Brownian $(W_t)_{t \in [0, T]}$ are not observed; however, the prices and the signals are continuously observed by the agent and they convey information about the true value of the drift. We use the Bayesian framework to update the estimation of the drift throughout the execution programme using the latest information available to the agent.

Let $\mathbb{F}^S = (\mathcal{F}^S(t))_{t \in [0, T]}$ be the filtration generated by $(S_t)_{t \in [0, T]}$, which is the appropriate filtration that represents the information revealed to the agent. Notice that $(W_t)_{t \in [0, T]}$ is not $\mathbb{F}^S$ -adapted. We introduce the process $(b_t)_{t \in [0, T]}$ defined as

$$b_t = \mathbb{E}[\mu | \mathcal{F}_t^S].$$

The process $(b_t)_{t \in [0, T]}$ is the latest estimated value of the drift μ given the information in the prices and the signals up to time t . Now consider a non-degenerate multivariate Gaussian prior, i.e.,

$$m_\mu(dz) = \frac{1}{(2\pi)^{\frac{d}{2}} |\Pi_0|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(z - b_0)^\top \Pi_0^{-1}(z - b_0)\right) dz,$$

where $b_0 \in \mathbb{R}^d$ and $\Pi_0 \in S_d^{++}(\mathbb{R})$. The authors in [Bismuth et al. \(2019\)](#) show that $\forall t \in [0, T]$,

$$b_t = \Pi(t) \left(\Sigma^{-1} (S_t - S_0) + \Pi_0^{-1} b_0 \right),$$

where $\Pi(t) = (\Pi_0^{-1} + t \Sigma^{-1})^{-1}$. They also show that there exists a Brownian motion $(\widehat{W}_t)_{t \in [0, T]}$ adapted to $\mathbb{F}^S$ and that has the same correlation structure as that of $(W_t)_{t \in [0, T]}$, and that the dynamics of S in (3.48) in the appropriate filtration are

$$dS_t = b_t dt + V d\widehat{W}_t = R(t) (\bar{S} - S_t) dt + V d\widehat{W}_t, \quad (3.49)$$

where we define

$$R(t) = -\Pi(t) \Sigma^{-1} \quad \text{and} \quad \bar{S} = S_0 - \Sigma \Pi_0^{-1} b_0. \quad (3.50)$$

Note that the dynamics of S in (3.49) are linear and resemble those of a multi-OU process with a time-dependent mean reversion matrix. This allows us to use the results of Section 3.1 to provide a solution to the problem of optimal execution with Bayesian learning of the drift.

3.3.2 Modelling framework and notations

An agent is in charge of a portfolio consisting of r assets and she wishes to liquidate her initial inventory, or execute a statistical arbitrage if the initial inventory is zero, within a given period of time $[0, T]$, where $T > 0$ is fixed. She holds an r -dimensional inventory described by the process $(q_t)_{t \in [0, T]} = (q_t^1, \dots, q_t^r)_{t \in [0, T]}^\top$ with dynamics

$$dq_t = v_t dt,$$

where the initial inventory $q_0 \in \mathbb{R}^r$ is known, and where $(v_t)_{t \in [0, T]} = (v_t^1, \dots, v_t^r)_{t \in [0, T]}^\top$ is the trading speed process for each asset.

The joint dynamics of prices and signals are in (3.49), where $S_0 \in \mathbb{R}^d$ is given and R and $\bar{S}$ are in (3.50).⁹ We define the dynamics for the amount $(X_t)_{t \in [0, T]}$ on the agent's cash account as

⁹ We do not consider the permanent price; as justified in more detail in [Bergault et al. \(2022\)](#).

$$dX_t = -v_t^\top \mathcal{J} S_t dt - v_t^\top \eta v_t dt,$$

where X_0 is given, $\eta \in \mathcal{S}_r^{++}(\mathbb{R})$ is the positive-definite and symmetric price impact parameter which models execution costs incurred by the trader, and $\mathcal{J}$ is an $r \times d$ matrix with $\mathcal{J}_{ij} = \mathbb{1}_{i=j}$.

As in the previous section, the agent maximises the performance criterion (3.36) and admissible strategies are in the set $\mathcal{A} = \mathcal{A}_0$, where for $t \in [0, T]$,

$$\mathcal{A}_t = \left\{ (v_s)_{s \in [t, T]}, \mathbb{R}^r\text{-valued, } \mathcal{F}^S\text{-adapted,} \right. \\ \left. \text{and satisfying a linear growth condition w.r.t } (S_s)_{s \in [t, T]} \right\}.$$

3.3.3 From HJB to DRE

The value function $\vartheta : [0, T] \times \mathbb{R} \times \mathbb{R}^r \times \mathbb{R}^d \rightarrow \mathbb{R}$ associated with the problem solves the HJB

$$0 = \partial_t w + (\bar{S} - S)^\top R^\top \nabla_S w + \frac{1}{2} \text{Tr} \left(\Sigma D_{SS}^2 w \right) + \sup_{v \in \mathbb{R}^d} \left\{ v^\top \nabla_q w - (v^\top \mathcal{J} S + v^\top \eta v) \partial_x w \right\},$$

for all $(t, x, q, S) \in [0, T] \times \mathbb{R} \times \mathbb{R}^r \times \mathbb{R}^d$ with terminal condition

$$w(T, x, q, S) = -\exp(-\gamma(x + q^\top \mathcal{J} S - q^\top \Gamma q)), \quad \forall (x, q, S) \in \mathbb{R} \times \mathbb{R}^r \times \mathbb{R}^d.$$

Similar to the previous section, one consider the two ansatzs

$$w(t, x, q, S) = -\exp(-\gamma(x + q^\top \mathcal{J} S + \theta(t, q, S))), \\ \theta(t, q, S) = q^\top A(t)q + q^\top B(t)S + S^\top C(t)S + D(t)^\top q + E(t)^\top S + F(t),$$

to obtain an ODE system in $A \in C^1([0, T], \mathcal{S}_r(\mathbb{R}))$, $B \in C^1([0, T], M_{r,d}(\mathbb{R}))$, $C \in C^1([0, T], \mathcal{S}_d(\mathbb{R}))$, $D \in C^1([0, T], \mathbb{R}^r)$, $E \in C^1([0, T], \mathbb{R}^d)$, and $F \in C^1([0, T], \mathbb{R})$:

$$\begin{cases} \dot{A}(t) = \frac{\gamma}{2}(B(t) + \mathcal{J})\Sigma(B(t)^\top + \mathcal{J}) - A(t)\eta^{-1}A(t) \\ \dot{B}(t) = (B(t) + \mathcal{J})R(t) + 2\gamma(B(t) + \mathcal{J})\Sigma C(t) - A(t)\eta^{-1}B(t) \\ \dot{C}(t) = R(t)^\top C(t) + C(t)R(t) + 2\gamma C(t)\Sigma C(t) - \frac{1}{4}B(t)^\top \eta^{-1}B(t) \\ \dot{D}(t) = -(B(t) + \mathcal{J})R(t)\bar{S} + \gamma(B(t) + \mathcal{J})\Sigma E(t) - A(t)\eta^{-1}D(t) \\ \dot{E}(t) = -2C(t)R(t)\bar{S} + R(t)^\top E(t) + 2\gamma C(t)\Sigma E(t) - \frac{1}{2}B(t)^\top \eta^{-1}D(t) \\ \dot{F}(t) = -\bar{S}^\top R(t)^\top E(t) - \text{Tr}(\Sigma C(t)) + \frac{\gamma}{2}E(t)^\top \Sigma E(t) - \frac{1}{4}D(t)^\top \eta^{-1}D(t), \end{cases} \quad (3.52)$$

with terminal conditions $A(T) = -\Gamma$ and $B(T) = C(T) = D(T) = E(T) = F(T) = 0$.

The ODE system (3.52) can be solved sequentially. First, one solves the subsystem in (A, B, C) , then the subsystem in (D, E) becomes linear, and finally F is obtained

86 3 Solvability of Riccati equations and applications to algorithmic trading with an integration. Note that the subsystem of ODEs in (A, B, C) can be written as the matrix DRE in $P(t) = \begin{pmatrix} A(t) & \frac{1}{2}B(t)^\top \\ \frac{1}{2}B(t) & C(t) \end{pmatrix}$:

$$\dot{P}(t) = Q + Y(t)^\top P(t) + P(t) Y(t) + P(t) U P(t), \quad (3.53)$$

with terminal condition $P(T) = \begin{pmatrix} -\Gamma & 0 \\ 0 & 0 \end{pmatrix}$ where

$$Q(t) = \begin{pmatrix} \frac{1}{2}\gamma\Sigma^r & \frac{1}{2}\mathcal{J}R(t) \\ \frac{1}{2}R(t)^\top\mathcal{J}^\top & 0 \end{pmatrix}, \quad Y(t) = \begin{pmatrix} 0 & 0 \\ \gamma\Sigma\mathcal{J}^\top & R(t) \end{pmatrix}, \quad U = \begin{pmatrix} -\eta^{-1} & 0 \\ 0 & 2\gamma\Sigma \end{pmatrix}.$$

Assumptions 3.1.1 and 3.1.2 hold, in particular, the matrices $R(s)$ and $R(u)$ commute for all $(s, u) \in [0, T]$. The DRE (3.53) is similar to the DRE (3.46), which is a special case of the family of problems introduced in Section 3.1. Thus, Theorems 3.1.1 and 3.1.2 solve the problem of optimal trading with Bayesian learning of the drift, in particular (3.53) admits a unique solution. Finally, in practice, efficient ODE solving techniques rapidly compute the solution P to the DRE (3.53).

Conclusions

We obtained an existence and uniqueness result for a type of matrix DREs with indefinite matrix coefficients. These equations are a generalisation of the DRE obtained in Chapter 2. The result is used to solve two algorithmic trading problems for liquidity providers and takers. The first problem is multi-asset market making, and we derived an approximation strategy which uses signals to enhance trading performance and a trading venue to hedge risk. The second problem is multi-asset optimal execution and statistical arbitrage, where the agent updates her estimation of the drift throughout the execution programme.

Decentralised Finance and Automated Market Making

Chapter 4

Introduction

4.1 Automated Market Making

The emergence of decentralised finance (DeFi) ecosystems poses great challenges to traditional financial services based on intermediaries. DeFi is a collection of immutable programs that are referred to as *smart contracts*. A smart contract runs on a peer-to-peer permissionless network and encodes the rules that govern how users interact without relying on intermediaries such as brokers or banks.

Within DeFi, automated market makers (AMMs) are trading venues in which the rules to clear demand and supply depart considerably from those of the matching engines in traditional limit order books (LOBs). Currently, the majority of AMMs are constant function market makers (CFMMs), where liquidity providers (LPs) deposit their assets in the pool, and liquidity takers (LTs) exchange assets directly with the pool.

CFMMs rely on a deterministic trading function and a set of rules to determine how liquidity takers and makers interact with the pool. We refer to the rule that determine the state of the pool after an LT transaction as the *LT trading condition*, and to the rule that determine the state of the pool after an LP operation as *the LP trading condition*. In particular, the trading function and the LT trading condition determine reference exchange rates (akin to the midprice in an LOB) and execution exchange rates (akin to the prices received by liquidity taking orders that walk the book) as a function of the quantity of the assets in the pool. On the other hand, the trading function and the LP trading condition determine the quantity of assets that LPs must provide in both assets. In the remaining chapters of this thesis, we focus on CFMMs.

We consider an asset Y that is valued in terms of the reference asset X and we denote by Z the (exchange) rate implied by the available liquidity in the pool.¹ In the remainder of this section, q^X and q^Y denote the quantity of asset X and Y in the pool, and x and y denote the quantity of asset X and Y held by an LP or exchanged by an LT. Below, we show that for LTs, the LT trading condition and the available liquidity in the pool determine the marginal exchange rate in the pool and the execution costs. For LPs, the LP trading condition and the exchange rate determine the quantity of assets X and Y

¹ Liquidity pooling can be generalised to pools with more than two assets.

they hold in the pool, i.e., the quantity they deposit when they provide liquidity or the quantity they withdraw when they remove it.

LT trading condition

Consider a liquidity pool that consists of quantity q^X of asset X and quantity q^Y of asset Y . A CFMM is characterised by a trading function $f(q^X, q^Y)$ and the LT trading condition is

$$f(q^X, q^Y) = \kappa^2, \quad (4.1)$$

where $\kappa > 0$ is the *depth* of the pool. The trading function is differentiable, has convex level sets, and is increasing in both q^X and q^Y . It links the state of the pool before and after a liquidity taking trade is executed.

LT transactions involve exchanging a quantity y of asset Y for a quantity x of asset X , and vice-versa. The quantity x is determined by the trading function and the liquidity (q^X, q^Y) in the pool:²

$$f(q^X, q^Y) = f(q^X + x, q^Y - y) = \kappa^2. \quad (4.2)$$

The value of κ is constant before and after the LT trade is executed, so the trading condition in (4.1) defines a level curve. For a value of the depth κ , we define the *level function* φ so that

$$f(q^X, q^Y) = \kappa^2 \iff q^X = \varphi(q^Y).$$

We assume that the level function is differentiable, convex, and decreasing.³ If an LT wishes to sell a quantity y of asset Y , she receives $x = y \tilde{Z}(y)$ of asset X in exchange. Here, $\tilde{Z}(y)$, with units X/Y , is the exchange rate received by the agent when trading y . Therefore we write

$$q^X - x = \varphi(q^Y + y) \implies \varphi(q^Y) - y \tilde{Z}(y) = \varphi(q^Y + y),$$

so

$$\tilde{Z}(y) = \frac{\varphi(q^Y) - \varphi(q^Y + y)}{y}, \quad (4.3)$$

and for an infinitesimal quantity y we write

$$Z = \lim_{y \rightarrow 0} \tilde{Z}(y) = -\varphi'(q^Y). \quad (4.4)$$

² Here, the LT pays x and receives y .

³ To ensure that the pool cannot be depleted by LT trades, the level function should also satisfy the limits $\lim_{q^Y \rightarrow 0} \varphi(q^Y) = +\infty$ and $\lim_{q^Y \rightarrow +\infty} \varphi(q^Y) = 0$.

We refer to Z as the *marginal rate* of the AMM, which is equivalent to the midprice in an LOB. The marginal rate Z is a reference exchange rate – the difference between its value and the execution rate is similar to the difference between the LOB midprice and the average price obtained by a liquidity taking order that crosses the spread and walks the book when it is filled.

The level function φ is decreasing, thus the rate in (4.3) received by an LT deteriorates as the size of the trade increases.⁴ The formulae (4.3) and (4.4) encode all the information needed by an LT to interact with an AMM. For a trade of size y , the distance between the marginal rate Z and the execution rate $\tilde{Z}(y)$ is the *unitary execution cost* of the AMM, specifically

$$\text{Execution cost} = |Z - \tilde{Z}(y)|. \quad (4.5)$$

LP trading condition

LPs interact with the pool by depositing assets in the pool or withdrawing assets from the pool. The LP trading condition requires that LP operations do not impact the rate Z . However, LP operations do change the depth κ of the pool. More precisely, let (q^X, q^Y) be the state of the pool before an LP deposits liquidity (x, y) ,⁵ so $(q^X + x, q^Y + y)$ is the state of the pool after the deposit, which increases the depth of the pool because

$$f(q^X + x, q^Y + y) = \bar{\kappa}^2 > f(q^X, q^Y) = \kappa^2;$$

recall that the trading function is increasing.

Next, let φ be the level function corresponding to the depth κ and let ϕ be the level function corresponding to the depth $\bar{\kappa}$. The LP trading condition requires that when an LP deposits (x, y) such that $f(q^X + x, q^Y + y) = \bar{\kappa}^2$, then the following equality must hold

$$Z = -\varphi'(q^Y) = -\phi'(q^Y + y).$$

In order to enforce this property, we require the trading function $f(q^X, q^Y)$ to be homothetic. In practice, LPs usually invest an initial wealth $\tilde{x}$ in terms of the reference asset X , so the amounts x and y are obtained from the LP trading condition and the equality $\tilde{x} = x + Z y$. In contrast, LTs usually specify a quantity x to trade, and receive or pay a quantity y which is given by the LT trading condition.

Difference between AMMs and LOBs

CFMMs and LOBs differ in a number of other aspects including accessibility and the way LPs are compensated. By design, AMMs are permissionless, so anyone can

⁴ The trading function f is increasing in q^X and q^Y and $\partial_y f(q^X, q^Y) = \partial_y f(\varphi(q^Y), q^Y) = 0$ so $\varphi'(q^Y) = -\frac{\partial_y f(q^X, q^Y)}{\partial_x f(q^X, q^Y)} < 0$.

⁵ Here, $x \geq 0$ and $y \geq 0$.

participate in the market.⁶ The key difference between the traditional and the new venues is that in LOBs, market makers post limit orders above and below the midprice to earn the spread on roundtrip trades, while in AMMs, LPs earn fees paid by LTs, when their liquidity is used. In LOBs, market makers are not present in the book after all their orders are either executed or cancelled. On the other hand, in AMMs, the posted liquidity remains in the pool until it is withdrawn by the LP.

In some AMMs, LPs do not receive payments directly on their accounts and their holdings rest in the pool. Only when the LP removes her liquidity, do the accumulated fees are paid into her account and any capital gains or losses are realised. In others, the fees paid by LTs are added to the stock of liquidity of LPs instead of accruing in separate accounts, so the number of assets owned by LPs increases.

LPs in AMMs are compensated in two ways. One, they are rewarded with the proceeds from fees that LTs pay for every trade. Two, when a liquidity taking order is filled by the pool, the taker receives a quantity of one asset and pays a quantity of the other asset that is automatically deposited in the pool. Thus, the execution costs in (4.5) incurred by LTs stay in the pool.

The execution cost (4.5) in CFMMs is similar to the cost of “walking the book” when trading in LOBs, sometimes referred to as the temporary price impact. The difference between the temporary price impact and the CFMM execution cost is that in the CFMM we have a deterministic closed-form expression for the execution cost, both of which are estimated by LTs to estimate execution costs. On the other hand, in LOBs, traders usually rely on historical data analysis and assumptions to obtain an estimate of the execution costs. In LOBs, it is generally assumed that temporary impact is a linear function of the speed of trading where the slope of the function is assumed to be fixed; see [Cartea et al. \(2015\)](#) and [Guéant \(2016\)](#).

As in LOBs, there is uncertainty in execution costs received by LTs. On blockchains, rate and liquidity updates occur at the block validation frequency. For instance, in the *Ethereum* network hosting the most popular AMMs, a block is validated every 13 seconds, on average. Hence, the most up-to-date information available is the previous block’s rate and liquidity data. Transactions are grouped and executed in a block which is externally validated. Within the block, the transactions form a queue that determines the priority with which they are executed. Market participants can pay *tips* to gain priority in the block. Finally, every transaction sent on the network pays *gas fees*. Gas is a unit that measures computational effort of network transactions, and all market participants pay gas fees to the network – fees are proportional to the algorithmic complexity of the functions a trader invokes in the AMM. Consequently, there is randomness regarding the exact execution costs, because gas fees depend on energy prices and peer-to-peer network load.

⁶ This is true for AMMs running on permissionless peer-to-peer networks. If the same AMMs were to be implemented by a central authority for fiat currencies or stocks, then one expects stricter participation rules.

Constant product market makers

A popular choice of CFMM is the constant product market maker (CPMM), such as Uniswap v2, where the trading function is

$$f(q^X, q^Y) = q^X \times q^Y,$$

so the level function is

$$\varphi(q^Y) = \kappa^2 / q^Y.$$

When an LT trades y , the execution rate (4.3) is

$$\tilde{Z}(y) = \frac{1}{y} \left(\frac{\kappa^2}{q^Y} - \frac{\kappa^2}{q^Y + y} \right), \quad (4.6)$$

and the marginal rate (4.4) is

$$Z = -\varphi'(q^Y) = \kappa^2 / q^{Y2} = q^X / q^Y. \quad (4.7)$$

In CPMMs, the LP trading condition results in the equality

$$Z = \frac{q^X}{q^Y} = \frac{q^X + x}{q^Y + y}$$

when an LP deposits (x, y) in the pool. For the equality to hold, there exists ρ such that $x = \rho q^X$ and $y = \rho q^Y$, i.e., liquidity provision and withdrawal by LPs in a CPMM (without concentrated liquidity) is performed in fractions of the pool liquidity and the depth changes from κ to $(1 + \rho) \kappa$.

Figure 4.1 depicts the geometry of a CPMM. The level function indicates the various combinations of quantity q^X and quantity q^Y that lead to the same pool depth. Figure 4.1 shows the level function for two values of the depth. At point B in the figure, the intuition behind the equality

$$-\varphi'(q^Y) = q^X / q^Y$$

is that the slope of the level function is equal to the negative of the slope of the ray (OB) , which also holds for the second level curve. Thus, the slope of the level function, i.e., the rate Z , remains unchanged along a given ray. This holds in the general case of CFMMs; recall that the trading function is homothetic.

Recent AMM designs introduced the concentrated liquidity feature, which allow LPs to specify a specific range of exchange rates in which their liquidity is active. In the next subsection, we describe how LPs and LTs interact with a pool that implements concentrated liquidity (CL).

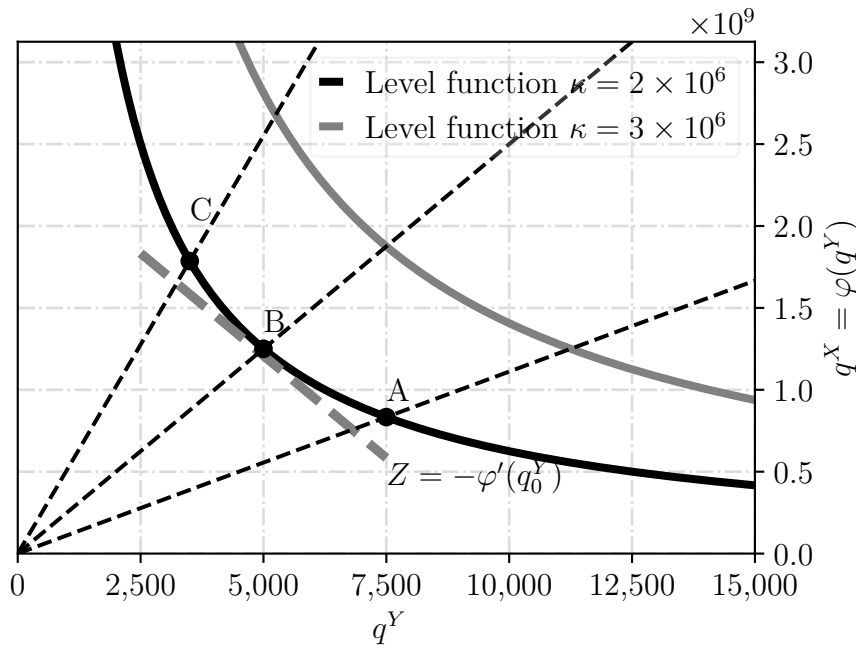


Fig. 4.1: Geometry of CPMs: level function $\varphi(q^Y) = q^X$ for two values of the pool depth.

4.1.1 Concentrated Liquidity

In CPMs without CL, liquidity is provided at all rates, and the LP trading condition ensures that the quantity of assets deposited or withdrawn by LPs is a fraction of the available liquidity. In pools with CL, LPs provide liquidity that is only active when the rate Z is within a range of rates $(Z^\ell, Z^u]$. The boundaries Z^ℓ and Z^u take values in a finite set of values $\{Z^1, \dots, Z^N\}$ called *ticks*.⁷ The range between two consecutive ticks defines the smallest range available for LPs. Every tick range $(Z^i, Z^{i+1}]$ has its own depth $\kappa^{i,i+1}$ which is used to define the execution cost (4.5) of LT trades when the marginal rate Z is in that tick range. Thus, the depth κ can change as a result of the marginal rate Z crossing the boundary of a tick range.

As in traditional CPMs, the LP and LT trading conditions in CPMs with CL determine execution costs for LTs, and the amount of fees that the liquidity of an LP earns in proportion to other LP positions in the pool. Moreover, the LP trading condition in CPMs with CL computes the amounts of tokens that an LP deposits as a function of her range boundaries. Next, we describe the geometry of CPMs with CL and derive both trading conditions.

Geometry of CPMs with CL

Here, we derive and study the LP trading condition for CPMs with CL. Assume the pool consists of the liquidity of one LP who deposits her assets in the tick range

⁷ In contrast, ticks in LOBs stand for the smallest possible price increment.

$[Z^A, Z^B]$ and assume the current rate Z is in the tick range. CL pools obey the constant product formula between two consecutive ticks. However, the assets deposited by the LP must consist of a quantity q^Y to cover rate movements from the current rate to the rate Z^B , and a quantity q^X to cover rate movements from the current rate to the rate Z^A ; see Figure 4.2.⁸

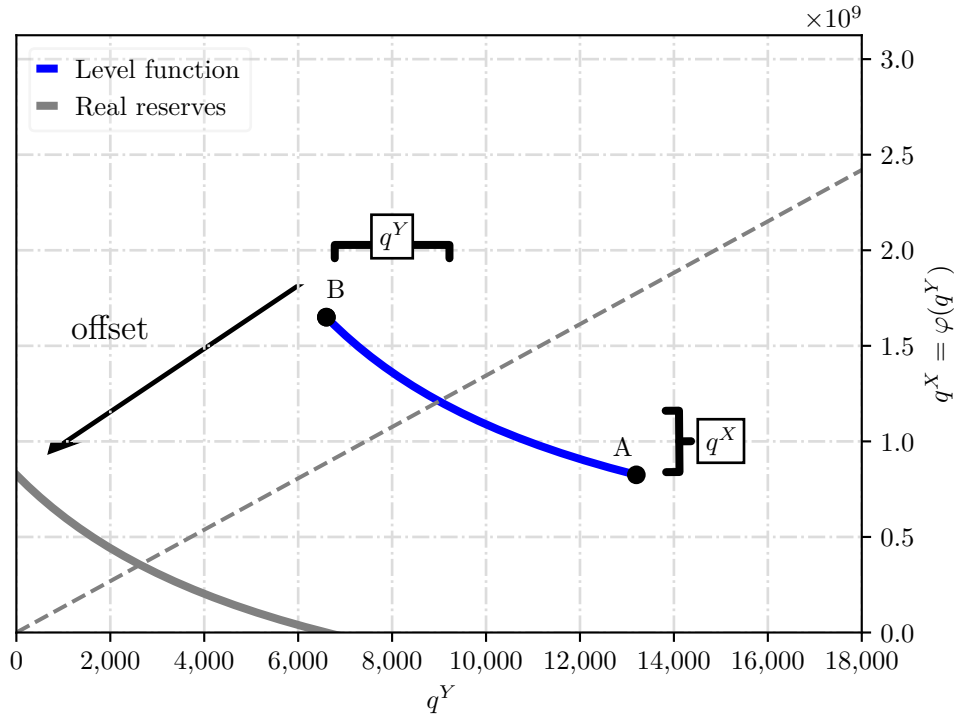


Fig. 4.2: Geometry of CPMs with CL: quantity of assets that the LP provides in the tick range $[Z^A, Z^B]$.

To determine the quantities (q^X, q^Y) to provide liquidity in the tick range $[Z^A, Z^B]$, one offsets the level function segment from the “virtual” assets coordinate system to the real assets coordinate system, where the quantity of asset Y at point B is zero, and the quantity of asset X at point A is zero; see Figure 4.2. The pool rate in the tick range $[Z^A, Z^B]$ obeys the traditional constant product formula, so the *virtual* quantity of assets X and Y when the rate is Z^A verifies

$$q^X q^Y = \kappa^2 \quad \text{and} \quad Z^A = q^X / q^Y,$$

and when the rate is Z^B , it verifies

$$q^X q^Y = \kappa^2 \quad \text{and} \quad Z^B = q^X / q^Y.$$

Thus, the virtual (q^X, q^Y) coordinates of A and B in Figure 4.2 are $(\kappa \sqrt{Z^A}, \kappa / \sqrt{Z^A})$ and $(\kappa \sqrt{Z^B}, \kappa / \sqrt{Z^B})$, respectively. Next, note that the algebraic formula to offset the

⁸ When the rate exits the range, the position consists of only one of the two assets, because the other one is completely depleted.

level curve $q^X q^Y = \kappa^2$ is $(q^X + x)(q^Y + y) = \kappa^2$. Thus, the quantities to provide by the LP verify the key formula

$$\left(q^X + \kappa\sqrt{Z^A}\right) \left(q^Y + \kappa/\sqrt{Z^B}\right) = \kappa^2,$$

which describes the behaviour of real reserves in the arc AB .

In this example, when the current rate $Z \in (Z^A, Z^B]$, the quantities of assets provided are

$$q^X = \kappa \left(Z^{1/2} - (Z^A)^{1/2} \right) \quad \text{and} \quad q^Y = \kappa \left(Z^{-1/2} - (Z^B)^{-1/2} \right), \quad (4.8)$$

so

$$q^X = q^Y \frac{Z^{1/2} - (Z^A)^{1/2}}{Z^{-1/2} - (Z^B)^{-1/2}}.$$

Thus, when the LP specifies a quantity of one of the assets and a liquidity range, the equation above determines the quantity of the second asset. Moreover, if the LP chooses $Z^A = 0$ and $Z^B = +\infty$, then $q^X = q^Y Z$, which is equivalent to a CPMM without CL. Finally, when the rate is outside the tick range, one of the two assets is depleted. More precisely, when $Z \leq Z^A$ the position only consists of

$$q^Y = \kappa \left((Z^A)^{-1/2} - (Z^B)^{-1/2} \right),$$

and when $Z > Z^B$ the position only consists of

$$q^X = \kappa \left((Z^B)^{1/2} - (Z^A)^{1/2} \right).$$

Now, assume a second LP provides liquidity with depth $\bar{\kappa}$ in the same tick range. She deposits $(\bar{q}^X, \bar{q}^Y)$ so the tick range $[Z^A, Z^B]$ consists of $(q^X + \bar{q}^X, q^Y + \bar{q}^Y)$. Use equation (4.8) to write the depth in the tick range which results from the combination of the two positions as

$$\frac{q^X + \bar{q}^X}{Z^{1/2} - (Z^A)^{1/2}} = \kappa + \bar{\kappa}.$$

So one simply sums the depths of the single liquidity positions in the same tick range to obtain the total depth of the liquidity in that tick range; see Figure 4.3.

Let $[Z^B, Z^C]$ be the tick range adjacent to $[Z^A, Z^B]$. Now assume the second LP provides liquidity $(\bar{q}^X, \bar{q}^Y) = (\bar{q}_{[A,B]}^X + \bar{q}_{[B,C]}^X, \bar{q}_{[A,B]}^Y + \bar{q}_{[B,C]}^Y)$ with depth $\bar{\kappa}$ in both tick ranges, i.e., she spreads her liquidity over the range $[Z^A, Z^C]$. Use the analysis above to obtain the quantity of assets she holds in the tick range $[Z^A, Z^B]$ and write

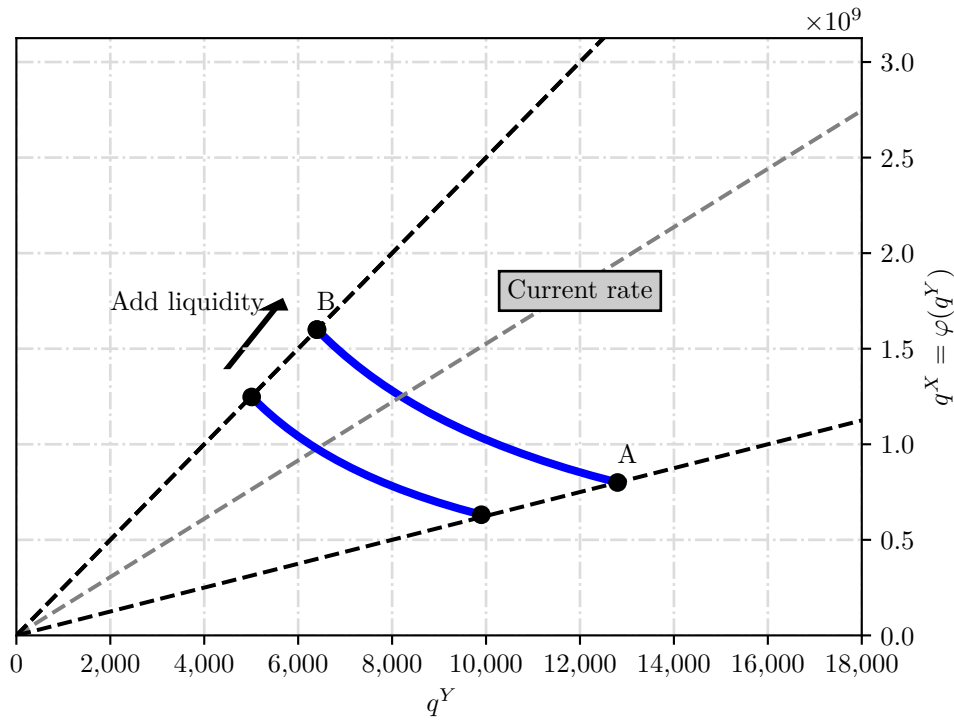


Fig. 4.3: Geometry of CPMMs with CL: changes in the level function in the virtual assets coordinate system when adding liquidity.

$$\bar{q}_{[A,B]}^X = \bar{\kappa} \left(Z^{1/2} - (Z^A)^{1/2} \right) \quad \text{and} \quad \bar{q}_{[A,B]}^Y = \bar{\kappa} \left(Z^{-1/2} - (Z^B)^{-1/2} \right).$$

Note that the rate Z is not in the tick range $[Z^B, Z^C]$, so the LP only holds asset Y and we write

$$\bar{q}_{[B,C]}^X = 0 \quad \text{and} \quad \bar{q}_{[B,C]}^Y = \bar{\kappa} \left((Z^B)^{-1/2} - (Z^C)^{-1/2} \right).$$

Thus, the depth of the liquidity in the tick range $[Z^A, Z^B]$ becomes $\kappa + \bar{\kappa}$ and the depth of the liquidity in the tick range $[Z^B, Z^C]$ is $\bar{\kappa}$. It is thus simple to track the depth of the liquidity in each individual tick range by summing the depth of LP who provided liquidity in a range comprising that tick range.

Figure 4.4 depicts the geometry of CPMMs with CL for two adjacent tick ranges, each with a different value of the depth κ . In CPMMs without CL, the level function is a hyperbola, and in CPMMs with CL, it is a series of adjacent segments of hyperbolas, each corresponding to a certain value of the depth.

LP trading condition

Here, we formalise the LP trading condition and introduce the notation used in the following chapters. Let Z be the rate observed in the pool, and let M be the number of LPs with liquidity resting in the pool. We denote the *depth* of the j th LP's liquidity

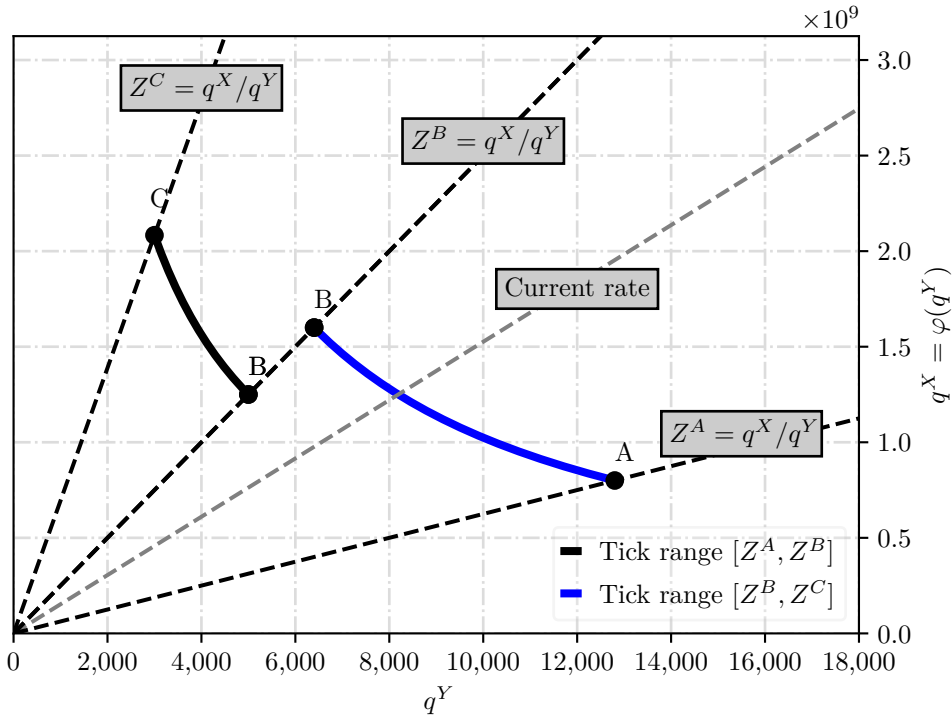


Fig. 4.4: Geometry of CPMMs with CL: two adjacent tick ranges $[Z^B, Z^C]$ and $[Z^A, Z^B]$ with different liquidity depth.

(x, y) deposited in the range $(Z^\ell, Z^u]$ by $\tilde{\kappa}^{\ell, u, j}$. The depth $\tilde{\kappa}^{\ell, u, j}$ verifies the following key formulae that define the LP trading condition in pools with CL:

$$\begin{cases} x = 0 & \text{and } y = \tilde{\kappa}^{\ell, u, j} \left((Z^\ell)^{-1/2} - (Z^u)^{-1/2} \right) & \text{if } Z \leq Z^\ell, \\ x = \tilde{\kappa}^{\ell, u, j} \left(Z^{1/2} - (Z^\ell)^{1/2} \right) & \text{and } y = \tilde{\kappa}^{\ell, u, j} \left(Z^{-1/2} - (Z^u)^{-1/2} \right) & \text{if } Z^\ell < Z \leq Z^u, \\ x = \tilde{\kappa}^{\ell, u, j} \left((Z^u)^{1/2} - (Z^\ell)^{1/2} \right) & \text{and } y = 0 & \text{if } Z > Z^u, \end{cases} \quad (4.9)$$

where, when $Z \leq Z^\ell$, the LP provides only asset Y, i.e., $x = 0$, and when $Z > Z^u$, the LP provides only asset X, i.e., $y = 0$. The quantity of assets (x, y) in (4.9) results from the formula

$$\left(x + \tilde{\kappa}^{\ell, u, j} \sqrt{Z^\ell} \right) \left(y + \tilde{\kappa}^{\ell, u, j} \frac{1}{\sqrt{Z^u}} \right) = \left(\tilde{\kappa}^{\ell, u, j} \right)^2, \quad (4.10)$$

which defines the geometry of the LP's liquidity in the pool.

The depth $\tilde{\kappa}^{\ell, u, j}$ stays constant until the LP withdraws her liquidity or modifies her position. When an LP withdraws her liquidity, the equations in (4.9) and the prevailing exchange rate Z determine the quantity of each asset received by the LP. Here, we refer to the rate Z in the pool when the LP posts her liquidity as the *position rate*. In particular, the position rate is the value of the marginal rate Z used in equation (4.9) to determine the assets deposited by the LP in the pool.

We denote the depth of the liquidity available in the pool in the range $(Z^i, Z^{i+1}]$ between two consecutive ticks by $\kappa^{i,i+1}$. A pool is characterised by the marginal rate Z and the values $\kappa^{i,i+1}$:

$$\kappa^{i,i+1} = \sum_{j \in \{1, \dots, M\}} \sum_{\substack{(\ell, u) \in \{1, \dots, N\}^2 \\ \ell < u}} \tilde{\kappa}^{\ell, u, j} \mathbb{1}_{Z^\ell \leq Z^i < Z^{i+1} \leq Z^u}, \quad (4.11)$$

where M is the number of liquidity providers in the pool and $\tilde{\kappa}_j^{\ell, u, j} \geq 0$ is the depth of the liquidity posted by the j th LP in the range $(Z^\ell, Z^u]$.

The range between two consecutive ticks that contains the rate Z is referred to as the *active* tick range. The value of the pool *depth* κ used in the trading function, and which defines the marginal and execution rates, is the depth (4.11) of the liquidity in the active tick range.⁹

Consequently, in CPMMs with CL, one must discern between two types of depths. First, the pool depth κ which defines the constant product formula (4.2) and the rates (4.4) and (4.3). Second, the depth $\tilde{\kappa}^{\ell, u, j}$ of the liquidity held by the j th LP for some values of ℓ and u . Thus, in pools with CL, the depth of the pool and the depth of LPs' liquidity are local properties which depend on the active tick range. In contrast, in pools without CL, the depth of the pool and LPs' liquidity are global properties.

The transaction fees paid by LTs are distributed amongst LPs in the same proportion as their contribution to the depth in the active range. More precisely, consider an LP with liquidity resting in the range $(Z^\ell, Z^u]$. When the active tick range is $(Z^i, Z^{i+1}]$ and $(Z^i, Z^{i+1}] \subset (Z^\ell, Z^u]$, then for a transaction fee π paid by LTs, the j th LP who posted liquidity with depth $\tilde{\kappa}^{\ell, u, j}$ in $(Z^\ell, Z^u]$ when the pool's depth in the tick range is $\kappa = \kappa^{i,i+1}$ receives

$$\tilde{\pi}^{\ell, u, j} = \frac{\tilde{\kappa}^{\ell, u, j}}{\kappa} \pi \mathbb{1}_{Z^\ell < Z \leq Z^u}. \quad (4.12)$$

In CPMMs without CL, this remuneration is added to the liquidity in the pool, because all LPs provide liquidity in the same range $(0, +\infty)$ and hold portions of the pool in the proportion of their contribution to it. However, in CPMMs with CL, fee income is not automatically reinvested in the pool because LPs can provide liquidity in various ranges simultaneously, so the distribution and reinvestment of fees across different ranges is a choice made by the LP. In the model developed below, we assume that fee income accumulates in accounts in X and Y outside the pool, owned by the LP, and these fees are paid by the LT in either asset depending on the direction of the transaction.

To better understand the mechanisms behind CL and formulae (4.9) and (4.10), assume that the LP chooses values δ^ℓ and δ^u to determine the liquidity provision range $(Z^\ell, Z^u]$ where Z^ℓ and Z^u are given by

⁹ When the rate Z crosses a tick as a result of an LT transaction, the order is executed as two separate transactions; if it crosses multiple ticks, the order is executed as multiple transactions. In particular, each transaction will use the depth κ of each tick range.

$$\begin{cases} (Z^u)^{1/2} = Z^{1/2} / (1 - \delta^u/2), \\ (Z^\ell)^{1/2} = Z^{1/2} (1 - \delta^\ell/2). \end{cases}$$

The motivation behind this change of variables is twofold. First, the formulae for the position depth, the quantity of assets to deposit in the pool, and the LP's wealth dynamics simplify significantly and help to clarify the mechanics of liquidity provision in pools with CL; see Chapter 7. Second, for small values of δ^ℓ and δ^u , the relative width is

$$\frac{Z^u - Z^\ell}{Z} = (1 - \delta^u/2)^{-2} - (1 - \delta^\ell/2)^2 \approx \delta^\ell + \delta^u.$$

Thus, $\delta^\ell + \delta^u$ is a first-order approximation of the relative width of the liquidity position.

Next, consider an initial wealth $\tilde{x} = x + Z y$ deposited by the LP in the pool in the range $(Z^\ell, Z^u] \ni Z$. Then, with (4.9) one obtains the depth of the position, which is given by

$$\tilde{\kappa} = 2\tilde{x} Z^{-1/2} / (\delta^\ell + \delta^u). \quad (4.13)$$

Now, consider two LPs with the same initial wealth $\tilde{x}$. One LP provides liquidity over the one-tick range $(Z^i, Z^{i+1}]$ with depth κ_1 , and the other LP provides liquidity over the two-tick range $(Z^i, Z^{i+2}]$ with depth κ_2 . Let $\delta^\ell + \delta^u$ be the (approximate) relative width of a tick range, i.e.,

$$Z^{i+1} - Z^i = (\delta^\ell + \delta^u) Z.$$

Note that $2(\delta^\ell + \delta^u)$ is the (approximate) relative width of two tick ranges, and use (4.13) to obtain $\tilde{\kappa}_1/\tilde{\kappa}_2 \approx 2$.¹⁰

Thus, the depth of the position of the LP who concentrates her liquidity over one tick range is (approximately) twice the depth of the liquidity held by the second LP. However, when the rate Z is in the tick range $(Z^{i+1}, Z^{i+2}]$, the first LP's liquidity is inactive and does not earn fees, and only the second LP's liquidity facilitates trades. Figure 4.5 shows the position depth of an LP who uses the same wealth to provide liquidity over ranges of 1, 3, and 5 ticks.

Decreasing the value of the position depth $\tilde{\kappa}$ reduces LP fee revenue, because the LP's holdings represent a smaller portion in the active tick range; see (4.12). Thus, a highly concentrated position increases fee revenue but also increases the risk of the position ceasing to be active. We refer to this risk as *concentration risk*; see Chapter 7 for more details. On the other hand, a large position width, i.e., a lower position depth $\kappa^{\ell,u}$, collects less fee revenue per LT transaction, but reaps earnings from a larger number of LT transactions as the liquidity is spread out over more tick ranges.

In the next subsection, we describe the Uniswap v3 exchange protocol and the data that we use to analyse the performance of the liquidity taking and liquidity provision strategies that we propose in Chapters 5 and 7.

¹⁰ This approximation is not precise for large ranges. In particular, in the limit case where the LP deposits liquidity in the range $(0, +\infty)$, i.e., when $\delta^\ell = \delta^u = 2$, the depth of the position is $\tilde{x} Z^{-1/2} / 2$.

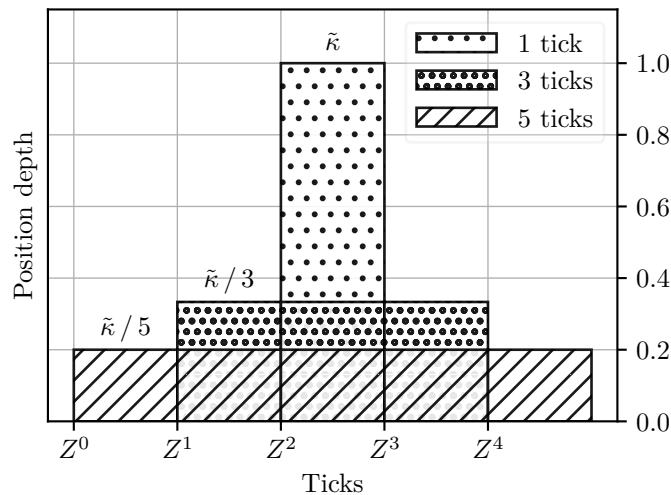


Fig. 4.5: Position depth for three LP ranges. The first is concentrated over a range of one tick, the second over a range of three ticks, and the last over a range of five ticks.

4.2 Uniswap v3

Uniswap v3 is a CPMM with CL running on the Ethereum blockchain, which is a type of distributed ledger technology that validates and stores transactions. It is implemented as a smart contract, which is a permissionless and immutable program running on the Ethereum peer-to-peer network. At present, it is the most liquid CPMM implementing the CL feature.

To analyse the results of the algorithmic trading strategies we derive, we use LP and LT transaction data from two Uniswap v3 pools for the pair ETH/USDC, and one pool for the pair ETH/DAI between inception of the pools on 5 May 2021 and 01 January 2023. The ticker ETH represents the cryptocurrency *Ether*, which is the native cryptocurrency of the Ethereum blockchain. The ticker USDC represents *USD coin*, a cryptocurrency fully backed by U.S. Dollars (USD); and DAI represents the cryptocurrency *Dai*, which tracks parity with the U.S. Dollar.

When LTs interact with a Uniswap v3 pool, a portion of the quantity sent to the pool, called *fee tier*, is deducted and distributed to LPs. The fee tier of the first ETHUSDC pool we consider is 0.05%, and the second is 0.3%. The fee tier of the ETHDAI pool we consider is 0.3%. For the ETHUSDC pools, the unit of the depth κ is $\sqrt{\text{ETH} \cdot \text{USDC}}$, of x is USDC, of y is ETH, and the marginal rate, the execution rate, and the unitary execution cost are all in USDC/ETH; similarly for the ETHDAI pool. For ease of reading, in the remainder of this thesis we omit the units of κ . Tables 4.1 and 4.2 show descriptive statistics of LP and LT activity in the three pools we consider. Trading activity is substantially higher in the ETHUSDC 0.05% and the pool has more depth.

Tracking the marginal rate and the liquidity around it is critical for an agent interacting with Uniswap v3. To obtain historical values of the depth in every tick range and analyse liquidity provision and consumption activity, we reconstruct the supply and demand of

	ETHUSDC 0.05%		ETHUSDC 0.3%	
	LT	LP	LT	LP
Number of instructions	3,171,530	48,305	216,739	31,972
Average daily number of instructions	6,512	99	445	65
Total USD volume	$\approx \$ 282 \times 10^9 \approx \$ 322 \times 10^9$		$\approx \$ 35.5 \times 10^9 \approx \$ 56.2 \times 10^9$	
Average daily USD volume	\$ 579,014,046	\$ 661,842,301	\$ 73,019,394	\$ 115,547,567
Average LT transaction or LP operation size	\$ 88,909	\$ 6,672,543	\$ 164,070	\$ 1,760,029
Average trading frequency	13 seconds	14 minutes	3 minutes	22 minutes
Median LP holding time	86 minutes		1.5 days	
Average pool depth	19,788,327 $\sqrt{\text{ETH} \cdot \text{USDC}}$		12,787,545 $\sqrt{\text{ETH} \cdot \text{USDC}}$	

Table 4.1: LT and LP activity details in the ETH/USDC 0.05% and ETH/USDC 0.3% pools between 5 May 2021 and 01 January 2023.

	ETHDAI 0.3%	
	LT	LP
Number of instructions	109,976	7,860
Average daily number of instructions	225	16
Total USD volume	$\approx \$ 8.86 \times 10^9 \approx \$ 4.98 \times 10^9$	
Average daily USD volume	\$ 18,202,784	\$ 10,222,110
Average LT transaction or LP operation size	\$ 80,606	\$ 633,354
Average trading frequency	6 minutes	89 minutes
Median LP holding time	86 minutes	
Average pool depth	19,788,327 $\sqrt{\text{ETH} \cdot \text{DAI}}$	

Table 4.2: LT and LP activity details in the ETHDAI 0.3% pools between 5 May 2021 and 01 January 2023.

liquidity since the inception of Uniswap v3 in May 2021 using historical LP operations. For example, Figure 4.6 shows the amount of liquidity, given by the depth κ_i , available at each tick range at 02:00, 12:00, and 18:00, on 14 April 2022 and 15 April 2022 for the ETHUSDC 0.05% pool.

The bars in the panels of Figure 4.6 represent a tick range. Whenever the marginal rate is in a given tick range, the liquidity of that range defines the depth κ used by the trading

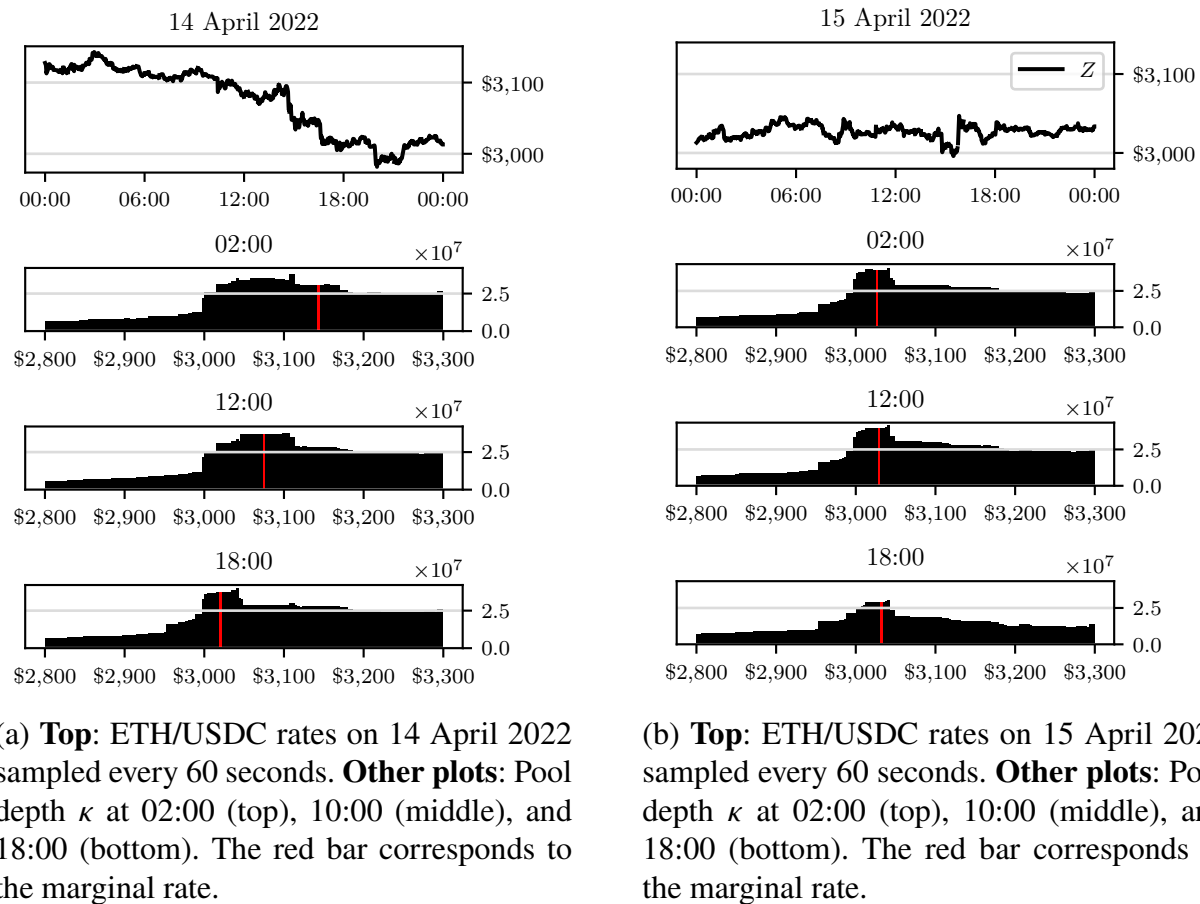


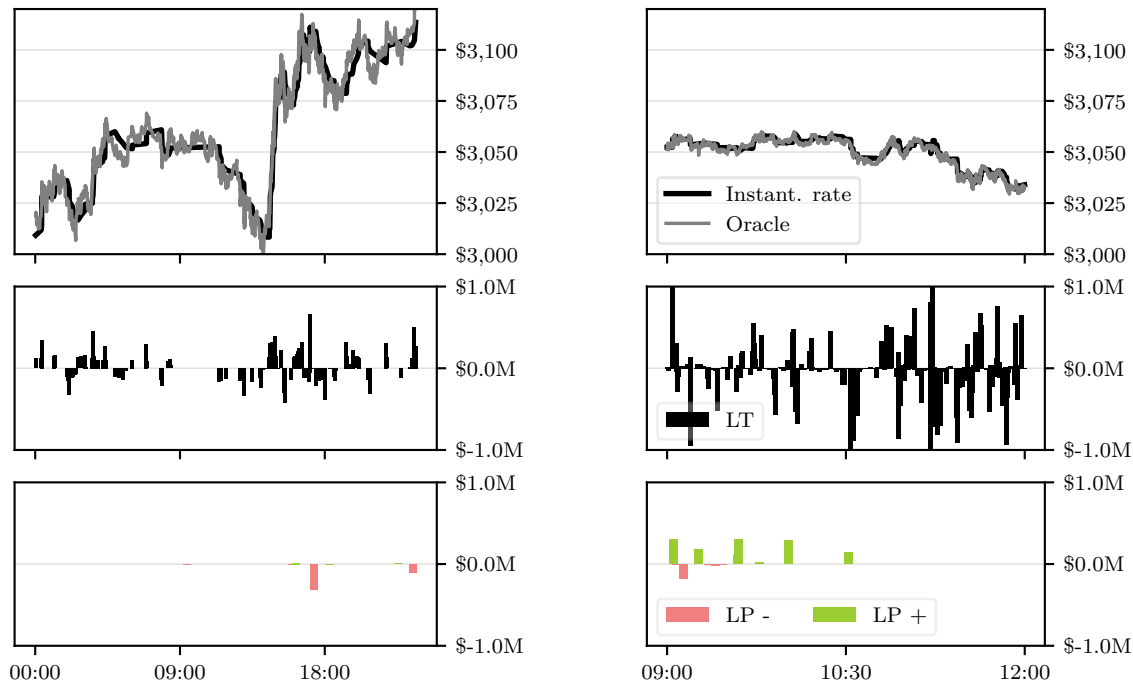
Fig. 4.6: Rate and LP dynamics in AMMs.

function, and hence the execution rate (4.3). When the volume of an LT transaction is large enough to make the marginal rate cross a tick where the level of liquidity changes, Uniswap v3 treats it as multiple trades, each with a different κ_i ; similar to a market order walking the levels of an LOB.

In Figure 4.6, most of the liquidity is concentrated around the marginal rate and the depth is the same over a large range around the rate. The behaviour of LPs is different when volatility is high or low. When Z is volatile, such as on 14 April on the left panel of Figure 4.6, LPs provide liquidity over a wider range around the rate to earn fees that anticipate large swings in Z . On the other hand, when markets are less volatile, such as on 15 April in the right panel of Figure 4.6, LPs concentrate their liquidity provision more tightly around the rate Z .

Next, we use transaction data from the LOB-based exchange Binance for the same pairs to compare the dynamics in the two trading venues. Figure 4.7 shows the Binance quoted rate and the AMM rate, and liquidity provision activity for the ETHUSDC 0.05% and ETHDAI 0.3% pools between 09:00 and 12:00 on 13 April 2022.

Figure 4.7 shows that Binance rates are more volatile and lead the rates in the AMM. This is not by design, it is a consequence of the higher liquidity in Binance. At present, it is crucial to consider the rate from a more liquid venue when trading in an AMM.



(a) **Top:** marginal rate Z in the ETHDAI 0.3% pool and oracle S between April 13 00:00 and April 13, 2022 12:00. **Middle:** LT Transaction size. **Bottom:** LP transaction size.

(b) **Top:** marginal rate Z in the ETHUSDC 0.05% pool and oracle S between April 13 09:00 and April 13, 2022 12:00. **Middle:** LT Transaction size. **Bottom:** LP transaction size.

Fig. 4.7: Rate and LP dynamics in AMMs.

In the remainder of this thesis, we call the leading exchange rate from another trading venue the *oracle*, which in this case is the Binance quoted rate. Similar to the levels of activity reported in Table 4.1 and Tables 4.2, Figure 4.7 shows that there is more LT than LP activity measured by the frequency of instructions and the size of the orders. During the periods with little trading activity in the ETHDAI 0.3% pool, the oracle rate plays a central role to attract LT activity in the AMM whenever the difference between the marginal rate and the oracle rate is significant (recall that only liquidity taking trades can change the rate of the pool). The widening of the difference between the two rates triggers LT activity which drives the two exchange rates to converge; i.e., arbitrageurs keep markets in check.

4.3 Contributions

To the best of our knowledge, the work of Chapter 5 is the first to study optimal execution in a CFMM, and the work of Chapters 6 and 7 is the first to characterise the wealth of LPs in CFMMs in a continuous-time setup, and to use stochastic optimal control tools to derive closed-form strategies in CPMMs with CL. Thus, the main contribution of

the next chapters is to provide new optimal trading strategies for LTs and LPs that trade in CFMMs or CPMMs with CL.

The work in Chapter 5 proposes an optimal trading model for LTs in CFMMs. We show that the convexity of the trading function is a good first-order approximation of the execution costs incurred by LTs in these venues, and use a framework similar to that in [Almgren and Chriss \(2001\)](#) to derive an optimal strategy to trade in CFMMs. At present, the AMM prices are not efficient and it is key to take into account the information from the oracle price.

Chapter 6 introduces a new comprehensive metric of predictable loss (PL) for liquidity providers in constant function automated market makers. PL compares the value of the LP's holdings in the liquidity pool (assuming no fee revenue) with that of a self-financing portfolio that replicates the LP's holdings and invests in a risk-free account. We provide closed-form formulae for PL, and show that the losses stem from two sources: the *convexity cost*, which depends on liquidity taking activity and the convexity of the pool's trading function; the *opportunity cost*, which is due to locking the LP's assets in the pool.

Chapter 7 derives continuous-time dynamics of the wealth of LPs, and provides dynamic liquidity provision strategies. More precisely, for LPs in constant product market makers with concentrated liquidity, we derive a closed-form strategy that dynamically adjusts the range around the exchange rate as a function of market trend, volatility, and liquidity taking activity in the pool. We prove that the profitability of liquidity provision depends on the tradeoff between PL and fee income. Finally, we use Uniswap v3 data to show that LPs have traded at a significant loss, and to show that the out-of-sample performance of our strategy is considerably superior to the historical performance of LPs in the pool we consider.

Literature review

Early work on microstructural aspects of AMMs can be found in [Lipton and Treccani \(2021\)](#) and [Chiu and Koepl \(2019\)](#). Optimal execution in AMMs considers new market impact models that use endogenous dynamics of price and liquidity. Early work on dynamic market impact models is in [Almgren \(2012\)](#) and [Gatheral and Schied \(2013\)](#); see also [Graewe et al. \(2018\)](#), which uses a price-dependent impact function, [Cheridito and Sepin \(2014\)](#), which uses stochastic volatility and temporary price impact, and [Fouque et al. \(2021\)](#), where the authors introduce fast mean-reverting stochastic price impact. The work in Chapter 7 is related to the algorithmic trading and optimal market making literature. Early works on liquidity provision are [Ho and Stoll \(1983\)](#) and [Avellaneda and Stoikov \(2008\)](#). Their framework has been extended in many directions; see [Cartea et al. \(2014, 2017\)](#); [Guéant \(2017\)](#); [Cartea and Wang \(2020\)](#); [Bergault et al. \(2021\)](#). We refer the reader to [Cartea et al. \(2015\)](#) and [Guéant \(2016\)](#) for a comprehensive review of algorithmic trading models for takers and makers of liquidity. Also, our work uses the expected changes in the exchange rates as a way to

improve performance; see [Cartea et al. \(2018a\)](#); [Barger and Lorig \(2019\)](#); [Cartea and Wang \(2020\)](#); [Donnelly and Lorig \(2020\)](#); [Forde et al. \(2022\)](#); [Bergault et al. \(2022\)](#); [Drissi \(2022\)](#) for the use of signals in algorithmic trading.

Early works on AMMs can be found in [Chiu and Koepl \(2019\)](#); [Lipton and Treccani \(2021\)](#); [Lipton and Hardjono \(2021\)](#); [Lipton and Sepp \(2021\)](#). Further, [Cartea et al. \(2022e\)](#), [Cartea et al. \(2023\)](#), and [Jaimungal et al. \(2023\)](#) show how to optimally trade a large position and execute statistical arbitrages using signals in CPMMs with CL, [Berg et al. \(2022\)](#) empirically studies inefficiencies in AMMs, [Bichuch and Feinstein \(2022\)](#) introduces an axiomatic framework for AMMs and exchange rates, and [Miori and Cucuringu \(2022\)](#) uses data-driven methods to characterise the Uniswap v3 ecosystem. Finally, algorithmic trading in AMMs is subject to important delays, and we refer the reader to the work in [Cartea et al. \(2021b\)](#); [Cartea and Sánchez-Betancourt \(2021, 2022b\)](#) for extensions of algorithmic trading models with stochastic delay.

[Neuder et al. \(2021\)](#) is the first work describing a trading strategy to provide liquidity in CPMMs with CL. The authors propose strategies where the agent maximises the expected utility of fees. The rates are discretised in bins and a Markov chain describes transition probabilities and rewards. Other works include [Heimbach et al. \(2022\)](#) where the authors discuss the tradeoff between risks and returns that LPs face in Uniswap v3, and [Fan et al. \(2022\)](#) where the authors show how LPs can exploit their beliefs on future rates. [Heimbach et al. \(2021\)](#) study how LPs profit from their activity.

Chapter 5

Optimal liquidity taking

The results of this chapter and further extensions have been obtained in collaboration with Álvaro Cartea and Marcello Monga, and are in [Cartea et al. \(2022e\)](#).

In this chapter, we use Uniswap v3 data to show that the execution costs incurred by LT trades can be approximated with the convexity of the level function in CPMMs. Next, we derive an optimal trading strategy for LTs interacting with a CFMM with a strictly convex level function. Finally, in the specific case of CPMMs, we showcase the use of the trading strategy.

5.1 Convexity and execution costs

Here, we use Uniswap v3 data described in Section 4.3 to analyse the execution costs implied by the depth of the pool and the CFMM trading condition (4.1) in the specific case of CPMMs; see (4.2). Figure 5.1 uses LT transactions and historical values of the depth for the ETHUSDC 0.05% pool to compute the historical execution costs, which are defined as the difference between the execution rate (4.6) and the marginal rate Z (4.7). The execution costs are shown as a function of the transaction size y and the pool depth κ for all transactions between 1 March and 31 March 2022. Note that for the same transaction size, it is cheaper to trade in a more liquid pool, i.e., trade when the pool depth κ is higher. Clearly, execution costs are a function of the trade size y and the depth κ .

To further study the relation between execution costs, the pool depth, and the rate Z , we analyse the geometry of the constant product trading function. Figure 5.2 shows the level function φ for some value of the depth κ . Every point on the curve gives possible values for q^X and q^Y that would result in the same pool depth κ , where q^X and q^Y are the quantity of asset X and that of asset Y in the pool. The point O corresponds to the current pool quantities q^X and q^Y . The slope of the tangent at that point gives the current marginal rate $Z = -\varphi'(q^Y)$.

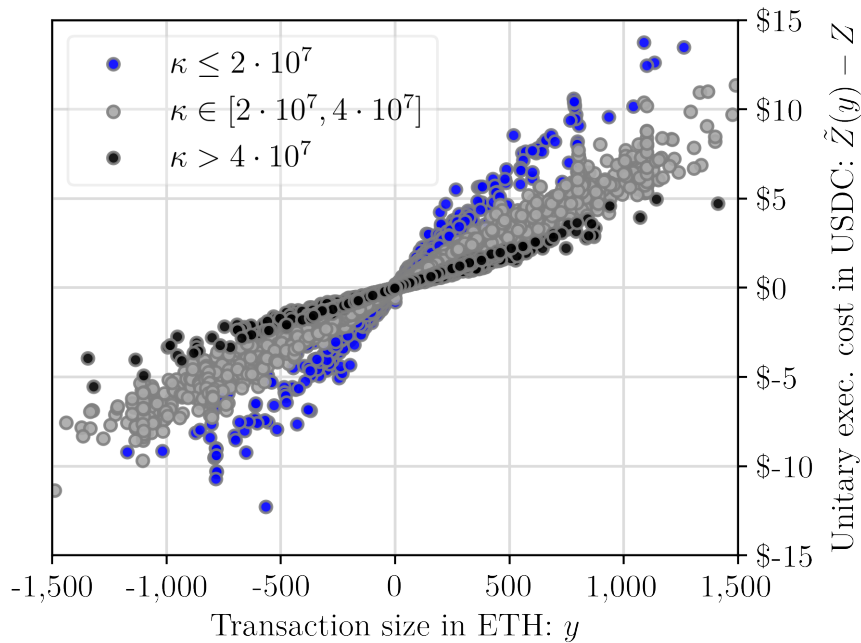


Fig. 5.1: Execution costs for all transactions between 1 March and 31 March, 2022 in the ETHUSDC 0.05% pool. Negative transaction sizes corresponds to buying ETH, and positive sizes to selling ETH. The execution given for the following ranges of the depth κ : $[0, 2 \times 10^7]$, $[2 \times 10^7, 4 \times 10^7]$, and $[4 \times 10^7, +\infty]$.

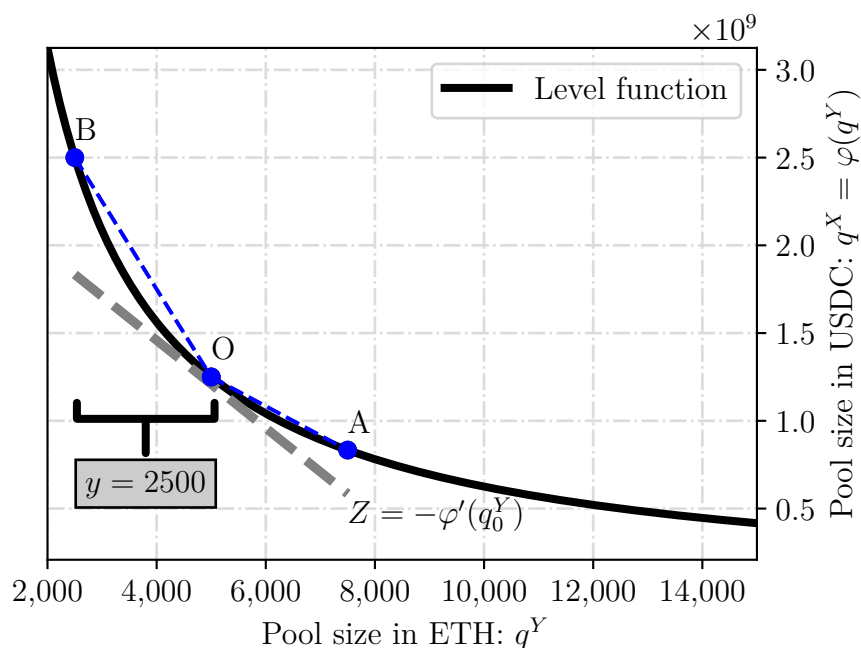


Fig. 5.2: Geometry of the constant product trading function. The figure shows the function $\varphi(q^Y) = q^X$ where q^X is the quantity of USDC, and q^Y is the quantity of ETH in the pool.

A change from O to A in Figure 5.2 is the result of an LT selling 2,500 ETH. A change from O to B is the result of an LT buying 2,500 ETH. The new rates, after these

transactions, are given by the slopes at the new points A and B , respectively. When an LT sells $y = 2,500$ ETH, the unitary execution rate q^X/q^Y is given by the slope of the line (OA) . Similarly, the slope of the line (OB) gives the unitary rate for buying y . On the other hand, the execution cost is given by the difference between the slope of the lines and the slope of the tangent at point O ; the magnitude of this difference depends on the curvature of φ in the neighbourhood of O .

The curvature in the neighbourhood of O is clearly proportional to the convexity of the level function and can be approximated using a second-order Taylor polynomial $\frac{1}{2} \varphi''(q_O^Y) y^2$. A higher degree of convexity, i.e., more curvature around point O , does not change the slope of the tangent at point O , but changes the slopes of the lines (OA) and (OB) . The convexity of the CPMM level function is given by

$$\varphi''(q^Y) = \frac{2\kappa^2}{q^{Y3}} = \frac{2Z^{3/2}}{\kappa}.$$

Therefore, the execution rate obtained for buying or selling y is always less advantageous than the marginal rate Z because the level function φ is (strictly) convex. Clearly, as the depth κ increases, the convexity of the level function is less pronounced. For orders of “small size” we approximate the execution rate with

$$\tilde{Z}(y) = Z - \frac{Z^{3/2}}{\kappa} y. \quad (5.1)$$

To show the extent to which the convexity of the level function captures the execution costs incurred by LTs, we use LT transaction data from the ETHUSDC 0.05% pool from inception on 5 May 2021 until 5 May 2022, approximately 1,757,000 LT transactions, to compare execution costs incurred by LTs with the approximation in (5.1). Figure 5.3 shows a scatter plot of the observed execution costs from transaction data and the approximation $(Z^{3/2}/\kappa) y$, and strongly suggests that the convexity of the level function φ can be used to model the execution costs incurred when interacting in CPMMs and more generally in CFMMs.

Due to the convexity of the level function in CPMMs, and more generally in CFMMs, it is sub-optimal to execute large orders in one trade. An optimal trading framework, similar to those developed for traditional LOB-based markets, should balance the trade-off between execution costs and rate risk. In an optimal trading framework, the investor controls the speed v_t at which she sends orders to the AMM. Now, assume that the investor trades the quantity $y_t = v_t \Delta t$, where Δt is a fixed time-step that determines the investor's frequency of trading and v_t is fixed during the time interval Δt . The execution rate for y in a CFMM is

$$\tilde{Z}(y) = Z - \xi(Z, \kappa) v_t \Delta t,$$

where $\xi : \mathbb{R}^+ \times \mathbb{R}^+ \mapsto \mathbb{R}$ is the CFMM's continuous convexity cost function of the rate Z and the depth κ .

Thus, to reflect the investor's pace of trading we write the investor's execution rate, in a continuous-time setup, as

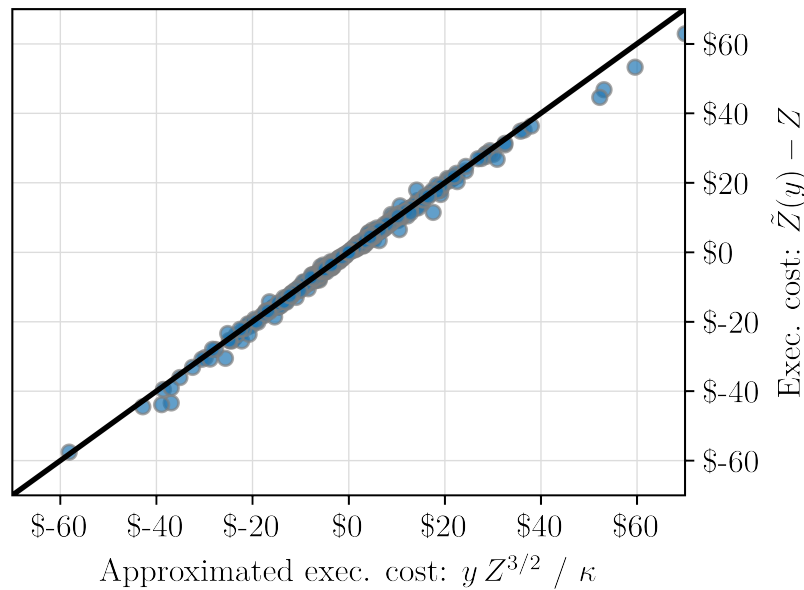


Fig. 5.3: Scatter plot of transaction costs and the approximation (5.1) for all transactions in the ETH/USDC pool between 5 May 2021 and 5 May 2022.

$$\tilde{Z}_t = Z_t - \eta \xi(Z_t, \kappa_t) \nu_t, \quad (5.2)$$

where the parameter η scales the execution costs according to the investor's trading frequency.

Section 5.2 proposes a trading model in CFMMs where execution costs are given by the convexity of the level function, where the depth κ is stochastic, and where it is assumed that there is an oracle rate S .

5.2 The optimal trading problem

We consider the problem of an investor who trades in a CFMM with CL and a strictly convex level function, and wishes to exchange a large position in asset Y into asset X . The investor uses rate information from the pool and from another more liquid exchange in which the oracle rate S is the price of Y in terms of that of X . Moreover, the CFMM pool is liquid and LP activity and LT activity are high, so the pool depth κ is random.

We fix a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P}; \mathcal{F} = (\mathcal{F}_t)_{t \in [0, T]})$ satisfying the usual conditions, where $\mathcal{F}$ is the natural filtration generated by the collection of observable stochastic processes that we define below.

The investor must liquidate a position $\tilde{y}_0$ in asset Y over the period of time $[0, T]$, and her wealth is valued in terms of asset X . We introduce the oracle rate process $(S_t)_{t \in [0, T]}$ with dynamics

$$dS_t = \sigma S_t dW_t, \quad (5.3)$$

where the volatility parameter σ is a nonnegative constant, and $(W_t)_{t \in [0, T]}$ is a standard Brownian motion.

Price formation is in an alternative venue but we consider that the CFMM is active and uses concentrated liquidity, so the innovations in the depth occur often as a result of LP and LT activity. Thus κ is random and its dynamics are described by the process $(\kappa_t)_{t \in [0, T]}$, and we write

$$d\kappa_t = \varsigma \kappa_t dL_t ,$$

where $(W_t)_{t \in [0, T]}$ and $(L_t)_{t \in [0, T]}$ are independent standard Brownian motions and ς is the volatility of the depth κ .

When an LT sends a trade to the CFMM, we write the difference between the execution rate $\tilde{Z}$ and the marginal rate Z as in (5.2), i.e.,

$$\tilde{Z} - Z = -\eta \xi(Z, \kappa) \nu , \quad (5.4)$$

where ξ is the convexity cost function. The term on the right-hand side of (5.4) is the rate impact function of the AMM, which is determined by the convexity of the level function φ , and depends on the marginal rate Z and the depth κ of the pool at the time the liquidity taking order is executed.¹

Recall that the level function is strictly convex, so ξ is strictly positive. The investor trades at the speed $(\nu_t)_{t \in [0, T]}$, so her inventory $(\tilde{y}_t)_{t \in [0, T]}$ evolves as

$$d\tilde{y}_t = -\nu_t dt , \quad (5.5)$$

where, for simplicity, the pool's fee tier is zero. We do not restrict the speed in (5.5) to be positive. When the initial inventory is $\tilde{y}_0 > 0$ (resp. $\tilde{y}_0 = 0$) the investor executes a liquidation (resp. speculation) programme.

The dynamics of the investor's holdings $(\tilde{x}_t)_{t \in [0, T]}$ in asset X are given by

$$d\tilde{x}_t = \tilde{Z}_t \nu_t dt = (Z_t - \eta \xi(Z_t, \kappa_t) \nu_t) \nu_t dt .$$

Here, the execution cost is stochastic and its dynamics are known; see [Barger and Lorig \(2019\)](#) and [Fouque et al. \(2021\)](#) for similar frameworks where the price impact of trading is stochastic.

In our model, the oracle rate is the most efficient rate so we consider the following dynamics for $(Z_t)_{t \in [0, T]}$:

$$dZ_t = \beta (S_t - Z_t) dt + \gamma Z_t dB_t , \quad (5.6)$$

where $\beta > 0$ is the mean-reverting parameter, $\gamma > 0$ quantifies the dispersion induced by the trading flow that drives the AMM marginal rate Z_t away from the oracle rate S_t , and $(B_t)_{t \in [0, T]}$ is a standard Brownian motion independent of $(W_t)_{t \in [0, T]}$ and $(L_t)_{t \in [0, T]}$.

¹ The key difference between the functional form of the execution costs in (5.4) and those in the equity LOB literature is that in general, the price impact functions proposed for LOB models do not depend on the price of the asset, see [Cartea et al. \(2015\)](#).

The investor maximises her expected terminal wealth in units of X while penalising inventory in Y . The set of admissible strategies is

$$\mathcal{A}_t = \left\{ (v_s)_{s \in [t, T]}, \mathbb{R}\text{-valued, } \mathcal{F}\text{-adapted, and } \int_t^T |v_s|^2 ds < +\infty, \mathbb{P}\text{-a.s.} \right\}, \quad (5.7)$$

and we write $\mathcal{A} := \mathcal{A}_0$. Let $v \in \mathcal{A}$, the performance criterion of the investor who trades at speed v is the function $u^v: [0, T] \times \mathbb{R}^5 \rightarrow \mathbb{R}$ given by

$$u^v(t, \tilde{x}, \tilde{y}, Z, S, \kappa) = \mathbb{E}_{t, \tilde{x}, \tilde{y}, Z, S, \kappa} \left[\tilde{x}_T^v + \tilde{y}_T^v Z_T - \alpha (\tilde{y}_T^v)^2 - \phi \int_t^T (\tilde{y}_s^v)^2 ds \right], \quad (5.8)$$

and the investor's value function $u: [0, T] \times \mathbb{R}^5 \rightarrow \mathbb{R}$ is

$$u(t, \tilde{x}, \tilde{y}, Z, S, \kappa) = \sup_{v \in \mathcal{A}} u^v(t, \tilde{x}, \tilde{y}, Z, S, \kappa).$$

The first term on the right-hand side of (5.8) represents the investor's holdings in asset X at the end of the trading window. The second term represents the investor's earnings from liquidating her remaining inventory at the final time T at rate Z_T . The third term is the 'cost' of liquidating the final inventory $\tilde{y}_T$ at time T . Finally, the last term on the right-hand side of (5.8) represents a running inventory penalty where the parameter $\phi \geq 0$ quantifies the urgency of the investor to liquidate inventory; the units of ϕ are such that the penalty is in units of X .

The value function solves the HJB equation

$$\begin{aligned} 0 = & \partial_t w - \phi \tilde{y}^2 + \beta (S - Z) \partial_Z w + \frac{1}{2} \gamma^2 Z^2 \partial_{ZZ} w + \frac{1}{2} \sigma^2 S^2 \partial_{SS} w \\ & + \frac{1}{2} \varsigma^2 \kappa^2 \partial_{\kappa\kappa} w + \sup_{v \in \mathbb{R}} \left((v Z - \eta \xi(Z, \kappa) v^2) \partial_{\tilde{x}} w - v \partial_{\tilde{y}} w \right), \end{aligned}$$

with terminal condition

$$w(T, \tilde{x}, \tilde{y}, Z, S, \kappa) = \tilde{x} + \tilde{y} Z - \alpha \tilde{y}^2.$$

To reduce the dimensionality of the problem, we use the ansatz

$$w(t, \tilde{x}, \tilde{y}, Z, S, \kappa) = \tilde{x} + \tilde{y} Z + \theta(t, \tilde{y}, Z, S, \kappa),$$

to obtain the PDE

$$\begin{aligned} 0 = & \partial_t \theta - \phi \tilde{y}^2 + \beta (S - Z) (\tilde{y} + \partial_Z \theta) + \frac{1}{2} \gamma^2 Z^2 \partial_{ZZ} \theta + \frac{1}{2} \sigma^2 S^2 \partial_{SS} \theta \\ & + \frac{1}{2} \varsigma^2 \kappa^2 \partial_{\kappa\kappa} \theta + \sup_{v \in \mathbb{R}} \left(-\eta \xi(Z, \kappa) v^2 - v \partial_{\tilde{y}} \theta \right), \end{aligned} \quad (5.9)$$

defined over $[0, T) \times \mathbb{R}^4$, with terminal condition

$$\theta(T, \tilde{y}, Z, S, \kappa) = -\alpha \tilde{y}^2 .$$

Note that, for $x \in [t, T]$, the optimal trading speed in feedback form is

$$v_s^* = -\frac{\partial_{\tilde{y}_s} \theta}{2\eta\xi(Z_s, \kappa_s)} , \quad (5.10)$$

so (5.9) becomes

$$\begin{aligned} \partial_t \theta = & -\phi \tilde{y}^2 + \beta (S - Z) (\tilde{y} + \partial_Z \theta) + \frac{1}{2} \gamma^2 Z^2 \partial_{ZZ} \theta + \frac{1}{2} \sigma^2 S^2 \partial_{SS} \theta \\ & + \frac{1}{2} S^2 \kappa^2 \partial_{\kappa\kappa} \theta - \frac{1}{4\eta\xi(Z, \kappa)} \partial_{\tilde{y}} \theta^2 . \end{aligned} \quad (5.11)$$

Finally, use the ansatz

$$\theta(t, \tilde{y}, Z, S, \kappa) = \theta_2(t, Z, S, \kappa) \tilde{y}^2 + \theta_1(t, Z, S, \kappa) \tilde{y} + \theta_0(t, Z, S, \kappa) ,$$

to obtain the system of PDEs

$$\left\{ \begin{aligned} -\partial_t \theta_2 = & -\phi + \beta (S - Z) \partial_Z \theta_2 + \frac{1}{2} \gamma^2 Z^2 \partial_{ZZ} \theta_2 + \frac{1}{2} \sigma^2 S^2 \partial_{SS} \theta_2 \\ & + \frac{1}{2} S^2 \kappa^2 \partial_{\kappa\kappa} \theta_2 + \frac{1}{\eta\xi(Z, \kappa)} \theta_2^2 , \\ -\partial_t \theta_1 = & \beta (S - Z) (1 + \partial_Z \theta_1) + \frac{1}{2} \gamma^2 Z^2 \partial_{ZZ} \theta_1 + \frac{1}{2} \sigma^2 S^2 \partial_{SS} \theta_1 \\ & + \frac{1}{2} S^2 \kappa^2 \partial_{\kappa\kappa} \theta_1 + \frac{1}{\eta\xi(Z, \kappa)} \theta_2 \theta_1 , \\ -\partial_t \theta_0 = & \beta (S - Z) \partial_Z \theta_0 + \frac{1}{2} \gamma^2 Z^2 \partial_{ZZ} \theta_0 + \frac{1}{2} \sigma^2 S^2 \partial_{SS} \theta_0 \\ & + \frac{1}{2} S^2 \kappa^2 \partial_{\kappa\kappa} \theta_0 + \frac{1}{4\eta\xi(Z, \kappa)} \theta_1^2 , \end{aligned} \right.$$

defined over $[0, T) \times \mathbb{R}^3$, with terminal condition

$$\theta_2(T, Z, S, \kappa) = -\alpha , \quad \theta_1(T, Z, S, \kappa) = 0 , \quad \theta_0(T, Z, S, \kappa) = 0 .$$

The optimal strategy in feedback form (5.10) is given by

$$v_s^* = -\frac{1}{2\eta\xi(Z_s, \kappa_s)} (2\theta_2(s, Z_s, S_s, \kappa_s) \tilde{y}_s + \theta_1(s, Z_s, S_s, \kappa_s)) .$$

The equation in θ_2 is a semilinear PDE which we cannot solve in closed-form. Thus, one should approximate the solution with a numerical scheme with a linearisation technique such as Picard iterations. It is beyond the scope of this work to rigorously characterise the optimal strategy in this general framework.

The case of constant convexity cost and constant pool depth

In the case where the convexity cost and the pool depth are fixed prior to the trading window, i.e., when $\xi(Z_s, \kappa_s) = \xi(Z_0, \kappa_0) = \xi$ and $\kappa_s = \kappa_0 = \kappa$ for all $s \in [t, T]$, one obtains a closed-form solution. The HJB (5.11) becomes

$$\begin{aligned} \partial_t \theta = & -\phi \tilde{y}^2 + \beta (S - Z) (\tilde{y} + \partial_Z \theta) + \frac{1}{2} \gamma^2 Z^2 \partial_{ZZ} \theta \\ & + \frac{1}{2} \sigma^2 S^2 \partial_{SS} \theta - \frac{1}{4\eta\xi} \partial_{\tilde{y}} \theta^2, \end{aligned} \quad (5.12)$$

defined over $[0, T) \times \mathbb{R} \times \mathbb{R} \times \mathbb{R}$, with terminal condition

$$\theta(T, \tilde{y}, Z, S) = -\alpha \tilde{y}^2.$$

The HJB (5.12) can be further studied with the ansatz

$$\begin{aligned} \theta(t, \tilde{y}, Z, S) = & A(t) \tilde{y}^2 + B(t) Z \tilde{y} + C(t) \tilde{y} S + D(t) \tilde{y} + E(t) Z^2 \\ & + F(t) S^2 + G(t) Z S + H(t) Z + I(t) S + J(t), \end{aligned}$$

which leads to the system of ODEs

$$\begin{cases} A'(t) &= \phi - \frac{A(t)^2}{\eta\xi}, \\ B'(t) &= \beta + \beta B(t) - \frac{A(t)B(t)}{\eta\xi}, \\ C'(t) &= -\beta - \beta B(t) - \frac{A(t)C(t)}{\eta\xi}, \\ D'(t) &= -\frac{A(t)D(t)}{\eta\xi}, \\ E'(t) &= -(\gamma^2 - 2\beta)E(t) - \frac{B(t)^2}{4\eta\xi}, \\ F'(t) &= -\beta G(t) - \sigma^2 F(t) - \frac{C(t)^2}{4\eta\xi}, \\ G'(t) &= -2\beta E(t) + \beta G(t) - \frac{B(t)C(t)}{2\eta\xi}, \\ H'(t) &= \beta H(t) - \frac{B(t)D(t)}{2\eta\xi}, \\ I'(t) &= -\beta H(t) - \frac{C(t)D(t)}{2\eta\xi}, \\ J'(t) &= -\frac{D(t)^2}{4\eta\xi}, \end{cases} \quad (5.13)$$

on $[0, T)$ with terminal conditions

$$A(T) = -\alpha, B(T) = C(T) = D(T) = E(T) = F(T) = G(T) = H(T) = I(T) = J(T) = 0.$$

The system of ODEs (5.13) admits the unique solution

$$\left\{ \begin{array}{l} A(t) = \sqrt{\phi\eta\xi} \tanh\left(\frac{\sqrt{\phi}}{\sqrt{\eta\xi}}(T-t)t + \operatorname{arctanh}\left(-\frac{\alpha}{\sqrt{\phi\eta\xi}}\right)\right), \\ B(t) = -\int_t^T \beta \exp\left(\int_s^t \left(\beta - \frac{1}{\eta\xi}A(u)\right) du\right) ds, \\ C(t) = -B(t), \\ E(t) = \int_t^T \exp\left(-(\gamma^2 - 2\beta)(t-s)\right) \frac{1}{4\eta\xi} B(s)^2 ds, \\ F(t) = \int_t^T \exp\left(-\sigma^2(t-s)\right) \left(\beta G(s) + \frac{1}{4\eta\xi} C(s)^2\right) ds, \\ G(t) = \int_t^T \exp\left(\beta(t-s)\right) \left(2\beta E(s) - \frac{1}{2\eta\xi} B(s)^2\right) ds, \\ H(t) = D(t) = I(t) = J(t) = 0. \end{array} \right.$$

The optimal trading strategy in feedback form (5.10) over $[t, T] \times \mathbb{R}^3$ is now given by

$$v_s = -\frac{1}{\eta\xi}A(s)\tilde{y}_s + \frac{1}{2\eta\xi}B(s)(S_s - Z_s). \quad (5.14)$$

In the simple case of constant execution costs and constant pool depth, we obtain a classical solution to the problem so standard results apply. In particular, one only needs to verify that the optimal control is admissible, which is clear from the optimal control (5.14) and the admissible set (5.7).

5.3 Performance analysis

This section showcases the use of the strategy described in Section 5.2 in the case of a CPMM, constant convexity cost, and constant pool depth; see [Cartea et al. \(2022e\)](#) for an approximation strategy that adapts the convexity to observed prices and for an in-depth performance analysis of optimal execution in AMMs. In that case, the execution costs write

$$\tilde{Z}_t = Z_t - \frac{\eta}{\kappa} Z_t^{3/2} v_t. \quad (5.15)$$

Here, an investor holds an initial inventory $\tilde{y}_0$ which she wishes to liquidate throughout a trading window $[0, T]$, where $T > 0$. We use the Uniswap v3 data described in Section 4.3 for the liquid ETHUSDC 0.05% pool and the illiquid ETHDAI 0.3% pool. Here, we do not account for AMM and gas fees and assume that the orders sent by the investor do not impact the dynamics of the pools.

We use in-sample data to estimate model parameters and use out-of-sample data to execute the strategies. For in-sample data, we use a window of 24 hours prior to the start of the trading programme for both pools. For out-of-sample data, we use windows of 2 and 12 hours when the investor trades in the liquid and illiquid pools, respectively. We use the rates of the AMM and Binance, and execution costs are those the trades would have received in Uniswap v3.

In our example, the investor starts trading at noon on 16 March 2022, so she uses the data between noon 15 March 2022 and noon 16 March to estimate model parameters. We use the discrete counterpart of the dynamics (5.3) and (5.6), i.e.,

$$\begin{aligned}\Delta \log S_t &= -\frac{\sigma^2}{2} \Delta t + \sigma \sqrt{\Delta t} \nu_t, \\ \Delta \log Z_t &= -\frac{\gamma^2}{2} \Delta t + \beta \left(\frac{S_t - Z_t}{Z_t} \right) \Delta t + \gamma \sqrt{\Delta t} \epsilon_t,\end{aligned}\quad (5.16)$$

where $\{\epsilon_t, \nu_t\}$ are error terms.

For the 24 hours before noon 16 March 2022, there are, on average, one liquidity taking order every 13 seconds in the liquid pool and one every 360 seconds in the illiquid pool; i.e., the time steps in the regressions (5.16) are $\Delta t = 13$ seconds for the ETHUSDC 0.05% pool and $\Delta t = 360$ seconds for the ETHDAI 0.3% pool. Table 5.1 shows parameter estimates.

	ETHUSDC 0.05%	ETHDAI 0.3%
$\hat{\sigma}$	0.045 day ^{-1/2}	0.053 day ^{-1/2}
$\hat{\gamma}$	0.034 day ^{-1/2}	0.027 day ^{-1/2}
$\hat{\beta}$	657.9 day ⁻¹	14.78 day ⁻¹

Table 5.1: Parameter estimates for dynamics of Z and S with data between noon 15 March 2022 and noon 16 March 2022.

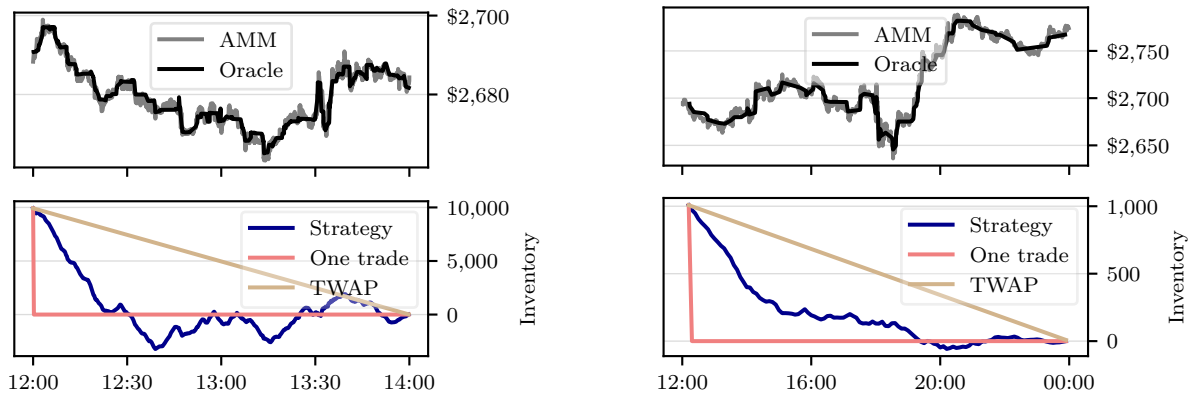
The parameter η of the execution costs in (5.15) is also set to 13 seconds = 17.3×10^{-5} days and 360 seconds = 41×10^{-4} days for the liquid and illiquid pool, respectively. The number of observed transactions in the in-sample data is approximately 238,039 ETH and 4,031 ETH in the liquid and illiquid pool, respectively. Thus the investors' target is to liquidate 14,877 and 1,007 units of ETH within 2 and 12 hours in the ETHUSDC and ETHDAI pools, respectively.

For executions costs, the investor uses the observed convexity of the level function at time $t = 0$. More precisely, the execution cost parameter is set to $\xi = Z_0^{3/2} / \kappa_0$, where Z_0 and κ_0 are the marginal rate and the pool depth observed at time $t = 0$, respectively. Note that in LOB optimal execution models that consider linear transaction costs, the price impact parameter corresponds to $\eta \xi$ in our case. Table 5.2 summarises all the parameters used to run the strategy.

	ETHUSDC 0.05%	ETHDAI 0.3%
κ_0	22,561,783	1,666,175
$\tilde{y}_0$	14,877 ETH	1,007 ETH
$\tilde{S}_0$	2,689.2 USDC	2,686.09 DAI
$\tilde{Z}_0$	2,690.77 USDC	2,694.04 DAI
η	17.3×10^{-5} days	41×10^{-4} days
ξ	0.00673 USDC	0.0838 USDC

Table 5.2: Values of model parameters.

Figure 5.4 shows the marginal and oracle rates and the inventories of the strategies during the execution, for both ETHUSDC and ETHDAI pools. Figure 5.4 clearly showcases the difference between the strategies. In particular, the liquidation strategy is adaptive and trades on the difference between the two rates S and Z during the liquidation programme. Figure 5.5 shows how the difference $S_t - Z_t$ drives the trading speed v_t . The oracle rate is used as a predictive signal for future moves of the marginal rate.



(a) **Top:** Out-of-sample marginal and oracle rates for the pair ETH/USDC. **Bottom:** Inventory process $\tilde{y}$ for the optimal, TWAP, and single order strategies.

(b) **Top:** Out-of-sample marginal and oracle rates for the pair ETH/DAI. **Bottom:** Inventory process $\tilde{y}$ for the optimal, TWAP, and single order strategies.

Fig. 5.4: Liquidation strategies starting at noon on 16 March 2022.

Conclusions

In this chapter, we used Uniswap v3 data to analyse execution costs of CPMMs. We proposed a model for optimal trading in CFMMs with convex trading functions where we assumed that prices are formed in an alternative venue and the pool depth is stochastic. Next, we derived closed-form strategies in the case where the agent fixes the

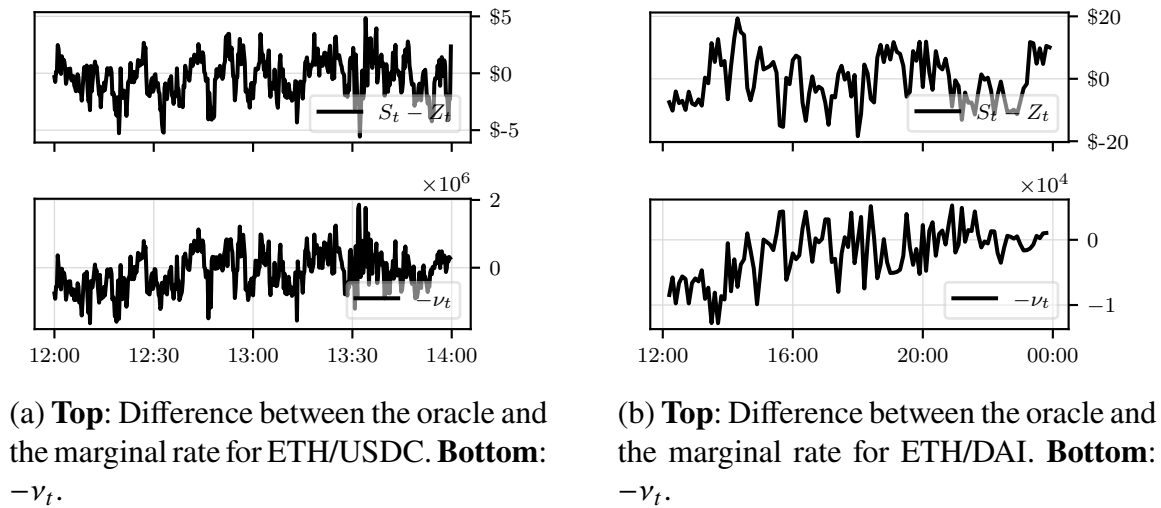


Fig. 5.5: Trading speed.

value of convexity costs and the pool depth at the start of her trading window. Finally, we used in-sample estimation of model parameters and out-of-sample market data to showcase the use of the strategy.

Chapter 6

Predictable losses of liquidity provision

The results of this chapter and further extensions have been obtained in collaboration with Álvaro Cartea and Marcello Monga, and are in [Cartea et al. \(2022f\)](#).

In this chapter, we prove that if LPs do not collect enough fees for making liquidity, their business is not viable in CFMMs because they would provide liquidity to the market at a loss. We use two measures to characterise the losses due to liquidity provision in AMMs. The first measure is impermanent loss (IL), which is commonly used in the literature. The second measure, new in the literature, is *predictable loss* (PL), for which we give a closed-form formula for CFMMs and for CPMMs with CL. IL quantifies the loss of value when depositing assets in a CFMM pool instead of holding the assets outside the pool. We prove that IL is nonpositive for AMMs that operate as a CFMM, and it is negative (almost surely) for CPMMs, i.e., LPs trade at a loss. On the other hand, PL quantifies the loss of value when depositing one's assets in a CFMM pool instead of holding a self-financing portfolio outside the pool. One part of the portfolio replicates the holdings of the LP in the pool, and the other part is invested in a risk-free account. We prove that PL is a negative (i.e., LP provides liquidity at a loss) and a predictable component in the dynamics of the LP position value in the pool. The losses stem from two sources. One source is the *convexity cost*; the magnitude of this cost is a function of liquidity taking activity and the convexity of the CFMM's trading function. The other source is the *opportunity cost*, which is incurred by LPs who deposit assets in the pool instead of investing them in the risk-free asset. Finally, we derive a closed-form formula for PL in a CPMM with CL.

Throughout the chapter, we consider a liquidity pool that consists of quantity q^X of asset X and quantity q^Y of asset Y . We focus on constant function market makers (CFMM), the most popular type of AMMs. Recall that a CFMM is characterised by a trading function $f(q^X, q^Y)$ and a trading condition $f(q^X, q^Y) = \kappa^2$, where $\kappa > 0$ is the depth of the pool. The trading condition links the state of the pool before and after a liquidity taking trade is executed. For LTs, the trading function and the available liquidity in the pool determine the marginal rate Z and the execution costs. For LPs,

the trading function and Z determine the quantities of assets X and Y they hold in the pool, e.g., the quantities they deposit when they provide liquidity or the quantities they withdraw when they remove it; see Chapter 4.

In the remainder of this section, we assume that LPs do not receive fee payments. A key feature of CFMMs is that as the rate changes throughout the investment horizon, so do the quantities of asset X and Y held by the LP in the pool, hence, the value of the LP's holdings also change. We denote by (x_t, y_t) the quantities held by the LP at time t over the trading horizon $[0, T]$, and denote by $\alpha_t = x_t + y_t Z_t$ the value of the position at time t , which is marked-to-market in units of the reference asset X . At the beginning of the trading horizon, the LP deposits quantities (x_0, y_0) in a CFMM pool, so the value of her initial position is

$$\alpha_0 = x_0 + y_0 Z_0,$$

and at the terminal time T , the LP withdraws her liquidity from the pool, and the value of her terminal position is

$$\alpha_T = x_T + y_T Z_T. \quad (6.1)$$

Here, we compare the evolution of the value of the LP's position in the pool with the evolution of two self-financing portfolios invested in an alternative venue. Both portfolios are initiated with the same quantities as those the LP deposits in the pool. The first portfolio executes a buy-and-hold strategy; i.e., the quantities in the portfolio are not changed during the LP's trading horizon. In the second portfolio, as the rate in the pool changes, the portfolio is rebalanced to replicate the holdings in assets X and Y , and the remaining wealth is invested in a risk-free account.

We compare the performance of both portfolios with that of the LP's investment in the pool in the absence of fee payments to the LP. Specifically, we compute IL with the buy-and-hold portfolio and PL with the dynamic portfolio. Both metrics show that liquidity provision in pools is a loss leading activity in CPMMs, and more generally in CFMMs.

The remainder of the chapter proceeds as follows. Section 6.1 discusses impermanent loss (IL), which is an existent measure of LPs losses when interacting with a CPMM pool, and generalises the concept of IL to the general case of CFMMs. Section 6.2 introduces PL as a new measure of the loss incurred by LPs in CFMMs.

6.1 Impermanent loss

In this subsection, we assume that X is the LP's reference asset or numeraire which is used to value her wealth. The literature on CFMMs characterises the risk of providing liquidity in a CFMM with IL, which refers to the loss in value when depositing one's assets in a pool, instead of holding the assets outside the pool. Here, we first characterise the trading function of CFMMs, after which we describe IL in a CFMM and then focus on CPMMs.

The next proposition shows that the level function of a CFMM is convex.

Proposition 6.1.1 Let Z be the marginal rate of a CFMM with level function $\varphi \in C^1(\mathbb{R}_{++})$. The pool provides quantities q^X and q^Y of assets X and Y . Assume there is no arbitrage. Then, φ is convex and $Z = -\varphi'(q^Y)$. $\square$

Proof 6.1.1 The execution prices $Z^{\text{bid}}(y)$ and $Z^{\text{ask}}(y)$ in a CFMM with level function φ for a trade of size y in a pool with liquidity (q^X, q^Y) are given by

$$Z^{\text{bid}}(y) = \frac{\varphi(q^Y) - \varphi(q^Y - y)}{y} \leq Z \leq \frac{\varphi(q^Y + y) - \varphi(q^Y)}{y} = Z^{\text{ask}}(y), \quad (6.2)$$

where Z is the marginal rate, and the inequalities hold because of the no-arbitrage condition. Now, let $y \rightarrow 0$ to obtain the equality $Z = -\varphi'(q^Y)$. Next, the inequalities in (6.2) also show that the level function φ is convex because

$$\frac{\varphi(q^Y) - \varphi(q^Y - y)}{y} \leq -\varphi'(q^Y) \leq \frac{\varphi(q^Y + y) - \varphi(q^Y)}{y}.$$

We do not require the level function φ to be strictly convex. In fact, CFMMs with zero slippage such as constant sum market makers do exist, but the pool's exchange rate is pegged to the rate of an alternative venue so liquidity of the pool is not depleted by LTs. However, if the alternative venue has non zero execution costs as it is the case in LOBs, then one expects the level function of a CFMM to be strictly convex, as it is the case for most CFMMs. In this chapter, we assume that φ is strictly convex.

Next, assume that one LP is the only liquidity provider in a CFMM with initial quantities (x, y) at time $t = 0$, and the LP holds her position over a period $[0, T]$.¹ However, instead of making liquidity in the pool, the LP can invest the quantities (x, y) over the same period in an alternative venue that does not operate a CFMM. The following proposition characterises IL as the difference between the two strategies in the general case of CFMMs and in the particular case of CPMMs.

Proposition 6.1.2 Impermanent loss. The pool consists of only the assets of one LP who provides liquidity (x, y) over a period $[0, T]$. The LP receives zero fees for providing liquidity. Then, it is more profitable for the LP to hold the assets outside the pool than to provide liquidity in a CFMM without CL, with a convex trading function. Moreover, the IL in CPMMs is given by

$$-\kappa \sqrt{Z} \left(1 - \sqrt{\frac{Z_T}{Z}} \right)^2 \leq 0.$$

Proof 6.1.2 Assume an agent holds (x_0, y_0) in assets X and Y at time $t = 0$. Consider two strategies, one where the agent provides liquidity in a pool, and one where the agent holds her assets outside the pool.

¹ In this case, $(x, y) = (q^X, q^Y)$ are the quantities held by the LP in the pool.

Denote the LP's terminal position value in the pool by α_T and the terminal value of the alternative portfolio that holds the assets outside the pool by $\bar{\alpha}_T$. Write

$$\begin{cases} \bar{\alpha}_T = x + Z_T y, \\ \alpha_T = x_T + Z_T y_T, \end{cases} \implies \begin{cases} \bar{\alpha}_T = \varphi(y) - \varphi'(y_T) y, \\ \alpha_T = \varphi(y_T) - \varphi'(y_T) y_T, \end{cases}$$

where x_T and y_T are the terminal liquidity in the pool.

IL is given by

$$\alpha_T - \bar{\alpha}_T = -(\varphi(y) - \varphi(y_T) - \varphi'(y_T)(y - y_T)). \quad (6.3)$$

The convexity of the level function shows that $\alpha_T - \bar{\alpha}_T \leq 0$, and one can approximate IL for small variations of the rate as

$$\alpha_T - \bar{\alpha}_T \approx -\frac{1}{2} \varphi''(y_T) (y - y_T)^2.$$

In a CPMM, the IL in (6.3) is explicitly given by

$$\begin{aligned} \alpha_T - \bar{\alpha}_T &= -(\varphi(y) - \varphi(y_T) - \varphi'(y_T)(y - y_T)) \\ &= \left(\kappa \sqrt{Z} - \kappa \sqrt{Z_T} + \kappa \left(Z_T \sqrt{\frac{1}{Z}} - \sqrt{Z_T} \right) \right) \\ &= -\kappa \sqrt{Z} \left(1 - \sqrt{\frac{Z_T}{Z}} \right)^2, \end{aligned}$$

which is equal to zero when $Z_T = Z$, hence the loss is called “impermanent”. $\square$

Corollary 6.1.1 Assume the marginal rate Z is a continuous random variable, then IL is strictly negative almost surely in CPMMs without CL. $\square$

IL scales with the ratio Z_T/Z , so the likelihood of incurring large losses increases with the volatility of the rate. Also, LPs are exposed to IL whether the rate Z increases or decreases. IL is not an appropriate measure to characterise the losses of LPs, as the alternative buy-and-hold portfolio is not exposed to the same market risk throughout the period $[0, T]$. In the next section, we compare the position value of the LP with the value of a dynamic portfolio that tracks the holdings of the LP in the pool.

6.2 Predictable loss

6.2.1 PL in CFMMs

We assume that there exists a riskless asset B that yields the risk-free rate $r \geq 0$, $(P_t)_{t \in [0, T]}$ denotes the marginal rate to exchange asset X for asset B , which we consider

independent of the quantity $(y_t)_{t \in [0, T]}$ held by the LP in the pool.² Thus, the marginal rate to exchange Y for B is $(Z_t P_t)_{t \in [0, T]}$. Here, both assets X and Y are risky and the LP marks-to-market her wealth in terms of B .

Assume that the LP is the only liquidity provider in a CFMM pool with initial quantities (x_0, y_0) of asset X and Y and that the LP does not receive fees for making liquidity. The LP continuously rebalances an alternative portfolio $\bar{\alpha}$, valued in terms of B , where $\bar{\alpha}_0 = \alpha_0$, to track the quantities (x_t, y_t) . For each rebalancing time t , the difference between the two self-financing portfolios $\bar{\alpha}_t - \alpha_t$ is positive because of the loss in value of the LP's position in the pool discussed in Section 6.1. Here, the LP converts the remaining wealth in the riskless asset B and deposits the cash $(\bar{\alpha}_t - \alpha_t) P_t$ in a risk-free account that earns the risk-free rate $r \geq 0$; see the proof of Proposition 6.2.1 for more details.

We assume that exchanging Y and X for B is frictionless, i.e., one exchanges X and Y at the rates P_t and $Z_t P_t$, respectively, with no other costs. Next, we characterise the continuous-time dynamics of $\bar{\alpha}$ and define the PL process

$$(L_t)_{t \in [0, T]} = (\alpha_t - \bar{\alpha}_t)_{t \in [0, T]}, \quad (6.4)$$

where α is the LP's position value and $\bar{\alpha}$ is the value of the alternative replicating portfolio. The proposition below provides a closed-form formula for PL in a CFMM and shows it is a function of the convexity of the level function and the liquidity taking activity in the pool.

Proposition 6.2.1 Predictable loss. The liquidity in a CFMM (without CL) is provided by one LP. The LP provides quantities (x_0, y_0) of the pair of assets X and Y and withdraws her liquidity at the terminal time T , so $(y_t)_{t \in [0, T]}$ and $(x_t)_{t \in [0, T]}$ are the quantities she holds in the pool over the period $[0, T]$. The LP receives zero fees for providing liquidity. Let $\varphi \in C^3(\mathbb{R}_{++})$ be the strictly convex level function of the CFMM. Next, denote the position value process in terms of the numeraire B as $(\alpha_t)_{t \in [0, T]}$, and let the process $(\bar{\alpha}_t)_{t \in [0, T]}$ describe the value, in terms of the numeraire B , of a portfolio which is continuously rebalanced in an alternative venue to track the LP's holdings x_t and y_t in the pool, and that invests any remaining wealth in a risk-free account. Denote the PL process as $(L_t)_{t \in [0, T]}$, which is valued in terms of B . Then L is decreasing, with $L_0 = 0$, and is given by

$$L_t = - \underbrace{\frac{1}{2} \int_0^t P_s \varphi''(y_s) d\langle y, y \rangle_s}_{\text{Convexity cost} \geq 0} - \underbrace{\int_0^t \bar{x}_s P_s r ds}_{\text{Opportunity cost} \geq 0}, \quad (6.5)$$

where $\langle \cdot, \cdot \rangle$ denotes the quadratic variation operator, the process $\bar{x} = \bar{\alpha} - \alpha$ is increasing, and $\bar{x}_0 = 0$. In particular, PL in (6.5) satisfies

$$L_t \leq -\frac{1}{2} \int_0^t P_s \varphi''(y_s) d\langle y, y \rangle_s \leq 0. \quad (6.6)$$

² The processes $(Z_t = -\varphi'(y_t))_{t \in [0, T]}$ and $(x_t = -\varphi(y_t))_{t \in [0, T]}$ are also independent from $(P_t)_{t \in [0, T]}$.

Proof 6.2.1 Define an alternative self-financing portfolio $\underline{\alpha}$ which starts with the same initial wealth α_0 and tracks the holdings (x_t, y_t) in the pool. The dynamics of $\underline{\alpha}$ are

$$d\underline{\alpha}_t = x_t dP_t + y_t d(Z_t P_t) = x_t dP_t + y_t Z_t dP_t + y_t P_t dZ_t,$$

where y_t is the quantity of asset Y in the pool at time t , and $Z_t = -\varphi'(y_t)$ is the marginal rate. Next, the quantities Z_t and x_t in the pool are stochastic because they vary with the quantity y_t , so we use Ito's lemma to write the dynamics of the position value in terms of the numeraire B as

$$\begin{aligned} d\alpha_t &= d((x_t + y_t Z_t) P_t) \\ &= (x_t + y_t Z_t) dP_t + d(\varphi(y_t) + Z_t y_t) P_t \\ &= (x_t + y_t Z_t) dP_t + \varphi'(y_t) P_t dy_t + \frac{1}{2} P_t \varphi''(y_t) d\langle y, y \rangle_t \\ &\quad + y_t P_t dZ_t + Z_t P_t dy_t + P_t d\langle Z, y \rangle_t \\ &= (x_t + y_t Z_t) dP_t + \frac{1}{2} \varphi''(y_t) P_t d\langle y, y \rangle_t + y_t P_t dZ_t + P_t d\langle Z, y \rangle_t, \end{aligned}$$

where $\langle \cdot, \cdot \rangle$ denotes the quadratic variation operator. Next, use $Z_t = -\varphi'(y_t)$ and Ito's lemma to write

$$dZ_t = -\varphi''(y_t) dy_t - \frac{1}{2} \varphi'''(y_t) d\langle y, y \rangle_t,$$

and

$$d\langle Z, y \rangle_t = -\varphi''(y_t) d\langle y, y \rangle_t.$$

Note that $(\underline{\alpha}_t - \alpha_t)_{t \in [0, T]}$ is an increasing process because

$$d\alpha_t - d\underline{\alpha}_t = -\frac{1}{2} P_t \varphi''(y_t) d\langle y, y \rangle_t.$$

Next, define an alternative portfolio $\bar{\alpha}$, starting with initial wealth α_0 , which replicates $\underline{\alpha}$ and invests the remaining cash in the risk-free account, so the dynamics of $\bar{\alpha}$ are

$$d\bar{\alpha}_t = (\bar{\alpha}_t - \alpha_t) r dt + x_t dP_t + y_t Z_t dP_t + y_t P_t dZ_t,$$

where, as shown with $\underline{\alpha}$, $(\bar{\alpha}_t - \alpha_t)_{t \in [0, T]}$ is an increasing process, because $r \leq 0$.

Next, write the dynamics of PL

$$d\alpha_t - d\bar{\alpha}_t = -\frac{1}{2} P_t \varphi''(y_t) d\langle y, y \rangle_t - r \bar{x}_t P_t dt,$$

where $\bar{x}_t P_t = \bar{\alpha}_t - \alpha_t$. Finally, the inequality in (6.6) follows from

$$\varphi''(y_t) > 0, \quad P_t > 0, \quad d\langle y, y \rangle_t \geq 0, \quad \bar{x}_t = \bar{\alpha}_t - \alpha_t \geq 0, \quad \forall t \in [0, T], \quad \text{and } r \geq 0.$$

The PL in CFMMs is a negative and predictable component in the dynamics of the LP position value α . PL includes losses from two sources. One source is the losses in the value of the LP's position in the pool due to the convexity of the trading function and as a result of the intensity of liquidity taking activity, which is given by the quadratic variation of the quantity y . The other source of loss is the opportunity cost. This cost is incurred because the LP deposits her holdings in the pool so the value loss of the position is not invested in a risk-free account, or invested elsewhere outside the pool.

Note that PL in (6.5) can also be written as

$$L_t = - \underbrace{\frac{1}{2} \int_0^t P_s \frac{d\langle Z, Z \rangle_s}{\varphi''(y_s)}}_{\text{Convexity cost} \geq 0} - \underbrace{\int_0^t \bar{x}_s P_s r ds}_{\text{Opportunity cost} \geq 0}, \quad (6.7)$$

where now PL is a function of the quadratic variation of the marginal rate process Z . Both equations (6.5) and (6.7) show that the magnitude of PL depends on the convexity of the level function, but indicate opposing effects depending on which dynamics one assumes for y and Z . When one models the rate Z as an exogenous process with a constant quadratic variation $d\langle Z, Z \rangle_t = \zeta^2 dt$, the convexity in the pool reduces PL; see (6.7). In this case, the rate Z drives the trading flow in the pool which aligns the ratio between both assets, so a higher convexity reduces the gains of LTs and protects LPs. On the other hand, when one models the quantity y as an exogenous process with constant quadratic variation $d\langle y, y \rangle_t = \chi^2 dt$, the convexity in the pool increases PL; see (6.5). In this case, the changes in quantity y drive the rate Z , and a higher convexity increases the volatility of the rate Z ; recall that $d\langle Z, Z \rangle_t = \varphi''(y_t)^2 d\langle y, y \rangle_t$.

We call the combined effect of the convexity cost and the opportunity cost “predictable loss” because it is zero only if the process y of the quantity of asset Y in the pool or the process Z of the rate have zero quadratic variation, and it is a predictable loss component in the dynamics of the LPs' wealth. In particular, a zero PL is only possible if there is no randomness in liquidity taking activity. Finally, at the end of the next section we discuss the particular case of CPMMs with CL.

6.2.2 PL in CPMMs with CL

In CPMMs without CL, the expression for PL is a straightforward application of Proposition 6.2.1. Next, we discuss PL in the specific case of CPMMs with CL, which requires to account for the spread of the LP's position. Here, for simplicity, we consider that asset X is the numeraire and the LP values her wealth in terms of X , so only asset Y drives uncertainty in the LP's wealth. The next proposition gives a closed-form formula for PL in CPMMs with CL.

To characterise PL in CPMMs with CL, we first need to derive the exact dynamics of the value of the LP's holdings in a pool with CL, which has not been done in the

literature. To achieve this, we focus on the case where the rate Z is in the range of the LP at the beginning and at the end of her investment window.

Here, an LP provides liquidity in a range $(Z^\ell, Z^u]$ at time $t = 0$ and withdraws her liquidity at $t = T$. The LP's initial wealth in the pool is $\tilde{x}_0$, so equations (4.9) and the rate Z_0 determine the initial quantities (x_0, y_0) deposited by the LP in the pool. In particular, when $Z_0 \in (Z^\ell, Z^u]$, one solves

$$\begin{cases} x_0 &= \tilde{\kappa}_0 \left(Z_0^{1/2} - (Z^\ell)^{1/2} \right), \\ y_0 &= \tilde{\kappa}_0 \left(Z_0^{-1/2} - (Z^u)^{-1/2} \right), \\ \tilde{x}_0 &= x_0 + y_0 Z_0, \end{cases}$$

where $\tilde{\kappa}_0$ is the depth of the LP's liquidity position at time $t = 0$, to obtain

$$x_0 = \tilde{x}_0 \left(1 + \frac{Z_0^{1/2} \left((Z^u)^{1/2} - Z_0^{1/2} \right)}{(Z^u)^{1/2} \left(Z_0^{1/2} - (Z^\ell)^{1/2} \right)} \right)^{-1} \quad \text{and} \quad y_0 = \frac{\tilde{x}_0 - x_0}{Z_0}. \quad (6.8)$$

The holdings (x_0, y_0) define the LP's position depth $\tilde{\kappa}_0$. Although there is LT and LP activity in the pool, the depth $\tilde{\kappa}_0$ of the position is kept constant if the LP does not deposit nor withdraws assets in the position range. When the LP removes her liquidity at time $t = T$, the value of her holdings, marked-to-market in units of the reference asset X , in the case where $Z_T \in (Z^\ell, Z^u]$ is determined by

$$x_T = \tilde{\kappa}_0 \left(Z_T^{1/2} - (Z^\ell)^{1/2} \right) \quad \text{and} \quad y_T = \tilde{\kappa}_0 \left(Z_T^{-1/2} - (Z^u)^{-1/2} \right). \quad (6.9)$$

Thus, the change in the value of the LP's holdings between both ends of the trading window $[0, T]$ is

$$\alpha_T - \alpha_0 = x_T + y_T Z_T - x_0 - y_0 Z_0 = x_T + y_T Z_T - \tilde{x}_0.$$

However, The equations (6.8) and (6.9) prevent from obtaining continuous-time dynamics for the wealth of LPs. To circumvent this complexity, we introduce a change of variables, and consider that the LP chooses δ_0^u and δ_0^ℓ to determine the liquidity provision boundaries Z_t^u and Z_t^ℓ :

$$\begin{cases} (Z_0^u)^{1/2} &= Z_0^{1/2} / (1 - \delta_0^u), \\ (Z_0^\ell)^{1/2} &= Z_0^{1/2} (1 - \delta_0^\ell). \end{cases} \quad (6.10)$$

Next, use the change of variables (6.10) and equations (4.9) and (6.10) to write the depth $\tilde{\kappa}_0$ of the LP's position as

$$\tilde{\kappa}_0 = \tilde{x}_0 \left(\frac{1}{\delta_0^\ell + \delta_0^u} \right) Z_0^{-1/2}.$$

Next, we derive the change in the value of the LP's holdings as a function of the change in value of Z and $\sqrt{Z}$ between times 0 and T . Use equations (6.8) and (6.9) to write

$$\begin{aligned} \alpha_T - \alpha_0 &= x_T + y_T Z_T - \tilde{x}_0 \\ &= \tilde{x}_0 \left(\frac{1}{\delta_0^\ell + \delta_0^u} \right) \left(2 \frac{Z_T^{1/2} - Z_0^{1/2}}{Z_0^{1/2}} - \frac{Z_T - Z_0}{Z_0} (1 - \delta_0^u) \right). \end{aligned} \quad (6.11)$$

We assume that the rate process $(Z_t)_{t \in [0, T]}$ follows the geometric Brownian dynamics

$$dZ_t = \mu Z_t dt + \sigma Z_t dW_t, \quad (6.12)$$

where the volatility parameter σ is a nonnegative constant, the trend parameter μ is a constant, and $(W_t)_{t \in [0, T]}$ is a standard Brownian motion. Also

Proposition 6.2.2 An LP deposits quantities (x_0, y_0) in a CPMM with CL at time $t = 0$ and withdraws her liquidity at the terminal time $T > 0$. Let $(x_t)_{t \in [0, T]}$ and $(y_t)_{t \in [0, T]}$ be the quantities she holds in the pool over the period $[0, T]$. The LP receives zero fees and assume that the rate Z is always within the LP's range of liquidity. Additionally, consider that the LP concentrates liquidity in the pool according to the values $(\delta_t^\ell, \delta_t^u)$; see (6.10). Then, the quantities held by the LP in the pool at time t are

$$x_t = \frac{\delta_t^\ell}{\delta_t^\ell + \delta_t^u} \alpha_t \quad \text{and} \quad y_t = \frac{\delta_t^u}{Z_t (\delta_t^\ell + \delta_t^u)} \alpha_t, \quad (6.13)$$

in which case the dynamics of the value of her position

$$d\alpha_t = \alpha_t \left(\frac{1}{\delta_t^\ell + \delta_t^u} \right) \left(-\frac{1}{4} \sigma^2 dt + \mu \delta_t^u dt + \sigma \delta_t^u dW_t \right), \quad (6.14)$$

and the PL, see (6.4), is given by

$$L_t = - \underbrace{\frac{1}{4} \sigma^2 \int_0^t \frac{1}{\delta_s^\ell + \delta_s^u} \alpha_s ds}_{\text{Convexity cost} \geq 0} - \underbrace{r \int_0^t \bar{x}_s ds}_{\text{Opportunity cost} \geq 0} \leq 0. \quad (6.15)$$

Proof 6.2.2 Recall that X is the reference asset that earns the risk-free rate r , and it is used to value the wealth of the LP. At time $t \in [0, T]$, consider an LP who uses her wealth α_t to provide liquidity (x_t, y_t) in a range determined by the spreads $(\delta_t^\ell, \delta_t^u)$ defined in (6.10). Later, at time $t' > t$ the LP withdraws all her holdings $(x_{t'}, y_{t'})$ from

the pool. With equation (6.11), the change in the value of the LP's position, in units of X , between t and t' is given by

$$\alpha_{t'} - \alpha_t = \alpha_t \left(\frac{1}{\delta_t^\ell + \delta_t^u} \right) \left(2 \frac{Z_{t'}^{1/2} - Z_t^{1/2}}{Z_t^{1/2}} - \frac{Z_{t'} - Z_t}{Z_t} (1 - \delta_t^u) \right).$$

Now, to obtain the infinitesimal change in the value of the LP's position, we recall that Z follows the geometric Brownian motion in (6.12) and use Ito's lemma to write

$$dZ_t^{1/2} = \frac{1}{2} Z_t^{1/2} \mu dt + \frac{1}{2} Z_t^{1/2} \sigma dW_t - \frac{1}{8} Z_t^{1/2} \sigma^2 dt,$$

so the infinitesimal increments of the position value process $(\alpha_t)_{t \in [0, T]}$ are given by

$$d\alpha_t = \alpha_t \left(\frac{1}{\delta_t^\ell + \delta_t^u} \right) \left(-\frac{1}{4} \sigma^2 dt + \mu \delta_t^u dt + \sigma \delta_t^u dW_t \right). \quad (6.16)$$

The changes in the value of the position α in (6.16) result from depositing x_t units of the reference asset X , and y_t units of asset Y for each time t , where

$$x_t = \frac{\delta_t^\ell}{\delta_t^\ell + \delta_t^u} \alpha_t \quad \text{and} \quad y_t = \frac{\delta_t^u}{Z_t (\delta_t^\ell + \delta_t^u)} \alpha_t.$$

Instead of making liquidity in the pool, the LP can invest the quantity y_t in (6.13) in an alternative venue that does not operate a CPMM. Similarly to the case of CFMMs, we denote by $\bar{\alpha}$ the strategy that holds y_t as in the pool and the quantity $\bar{x}_t$ in the risk-free account.³ We have

$$\bar{x}_t = \bar{\alpha}_t - y_t Z_t \geq x_t \geq 0,$$

as explained above. The infinitesimal change in $\bar{\alpha}$ is given by

$$\begin{aligned} d\bar{\alpha}_t &= \alpha_t \left(\frac{1}{\delta_t^\ell + \delta_t^u} \right) \frac{dZ_t}{Z_t} + \bar{x}_t r dt \\ &= \alpha_t \left(\frac{1}{\delta_t^\ell + \delta_t^u} \right) (\mu \delta_t^u dt + \sigma \delta_t^u dW_t) + \bar{x}_t r dt, \end{aligned} \quad (6.17)$$

where, as in Proposition 6.2.1, market frictions are zero. The PL in CPMMs with CL is thus given by

$$L_t = -\frac{1}{4} \sigma^2 \int_0^t \frac{1}{\delta_s^\ell + \delta_s^u} \alpha_s ds - r \int_0^t \bar{x}_s ds \leq 0.$$

³ Here, because the numeraire is X , both asset X and the extra wealth are put in a risk-free account. Recall that in the general case, both X and Y are risky.

As above in CFMMs, PL is a predictable loss component and key to understanding the profitability of liquidity provision in liquidity pooling protocols with CL. Clearly, the magnitude of PL incurred by the LP's position increases with the volatility of the rate Z . Thus, one expects an optimal liquidity provision model to adjust the position spread as a function of volatility.

The dynamics in (6.14) and (6.15) show how the LP can reduce PL by increasing the spread of the position in the pool. If the spread is $\delta^\ell = 1$ and $\delta^u = 1$, and the risk-free rate is zero, the PL is minimal (but not zero) and we denote it by $L_t^{\min} = \alpha_t^{\min} - \bar{\alpha}_t^{\min}$. In this case, the LP provides liquidity in the maximal range $(0, +\infty)$, which decreases the magnitude of the first term on the right-hand side of (6.15), and the opportunity cost, given by the second term on the right-hand side of (6.15), is zero. Recall that PL is given by $L_t = \alpha_t - \bar{\alpha}_t$, so when $\delta^\ell = \delta^u = 1$ and $r = 0$ we have that $\mathbb{E}[L_T] \leq \mathbb{E}[L_T^{\min}]$, where

$$\mathbb{E}[L_T^{\min}] = -\frac{1}{8\hat{\mu}}\sigma^2\alpha_0(\exp(\hat{\mu}T) - 1) \leq 0, \text{ if } \hat{\mu} = \frac{\mu}{2} - \frac{\sigma^2}{8} \neq 0,$$

and

$$\mathbb{E}[L_T^{\min}] = -\sigma^2\alpha_0 T/8, \text{ if } \hat{\mu} = 0.$$

Note that the volatility σ is a negative component in the trend of α but not in that of $\bar{\alpha}$; see (6.14). When volatility becomes arbitrarily high, the LP expects to lose at least her initial wealth because

$$\lim_{\sigma \rightarrow +\infty} \mathbb{E}[L_T] \leq \lim_{\sigma \rightarrow +\infty} \mathbb{E}[L_T^{\min}] = -\alpha_0.$$

Next, we discuss further the PL incurred by the LP for various ranges of the drift of the rate and for long trading horizons (with $\delta^\ell = \delta^u = 1$ and $r = 0$). When the trend in the rate Z is $\mu \geq \sigma^2/4$, the value of the holdings α in the pool and the value of the alternative portfolio $\bar{\alpha}$ increase because of the positive trend in their dynamics. However, the value of the alternative portfolio $\bar{\alpha}$ increases faster than that of α , hence, the LP's trading horizon becomes arbitrarily large, we have that

$$\lim_{T \rightarrow +\infty} \mathbb{E}[L_T] = -\infty,$$

that is, the difference between the value of the alternative portfolio and that of the holdings in the pool is infinite; see first panel in Figure 6.1.

On the other hand, if $\mu < \sigma^2/4$, we have that

$$\lim_{T \rightarrow +\infty} \mathbb{E}[L_T] \leq \lim_{T \rightarrow +\infty} \mathbb{E}[L_T^{\min}] = \frac{\alpha_0\sigma^2}{4\mu - \sigma^2} < 0.$$

In particular, the expected minimum PL decreases when the trend is large and negative as both portfolios α and $\bar{\alpha}$ incur large losses; see second panel in Figure 6.1.

Finally, when $\mu = 0$, the expected minimum PL is

$$\mathbb{E} [L_T^{\min}] = \alpha_0 \left(\exp \left(-\frac{1}{8} \sigma^2 T \right) - 1 \right),$$

in which case the LP expects to lose her initial wealth when volatility or the time horizon becomes arbitrarily high.

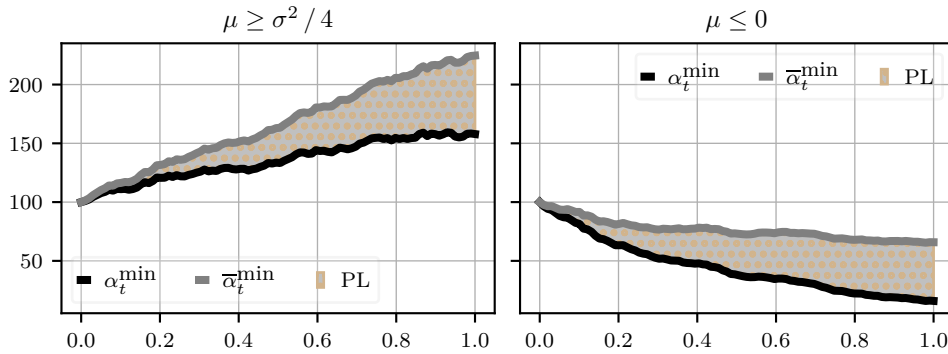


Fig. 6.1: Simulated paths for the position value α in (6.14) and the alternative portfolio $\bar{\alpha} = \alpha - L$ in (6.17) when $\delta^\ell = \delta^u = 1$ and $r = 0$. Left panel: $\mu = 0.02\%$ and $\sigma = 2\%$. Right panel: $\mu = -0.02\%$ and $\sigma = 3\%$.

Conclusions

In this chapter, we introduced predictable loss (PL) as a new measure of loss in liquidity provision activity in CFMMs and in CPMMs with CL. We provided closed-form formulae for PL and showed that the losses stem from two sources: the convexity cost, which depends on liquidity taking activity and the convexity of the pool's trading function; the opportunity cost, which is due to locking the LP's assets in the pool.

Chapter 7

Optimal liquidity provision

The results of this chapter and further extensions have been obtained in collaboration with Álvaro Cartea and Marcello Monga, and are in [Cartea et al. \(2022f\)](#).

In CPMMs with CL, we prove that the viability of liquidity provision depends on the tradeoff between the fees (provision fees minus gas fees), the risk of concentrating liquidity provision, and PL. Concentration risk refers to the risk that the LP faces when her position is concentrated in narrow ranges; the rate can exit the position and the LP stops collecting fees. We use the tools of stochastic optimal control to derive a dynamic strategy to provide liquidity in a CPMM with CL over a trading horizon. The LP maximises expected utility of terminal wealth, which consists of the accumulated trading fees and the gains and losses from the market making strategy. The model strategically targets ranges around the reference exchange rate to maximise fee collection and minimise PL. For the particular case of log-utility, we obtain the strategy in closed-form and show how the solution balances the opposing effects between PL and fee collection. When volatility increases, PL increases, so there is an incentive for the LP to widen the range of liquidity provision to reduce the strategy's exposure to PL. Also, when there is an increase in the potential fees that the LP can collect, the strategy balances two opposing forces. One, there is an incentive to increase fee collection by concentrating the liquidity of the LP in a tight range around the exchange rate of the pool. Two, there is a limit to how concentrated the LP posts liquidity in the pool because the LP does not collect fees if the exchange rate exits the LP's liquidity range. Finally, we show that in the extreme case of very high volatility, the LP must withdraw from the pool because the exposure to PL is too high.

We use Uniswap v3 data from the pool ETHUSDC (Ethereum and USD coin) between 5 May 2021 and 18 August 2022 to motivate our model and to test the performance of the strategy we derive. To illustrate the performance of the strategy we use in-sample data to estimate model parameters and out-of-sample data to test the strategy. Our analysis of the historical transactions in Uniswap v3 shows that LPs have traded at a significant loss, on average, in the ETHUSDC pool. We show that the out-of-sample

performance of our strategy is considerably superior to the average LP performance we observe in the ETHUSDC pool.

Fees collected from market making activity are not enough to cover PL and to cover the gas fees to operate in Uniswap v3. In an efficient market, one would expect LPs to be driven out of the market unless, everything else being equal, gas fees decrease and the fees paid by LTs increase. With the current fee structure and fee levels, LPs are overexposed to PL during periods of heightened volatility of the rate of the pool. Currently, the fee structure in CPMMs cannot lead to efficient and resilient markets because liquidity will be scarce during times of high volatility and, in the long term, LPs will withdraw from the market. A potential solution is one or a combination of the following two proposals. Increase the level of fixed fees paid by LTs, or implement a dynamic structure where fees are an increasing function of the PL; the PL function depends on the volatility of the rate, convexity of the pool's trading function, and opportunity cost of the LP's wealth.¹ Clearly, a change in fee level and fee structure will affect trading activity in AMMs. In particular, it will impact the demand for liquidity because it will be more expensive for LTs to trade, while, on the other hand, it will incentivise liquidity provision because compensation will increase, on average – it is a challenging problem to find a fee structure that provides an equilibrium where activity is not hindered, the market is resilient, and the rates of the pool are efficient.

The remainder of the chapter proceeds as follows. Section 7.1 derives the dynamics of the wealth for LPs that interact with a CPMM with CL. The wealth of LPs changes as a result of fee revenue and changes in the value of the holdings in the pool. Section 7.2 solves the optimal liquidity provision problem and provides a closed-form strategy. Finally, Section 7.2 showcases the performance of the liquidity provision strategy.

7.1 Wealth dynamics

In this section, we consider an LP with initial wealth $\tilde{x}_0$, in units of X , and an investment horizon $[0, T]$, with $T > 0$. She deposits quantities (x_0, y_0) in the range $(Z^\ell, Z^u]$ so the depth of her position is $\tilde{\kappa}_0$, and the value of her initial position, marked-to-market in units of X , is $\tilde{x}_0 = x_0 + y_0 Z_0$.² At the terminal time T , she withdraws her liquidity from the pool and collects the fees generated over the trading window. The LP's terminal wealth in terms of asset X is given by

$$\tilde{x}_T = \alpha_T + \pi_T,$$

where π_T denotes the revenue from fees and, as in Chapter 6, α_T denotes the terminal *position value*. Note that $\tilde{x}_0 = \alpha_0$ because at $t = 0$ the LP has not yet collected any fees.

¹ In LOBs, this dynamic adjustment of the fee is endogenous because liquidity providers increase the spread between the best bid and best ask when volatility of the midprice increases.

² In the remainder of this chapter, we drop the superscripts indicating the ticks defining the boundaries of the LP's position range.

At the terminal time T , the value of the LP's terminal position α_T is determined by the quantities x_T , y_T , and the pool rate Z_T ; see (6.1).

The holdings of the LP change because the rate Z changes throughout the investment window – not as a result of the LP rebalancing her holdings. On the other hand, the LP's accumulated fee income π_T depends on the taking activity that engages the LP's liquidity in the pool during the trading horizon $[0, T]$; i.e., the number and size of LT transactions at rates that lie within the LP's range during her investment horizon.

In the remainder of this section, we describe and characterise the dynamics of the LP's wealth as a result of the position value and of the fee income. In addition, we characterise PL for the case of CPMs with CL. We use Uniswap v3 to study both components of the terminal position of the LP's wealth and our results motivate the assumptions we make for our model in Section 7.2. The LP and LT data are from the pool ETHUSDC between the inception of the pool on 5 May 2021 and 18 August 2022. Section 4.3 describes data and Table 4.1 provides descriptive statistics.

7.1.1 Position value

In this subsection, we focus our analysis on the *position value*, which refers to the value of the LP holdings in the pool, and which we mark-to-market in units of the reference asset X . Here, an LP provides liquidity in a range $(Z^\ell, Z^u]$ at time $t = 0$ and withdraws her liquidity at $t = T$. The LP's initial wealth in the pool is $\tilde{x}_0$, so equations (4.9) and the rate Z_0 determine the initial quantities (x_0, y_0) deposited by the LP in the pool. In the two cases where liquidity is provided in a range that does not include the rate Z_0 , the initial holdings are given by

$$\begin{cases} x_0 = 0 & \text{and} & y_0 = \tilde{x}_0 / Z_0 & \text{if } Z_0 \leq Z^\ell, \\ x_0 = \tilde{x}_0 & \text{and} & y_0 = 0 & \text{if } Z_0 > Z^u, \end{cases} \quad (7.1)$$

and when $Z_0 \in (Z^\ell, Z^u]$, one solves

$$\begin{cases} x_0 &= \tilde{\kappa}_0 \left(Z_0^{1/2} - (Z^\ell)^{1/2} \right), \\ y_0 &= \tilde{\kappa}_0 \left(Z_0^{-1/2} - (Z^u)^{-1/2} \right), \\ \tilde{x}_0 &= x_0 + y_0 Z_0, \end{cases}$$

where $\tilde{\kappa}_0$ is the depth of the LP's liquidity in the pool at time t , to obtain

$$x_0 = \tilde{x}_0 \left(1 + \frac{Z_0^{1/2} \left((Z^u)^{1/2} - Z_0^{1/2} \right)}{(Z^u)^{1/2} \left(Z_0^{1/2} - (Z^\ell)^{1/2} \right)} \right)^{-1} \quad \text{and} \quad y_0 = \frac{\tilde{x}_0 - x_0}{Z_0}. \quad (7.2)$$

Thus, the LP uses her initial wealth $\tilde{x}_0$, the boundaries of the range Z^ℓ and Z^u , and the position rate Z_0 , from which she obtains the quantities (x_0, y_0) she deposits in the

pool and the depth $\tilde{\kappa}_0$ of her liquidity. Recall that $\tilde{\kappa}_0$ is constant until the LP decides to withdraw her liquidity. As an example, consider an LP who deposits her initial wealth $\tilde{x}_0 = 100$ in the range $(90, 110]$. Figure 7.1 shows the quantities the LP provides in each asset as a function of the position rate. When the rate is outside the range $(90, 110]$, the LP provides only one of the assets in the liquidity pool. When the rate is within the range $(90, 110]$, the LP provides both assets in proportions that depend on the position rate.

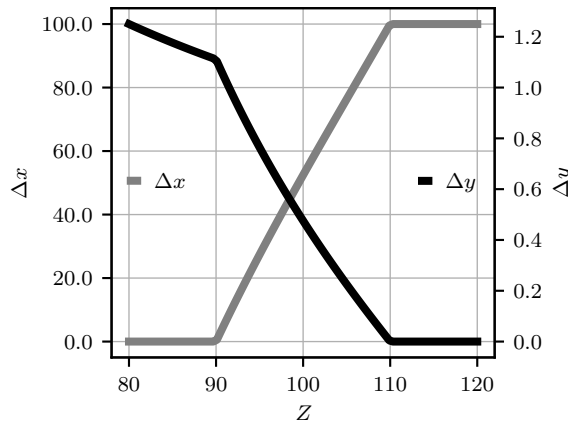


Fig. 7.1: Quantities x_0 and y_0 provided by an LP with initial wealth $\tilde{x}_0 = x_0 + y_0 Z_0$ in the range $(90, 110]$ as a function of the position rate Z_0 ; see equations (7.1) and (7.2).

The holdings (x_0, y_0) define the LP's position depth $\tilde{\kappa}_0$. Although there is LT and LP activity in the pool, the depth $\tilde{\kappa}_0$ of the position is kept constant, which assumes that the LP does not deposit nor withdraws assets in the range. However, LP activity may change the portion of the pool depth κ that the LP holds, i.e., $\tilde{\kappa}/\kappa$. When the LP removes her liquidity at time $t = T$, the value of her holdings, marked-to-market in units of the reference asset X , is determined by

$$\begin{cases} x_T = 0 & \text{and} & y_T = \tilde{\kappa}_0 \left((Z^\ell)^{-1/2} - (Z^u)^{-1/2} \right) & \text{if } Z_T \leq Z^\ell, \\ x_T = \tilde{\kappa}_0 \left(Z_T^{1/2} - (Z^\ell)^{1/2} \right) & \text{and} & y_T = \tilde{\kappa}_0 \left(Z_T^{-1/2} - (Z^u)^{-1/2} \right) & \text{if } Z^\ell < Z_T \leq Z^u, \\ x_T = \tilde{\kappa}_0 \left((Z^u)^{1/2} - (Z^\ell)^{1/2} \right) & \text{and} & y_T = 0 & \text{if } Z_T > Z^u. \end{cases} \quad (7.3)$$

The change in the value of the position between both ends of the trading horizon $[0, T]$ is

$$\alpha_T - \alpha_0 = x_T + y_T Z_T - x_0 - y_0 Z_0 = x_T + y_T Z_T - \tilde{x}_0. \quad (7.4)$$

Equations (7.3) and (7.4) define a payoff α_T in terms of asset X as a function of the terminal rate Z_T . Figure 7.2 shows the terminal relative payoff $\alpha_T/\tilde{x}_0$ as a function of the final rate for different position ranges, when the position rate is $Z_0 = 100$. Provision

of liquidity with wide spreads protects the LP from losses when $Z_T \leq Z_0$, and facilitates more gains when $Z_T \geq Z_0$. Figure 7.2 suggests that the LP's relative payoff $\alpha_T/\tilde{x}_0$ is concave in the rate Z_T .

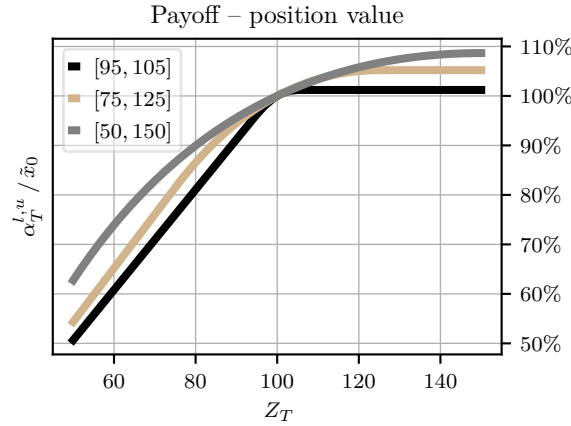


Fig. 7.2: Payoff (7.3) for an LP providing liquidity in the ranges $[95, 105]$, $[75, 125]$ and $[50, 150]$, around the position rate $Z_0 = 100$.

Wider spreads in the LP's concentrated liquidity protect the value of the LP's position because when rates drop and the spread is narrow, the position rapidly consists only in the less valuable asset Y and loses value linearly in Z . On the other hand, when the rate Z increases and the spread is narrow, the gains from rate appreciation are capped at Z^u ; see equation (7.3). Consequently, a wide spread protects the value of the LP's assets. We refer to the width of the LP's position as the *spread* because it shares similar properties with those of the spread of a market maker in LOBs. In LOBs, the spread of an LP refers to the distance between the limit orders posted on both sides of the midprice. In particular, in LOBs, market makers widen their spread when adverse selection risk increases, i.e., they post limit orders at deeper levels in the book.

To make markets optimally, the LP chooses δ_t^u and δ_t^ℓ which determine the liquidity provision boundaries Z_t^u and Z_t^ℓ :

$$\begin{cases} (Z_t^u)^{1/2} = Z_t^{1/2} / (1 - \delta_t^u), \\ (Z_t^\ell)^{1/2} = Z_t^{1/2} (1 - \delta_t^\ell). \end{cases} \quad (7.5)$$

In the remainder of this chapter we define the *spread* δ_t as

$$\delta_t = \delta_t^u + \delta_t^\ell, \quad (7.6)$$

and note that

$$\frac{Z_t^u - Z_t^\ell}{Z_t} = (1 - \delta_t^u)^{-2} - (1 - \delta_t^\ell)^2.$$

With the formulae for δ_t^u and δ_t^ℓ in (7.5) one can write the position value dynamics as a function of the dynamics of Z and $\sqrt{Z}$ when the rates Z_0 and Z_T are within the

position range. More precisely, use the second equation in (4.9) and (7.5) to write the depth $\tilde{\kappa}_0$ of the LP's position as

$$\tilde{\kappa}_0 = \tilde{x}_0 \left(\frac{1}{\delta_0^\ell + \delta_0^u} \right) Z_0^{-1/2}. \quad (7.7)$$

Next, use equations (7.2) and (7.3) to derive the change in position value between times 0 and T as

$$\begin{aligned} \alpha_T - \alpha_0 &= x_T + y_T Z_T - \tilde{x}_0 \\ &= \tilde{x}_0 \left(\frac{1}{\delta_0^\ell + \delta_0^u} \right) \left(2 \frac{Z_T^{1/2} - Z_0^{1/2}}{Z_0^{1/2}} - \frac{Z_T - Z_0}{Z_0} (1 - \delta_0^u) \right). \end{aligned} \quad (7.8)$$

Equation (7.8) uses the formulae for the initial quantities (x_0, y_0) that the LP deposits in the pool and the terminal quantities (x_T, y_T) that the LP withdraws from the pool. Specifically, use definition (7.5) to write

$$x_0 = \frac{\delta_0^\ell}{\delta_0^\ell + \delta_0^u} \tilde{x}_0 \quad \text{and} \quad y_0 Z_0 = \frac{\delta_0^u}{\delta_0^\ell + \delta_0^u} \tilde{x}_0,$$

and use (7.3) and (7.7), so that

$$\begin{aligned} \begin{cases} x_T &= \tilde{\kappa}_0 \left(Z_T^{1/2} - (Z^\ell)^{1/2} \right), \\ y_T Z_T &= \tilde{\kappa}_0 \left(Z_T^{1/2} - Z_T (Z^u)^{-1/2} \right), \end{cases} \\ \Rightarrow \begin{cases} x_T &= \tilde{x}_0 \left(\frac{1}{\delta_0^\ell + \delta_0^u} \right) \left(2 \frac{Z_T^{1/2}}{Z_0^{1/2}} - (1 - \delta_0^\ell) \right), \\ y_T Z_T &= \tilde{x}_0 \left(\frac{1}{\delta_0^\ell + \delta_0^u} \right) \left(\frac{Z_T^{1/2}}{Z_0^{1/2}} - \frac{Z_T}{Z_0} (1 - \delta_0^u) \right). \end{cases} \end{aligned}$$

In our model, we use (7.8) to characterise the changes in the wealth of the LP as a result of changes in the value of her position. Clearly, the risk stems from the volatility of Z in (7.8).

7.1.2 Fee income

In this subsection, we first show that fee income is subject to a tradeoff between the spread δ of the LP's position (or equivalently the value of $\tilde{\kappa}$) and the volatility of Z . Next, we show how the LP can skew her liquidity around Z as a function of the trend and the volatility of the rate. Finally, we derive the fee income dynamics that we use in our model.

First, in the three cases of (4.9), increasing the spread of the LP's position reduces the depth $\tilde{\kappa}$ of the LP's position in the pool. Recall that the LP fee income is proportional to

$\tilde{\kappa}/\kappa$, where κ is the pool depth; see (4.12). Thus, decreasing the value of $\tilde{\kappa}$ potentially reduces LP fee income because the LP holdings represent a smaller portion of the pool depth around the rate Z . Figure 7.3 shows the value of $\tilde{\kappa}$ as a function of the spread δ .

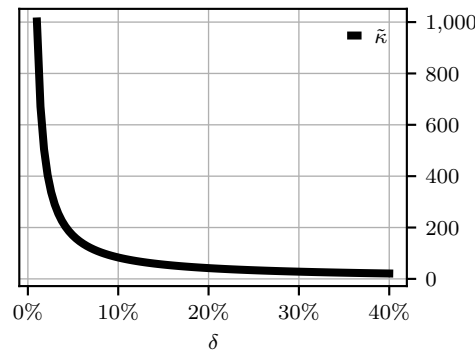


Fig. 7.3: Value of the LP's depth κ in the pool as a function of the spread δ .

However, although narrow ranges increase the potential fee income, they also increase *concentration risk*, i.e., the risk of the rate Z exiting the range posted by the LP; when the rate exits the range, the LP position no longer earns fees. On the other hand, a wide spread, i.e., a lower value of the depth $\tilde{\kappa}$, decreases fee income per LT transaction, but reaps earnings from a larger number of LT transactions as the position is active for longer periods of time (i.e., it takes longer, on average, for Z to exit the LP's range). Thus, the LP must strike a balance between maximising the depth $\tilde{\kappa}$ around the rate and minimising the concentration risk, which depends on the volatility of the rate Z .

Next, we study the effect of asymmetry in the range of the position. We define the *asymmetry* of a position as

$$\rho_t = \delta_t^u / (\delta_t^u + \delta_t^\ell) = \delta_t^u / \delta_t, \quad (7.9)$$

where δ_t^u and δ_t^ℓ are defined in (7.5). In one extreme, when the asymmetry $\rho \rightarrow 0$, then $Z^u \rightarrow Z$ and the position consists of only asset X , and in the other extreme, when $\rho \rightarrow 1$, then $Z^\ell \rightarrow Z$ and the position consists of only asset Y .

We use Uniswap v3 data (see Section 4.3 for details) to estimate the realised trend μ in the pool ETHUSDC over a rolling window of $T = 5$ minutes.³ Next, for any time t , the fee income for different LP positions is computed for various values of the spread δ and for various values of the asymmetry ρ .

Figure 7.4 shows the distribution of the fee income as a function of the asymmetry in the position. We consider an investment window $T = 5$ minutes and we normalise the fee income by the maximum ex-post fee income. Figure 7.4 suggests that it is (on average) optimal, for LPs to provide liquidity in symmetric positions. For large position

³ The values of the trend in this section are normalised to reflect daily estimates. In particular, we use $\mu = \tilde{\mu} / \Delta t$ where $\tilde{\mu}$ is the average of the observed log returns and Δt is the observed average LT trading frequency.

spreads ($\delta = 5\%$), fee income is only impacted when the asymmetry is extreme. Thus, multiple values of the asymmetry are optimal. On the other hand, for small position spreads ($\delta = 0.1\%$), the position's asymmetry plays an important role.

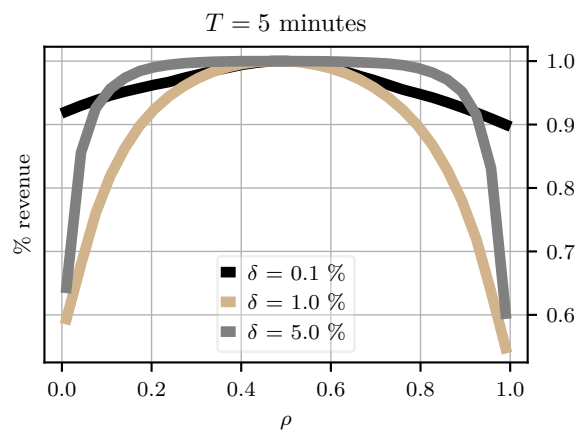


Fig. 7.4: Distribution of the fee income as a function of the asymmetry ρ in (7.9), for different values of the spread δ in (7.6) when $T = 5$ minutes. The fee income is the percentage of the maximum ex-post fee income.

Next, we investigate the impact of the trend μ of the rate Z on the optimal asymmetry value. We use rate returns over the investment window to estimate μ . The values of the trend in this section are normalised to reflect daily estimates. In particular, we use $\mu = \tilde{\mu} / \Delta t$ where $\tilde{\mu}$ is the average of the observed log returns and Δt is the observed average LT trading frequency. We estimate the distribution of fee income as a function of the asymmetry in the position range for multiple values of the trend. Figure 7.5 suggests that for small spreads ($\delta = 0.1\%$), when the trend is strong and the rate can quickly exit the position range, it is optimal on average to post liquidity with extreme asymmetry to capture more LT transactions. When the spread is $\delta = 1\%$, the optimal asymmetry deviates a little from 0.5, and when the spread is large ($\delta = 5\%$), the asymmetry has a smaller effect.

Finally, for each value of the realised trend μ during the investment horizon, and for each spread δ , we record the asymmetry that maximises fee income. Figure 7.6 shows the optimal (on average) asymmetry ρ as a function of the spread δ of the position for multiple values of the realised trend μ . Figure 7.6 suggests that the optimal asymmetry is sensitive to the spread of the position. Narrow spreads require more asymmetry when the trend is not zero. Thus, an optimal liquidity provision model should take into account the trend to adjust the asymmetry ρ of the LP positions.

Figure 7.6 suggests that there exists a preferred asymmetry of the position for a given value of the spread δ and a given value of the trend μ . In our model of Section 7.2, the LP either holds a belief or estimates the trend μ before the start of her investment window, and controls the spread $\delta = \delta^u + \delta^\ell$ of her position. Thus, she strategically chooses the asymmetry of her position as a function of δ and μ . We approximate the relationship exhibited in Figure 7.6 with the asymmetry function

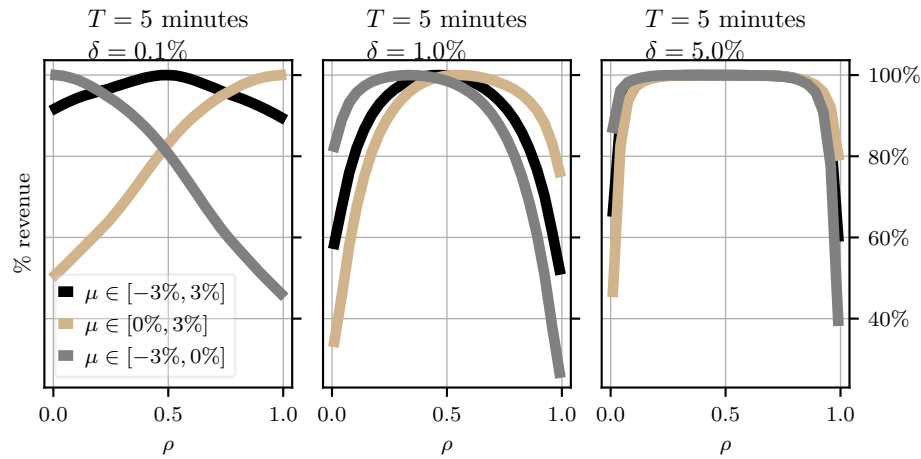


Fig. 7.5: Distribution of the fee income as a function of the asymmetry ρ ; see (7.9), for different values of the spread δ ; see (7.6), when $T = 5$ minutes. The fee income is a percentage of the maximum ex-post fee income.

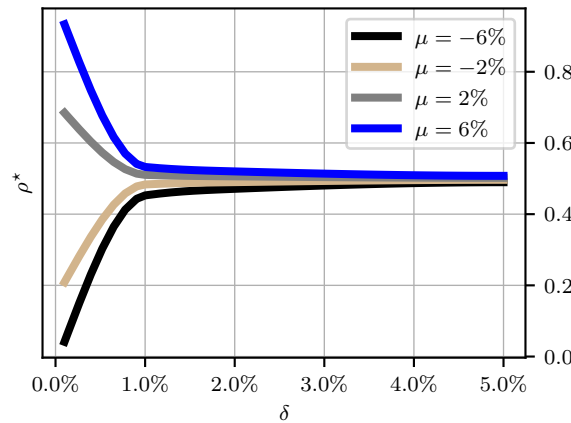


Fig. 7.6: Optimal position asymmetry ρ^* in (7.9) as a function of the spread δ of the position, for multiple values of the trend μ . The asymmetry ρ^* is the value of ρ that maximises fee income for observed values of (δ, μ) .

$$\rho = \rho(\delta, \mu) = \frac{1}{2} + \frac{\mu}{\delta} = \frac{1}{2} + \frac{\mu}{\delta^u + \delta^\ell}. \quad (7.10)$$

The dynamics of the fee income in our model of Section 7.3 uses a fixed depth κ and assumes that the pool generates fee income for all LPs at an marginal *pool fee rate* Π ; clearly, these fees are paid by LTs who interact with the pool. The value of Π represents the instantaneous profitability of the pool, and it is equivalent to the size of market orders and their arrival rate in LOBs, which are some of the key drivers of the profitability, on average, of a market making strategy based on roundtrip trades.

To analyse the dynamics of the pool fee rate Π , we use historical LT transactions as a measure of activity and to estimate the total fee income generated by the pool. Figure 7.7 shows the estimated fee rate Π in the ETHUSDC pool. For any time t , we use

$$\Pi_t = 0.3\% \frac{V_t}{2\kappa Z_t^{1/2}},$$

where V_t is the volume of LT transactions one day before t , $2\kappa Z_t^{1/2}$ is the pool value at time t , and 0.3% is the fixed fee of the pool.⁴ Figure 7.7 suggests that the pool fee rate Π generated by liquidity taking activity in the pool is stochastic and mean reverting. Here, we assume that Π is independent of the rate Z over the time scales we consider.

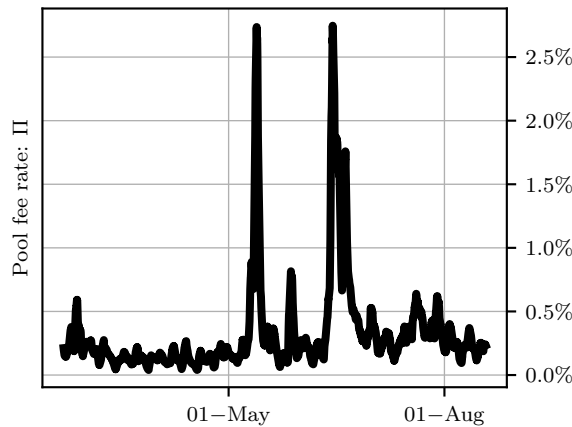


Fig. 7.7: Estimated pool fee rate from February to August 2022 in the ETHUSDC pool. For any time t , the estimate of the pool fee is the total fee income, as a percentage of the total pool size, paid by LTs on the period $[t - 1 \text{ day}, t]$. The pool size at time t is $2\kappa Z_t^{1/2}$, where Z_t is the active rate in the pool at time t .

	$\Delta t = 1 \text{ minute}$	$\Delta t = 5 \text{ minutes}$	$\Delta t = 1 \text{ hour}$	$\Delta t = 1 \text{ day}$
Correlation	-2.1%	-2.4%	-2.6%	-10.9%

Table 7.1: Correlation of the returns of the rate Z and the fee rate Π , i.e., $(Z_{t+\Delta t} - Z_t) / Z_t$ and $(\Pi_{t+\Delta t} - \Pi_t) / \Pi_t$ for $\Delta t = 1 \text{ minute}$, five minutes, one hour, and one day, using data of the ETHUSDC pool between 5 May 2021 and 18 August 2022.

In our model, the LP continuously adjusts her position around the current rate Z , so we derive continuous-time dynamics of (4.12), conditional on the rate not exiting the LP's range, and write

$$d\pi_t = \underbrace{(\tilde{\kappa}_t / \kappa)}_{\text{Position depth}} \underbrace{(\Pi_t)}_{\text{Fee rate}} \underbrace{(2\kappa Z_t^{1/2})}_{\text{Pool size}} dt, \quad (7.11)$$

⁴ The value of a CPMM pool is given by the active pool depth. In particular, the size of a pool with quantity x of asset X and quantity y of asset Y is $x + Z y = 2x = 2\kappa Z^{1/2}$ units of X because $x y = \kappa^2$ and $Z = x/y \implies x^2 = \kappa^2 Z$.

where π is the LP's fee income for providing liquidity with depth $\tilde{\kappa}$ in the pool. The fee income is proportional to the pool size, i.e., proportional to $2\kappa Z_t^{1/2}$.

In practice, the LP chooses a trading frequency, so the LP faces the risk that the rate exits the spread around Z in between the times the LP repositions her liquidity. Clearly, the continuous-time dynamics in (7.11) do not take into account concentration risk. In practice, narrow spreads generate less fee income because the rate Z may exit the range of the LP's liquidity, especially in volatile markets. Thus, we introduce a concentration cost to reduce the fees collected by the LP. The concentration costs increase (decrease) when the spread narrows (widens), so we modify the dynamics of the fees collected by the LP in (7.11) to account for the concentration cost and write

$$d\pi_t = 2\tilde{\kappa}_t \Pi_t Z_t^{1/2} dt - \gamma \left(\frac{1}{\delta_t^\ell + \delta_t^u} \right)^2 \tilde{x}_t^2 dt, \quad (7.12)$$

where $\gamma > 0$ is the concentration cost parameter and $\tilde{x}_t$ is the wealth invested by the LP in the pool at time t .

Figure 7.8 compares the fee income from (7.11) and from (7.12), which includes the concentration cost, as a function of the spread of the LP's position. We simulate a driftless rate Z with $\sigma = 1\%$ and $\sigma = 3\%$ and assume that the LP adjusts her liquidity with a frequency of $\Delta t = 1$ minute. The fee income corresponding to every spread is normalised by the maximum ex-post fee income. The figure shows that the concentration cost term in (7.12) captures the loss in fee revenue because of concentration risk. In addition, LPs should adapt the concentration cost parameter γ to the volatility of the rate Z . The LP can also choose the value of γ based on her beliefs about the future realised volatility.

In the next section, we derive an optimal liquidity provision strategy, and prove that the profitability of liquidity provision is subject to a tradeoff between fee revenue, PL, and gas fees.

7.2 The optimal LP problem

We consider the problem of an LP who wants to provide liquidity in a CPMM with CL during an investment window $[0, T]$ with $T > 0$. The LP implements a self-financing strategy that constantly adjusts the spread δ , which requires depositing and withdrawing liquidity continuously in the pool. The LP holds assets X and Y in the pool and marks-to-market her holdings in terms of asset X . During the trading window, the LP's wealth changes when the value of her holdings in the pool change. Her positions in the pool accumulate earnings from fees that she collects every time she withdraws her liquidity throughout the trading horizon. Finally, we assume that the risk-free rate is zero, so the PL faced by the LP is the convexity cost because the opportunity cost would be zero.

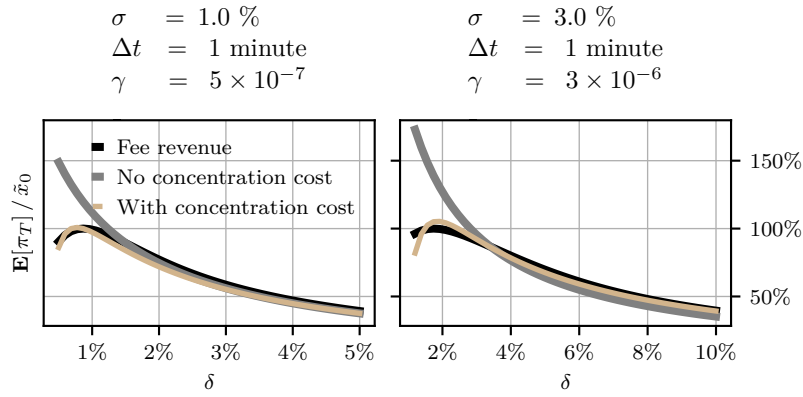


Fig. 7.8: Fee income based on the dynamics in (7.11), without concentration cost (i.e., $\gamma = 0$), and in (7.12), with concentration cost ($\gamma > 0$). We use simulations of the rate Z in (6.12) and Π which we assume follows the mean-reverting process $d\Pi_t = a(b - \Pi_t)dt + c dB_t$, where B_t is a standard Brownian motion. For the rate Z , we use $\mu = 0$, and $\sigma = 1\%$ and 3% . For the pool fee rate Π , the parameters are obtained using a maximum likelihood estimation from the fee rate of the ETHUSDC pool in Figure 7.7. We obtain $a = 2.71$, $b = 0.22\%$, and $c = 0.18$. For every spread value, we use (4.12) to obtain the fee income and use (7.11) to obtain the fee income.

We fix a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P}; \mathbb{F} = (\mathcal{F}_t)_{t \in [0, T]})$ satisfying the usual conditions, where $\mathbb{F}$ is the natural filtration generated by the collection of observable stochastic processes that we define in this section.

The rate process $(Z_t)_{t \in [0, T]}$ satisfies the dynamics

$$dZ_t = \mu Z_t dt + \sigma Z_t dW_t,$$

where σ is the volatility, μ is the trend, Z_0 is known, and $(W_t)_{t \in [0, T]}$ is a standard Brownian motion.

Throughout the trading window, the LP is continuously repositioning her liquidity in a range $(Z_t^\ell, Z_t^u]$ around the active rate Z_t . The two ends of the spread, $(\delta_t^\ell)_{t \in [0, T]}$ and $(\delta_t^u)_{t \in [0, T]}$, are given by

$$\begin{cases} (Z_t^u)^{1/2} = Z_t^{1/2} / (1 - \delta_t^u), \\ (Z_t^\ell)^{1/2} = Z_t^{1/2} (1 - \delta_t^\ell), \end{cases} \quad (7.13)$$

where the processes $(Z_t^\ell)_{t \in [0, T]}$ and $(Z_t^u)_{t \in [0, T]}$ define the dynamics of the position boundaries.

We introduce the process $(\tilde{\kappa}_t)_{t \in [0, T]}$ which models the depth of the LP's position and, similar to equation (7.7), its dynamics are given by

$$\tilde{\kappa}_t = \tilde{x}_t \left(\frac{1}{\delta_t^\ell + \delta_t^u} \right) Z_t^{-1/2}.$$

Next, define the dynamics of the position value process $(\alpha_t)_{t \in [0, T]}$ as the continuous-time version of equation (7.8). Recall that for any time t , the LP uses her wealth $\tilde{x}_t$ to provide liquidity. Use Ito's lemma to write

$$dZ_t^{1/2} = \frac{1}{2} Z_t^{1/2} \mu dt + \frac{1}{2} Z_t^{1/2} \sigma dW_t - \frac{1}{8} Z_t^{1/2} \sigma^2 dt$$

and

$$\begin{aligned} d\alpha_t &= \tilde{x}_t \left(\frac{1}{\delta_t^\ell + \delta_t^u} \right) \left(-\frac{1}{4} \sigma^2 dt + \mu \delta_t^u dt + \sigma \delta_t^u dW_t \right) \\ &= L_t + \tilde{x}_t \left(\frac{1}{\delta_t^\ell + \delta_t^u} \right) (\mu \delta_t^u dt + \sigma \delta_t^u dW_t), \end{aligned}$$

where $(L_t)_{t \in [0, T]}$ is the PL process. (Recall that the risk-free rate is zero.)

Throughout the investment window, the LP's position accumulates fee earnings, and we consider that the depth κ of the pool is constant. Liquidity taking activity in the pool generates fee income that is proportional to the pool size, i.e., proportional to $2\kappa Z_t^{1/2}$; see discussions of Subsection 7.1.2. The proportion of the pool that is generated as fees, which we denote as the pool fee rate, is given by the process $(\Pi_t)_{t \in [0, T]}$; see (7.12). Thus, for each time t , the pool generates a quantity $2\kappa Z_t^{1/2} \Pi_t$ in terms of asset X , which is distributed to LPs proportionally to the liquidity $\tilde{\kappa}_t$ they hold in the pool, so the fee income $(\pi_t)_{t \in [0, T]}$ of the agent evolves as

$$\begin{aligned} d\pi_t &= \frac{\tilde{\kappa}_t}{\kappa} \left(2\kappa \Pi_t Z_t^{1/2} \right) dt - \gamma \left(\frac{1}{\delta_t^\ell + \delta_t^u} \right)^2 \tilde{x}_t^2 dt \\ &= \left(\frac{2}{\delta_t^\ell + \delta_t^u} \right) \Pi_t \tilde{x}_t dt - \gamma \left(\frac{1}{\delta_t^\ell + \delta_t^u} \right)^2 \tilde{x}_t^2 dt, \end{aligned}$$

where $\gamma \geq 0$ is the concentration cost parameter; see Subsection 7.1.2.

The dynamics of the LP's wealth consist of the fees earned and the value of the position in the pool. We introduce the total wealth process $(\tilde{x}_t = \alpha_t + \pi_t)_{t \in [0, T]}$, with $\tilde{x}_0 > 0$ known. Thus, the dynamics of the LP's total wealth are given by

$$\begin{aligned} d\tilde{x}_t &= \tilde{x}_t \left(\frac{1}{\delta_t^\ell + \delta_t^u} \right) \left[-\frac{\sigma^2}{4} dt + \mu \delta_t^u dt + 2\Pi_t dt + \sigma \delta_t^u dW_t \right] \\ &\quad - \gamma \left(\frac{1}{\delta_t^\ell + \delta_t^u} \right)^2 \tilde{x}_t^2 dt. \end{aligned} \tag{7.14}$$

Next, the asymmetry of the LP's position depends on μ and δ , hence we denote it by $\rho_t = \rho(\mu, \delta_t)$, where $\rho \in C^1(\mathbb{R} \times \mathbb{R}, \mathbb{R})$ is a deterministic function; see Subsection 7.1.2. With the position spread

$$\delta_t = \delta_t^u + \delta_t^\ell$$

and the position asymmetry

$$\delta_t^u / \delta_t = \rho(\mu, \delta_t)$$

in (7.14), we write the dynamics of the LP's total wealth as

$$d\tilde{x}_t = \frac{1}{\delta_t} \left(2\Pi_t - \frac{\sigma^2}{4} \right) \tilde{x}_t dt + \mu \rho(\delta_t, \mu) \tilde{x}_t dt + \sigma \rho(\delta_t, \mu) \tilde{x}_t dW_t - \frac{\gamma}{\delta_t^2} \tilde{x}_t dt.$$

In the remainder of this section, we assume that the process $(\Pi_t - \eta)_{t \in [0, T]}$ follows Cox–Ingersoll–Ross-type dynamics, where $\eta = \varepsilon - \frac{1}{2}\mu^2 + \frac{\sigma^2}{2} \left(\mu + \frac{1}{4} \right)$ and we write the dynamics of the fee rate as

$$d\Pi_t = \Gamma \left(\bar{\Pi} + \eta - \Pi_t \right) dt + \psi \sqrt{\Pi_t - \eta} dB_t. \quad (7.15)$$

Here, $\varepsilon > 0$ is an arbitrarily small constant, $\Gamma > 0$ denotes the mean reversion speed, $\bar{\Pi} + \eta > 0$ is the long-term mean, $\psi > 0$ is the volatility parameter, $(B_t)_{t \in [0, T]}$ is a Brownian motion independent of $(W_t)_{t \in [0, T]}$, and $\Pi_0 - \eta > 0$ is known. From (7.15) it follows that

$$2\Pi_t + \mu^2 - \sigma^2 \left(\mu + \frac{1}{4} \right) \geq 2\varepsilon, \quad \forall t \in [0, T], \quad (7.16)$$

which ensures that the spread δ is well defined in the dynamic strategy we derive next. Financially, the expression in (7.16) guarantees that the fee income is larger than the PL faced by the LP; this expression is further discussed below.

7.2.1 Performance criterion and value function

The LP maximises the expected utility of her terminal wealth in units of X by controlling the spread δ of her position, where the set of admissible strategies is

$$\mathcal{A}_t = \left\{ (\delta_s)_{s \in [t, T]}, \mathbb{R}\text{-valued, } \mathbb{F}\text{-adapted, and } \int_t^T |\delta_s|^2 ds < +\infty \mathbb{P}\text{-a.s.} \right\}$$

with $\mathcal{A} := \mathcal{A}_0$.

Let $\delta \in \mathcal{A}$. The performance criterion of the LP is a function $u^\delta: [0, T] \times \mathbb{R} \times \mathbb{R}_{++} \times (\eta, \infty) \rightarrow \mathbb{R}$ given by

$$u^\delta(t, \tilde{x}, Z, \Pi) = \mathbb{E}_{t, \tilde{x}, Z, \Pi} \left[U \left(\tilde{x}_T^\delta \right) \right],$$

where U is a concave utility function, and the value function $u: [0, T] \times \mathbb{R} \times \mathbb{R}_{++} \times (\eta, \infty) \rightarrow \mathbb{R}$ of the LP is⁵

$$u(t, \tilde{x}, Z, \Pi) = \sup_{\delta \in \mathcal{A}} \{ u^\delta(t, \tilde{x}, Z, \Pi) \}. \quad (7.17)$$

⁵ Here, $\mathbb{R}_{++} = (0, \infty)$.

7.2.2 The dynamic programming equation

The value function in (7.17) admits the dynamic programming principle, thus on $[0, T) \times \mathbb{R} \times \mathbb{R}_{++} \times (\eta, \infty)$ it satisfies the Hamilton–Jacobi–Bellman (HJB) equation

$$0 = \partial_t w + \frac{1}{2} \sigma^2 Z^2 \partial_{zz} w + \mu Z \partial_z w + \Gamma (\bar{\Pi} + \eta - \Pi) \partial_{\Pi} w + \frac{1}{2} \psi^2 (\Pi - \eta) \partial_{\Pi\Pi} w \\ + \sup_{\delta \in \mathbb{R}} \left(\frac{1}{\delta} \left(2\Pi - \frac{\sigma^2}{4} \right) \tilde{x} \partial_{\tilde{x}} w + \mu \rho(\delta, \mu) \tilde{x} \partial_{\tilde{x}} w + \sigma^2 \rho(\delta, \mu)^2 \tilde{x}^2 \partial_{\tilde{x}\tilde{x}} w \right. \\ \left. - \frac{\gamma}{\delta^2} \tilde{x} \partial_{\tilde{x}} w + \sigma^2 \rho(\delta, \mu) \tilde{x} Z \partial_{\tilde{x}Z} w \right),$$

with terminal condition

$$w(T, \tilde{x}, Z, \Pi) = U(\tilde{x}), \quad \forall (\tilde{x}, Z, \Pi) \in \mathbb{R} \times \mathbb{R}_{++} \times (\eta, \infty). \quad (7.21)$$

Assume the asymmetry function ρ is as in (7.10), then the optimal spread, in feedback form, is given by

$$\delta_t^* = \frac{2\gamma - 2\sigma^2 \mu^2 \tilde{x} \frac{\partial_{\tilde{x}\tilde{x}} w}{\partial_{\tilde{x}} w}}{2\Pi_t - \frac{\sigma^2}{4} + \mu^2 + \sigma^2 \mu \tilde{x} \frac{\partial_{\tilde{x}\tilde{x}} w}{\partial_{\tilde{x}} w} + \sigma^2 \mu Z \frac{\partial_{\tilde{x}Z} w}{\partial_{\tilde{x}} w}}.$$

When the utility function U is logarithmic, one obtains the optimal spread δ^* in closed-form, which we use to study the performance of the strategy in Section 7.3. The proposition below provides a candidate solution to the LP's problem for logarithmic utility.

Proposition 7.2.1 Assume the asymmetry function ρ is as in (7.10) and that $U(x) = \log(x)$. Then the HJB in (7.18) with terminal condition (7.21) admits the solution

$$w(t, \tilde{x}, \Pi) = \log(\tilde{x}) + \left(\frac{1}{4} \frac{(2\eta - \varepsilon)^2}{\gamma + \sigma^2 \mu^2} + \frac{1}{2} \mu - \frac{\sigma^2}{4} \right) (T - t) + \psi^2 \eta \int_t^T C(u) du + C(t) \Pi^2 \\ + \Pi \int_t^T \exp(\Gamma(t - u)) \left(\left(-2\Gamma(\eta + \bar{\Pi}) - \psi^2 \right) C(u) + \frac{2\eta - \varepsilon}{2(\gamma + \sigma^2 \mu^2)} \right) du, \quad (7.22)$$

where

$$C(t) = \frac{1}{8\Gamma(\gamma + \sigma^2 \mu^2)} (1 - \exp(-2\Gamma(T - t))).$$

Proof 7.2.1 To study the HJB (7.18), use the ansatz

$$w(t, \tilde{x}, Z, \Pi) = \log(\tilde{x}) + \theta(t, Z, \Pi),$$

to obtain the new HJB

$$\begin{aligned} 0 = & \partial_t \theta + \frac{1}{2} \sigma^2 Z^2 \partial_{zz} \theta + \mu Z \partial_z \theta + \Gamma (\bar{\Pi} + \eta - \Pi) \partial_{\Pi} \theta + \frac{1}{2} \psi^2 (\Pi - \eta) \partial_{\Pi\Pi} \theta \\ & + \sup_{\delta \in \mathbb{R}} \left(\frac{1}{\delta} \left(2\Pi - \frac{\sigma^2}{4} \right) + \mu \rho(\delta, \mu) - \sigma^2 \rho(\delta, \mu)^2 - \frac{\gamma}{\delta^2} \right), \end{aligned}$$

with terminal condition

$$\theta(T, Z, \Pi) = 0, \quad \forall (Z, \Pi) \in \mathbb{R}_{++} \times (\eta, \infty). \quad (7.25)$$

First, notice that the supremum in the HJB (7.23) are attained for

$$\delta^* = \frac{2\sigma^2\mu^2 + 2\gamma}{2\Pi - \frac{\sigma^2}{4} + \mu^2 - \sigma^2\mu} = \frac{2\sigma^2\mu^2 + 2\gamma}{2\Pi - 2\eta + \varepsilon}.$$

Thus, (7.23) becomes

$$\begin{aligned} 0 = & \partial_t \theta + \frac{1}{2} \sigma^2 Z^2 \partial_{zz} \theta + \mu Z \partial_z \theta + \Gamma (\bar{\Pi} + \eta - \Pi) \partial_{\Pi} \theta + \frac{1}{2} \psi^2 (\Pi - \eta) \partial_{\Pi\Pi} \theta \\ & + \frac{\mu}{2} - \frac{\sigma^2}{4} + \frac{1}{4} \frac{(2\Pi - 2\eta + \varepsilon)^2}{\gamma + \sigma^2\mu^2}. \end{aligned}$$

We propose the ansatz

$$\theta(t, \Pi, Z) = A(t) Z^2 + B(t) \Pi Z + C(t) \Pi^2 + D(t) Z + E(t) \Pi + F(t). \quad (7.28)$$

Replace (7.28) in (7.23), collect the terms in Z , and obtain the system of ODEs

$$\begin{cases} A'(t) = -\sigma^2 A(t) - 2\mu A(t), \\ B'(t) = -\mu B(t) + \Gamma B(t), \\ C'(t) = 2C(t) \Gamma - \frac{1}{\gamma + \sigma^2\mu^2}, \\ D'(t) = -\mu D(t) - \Gamma (\eta + \bar{\Pi}) B(t), \\ E'(t) = -2\Gamma (\eta + \bar{\Pi}) C(t) - \psi^2 \Pi C(t) + \Gamma E(t) + \frac{1}{2} \frac{2\eta - \varepsilon}{\gamma + \sigma^2\mu^2}, \\ F'(t) = -\frac{1}{4} \frac{(2\eta - \varepsilon)^2}{\gamma + \sigma^2\mu^2} - \frac{\mu}{2} + \frac{\sigma^2}{4} + \psi^2 \eta C(t), \end{cases}$$

with terminal conditions $A(T) = B(T) = C(T) = D(T) = E(T) = F(T) = 0$.

The system of ODEs admits the unique solution

$$\begin{cases} A(t) = 0, \\ B(t) = 0, \\ C(t) = \frac{1 - \exp(-2\Gamma(T-t))}{2\Gamma(\gamma + \sigma^2\mu^2)}, \\ D(t) = 0, \\ E(t) = \int_t^T \exp(\Gamma(t-u)) \left((-2\Gamma(\eta + \bar{\Pi}) - \psi^2) C(u) + \frac{2\eta - \varepsilon}{2(\gamma + \sigma^2\mu^2)} \right) du, \\ F(t) = \left(\frac{1}{4} \frac{(2\eta - \varepsilon)^2}{\gamma + \sigma^2\mu^2} + \frac{\mu}{2} - \frac{\sigma^2}{4} \right) (T-t) + \psi^2 \eta \int_t^T C(u) du, \end{cases}$$

which proves the result. $\square$

Theorem 7.2.1 Assume the asymmetry function ρ is as (7.10) and that $U(x) = \log(x)$. Then, the solution in Proposition 7.2.1 is the unique solution to the optimal control problem (7.17), and the optimal spread is given by

$$\delta_t^* = \frac{2\gamma + 2\sigma^2\mu^2}{2\Pi_t + \mu^2 - \sigma^2\left(\mu + \frac{1}{4}\right)}. \quad (7.29)$$

Proof 7.2.2 Proposition 7.2.1 provides a classical solution to (7.18). Therefore, showing that (7.29) is an admissible control is enough to prove that (7.22) is the value function (7.17); see [Cartea and Jaimungal \(2016b\)](#) for a similar proof. Specifically, from the feedback form of the optimal control δ^* in (7.25) and from the dynamics for Π in (7.15) we obtain that

$$0 < \delta_t^* \leq \frac{2\sigma^2\mu^2 + 2\gamma}{\varepsilon} \quad \forall t \in [0, T],$$

thus δ^* is an admissible control. $\square$

7.2.3 Discussion: optimal strategy

In this section, we study how the strategy depends on various model parameters; e.g., trend μ and volatility σ of the pool's rate, fees paid to the LP, and terminal date T .

Use $\delta_t = \delta_t^\ell + \delta_t^u$ and $\rho(\delta_t, \mu) = \delta_t^u / \delta_t$, to write

$$\delta_t^{u*} = \frac{\gamma + \sigma^2\mu^2}{2\Pi_t + \mu^2 - \sigma^2\left(\mu + \frac{1}{4}\right)} + \mu \quad \text{and} \quad \delta_t^{\ell*} = \frac{\gamma + \sigma^2\mu^2}{2\Pi_t + \mu^2 - \sigma^2\left(\mu + \frac{1}{4}\right)} - \mu. \quad (7.30)$$

Next, when $\mu = 0$, one has $\delta_s^u = \delta_s^\ell$, so the position around the rate of the pool is approximately symmetric and the optimal control is

$$\delta_t^* = \frac{2\gamma}{2\Pi_t - \frac{\sigma^2}{4}}. \quad (7.31)$$

The position range is not symmetric around the position rate Z because $\delta_t^u = \delta_t^\ell$ does not imply that $Z - Z^\ell = Z^u - Z$; see (7.13). However, for small values of δ^ℓ and δ^u , one can write the approximation $Z - Z^\ell \approx Z^u - Z$, in which case the position is symmetric around the rate Z .

The condition in (7.16) guarantees that the control does not explode, and ensures that fee income is large enough for LP activity to be profitable (in the absence of fees). In particular, when $\mu = 0$, it ensures that $\Pi > \sigma^2/8 + \varepsilon$. Note that when $\sigma^2/4 \rightarrow \Pi$, the spread $\delta \rightarrow +\infty$. The spread $\delta = \delta^u + \delta^\ell$ does not exceed 2, and $\delta = 2$ when $\delta^\ell = \delta^u = 1$, so the condition $\delta \leq 2$ becomes

$$2\Pi - \gamma \geq \frac{\sigma^2}{4} \implies 2\Pi - \gamma \geq \text{PL}. \quad (7.32)$$

When $\delta^\ell = \delta^u = 1$, the LP provides liquidity in the maximum range $(0, +\infty)$, so the depth $\tilde{\kappa}$ is minimal, the PL in (6.15) is minimal, and the LP's position is equivalent to providing liquidity in CPMs without CL. The condition in (7.32) appears naturally in our model, and it imposes a minimum profitability level of the pool, so LP activity is viable – recall we assumed that the risk-free rate is zero. In particular, it ensures that the profitability $2\Pi - \gamma$, which is the pool fee rate adjusted by the concentration cost, is higher than the PL. When $\mu = 0$, LPs should use $\sigma^2/4$ as a simplified measure of PL, and check if fees cover this rule-of-thumb quantity before considering depositing their assets in the pool. Below we give another rule-of-thumb when $\mu \neq 0$. Finally, we interpret an optimal control $\delta^* > 2$ as an indicator of non-viable LP activity, because fees are not enough to compensate for the PL borne by the LP. Figure 7.9 shows the estimated pool fee rate and the estimated PL in the ETHUSDC pool (from January to August 2022). In particular, the CIR model captures the dynamics of $\Pi_t - \sigma^2/4$. Figure 7.9 lends strong support for the condition in (7.16) when fee income is reasonable.⁶

When $|\mu| > 0$, the controls in (7.30) must obey the inequalities

$$\begin{cases} 0 \leq \delta_t^\ell \leq 1, \\ 0 \leq \delta_t^u \leq 1, \end{cases}$$

which simplify to the condition

$$2|\mu| \leq \delta_t \leq 2 - 2|\mu|, \quad (7.33)$$

hence $\mu \in [-0.5, 0.5]$. If μ is outside this range, concentration risk and exposure to PL are too high, thus, it is optimal for the LP to withdraw her holdings from the pool. Recall that μ is a daily estimate of the trend in percentage of the rate Z , so values of μ outside the range $[-0.5, 0.5]$ are unlikely. Moreover, when $\mu = -0.5$, the rate Z has

⁶ There is another ETHUSDC pool with a 0.05% fee tier in Uniswap v3. In this extreme case where volatility is high and fees are low, LP activity consistently generates large losses, and the condition in (7.16) only holds 45% of the time between January and August 2022.

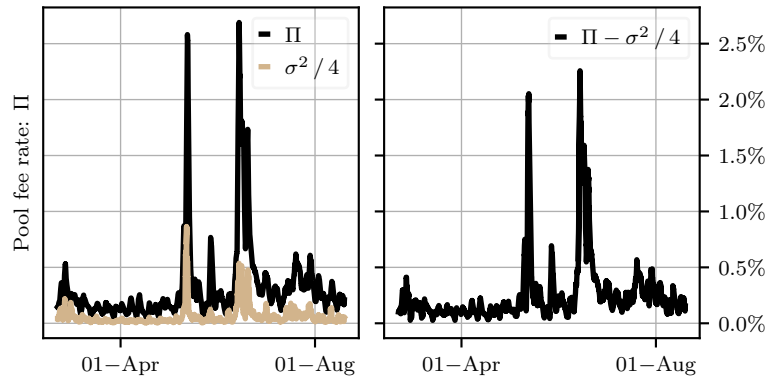


Fig. 7.9: Estimated pool fee rate from February to August 2022 in the ETHUSDC pool. The pool fee rate is estimated for any time t as the total fee income, as a percentage of the total pool size, paid by LTs in the period $[t - 1 \text{ day}, t]$. The pool size at time t is given by $2\kappa Z_t^{1/2}$ where Z_t is the active rate in the pool at time t .

a large negative trend and the optimal range is $(0, Z]$, i.e., the largest possible range to the left of Z . When $\mu = 0.5$, the rate Z has a large positive trend and the optimal range is $(Z, +\infty)$, which is the largest possible rate to the right of Z . Condition (7.33) is always verified when we study the performance of the strategy in the ETHUSDC pool. Finally, the profitability condition in (7.32) becomes

$$2\Pi_t - \gamma \geq \sigma^2 \left(\mu + \frac{1}{4} \right) + \mu^2 (\sigma^2 - 1),$$

so LPs should use this simplified measure of PL as a rule-of-thumb before considering depositing their assets if they hold a belief on the future returns of the rate Z .

Next, we show the dependence of the optimal range on the value of the concentration cost coefficient γ , the trend μ , the fee rate Π , and the volatility σ . The concentration cost coefficient γ scales the spread linearly in (7.30) and (7.31). Recall that the cost term penalises small spreads because there is a risk that the rate will exit the LP's range. Thus, large values of γ generate large values of the spread. Figure 7.10 shows how the optimal spread adjusts to the trend in the rate of the pool. Next, Figure 7.11 shows the optimal spread as a function of the pool fee rate Π . Large potential fee income pushes the strategy towards targeting more closely the marginal rate Z to profit from fees. Finally, Figure 7.12 shows that the optimal spread increases as the volatility of the rate Z increases. Note that

$$\frac{\partial \delta^{u*}}{\partial \sigma} = \frac{\partial \delta^{\ell*}}{\partial \sigma} = \frac{4\sigma \left(\mu^2 (2\Pi + \mu^2) + \gamma\mu + \gamma\frac{1}{4} \right)}{\left(2\Pi + \mu^2 - \sigma^2 \left(\mu + \frac{1}{4} \right) \right)^2} > 0 \quad \forall \mu \in [-0.5, 0.5],$$

shows that the optimal range is strictly increasing in the volatility σ of the rate Z , which one expects as increased activity that exposes the position value to more PL, and increases the concentration risk.

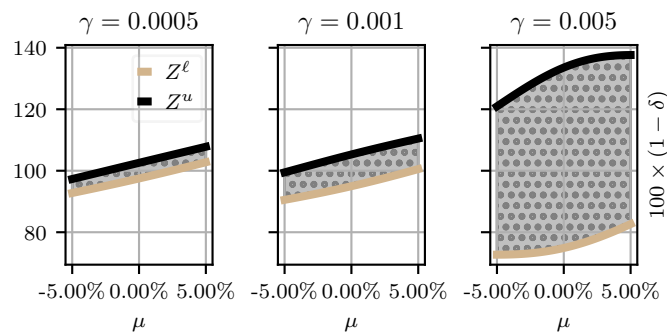


Fig. 7.10: Optimal LP position range $(Z^\ell, Z^u]$ as a function of the trend μ for different values of the concentration cost parameter γ , when $Z = 100$, $\Pi = 0.02$, and $\sigma = 0.02$.

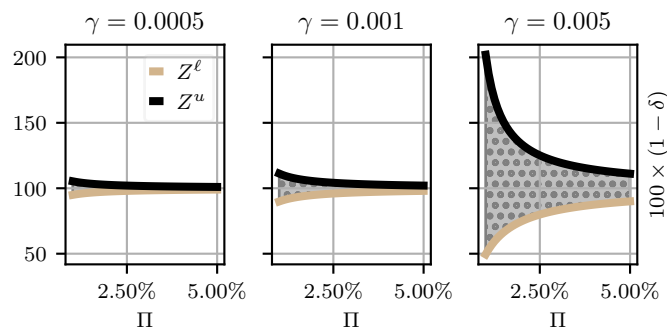


Fig. 7.11: Optimal LP position range $(Z^\ell, Z^u]$ as a function of the pool fee rate Π for different values of the concentration cost parameter γ , when $Z = 100$, $\sigma = 0.02$, and $\mu = 0$.

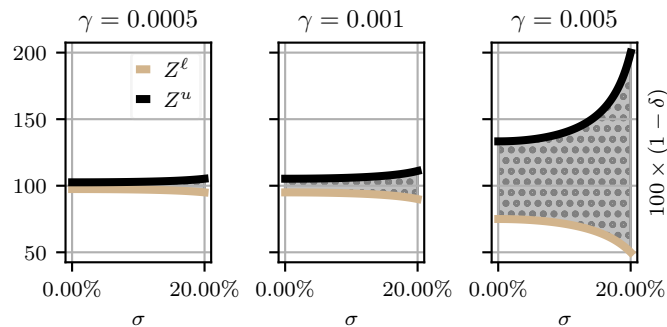


Fig. 7.12: Optimal LP position range $(Z^\ell, Z^u]$ as a function of the volatility σ for different values of the concentration cost parameter γ , when $Z = 100$, $\Pi = 0.02$, and $\mu = 0$.

Finally, the optimal spread does not depend on time or the terminal date T . Note that the LP marks-to-market her wealth in units of X , but does not penalise her holdings in asset Y . In particular, the LP's performance criterion does not include a running penalty or a final liquidation penalty (to turn assets into cash or into the reference asset). For example, if at the end of the trading window the holdings in asset Y must be

exchanged for X , then the optimal strategy would skew, throughout the trading horizon, the liquidity range to convert holdings in Y into X through LT activity.⁷

7.3 Performance analysis

7.3.1 Methodology

In this section, we use Uniswap v3 data between 1 January and 18 August 2022 (see data description in 4.3) to study the performance of the strategy of Section 7.2. We consider execution costs and discuss how gas fees and liquidity taking activity in the pool affect the performance of the strategy.⁸

Our strategy in Section 7.2 is solved in continuous time. In our performance study, we discretise the trading window in one-minute periods when the optimal spread is fixed at the beginning of each time-step. More precisely, let t_i be the times where the LP interacts with the pool, where $i \in \{1, \dots, N\}$ and $t_{i+1} - t_i = 1$ minute. For each time t_i , the LP uses the optimal strategy in (7.30) based on information available at time t_i , and she fixes the optimal spread of her position throughout the period $[t_i, t_{i+1}]$; recall that the optimal spread is not a function of time.

To determine the optimal spread (7.30) of the LP's position at time t_i , we use in-sample data $[t_i - 1 \text{ day}, t_i]$ to estimate the parameters. The volatility σ of the rate Z is given by the standard deviation of one-minute log returns of the rate Z , which is multiplied by $\sqrt{1440}$ to obtain a daily estimate. The pool fee rate Π_t is given by the total fee income generated by the pool during the in-sample period, divided by the pool size $2\kappa Z_t^{1/2}$ observed at time t , where κ is the observed active depth at time t . We use a value $\gamma = 5 \times 10^{-7}$ for the concentration cost term, which penalises small spreads as Figure 7.13 shows. Finally, prediction of the future trend in the rate Z is out of the scope of this chapter, thus we set $\mu = 0$ and $\rho = 0.5$.

To compute the LP's performance as a result of changes in the value of her holdings in the pool (position value), and as a result of fee income, we use out-of-sample data $[t_i, t_{i+1}]$. For the position value, we use equation (7.8) to determine the one-minute out-of-sample period. For fee income, we use LT transactions in the pool at rates included in the range $(Z_t^\ell, Z_t^u]$ and equation (4.12). The income from fees accumulates in a separate account in units of X with zero risk-free rate.⁹ At the end of the out-of-sample window, the LP withdraws her liquidity and earns the accumulated fees, and we repeat the in-sample estimation and out-of-sample liquidity provision described above.

⁷ In LOBs, one usually assumes that final inventory is liquidated with adverse price impact and that there is a running inventory penalty, thus market making strategies in LOBs depend on the terminal date T .

⁸ In practice, LPs pay gas fees when using the Ethereum network to deposit liquidity, withdraw liquidity, and adjust their holdings. Gas is paid in Ether, the native currency of the Ethereum network, and measures the computational effort of the LP operation; see [Cartea et al. \(2022e\)](#).

⁹ In practice, fees accumulate in both assets X and Y .

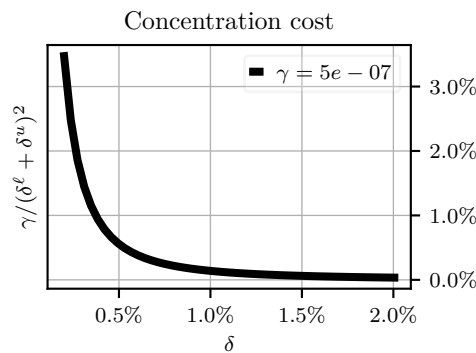


Fig. 7.13: Concentration cost term γ/δ^2 where $\delta = \delta^u + \delta^\ell$ as a function of the position spread when $\gamma = 5 \times 10^{-7}$.

Thus, at times t_i , where $i \in \{1, \dots, N-1\}$, the LP consecutively withdraws and deposits liquidity in different ranges.

Between two consecutive operations (i.e, reposition liquidity provision), the LP may need to take liquidity in the pool to adjust her holdings in asset X and Y . In that case, we use results in [Cartea et al. \(2022e\)](#) to compute execution costs.¹⁰ In particular, we consider execution costs when the LP trades asset Y in the pool to adjust her holdings between two consecutive operations. More precisely, we consider that for every quantity y of asset Y bought or sold in the pool, a transaction cost $y Z_t^{3/2}/\kappa$ is incurred. We assume that the LP's taking activity does not impact the dynamics of the pool. Finally, we obtain 331,858 individual LP operations from 1 January to 18 August 2022.

7.3.2 Benchmark

We compare the performance of our strategy with the performance of LPs in the pool we consider. We select operation pairs that consist of first providing and then withdrawing the same depth of liquidity $\tilde{\kappa}$ at two different points in time by the same LP.¹¹ The operations that we select represent approximately 66% of all LP operations. Figure 7.14 shows the distribution of key variables that describe how LPs provide liquidity. The figure shows the distribution of the number of operations per LP, the changes in the position value, the length of time the position is held in the pool, and the position spread.

Finally, Table 7.2 shows the average and standard deviation of the distributions in Figure 7.14. Notice that the bulk of liquidity is deposited in small ranges, and positions are held for short periods of time; 20% of LP positions are held for less than five minutes

¹⁰ The authors in [Cartea et al. \(2022e\)](#) show that execution costs in the pool are a closed-form function of the rate Z , the pool depth κ , and the transaction size.

¹¹ In blockchain data, every transaction is associated to a unique wallet address.

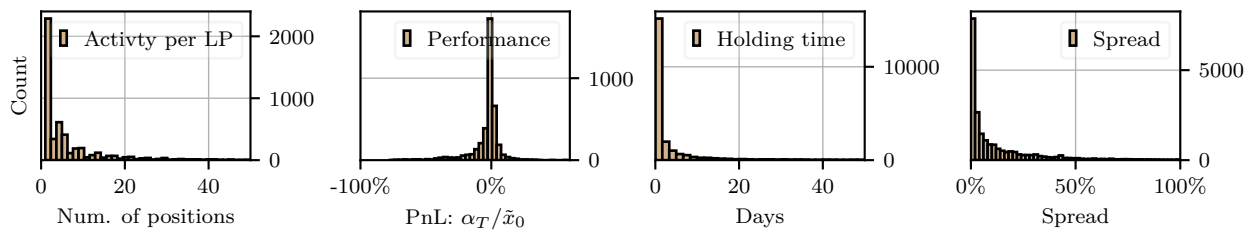


Fig. 7.14: From left to right: distribution of the number of operations per LP, changes in the holdings value as a percentage of initial wealth, position hold time, and position spread. ETHUSDC pool with selected operations from 5,156 LPs between 5 May 2021 and 18 August 2022.

and 30% for less than one hour. Table 7.2 also shows that, on average, the performance of the LP operations in the pool and the period we consider is -1.49% per operation.

	Average Standard deviation	
Number of transactions per LP	11.5	40.2
Position value performance ($\alpha_T/\tilde{x}_0 - 1$)	-1.64%	7.5%
Fee income ($\pi_T/\tilde{x}_0 - 1$)	0.155%	0.274%
Hold time	6.1 days	22.4 days
Spread	18.7%	43.2%

Table 7.2: LP operations statistics in the ETHUSDC pool using operation data of 5,156 different LPs between 5 May 2021 and 18 August 2022. Performance includes transaction fees and excludes gas fees. The position value performance and the fee income are not normalised by the hold time.

7.3.3 Performance results

This subsection focuses on the performance of our strategy when gas fees are zero – at the end of the section we discuss the profitability of the strategy when gas fees are included. Figure 7.15 shows the distribution of the optimal spread (7.31) obtained by the LP. The bulk of liquidity is deposited in ranges with a spread below $\delta = 1\%$. Table 7.2 compares the average performance of the components of the optimal strategy with the performance of LP operations observed in the ETHUSDC pool.¹² Table 7.3 suggests that the position of the LP loses value in the pool (on average) because of PL;

¹² In particular, performance is given for the selected operations shown in Figure 7.14.

however, the fee income would cover the loss, on average, if one assumes that gas fees are zero. Finally, the results show that our optimal strategy significantly improves the performance of LP activity in the pool and the performance of the assets themselves.

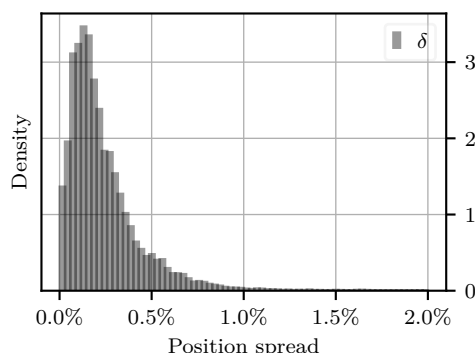


Fig. 7.15: Distribution of the position spread δ .

	Position value performance per operation	Fee income per operation	Total performance per operation (with transaction costs, without gas fees)
Optimal strategy	-0.015% (0.0951%)	0.0197% (0.005%)	0.0047% (0.02%)
Market	-0.0024% (0.02%)	0.0017% (0.005%)	-0.00067% (0.02%)
Hold	n.a.	n.a.	-0.00016% (0.08%)

Table 7.3: **Optimal strategy**: Mean and standard deviation of the one-minute performance of the LP strategy (7.31) and its components. **Market**: Mean and standard deviation of one-minute performance of LP activity in the ETHUSDC pool using data between 1 January and 18 August 2022. **Hold**: Mean and standard deviation of the one-minute performance of holding the assets. In all cases, the performance is net of transaction costs (pool fee and execution cost), but does not include gas fees.

The results of Table 7.3 do not consider gas fees. Gas cost is a flat fee, so it does not depend on the position spread or size of transaction. If the activity of the LP does not affect the pool and if the fees collected scale with the wealth that the LP deposits in the pool, then the LP should consider an initial amount of X and Y that would yield fees enough to cover the flat gas fees. An estimate of the average gas cost gives an estimate of the minimum amount of initial wealth for a self-financing strategy to be profitable. Recall that, at any point in time t , the LP withdraws her liquidity, adjusts her holdings, and then deposits new liquidity. In the data we consider, the average gas

fee is 30.7 USD to provide liquidity, 24.5 USD to withdraw liquidity, and 29.6 USD to take liquidity. Average gas costs are obtained from blockchain data which record the gas used for every transaction, and from historical gas prices. The LP pays a flat fee of 84.8 USD per operation when implementing the strategy in the pool we consider, so the LP strategy is profitable, on average, if the initial wealth deposited in the pool is greater than 1.8×10^6 USD.

Fee income of the LP strategy is limited by the volume of liquidity taking activity in the pool, so one should not only consider increasing the initial wealth to make the strategy more profitable. There are 4.7 LT transactions per minute and the average volume of LT transactions is 477,275 USD per minute, so if the LP were to own 100% of the available liquidity, the average fee income per operation would be 1,431 USD.

Finally, our analysis did not take into account the impact of liquidity provision on liquidity taking activity, however, we expect liquidity provision in CPMMs with CL to be profitable in pools with low volatility asset pairs, with increased liquidity taking activity, and when the gas fee cost to interact with the liquidity pools is low. These conditions ensure that the fees paid to LPs in the pool, adjusted by gas fees and concentration cost, exceed PL so liquidity provision is viable.

Conclusions

This chapter discussed the microstructural properties for liquidity takers and providers in CFMMs. Next, we derived the optimal strategy to provide liquidity in a CPMM with CL when the LP maximises expected utility of terminal wealth. This strategy is found in closed-form for log-utility of wealth, and it shows that liquidity provision is subject to a profitability condition. In particular, the potential gains from fees, net of gas fees and concentration costs, must exceed PL. Our model shows that the LP strategically adjusts the spread of her position around the reference exchange rate. The spread depends on various market features including the trend of the rate, the volatility of the rate, and the liquidity taking activity in the pool. Finally, our analysis of Uniswap v3 data showed that LPs trade at a loss in the pool we consider, and that our strategy significantly improves performance.

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